The Rate–Distortion Theory of Bounded Sequential Transduction: A Comparative Syntax for Finite-State Approximation

Abstract

We develop a comparative framework for finite-state approximation of sequential objects: a rate–distortion theory of bounded sequential transduction in which the rate is a state budget M and the distortion is a regime-specific task loss. Three regimes share one syntax — histories, finite-index right congruences, quotient machines, stationary aggregation, and variational gaps — and differ in their semantic objects. Commitment targets deterministic specifications through deterministic Mealy machines and Myhill–Nerode residuals. Retention targets stochastic controlled processes through stationary controlled unifilar causal machines (controlled ϵ-machines, whose state update may read the realized output) and their lumpable quotients. Grounding targets stochastic channels through linear finite-rank Hankel realizations, whose resource is operator rank rather than quotient index; a closed-form finite-budget characterization of exact symbolic deterministic grounding remains open, but an observable deterministic floor, its finite-state limit, and certified finite-horizon and finite-section intervals are established. Every approximation claim carries an explicit type signature fixing regime, feasible set, aggregation, support, protocol, and analytical bridge, so that statements from different regimes are not conflated.

The principal exact results are as follows. The retention full-KL gap admits an exact finite-state information-bottleneck decomposition over stationary lumpable quotients, with a controlled analogue, conditional on the input, for general unifilar machines, and a zero-retention threshold at the stationary support size of distinct predictive states in the input-driven theory and at the coarsest unifilar-lumpable refinement of the predictive-kernel partition in general. For finite output alphabets a global spectral converse holds in probability coordinates with sharp constant 1; under uniform interiority a corresponding bound holds in a fixed minimal natural-parameter chart, and the interiority hypothesis cannot be removed. The unrestricted linear finite-rank Hankel relaxation gap is $\Delta_{\mathrm{grd}}^{\mathrm{unres}}(M) = \sigma_{M+1}(H_\nu)$ by Eckart–Young–Mirsky, while the Hankel-structured gap is bounded below by $\sigma_{M+1}(H_\nu)$ and equals it under an Adamjan–Arov–Krein Hardy-space embedding — unconditional for one-letter alphabets, conditional on a multiletter Nehari/AAK-type theorem otherwise, which is not established here. A superpolynomial, sub-exponential determinism gap separates formal Hankel rank from deterministic state complexity. Retention is NP-complete in the Gaussian quadratic restricted regime and NP-hard under a promise with APX-hardness for unrestricted full KL. Mistake complexity is $\Theta(M \log M)$ in the reset-word, persistent-stream, and active residual-identification protocols, with agnostic regret $\Theta(\sqrt{TM \log M})$.

The framework separates shared syntax from regime-specific analysis: a typed variational schema; structural monotonicity and zero thresholds; a conditional response-operator converse and Schatten template; adaptive oracle inequalities with a matching minimax lower bound under explicit floors; and a linear second-order Price-of-Safety surrogate with a Ky Fan majorization law. A vertex correspondence organizes retention, commitment, and grounding by Rényi/Schatten labels $\{0, 1, \infty\}$ as a formal consistency rather than a derivation, and no analytic theorem depends on it. An independence theorem shows that the three regimes' obstructions vary independently, so no cross-regime ordering holds. The average-case theory uses stationary Cesàro or discounted prefix aggregation throughout, finite prefixes not defining a probability measure over all lengths. The manuscript identifies a small set of sharply posed open research programs in place of a single undifferentiated open problem of exact symbolic deterministic grounding.

Contents

1 Introduction: The Bounded Sequential Generator and Its Regimes

This manuscript develops a comparative theory of finite-state approximation for sequential objects. The resource is memory, measured by the number of internal states or, equivalently, by the index of a finite right congruence on histories. The cost is a regime-specific distortion functional: Boolean mismatch for commitment, KL divergence for retention, operator-norm distance for linear grounding relaxations, and related quantitative costs.

The framework may be viewed as a rate–distortion theory of bounded sequential transduction. The rate is the finite-state budget M. The distortion is the task loss induced by the regime.

1.1 Bounded sequential generators

A **Bounded Sequential Generator** (BSG) is a deterministic finite-state transducer, i.e. a Mealy machine,

$$\mathcal{A} = (\mathcal{S}, s_0, \delta, \lambda), \qquad |\mathcal{S}| \le M,$$

with finite input alphabet $\mathcal{I}$, finite output alphabet $\mathcal{O}$, transition map

$$\delta : \mathcal{S} \times \mathcal{I} \to \mathcal{S},$$

and output map

$$\lambda : \mathcal{S} \times \mathcal{I} \to \mathcal{O}.$$

It defines a synchronous causal function

$$F_{\mathcal{A}} : \mathcal{I}^{\mathbb{N}} \to \mathcal{O}^{\mathbb{N}}.$$

The deterministic Mealy BSG is the machine model for the commitment regime. In that regime, the target is a deterministic specification, a safety property, or a deterministic strategic rule, and the question is whether a finite-state machine implements or enforces it exactly or approximately. The exact zero-error theory is governed by Myhill–Nerode residual equivalence.

1.2 Three approximation regimes

The manuscript studies three regimes. They share a finite-state syntax but differ in their semantic objects. The framework does not reduce the three regimes to a single machine category.

(1) **Commitment.** The target is a deterministic residual function. A history u determines a residual continuation map

$$F_u(w) = F(uw)_{|u|+1:|u|+|w|}.$$

Two histories are equivalent if they induce the same residual function. The minimal exact state complexity is the Myhill–Nerode index

$$\kappa_{\mathrm{det}}(F) = \mathrm{index}(\sim_F).$$

The exact commitment gap is a threshold phenomenon:

$$\Delta_{\mathrm{com}}(M) = 0 \quad \Longleftrightarrow \quad M \geq \kappa_{\mathrm{det}}(F).$$

(2) **Retention.** The target is a stochastic controlled process whose predictive structure is encoded by causal states. Each positive-mass causal state $s \in \mathcal{S}^+$ carries a predictive distribution P_s over future outputs. An M-state approximation merges causal states into at most M lumped machine states through a stationary lumpable quotient.

The full-KL retention gap[1] is

$$\Delta_{\mathrm{ret}}^{\mathrm{KL}}(M) = \min_{\substack{\phi:\mathcal{S}^+\to\mathcal{K} \text{ lumpable} \\ |\mathcal{K}|\leq M}} \sum_{k\in\mathcal{K}} \sum_{s:\phi(s)=k} \pi_s D_{\mathrm{KL}}(P_s \| \bar{P}_k),$$

where

$$\bar{P}_k = \frac{\sum_{s:\phi(s)=k} \pi_s P_s}{\sum_{s:\phi(s)=k} \pi_s}.$$

In synchronized machines, lumpable quotients of $\mathcal{S}^+$ correspond to finite-index right congruences on sufficiently long histories. In the general unifilar model the causal-state update reads the realized output, the feasible quotients are the unifilar-lumpable ones, and the input-driven synchronized subclass recovers right congruences on input histories.

[1] Here and throughout, "retention gap" is this manuscript's state-compression cost — the stationary Kullback–Leibler price of lumping predictive states into at most M blocks — instantiated as the $\Delta_{\mathrm{ret}}^{\mathrm{KL}}(M)$ of Definition 3.30, its controlled relative $\Delta_{\mathrm{ret}}^{\mathrm{KL,ctrl}}$ of Definition 5.31, and the quadratic surrogate $\Delta_{\mathrm{ret}}^{\mathrm{quad}}$ of Definition 3.31. It is unrelated to the same phrase in the thermodynamics-of-learning literature, where a retention gap is a value-side quantity L_{gen} for finite-state learning devices under task-distribution shift [40].

When the predictive family is restricted to common-covariance Gaussians, one obtains a quadratic restricted gap

$$\Delta_{\mathrm{ret}}^{\mathrm{quad}}(M).$$

The full-KL gap and the quadratic restricted gap are distinct objects.

Positioning. The retention chapter extends the causal-state program of computational mechanics [32] from output processes to controlled, input-driven transductions: the objects approximated are lumpable quotients of a *controlled* unifilar causal machine rather than partitions of a stationary process, and the feasibility notion of unifilar lumpability is correspondingly stricter than ordinary lumpability. Unlike the information-bottleneck-for-causal-states line [33, 34], which characterizes the optimal causal statistic, and unlike the Kullback–Leibler aggregation line for regular Markov chains [35, 36], which computes the optimal reduction, the manuscript tracks the full budget-M gap curve: the zero threshold at the stable kernel refinement, NP-hardness below it, and spectral lower certificates from the predictive covariance, rather than only the optimum.

(3) **Grounding.** The target is a stochastic channel or residual kernel. The spectral converse uses linear finite-rank realizations rather than deterministic Mealy machines. The relevant object is a Hankel operator H_ν built from an impulse-response function.

The unrestricted linear finite-rank Hankel relaxation gap is

$$\Delta_{\mathrm{grd}}^{\mathrm{unres}}(M) = \inf_{\operatorname{rank} B \le M} \|H_\nu - B\|_{\mathrm{op}},$$

and satisfies

$$\Delta_{\mathrm{grd}}^{\mathrm{unres}}(M) = \sigma_{M+1}(H_\nu)$$

by Eckart–Young–Mirsky [15, 16].

The Hankel-structured finite-rank realization gap is

$$\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) = \inf_{\substack{\operatorname{rank} B \le M \\ B \ \text{Hankel}}} \|H_\nu - B\|_{\mathrm{op}},$$

and satisfies

$$\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) \ge \sigma_{M+1}(H_\nu).$$

Equality in the Hankel-structured problem requires additional structure (Theorems 7.9 and 7.11). The resource M here is a finite-rank budget on a bounded operator, not the index of a right congruence; Section 7.1 and the master scope of Section 4.1 record this distinction explicitly, and Definition 7.27 below supplies a quotient-typed *symbolic* grounding gap[2], over deterministic Mealy machines, whose resource is again a

[2]As used here, a "grounding gap" is a quotient-typed approximation cost: the discounted symbolic quantity $\Delta_{\mathrm{grd}}(M;\gamma)$ of Definition 7.27 together with its linear finite-rank relatives $\Delta_{\mathrm{grd}}^{\mathrm{unres}}(M)$ and $\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M)$ of Section 7. The term has no connection to its natural-language-processing sense, where a grounding gap is a language model's failure to establish conversational common ground or to connect its generated outputs to the external world [39].

right-congruence index and which is therefore the object genuinely comparable to commitment and retention. A closed-form finite-budget value for that symbolic gap remains open; certified bounds are given in Propositions 7.7, 7.31 and Theorem 7.29.

Positioning. The grounding chapter transports the Adamjan–Arov–Krein program for weighted automata [13, 14] from weighted languages to transductions. The object approximated is the Hankel operator of a channel on the joint alphabet rather than that of a weighted language; the unrestricted and Hankel-restricted feasible sets are separated explicitly, the Eckart–Young–Mirsky value being available for the former and only the singular-value lower bound for the latter; and the published one-letter results extend to the multiletter setting only conditionally (Theorem 7.11, Open Problem 7.13), with grounded finite-section certificates and the zero-threshold characterization $\mathrm{rank}(H) \leq M$ standing in for the exact minimization values of the one-letter theory.

The three regimes share a **typed finite-memory budget** rather than a single literal resource. For commitment and retention (and for the symbolic grounding gap of Definition 7.27), the budget M is the index of a finite right congruence on histories. For the *unrestricted linear* grounding relaxation, the budget M is instead the rank of a finite-rank operator approximating a Hankel operator. These two notions of budget are related — both cap the "resolution" of the approximant — but they are not asserted equivalent: a rank-M operator is not in general reachable through an M-state quotient, and an M-state quotient does not in general act on the joint alphabet the way a rank-M Hankel truncation does. Section 4.1 makes this typed picture precise as four pillars common to all three regimes.

1.3 The shared finite-state schema

The shared combinatorial resource is the index of a finite quotient of a regime-typed history system (Definition 2.6, developed in Section 2.5). Commitment quotients live on input histories $\mathcal{I}^*$ with a total right action; unifilar retention quotients live on feasible joint input–output histories with a pruned action, in the support-relative sense of Definition 2.7; grounding quotients live on the joint alphabet $\Sigma = \mathcal{I} \times \mathcal{O}$. For the commitment regime, and for the input-driven retention subclass, the classical form is the following; Remark 2.9 tabulates the three instantiations.

Let $\mathcal{I}^*$ denote the set of finite input histories. A right congruence $\sim$ on $\mathcal{I}^*$, in the standard sense of automata theory [2], is an equivalence relation satisfying

$$u \sim v \implies ux \sim vx \qquad \forall x \in \mathcal{I}.$$

Equivalently, the quotient $\mathcal{I}^*/\sim$ carries a well-defined deterministic transition

$$[u] \cdot x = [ux].$$

The index of $\sim$ is the number of equivalence classes. This index is the finite-state budget.

A target sequential object may be represented as a trajectory

$$\delta : \mathcal{I}^* \to \mathrm{ob}(\mathbf{R}),$$

where $\mathbf{R}$ is a regime-specific residual category. An M-state approximation factors δ through a finite quotient:

$$\mathcal{I}^* \longrightarrow \mathcal{I}^*/\!\sim \longrightarrow \mathbf{R}, \qquad \operatorname{index}(\sim) \le M.$$

The associated variational gap is

$$\Delta_{\mathbb{T}}(M) = \inf_{\substack{\sim,\Psi \\ \operatorname{index}(\sim)\le M}} \mathcal{L}(\delta, \Psi \circ Q).$$

Remark 1.1 (Scope of the Quotient Display)**.** The display above is the quotient-typed instance of the schema, and it is the literal resource for commitment, for retention, and for the symbolic grounding gap of Definition 7.27. The *unrestricted linear* grounding relaxation of Theorem 7.3 instead instantiates the schema with a finite-rank feasible set,

$$\Delta_{\mathbb{T}}(M) = \inf_{\operatorname{rank}(B)\le M} \mathcal{L}(\delta, B),$$

a genuinely different budget type. Section 4.1 records both instances as coordinates of one typed variational schema without asserting that the two budgets coincide.

Each regime attaches its own per-state semantic object:

- **commitment:** a residual function;
- **retention:** a predictive distribution or exponential-family parameter;
- **grounding:** an operator, kernel, or linear realization coordinate.

The shared theorems concern the quotient structure and the index of right congruences. The analytical engines—Myhill–Nerode theory, information bottleneck, Hankel approximation, game-theoretic commitment, online learning, and operator algebra—are regime-specific.

1.4 Stationary aggregation and type-correct costs

The average-case theory uses stationary aggregation. For each fixed length n, a stationary process assigns total mass one to the prefixes of length n:

$$\sum_{|u|=n} \mu(u) = 1.$$

Consequently,

$$\sum_{u\in\mathcal{I}^*} \mu(u) = \infty,$$

so the prefix weights do not define a probability measure on the whole free monoid $\mathcal{I}^*$. The theory uses stationary aggregation schemes.

Two canonical average-case aggregations are admitted:

(a) a stationary Cesàro average;
(b) a discounted prefix measure.

The worst-case aggregation remains the supremum over all finite histories:

$$\mathcal{L}_\infty(\delta, \widehat{\delta}) = \sup_{u \in \mathcal{I}^*} \mathcal{E}(\delta(u), \widehat{\delta}(u)).$$

The cost profunctor $\mathcal{E}$ is regime-specific. Commitment uses a Boolean or 0/1 equality cost. Retention uses KL divergence or a quadratic Fisher surrogate. Grounding uses operator-norm distance or a related linear relaxation cost. The structural Boolean cost $\{0, \infty\}$ is distinguished from the normalized operational cost $\{0, 1\}$.

1.5 Analytical boundaries

The framework separates shared syntax from regime-specific analysis.

- **Shared syntax versus regime-specific semantics.** The schema provides histories, right congruences, finite quotients, and variational gaps. It does not derive the Myhill–Nerode theorem, the Eckart–Young–Mirsky theorem, or the information-bottleneck identity from categorical axioms alone.
- **Exact theorems versus local theorems.** The commitment zero-error threshold, the full-KL finite-state information-bottleneck identity, and the unrestricted linear Hankel spectral converse are exact within their domains. The full-KL spectral lower bound is local and requires explicit regularity assumptions. The Hankel-structured finite-rank realization gap satisfies the unrestricted singular-value lower bound, but equality requires additional structure.
- **Converses versus achievability.** Spectral converses are lower bounds on approximation gaps. They become online regret upper rates only when paired with explicit approximation achievability bounds.
- **Linear relaxations versus discrete problems.** The linear Hankel gap and the linear Price-of-Safety surrogate are exact linear objects. They do not automatically solve the corresponding discrete right-congruence optimization problems.
- **Heuristics versus theorems.** The spectral-tail picture is a heuristic. The tail shape depends on aggregation, operator type, affine versus linear approximation, rank versus state count, compactness, domination, and quadratic-regime assumptions.
- **No unconditional cross-regime ordering.** The framework does not assert a universal inequality among grounding, retention, and commitment gaps. The Independence Theorem shows that the gaps can be varied independently.

1.6 Components of the framework

The framework comprises the following components.

Schatten conditional converse template. A task theory with a rank-controlled response operator A_δ and a Schatten-p domination inequality satisfies a unified converse:

$$\Delta_{\mathbb{T}}(M) \geq \frac{1}{c_p} \Phi^{(p)}_{r_{\mathbb{T}}(M)}(A_\delta),$$

where $r_{\mathbb{T}}(M)$ is the effective rank budget and $\Phi^{(p)}_r$ is the Schatten-p tail.

Grounding is the Schatten-∞ instance for the unrestricted linear finite-rank relaxation; retention is the Schatten-1 instance with effective rank $M-1$; commitment is the Boolean-rank vertex. Contextwise ℓ^p aggregation does not by itself imply Schatten-p domination.

Adaptive two-axis oracle inequality. Within each type-correct regime, offline approximation and online estimation compose into a bias–variance decomposition:

$$\mathrm{Reg}_T(\mathcal{A}_M) \leq A^{\mathsf{r}}_T(M) + \mathsf{E}^{\mathsf{r}}_M(T).$$

Adaptive aggregation over the state budget M yields an oracle inequality without knowledge of the optimal budget. For bounded agnostic losses, the adaptive penalty is

$$\Psi_M(T) = O\Big(\sqrt{T \ln(eM)} + \ln(eM)\Big).$$

Realizable 0/1 and mixable log-loss model selection have logarithmic overhead. Spectral tails enter online rates only under explicit spectral-rate assumptions.

Estimation rates may be supplied by a Littlestone bound, a multiclass Littlestone bound, a mixability bound, or a linear-regret bound. The oracle inequality does not assume a single estimation rate across regimes.

Linear Price-of-Safety surrogate. For a quadratic/Fisher retention embedding with state-indexed centered Gram matrix

$$G_{st} = \sqrt{\pi_s \pi_t}\, \langle y_s - \bar{y},\; y_t - \bar{y} \rangle,$$

and a safety partition inducing a pinching conditional expectation $\mathrm{Pinch}_{\mathcal{A}}$, the linear block-local Price-of-Safety surrogate is

$$\mathrm{PoS}^{\mathrm{loc}}_{\mathrm{lin}}(M) = \frac{1}{2} \|G\|_{(M-1)} - \frac{1}{2} \|\mathrm{Pinch}_{\mathcal{A}}\, G\|_{(M-1)}.$$

Pinching majorization gives nonnegativity:

$$\mathrm{PoS}^{\mathrm{loc}}_{\mathrm{lin}}(M) \geq 0.$$

The coupling bound

$$\mathrm{PoS}^{\mathrm{loc}}_{\mathrm{lin}}(M) \leq \frac{1}{2} \|G - \mathrm{Pinch}_{\mathcal{A}}\, G\|_{(M-1)}$$

is an upper bound, not an equality. This surrogate is exact for the linear block-local problem but is not the discrete right-congruence Price of Safety[3] without relaxation-error control.

[3]Here and throughout, "Price of Safety" is this manuscript's safety-constraint gap — the free optimum minus the safety-constrained optimum — instantiated as the discrete right-congruence quantity $\mathrm{PoS}_{\mathrm{quad}}$ and

Exponent vertex correspondence. The manuscript separates three exponent-like layers:

(1) Rényi order α;
(2) Schatten exponent p;
(3) aggregation or vertex label.

Retention is the $\alpha = 1$ KL vertex. Commitment is recovered from the $\alpha = 0$ support vertex after symmetrization and Boolean valuation. Grounding is organized by the $\alpha = \infty$ supremum vertex, but the exact unrestricted linear finite-rank grounding cost is operator-norm distance, not max-divergence.

The equality of vertex labels

$$\alpha = p \in \{0, 1, \infty\}$$

is a formal consistency, not a derivation of the Mirsky converses from Rényi identities.

Temporal protocols. On the deterministic Mealy classes $\mathcal{H}_M$, the reset-word, persistent-stream, and active residual-identification protocols each have mistake complexity $\Theta(M \log M)$; the persistent-stream and active lower bounds are explicit forcing constructions rather than Littlestone transfers, and agnostic regret is $\Theta(\sqrt{TM \log M})$.

1.7 Roadmap

Section 2 develops the shared finite-state schema. Section 3 defines specialized task theories. Section 4 states the structural meta-theorems. Section 5 develops retention. Section 6 develops commitment. Section 7 develops grounding. Section 8 develops the conditional Schatten-converse template. Section 9 develops the adaptive two-axis oracle inequality and the associated budget laws. Section 10 develops the linear Price-of-Safety surrogate and the pinching law. Section 11 develops the exponent vertex correspondence. Section 12 develops joint, comparative, interaction, and independence results. Section 13 develops the temporal axis. Section 14 states the spectral-tail heuristic. Section 15 states the Conditional Representation Theorem. Section 16 fixes the type signature that every approximation statement carries. Section 17 records epistemic boundaries. Section 18 concludes.

2 The Schema of Finite-State Approximation

This section develops the shared finite-state schema. The schema supplies a common syntax for all regimes: histories, enriched cost categories, finite right congruences, quotient machines, stationary aggregation, and profinite completion. The analytical meaning of the residual objects is supplied later by each regime.

its linear surrogate $\mathrm{PoS}^{\mathrm{loc}}_{\mathrm{lin}}$ of Section 10, and in mutual-information form as the $\mathrm{PoS}(M)$ of Corollary 12.12. It is unrelated to the same phrase in the safe-bandits literature, where it denotes the additional sample complexity incurred under stage-wise safety constraints in best-arm identification [38].

2.1 The Enriching Poset

All costs take values in the monoidal poset

$$\mathbf{V} = ([0, \infty], \geq, +, 0).$$

The unit is 0, and the order is reversed from the usual real order: a morphism $v \to v'$ exists iff $v \geq v'$ in the usual order. Thus 0 is terminal and ∞ is initial. A morphism $v \to v'$ records that the cost has decreased from v to v'.

The Boolean sub-poset is

$$\mathbf{V}_{\mathbb{B}} = \{0, \infty\} \subset [0, \infty].$$

It corresponds to exact equality versus definite mismatch. A normalized Boolean mismatch cost with values $\{0, 1\}$ is treated separately as an operational corollary. The structural zero-error theory uses $\{0, \infty\}$; the normalized decision-theoretic version uses $\{0, 1\}$.

The reversed-order convention is the standard Lawvere metric convention [37].

2.2 V-Categories and V-Profunctors

A **V-category R** consists of a set of objects $\mathrm{ob}(\mathbf{R})$ and hom-objects

$$\mathbf{R}(X, Y) \in \mathbf{V}$$

satisfying:

(i) **Identity.**

$$0 \geq \mathbf{R}(X, X),$$

hence $\mathbf{R}(X, X) = 0$.

(ii) **Composition.**

$$\mathbf{R}(X, Y) + \mathbf{R}(Y, Z) \geq \mathbf{R}(X, Z).$$

A **V-profunctor**

$$\mathcal{E} : \mathbf{R}^{\mathrm{op}} \times \mathbf{R} \to \mathbf{V}$$

assigns costs $\mathcal{E}(X, Y)$ with left and right actions satisfying

$$\mathbf{R}(X', X) + \mathcal{E}(X, Y) \geq \mathcal{E}(X', Y),$$

and

$$\mathcal{E}(X, Y) + \mathbf{R}(Y, Y') \geq \mathcal{E}(X, Y').$$

The profunctor $\mathcal{E}$ is the cost bimodule. It measures the mismatch between residual objects X and Y, while the left and right actions express compatibility with the internal geometry of $\mathbf{R}$.

Remark 2.1 (Discrete versus enriched residuals)**.** For retention and commitment, $\mathbf{R}$ is usually a discrete $\mathbf{V}$-category:

$$\mathbf{R}(X,Y) = \begin{cases} 0, & X = Y, \\ \infty, & X \neq Y. \end{cases}$$

Then the action inequalities impose no triangle inequality on $\mathcal{E}$. KL divergence is not a metric, and Boolean mismatch is not a metric.

For grounding, $\mathbf{R}$ may be a metric $\mathbf{V}$-category of operators, with

$$\mathbf{R}(X,Y) = \|X - Y\|_{\mathrm{op}}.$$

If $\mathcal{E}$ is the identity profunctor,

$$\mathcal{E}(X,Y) = \mathbf{R}(X,Y),$$

then the action inequalities reduce to the triangle inequality for the operator norm.

2.3 The History V-Category

Let $\mathcal{I}$ be a finite input alphabet. The **history V-category H** is the discrete $\mathbf{V}$-category on the free monoid $\mathcal{I}^*$:

- objects are finite words $u \in \mathcal{I}^*$;
- hom-objects are

$$\mathbf{H}(u,v) = \begin{cases} 0, & u = v, \\ \infty, & u \neq v. \end{cases}$$

Because $\mathbf{H}$ is discrete, any object-level map

$$\delta : \mathcal{I}^* \to \mathrm{ob}(\mathbf{R})$$

is automatically a $\mathbf{V}$-functor. No nontrivial morphism inequalities need to be checked.

The prefix structure of histories is not encoded in the hom-objects of $\mathbf{H}$. It is encoded externally in the right-congruence condition, which ensures that deterministic finite-state approximations evolve consistently under input extensions.

2.4 Task Theory with Stationary Aggregation

Definition 2.2 (Task Theory)**.** A **task theory** is a tuple

$$\mathbb{T} = (\mathbf{H}, \mathbf{R}, \mathcal{E}, \delta, \mathrm{Agg}, \mathcal{L})$$

where:

(1) $\mathbf{H}$ is the history $\mathbf{V}$-category.
(2) $\mathbf{R}$ is a $\mathbf{V}$-category of residual objects.

(3) $\mathcal{E} : \mathbf{R}^{\mathrm{op}} \times \mathbf{R} \to \mathbf{V}$ is a cost profunctor which is **reflexive**: $\mathcal{E}(X, X) = 0$ for every $X \in \mathbf{R}$. Reflexivity is not automatic for a $\mathbf{V}$-profunctor. In the Lawvere quantale $([0, \infty], \geq, +)$ the constant profunctor $\mathcal{E} \equiv 1$ on a one-object $\mathbf{R}$ satisfies every profunctor axiom, and for it the trajectory cannot be realized at any budget even though its Myhill–Nerode index is 1. Every cost used in this manuscript — KL divergence, operator norm, and the 0/1 commitment loss — is reflexive, so the axiom constrains nothing in the instances but is required for the generic statements of Meta-Theorem 4.1.
(4) $\delta : \mathbf{H} \to \mathbf{R}$ is the true trajectory.
(5) Agg is an aggregation scheme, one of:
   - a stationary Cesàro average;
   - a discounted prefix measure; or
   - the worst-case supremum $\mathcal{L}_\infty(\delta, \widehat{\delta}) = \sup_{u \in \mathcal{I}^*} \mathcal{E}(\delta(u), \widehat{\delta}(u))$.

   The first two are *average-case* schemes and require the stationary or discounted datum; the third requires none. A task theory carrying the worst-case scheme is called *uniform*. Results are stated against a declared scheme, since the Boolean dichotomy in particular takes different forms under worst-case and average-case aggregation (Meta-Theorem 4.7).
(6) $\mathcal{L}$ is the resulting cost functional.

Definition 2.3 (Stationary Cesàro aggregation)**.** Let U_t denote the stationary past at time t. For a cost process

$$c_t = \mathcal{E}(\delta(U_t), \widehat{\delta}(U_t)),$$

define

$$\overline{\mathcal{L}}_1(\delta, \widehat{\delta}) = \lim_{T \to \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[c_t],$$

whenever the limit exists.

For finite-state stationary ergodic processes and finite-state approximations, the limit exists under standard ergodicity assumptions. In non-ergodic settings, one may replace the limit by a limsup or by ergodic decomposition.

Definition 2.4 (Discounted prefix aggregation)**.** For $0 < \gamma < 1$, define a probability measure on $\mathcal{I}^*$ by

$$\mu_\gamma(u) = (1 - \gamma)\gamma^{|u|}\mathbb{P}(X_{1:|u|} = u).$$

Then

$$\mathcal{L}_1^\gamma(\delta, \widehat{\delta}) = \mathbb{E}_{u \sim \mu_\gamma} \mathcal{E}(\delta(u), \widehat{\delta}(u)).$$

The discounted prefix measure is a probability measure because

$$\sum_{u \in \mathcal{I}^*} \mu_\gamma(u) = (1 - \gamma) \sum_{n=0}^{\infty} \gamma^n \sum_{|u|=n} \mathbb{P}(X_{1:n} = u) = (1 - \gamma) \sum_{n=0}^{\infty} \gamma^n = 1.$$

The worst-case aggregation remains

$$\mathcal{L}_\infty(\delta, \widehat{\delta}) = \sup_{u \in \mathcal{I}^*} \mathcal{E}(\delta(u), \widehat{\delta}(u)).$$

It is not an average-case aggregation and should not be conflated with the stationary Cesàro or discounted prefix schemes.

Remark 2.5 (Finite Prefixes versus Stationary States)**.** Finite prefixes do not define a probability measure on all of $\mathcal{I}^*$. For stationary retention, the canonical object is the stationary state process (S_t) and its stationary support $\mathcal{S}^+$. Finite histories enter directly in synchronized machines, where S_t is a function of a finite suffix of the input past.

2.5 Regime-Typed History Actions

The finite-state resource shared by the three regimes is the index of a finite quotient of a history system. Which history system is regime-dependent: the regimes share the *form* of a finite-index right-action quotient, not a single literal free monoid. Forcing every regime onto $\mathcal{I}^*$ would misdescribe predictive machines whose state update reads the realized output.

Definition 2.6 (History System)**.** A **history system** is a tuple

$$\mathsf{H} = (H, h_\varnothing, \mathcal{A}, \cdot),$$

where H is a set of finite histories with root $h_\varnothing$, $\mathcal{A}$ is an event alphabet, and $\cdot$ is a partial right action

$$H \times \mathcal{A} \rightharpoonup H, \qquad (h, a) \mapsto h \cdot a,$$

defined exactly when a is *feasible* from h; write $\mathrm{Feas}(h) \subseteq \mathcal{A}$ for the set of feasible events. The action is *pruned*: whenever $h \cdot a$ is defined it lies in H. If $\mathrm{Feas}(h) = \mathcal{A}$ for every h, the action is *total*.

Definition 2.7 (Support-Relative Right Congruence)**.** Let $\mathsf{H} = (H, h_\varnothing, \mathcal{A}, \cdot)$ be a history system. A **support-relative right congruence** on H is an equivalence relation $\sim$ on H such that $u \sim v$ implies

(i) $\mathrm{Feas}(u) = \mathrm{Feas}(v)$; and
(ii) $u \cdot a \sim v \cdot a$ for every $a \in \mathrm{Feas}(u)$.

Its **index** is the number of equivalence classes. The quotient $H/\sim$ carries a well-defined partial transition $[u] \cdot a = [u \cdot a]$, with $\mathrm{Feas}([u]) = \mathrm{Feas}(u)$; clause (i) is exactly what makes the feasible-event set of a class well defined. For a total action clause (i) is vacuous and the definition reduces to Definition 2.11.

Remark 2.8 (Clause (i) Is Not Redundant)**.** Clause (ii) alone, read as a condition on commonly feasible events, does not give a quotient history system. Take $\mathcal{A} = \{a, b\}$ and two histories u, v with $\mathrm{Feas}(u) = \{a\}$ and $\mathrm{Feas}(v) = \{b\}$. The relation identifying u with v satisfies clause (ii) vacuously, there being no commonly feasible event, yet the class $[u]$ has no well-defined feasible set and the quotient carries no transition structure. This is the mechanism behind Remark 3.22 below.

Remark 2.9 (The Regimes as History Systems)**.** The principal instantiations are the following.

regime	H	$\mathcal{A}$	action
commitment	$\mathcal{I}^*$	$\mathcal{I}$	total
retention, unifilar	feasible joint histories	$\mathcal{I} \times \mathcal{O}$	pruned
retention, input-driven	$\mathcal{I}^*$	$\mathcal{I}$	total
grounding	Σ^*, $\Sigma = \mathcal{I} \times \mathcal{O}$	Σ	total

For commitment and grounding the action is total and Definition 2.11 applies verbatim. For unifilar retention the feasible joint histories form a pruned tree under the stationary law, and compatibility with the action is understood in the support-relative sense. The input-driven model of Definition 3.14 is the case in which the state process factors through the input projection, and the input-history syntax is recovered. The history **V**-category of Section 2.3 is built on whichever H the regime supplies; being discrete, it makes every object-level trajectory a **V**-functor regardless of which system is used.

Lemma 2.10 (Cofilteredness for Support-Relative Congruences)**.** *Let $\sim_1, \sim_2$ be finite-index support-relative right congruences on a history system* H*. Then $\sim_1 \cap \sim_2$ is one as well, refines both, and has index at most* $\operatorname{index}(\sim_1)\operatorname{index}(\sim_2)$*. The finite quotients of* H *therefore form a cofiltered system.*

Proof. The intersection of equivalence relations is an equivalence relation. If $u\,(\sim_1 \cap \sim_2)\,v$ then $\mathrm{Feas}(u) = \mathrm{Feas}(v)$ already from $\sim_1$, giving clause (i); and for a in that common feasible set, $u \cdot a \sim_i v \cdot a$ for $i = 1, 2$, giving clause (ii). The map $[u]_{\sim_1 \cap \sim_2} \mapsto ([u]_{\sim_1}, [u]_{\sim_2})$ is well defined and injective, which is the index bound. The argument is that of Lemma 2.16, with the feasible-set clause carried along. □

2.6 Right Congruences and Quotients

Definition 2.11 (Right Congruence)**.** A **right congruence** $\sim$ on **H** is an equivalence relation on $\mathcal{I}^*$ such that

$$u \sim v \implies uw \sim vw \quad \forall w \in \mathcal{I}^*.$$

Equivalently, it suffices to require

$$u \sim v \implies ux \sim vx \quad \forall x \in \mathcal{I}.$$

The **index** of $\sim$ is the number of equivalence classes.

This is the total-action instance of Definition 2.7; pruned retention history systems use the support-relative form.

The quotient $\mathbf{Q} = \mathbf{H}/\sim$ has objects $[u]$ and hom-objects

$$\mathbf{Q}([u], [v]) = \begin{cases} 0, & [u] = [v], \\ \infty, & \text{otherwise.} \end{cases}$$

The quotient map

$$Q : \mathbf{H} \to \mathbf{Q}, \qquad u \mapsto [u],$$

is a **V**-functor. The right-congruence condition ensures that the quotient carries a well-defined deterministic transition

$$[u] \cdot x = [ux].$$

Definition 2.12 (M-State Approximation)**.** An M**-state approximation** of a task theory $\mathbb{T}$ is a pair $(\sim, \Psi)$ where $\sim$ is a right congruence of index at most M and

$$\Psi : \mathbf{Q} \to \mathbf{R}$$

is a **V**-functor. The approximate trajectory is

$$\Psi \circ Q : \mathbf{H} \to \mathbf{R}.$$

Definition 2.13 (M-State Gap)**.** The M**-state gap** is

$$\Delta_{\mathbb{T}}(M) = \inf_{\substack{\sim, \Psi \\ \mathrm{index}(\sim) \leq M}} \mathcal{L}(\delta, \Psi \circ Q).$$

Remark 2.14 (Machine type depends on **R**)**.** The composite $\Psi \circ Q$ has deterministic state-transition structure, but its per-state content depends on **R**:

- **Commitment:** each state stores a residual function; the approximant is a deterministic Mealy BSG.
- **Retention:** each state stores a predictive distribution or kernel; the approximant is a unifilar lumped controlled predictor (Definition 3.21), and in the input-driven synchronized subclass a deterministic-transition, stochastic-emission machine, in the sense of Definition 3.15 and Remark 3.16.
- **Grounding spectral converse:** each state stores an operator or linear coordinate; the approximant is a linear finite-rank realization.

The shared theorems concern index$(\sim)$. The regime-specific theorems concern the semantic objects attached to the states.

Remark 2.15 (Stationary Retention Quotients)**.** For stationary controlled unifilar machines, the primary feasible objects are unifilar-lumpable quotients

$$\phi : \mathcal{S}^+ \to \mathcal{K}$$

(Definition 3.21). In a machine with joint-history synchronization depth L, each such quotient with block-uniform support induces a support-relative right congruence on feasible joint histories of length at least L by

$$u \sim_\phi v \iff \phi(\sigma(u)) = \phi(\sigma(v))$$

(Proposition 3.26); in the input-driven synchronized subclass this projects to the input-history right congruence of Proposition 3.23. Thus the finite-state budget remains an index resource, now on regime-typed histories. For non-synchronized stationary machines, finite-index right congruences on finite histories should be replaced by finite-state factors of the stationary state process.

2.7 Profinite Completion

The finite quotients of $\mathbf{H}$ form a cofiltered system; the support-relative analogue, for pruned history systems, is Lemma 2.10. Let $\mathcal{C}$ be the cofiltered poset of finite-index right congruences on $\mathcal{I}^*$, ordered by refinement:

$$\sim_1 \leq \sim_2 \quad \iff \quad \sim_1 \text{ refines } \sim_2 .$$

Lemma 2.16 ($\mathcal{C}$ Is Cofiltered)**.** *If $\sim_1$ and $\sim_2$ are finite-index right congruences on $\mathcal{I}^*$, then so is $\sim_1 \cap \sim_2$, and it refines both.*

Proof. The intersection is an equivalence relation. It is a right congruence: if $u \sim_1 v$ and $u \sim_2 v$, then for every $w \in \mathcal{I}^*$ we have $uw \sim_1 vw$ and $uw \sim_2 vw$, hence $uw\,(\sim_1 \cap \sim_2)\,vw$. The map

$$[u]_{\sim_1 \cap \sim_2} \mapsto \big([u]_{\sim_1},\, [u]_{\sim_2}\big)$$

is well defined and injective, so

$$\operatorname{index}(\sim_1 \cap \sim_2) \leq \operatorname{index}(\sim_1)\ \operatorname{index}(\sim_2) < \infty.$$

Thus $\sim_1 \cap \sim_2 \in \mathcal{C}$ refines both, so $\mathcal{C}$ is cofiltered. □

Definition 2.17 (Profinite completion)**.** The **profinite completion** of $\mathbf{H}$ is

$$\widehat{\mathbf{H}} = \varprojlim_{\sim \in \mathcal{C}} \mathbf{H}/\sim .$$

An object of $\widehat{\mathbf{H}}$ is a compatible family

$$(q_\sim)_\sim, \qquad q_\sim \in \mathbf{H}/\sim,$$

such that whenever $\sim_1$ refines $\sim_2$, the projection of $q_{\sim_1}$ to $\mathbf{H}/\sim_2$ equals $q_{\sim_2}$.

There is a canonical dense embedding

$$\iota : \mathbf{H} \hookrightarrow \widehat{\mathbf{H}}, \qquad u \mapsto ([u]_\sim)_\sim.$$

Injectivity is not formal: it holds because the free monoid $\mathcal{I}^*$ is *residually finite* with respect to finite-index right congruences. Given distinct $u \neq v$, set $N > \max\{|u|, |v|\}$ and let $\sim_N$ identify two words when they are equal, or when both have length at least N. This is a right congruence: if $w \sim_N w'$ and $x \in \mathcal{I}$ then either $w = w'$, whence $wx = w'x$, or both $|w|, |w'| \geq N$, whence both $|wx|, |w'x| \geq N$. It has finite index $|\mathcal{I}^{<N}| + 1$, and separates u from v. Hence the components of $\iota(u)$ and $\iota(v)$ differ at $\sim_N$.

The same separation argument applies to pruned history systems, once feasibility is carried along.

Proposition 2.18 (Residual Finiteness of Synchronized Unifilar History Trees)**.** *Let a unifilar machine (Definition 3.15) have joint-history synchronization, so that its feasible histories of length at least L form a pruned history system F carrying a state readout $\sigma : F \to \mathcal{S}^+$ with $\sigma(u \cdot a) = \tau(\sigma(u), a)$ on feasible events. Then F is residually finite with respect to finite-index support-relative right congruences: the canonical map from F into the inverse limit of its finite quotients is injective.*

Proof. Let $u \neq v$ in F. Choose $N > \max\{|u|, |v|\}$ and define $\sim_N$ by declaring $w \sim_N w'$ when either $w = w'$, or $|w|, |w'| \geq N$ and $\sigma(w) = \sigma(w')$.

Finite index. There are at most $\sum_{n<N} |\mathcal{I} \times \mathcal{O}|^n$ singleton classes of length below N and at most $|\mathcal{S}^+|$ state-labelled classes at length N or above.

Clause (i). Feasibility of (x, y) from a history w means $P^x_{\sigma(w)}(\{y\}) > 0$, so it is a function of $\sigma(w)$ alone. Two histories identified by $\sim_N$ are either equal or have equal σ, and in both cases have the same feasible events.

Clause (ii). In the identity case $w \cdot a = w' \cdot a$. Otherwise $|w|, |w'| \geq N$ and $\sigma(w) = \sigma(w')$, so for a feasible (x, y),

$$\sigma\big(w \cdot (x, y)\big) = \tau(\sigma(w), x, y) = \tau(\sigma(w'), x, y) = \sigma\big(w' \cdot (x, y)\big),$$

and both extensions have length at least $N + 1$, hence $w \cdot (x, y) \sim_N w' \cdot (x, y)$.

Since N exceeds $|u|$ and $|v|$, both are singleton classes of $\sim_N$ and are therefore separated, so the corresponding components of their images in the inverse limit differ. $\square$

For total-action regimes the completion is that of the free monoid; for synchronized unifilar retention it is the completion of the feasible joint-history tree, residually finite by Proposition 2.18.

The **profinite uniformity** on $\mathbf{H}$ is the initial uniformity with respect to the projections

$$\mathbf{H} \to \mathbf{H}/\sim .$$

Equivalently, its basic entourages are

$$U_\sim = \{(u, v) \in \mathcal{I}^* \times \mathcal{I}^* : u \sim v\},$$

where $\sim$ ranges over finite-index right congruences.

Remark 2.19 (Cofiltered limits and Lawvere's framework)**.** The enriched poset

$$\mathbf{V} = ([0, \infty], \geq, +, 0)$$

is complete symmetric monoidal closed in the sense of Lawvere [37]. The relevant enriched cofiltered limits therefore exist. Because $\mathbf{V}$ is posetal, the enriched inverse limit coincides with the ordinary compatible-family limit on objects, equipped with the induced profinite uniformity.

Remark 2.20 (Profinite uniformity for discrete residual categories)**.** Let $\mathbf{R}$ be a discrete $\mathbf{V}$-category. Uniform continuity of a trajectory

$$\delta : \mathbf{H} \to \mathbf{R}$$

in the profinite uniformity is equivalent to δ factoring through some finite quotient:

$$\mathbf{H} \longrightarrow \mathbf{H}/{\sim} \longrightarrow \mathbf{R}$$

for some finite-index right congruence $\sim$. Indeed, in a discrete $\mathbf{V}$-category the only entourages are those containing the diagonal, so uniform continuity requires a finite-index congruence $\sim$ with $u \sim v \Rightarrow \delta(u) = \delta(v)$, which is exactly factorization through $\mathbf{H}/{\sim}$.

Remark 2.21 (Compactness and convergence)**.** Each finite quotient $\mathbf{H}/\sim$ is a finite discrete space. The inverse limit $\widehat{\mathbf{H}}$ is compact in the profinite topology. This compactness underlies the profinite convergence meta-theorem: under compact-uniform hypotheses on $\mathbf{R}$ and $\mathcal{E}$, uniform continuity of δ is equivalent to continuous extension to $\widehat{\mathbf{H}}$, and hence to approximability by finite-index right congruences with vanishing gap.

The profinite completion is not an analytic Hilbert-space completion such as an L^2 completion. It is a logical or Stone-type completion generated by finite deterministic quotients.

3 Specialized Task Theories

The generic schema becomes operative after the residual category, cost profunctor, and aggregation scheme are specialized. This section defines the principal specialized task theories used later: separated task theories, compact uniform task theories, Hilbert-module task theories, stationary controlled causal machines, full-KL retention, and Gaussian quadratic restricted retention.

3.1 Separated Task Theories

Definition 3.1 (Separated Task Theory)**.** A task theory $\mathbb{T}$ is **separated** if:

$$\mathbf{R}(X,Y) = \mathbf{R}(Y,X) = 0 \implies X = Y,$$

and

$$\mathcal{E}(X,Y) = 0 \implies X = Y.$$

Definition 3.2 (Myhill–Nerode Equivalence)**.** For any task theory $\mathbb{T}$, the **Myhill–Nerode equivalence** $\sim_\delta$ is

$$u \sim_\delta v \iff \forall w \in \mathcal{I}^*, \quad \mathbf{R}(\delta(uw), \delta(vw)) = \mathbf{R}(\delta(vw), \delta(uw)) = 0.$$

The two-sided condition is used because $\mathbf{R}$ need not be symmetric. If $\mathbf{R}$ is symmetric, the condition reduces to

$$\mathbf{R}(\delta(uw), \delta(vw)) = 0 \quad \forall w.$$

The Myhill–Nerode equivalence is always a right congruence. Indeed, if $u \sim_\delta v$, then for every $x \in \mathcal{I}$ and every $w \in \mathcal{I}^*$,

$$\delta(uxw) = \delta(u(xw)), \qquad \delta(vxw) = \delta(v(xw)),$$

so the defining zero-distance condition for u and v applied to the continuation xw gives $ux \sim_\delta vx$.

In the commitment regime, $\sim_\delta$ is the usual Myhill–Nerode residual equivalence. In the retention regime, when predictive distributions are distinct on the stationary support, it coincides with causal-state equivalence. In grounding regimes, it may correspond to equality of residual kernels or impulse-response functionals, depending on the chosen residual category.

3.2 Compact Uniform Task Theories

Definition 3.3 (Compact Metrizable $\mathbf{V}$-Category)**.** A $\mathbf{V}$-category $\mathbf{R}$ is **compact metrizable** if $\mathrm{ob}(\mathbf{R})$ carries a compact metric topology and the hom-object map

$$\mathbf{R}(-,-) : \mathrm{ob}(\mathbf{R}) \times \mathrm{ob}(\mathbf{R}) \to [0, \infty]$$

is continuous in the metric topology, with the usual extended-real topology on $[0, \infty]$.

More generally, one may use a compact uniform space in place of a compact metric space. The profinite convergence meta-theorem is stated under the hypothesis that $\mathbf{R}$ is compact metrizable or compact uniform and that the cost profunctor $\mathcal{E}$ is uniformly continuous and separated.

Remark 3.4 (Separate Continuity Is Insufficient)**.** Separate continuity alone is insufficient for the profinite equivalence. The meta-theorem uses joint uniform continuity, or compact-uniform continuity, of the cost profunctor.

3.3 Hilbert-Module Task Theories and Linear Hankel Gaps

Definition 3.5 (Hilbert-Module Task Theory)**.** A **Hilbert-module task theory** is a separated task theory on a finite joint alphabet

$$\Sigma = \mathcal{I} \times \mathcal{O}$$

equipped with the following additional data:

(1) a Banach space B of residual observables;

(2) a bounded linear functional

$$\phi : B \to \mathbb{C};$$

(3) an impulse-response function

$$h : \Sigma^* \to \mathbb{C}$$

obtained from the true trajectory by

$$h(u) = \phi(\delta(u));$$

(4) a Hankel operator

$$H_\delta : \ell^2(\Sigma^*) \to \ell^2(\Sigma^*)$$

specified on the standard basis by

$$(H_\delta e_u)(v) = h(uv),$$

provided this densely defined formula extends to a bounded operator on $\ell^2(\Sigma^*)$. Boundedness is not automatic: for general h the vector $v \mapsto h(uv)$ need not be square summable, and even when it is, the induced matrix need not be bounded. A sufficient condition is the geometric decay hypothesis of Proposition 3.11, which yields boundedness by the Schur test;

(5) compactness of the resulting bounded operator H_δ.

A Hilbert-module task theory is said to be *spectrally admissible* when the Hankel operator exists as a bounded operator and is compact, i.e. when conditions (iv) and (v) hold. All singular-value statements below are asserted only for spectrally admissible task theories.

If the residual objects are operators, B may be an operator space or a C^*-algebra, and ϕ may be a bounded functional selecting the scalar observable relevant to the impulse response. The scalar impulse response h is the object seen by the Hankel operator. Distinct residual objects may induce the same impulse response; the Hankel theory below depends only on h unless additional domination hypotheses are supplied.

Remark 3.6 (Residual Norm versus Hankel Norm)**.** The norm used for residual-operator comparisons is the norm of B, while the Hankel gap uses the ℓ^2 operator norm of H_δ. No domination between the B-norm and the Hankel operator norm is assumed.

The model does not assume an unqualified domination hypothesis of the form

$$\left\| H_\delta - H_{\widehat{\delta}} \right\|_{\mathrm{op}} \le \mathcal{L}_\infty(\delta, \widehat{\delta}).$$

Such an inequality may hold under explicit decay or summability conditions, but it is not a general axiom.

Definition 3.7 (Unrestricted Linear Hankel Relaxation Gap)**.** The **unrestricted linear Hankel M-rank relaxation gap** is

$$\Delta_{\mathbb{T}}^{\mathrm{unres}}(M) = \inf_{\operatorname{rank} \widehat{H} \le M} \left\| H_\delta - \widehat{H} \right\|_{\mathrm{op}}.$$

The infimum is over bounded operators $\widehat{H}$ of rank at most M, not necessarily Hankel operators.

This is an operator-approximation gap. It is the object to which Eckart–Young–Mirsky [15, 16] applies directly.

Definition 3.8 (Hankel-Structured Finite-Rank Gap)**.** The **Hankel-structured finite-rank gap** is

$$\Delta_{\mathbb{T}}^{\mathrm{Hank,str}}(M) = \inf_{\substack{\operatorname{rank} \widehat{H} \le M \\ \widehat{H} \ \text{Hankel}}} \left\| H_\delta - \widehat{H} \right\|_{\mathrm{op}}.$$

The infimum is over Hankel operators of rank at most M.

Because the Hankel-structured feasible set is a subset of the unrestricted feasible set,

$$\Delta_{\mathbb{T}}^{\mathrm{Hank,str}}(M) \geq \Delta_{\mathbb{T}}^{\mathrm{unres}}(M).$$

Equality requires additional structure.

Definition 3.9 (Linear Finite-Rank Residual-Cost Gap)**.** The **linear finite-rank residual-cost gap** is

$$\Delta_{\mathbb{T}}^{\mathrm{lin}}(M) = \inf\left\{\mathcal{L}_\infty(\delta, \widehat{\delta}) : \operatorname{rank} H_{\widehat{\delta}} \leq M\right\}.$$

The residual-cost gap $\Delta_{\mathbb{T}}^{\mathrm{lin}}(M)$ and the unrestricted Hankel relaxation gap $\Delta_{\mathbb{T}}^{\mathrm{unres}}(M)$ are distinct objects. The former measures task loss of finite-rank residual approximants; the latter measures operator-norm approximation to H_δ. They are related only under explicit domination or achievability hypotheses.

Definition 3.10 (Weighted Residual Norm)**.** For $0 < \rho < 1$, define

$$\|d\|_{\infty,\rho} = \sup_{w\in\Sigma^*} \rho^{-|w|}|d(w)|.$$

Proposition 3.11 (Stable-Decay Hankel Domination)**.** *Suppose*

$$|h_\delta(w) - h_{\widehat{\delta}}(w)| \leq \varepsilon\rho^{|w|}$$

with

$$0 < \rho < \frac{1}{|\Sigma|}.$$

Then

$$\left\|H_\delta - H_{\widehat{\delta}}\right\|_{\mathrm{op}} \leq \frac{1}{1-|\Sigma|\rho}\varepsilon.$$

Equivalently, if

$$\left\|h_\delta - h_{\widehat{\delta}}\right\|_{\infty,\rho} \leq \varepsilon,$$

then

$$\left\|H_\delta - H_{\widehat{\delta}}\right\|_{\mathrm{op}} \leq \frac{1}{1-|\Sigma|\rho}\left\|h_\delta - h_{\widehat{\delta}}\right\|_{\infty,\rho}.$$

Proof. Let

$$A = H_\delta - H_{\widehat{\delta}}.$$

The Hankel matrix entries satisfy

$$|A(u,v)| = |h_\delta(uv) - h_{\widehat{\delta}}(uv)| \leq \varepsilon\rho^{|u|+|v|}.$$

By Schur's test,

$$\sup_u \sum_v |A(u,v)| \leq \varepsilon \sup_u \rho^{|u|} \sum_{v\in\Sigma^*} \rho^{|v|}.$$

Since $\rho^{|u|} \le 1$ and the number of words of length n is $|\Sigma|^n$,

$$\sum_{v \in \Sigma^*} \rho^{|v|} = \sum_{n=0}^{\infty} |\Sigma|^n \rho^n = \frac{1}{1 - |\Sigma|\rho}.$$

Thus

$$\sup_u \sum_v |A(u,v)| \le \frac{\varepsilon}{1 - |\Sigma|\rho}.$$

The same bound holds for columns. Schur's test therefore gives

$$\|A\|_{\mathrm{op}} \le \frac{\varepsilon}{1 - |\Sigma|\rho}.$$

□

Remark 3.12 (AAK versus EYM)**.** The default spectral converse for $\Delta_{\mathbb{T}}^{\mathrm{unres}}(M)$ is Eckart–Young–Mirsky:

$$\Delta_{\mathbb{T}}^{\mathrm{unres}}(M) = \sigma_{M+1}(H_\delta).$$

Adamjan–Arov–Krein applies only after a Hardy-space embedding has been separately established. The model does not assume such an embedding by default.

Remark 3.13 (Infinite support and boundedness)**.** Infinite support of an impulse response does not by itself imply that the associated Hankel operator is unbounded. Boundedness requires decay or summability conditions. For example, scalar impulse responses satisfying $|h(u)| \le C\rho^{|u|}$ with $0 < \rho < 1/|\Sigma|$ define bounded Hankel-type operators by the Schur-test argument above. Bounded symbolic impulse responses that fail to decay may or may not define bounded Hankel operators; boundedness is checked case by case.

3.4 Stationary Controlled Causal Machines

Definition 3.14 (Stationary Controlled Causal Machine)**.** A **stationary controlled causal machine** consists of:

(1) a finite input alphabet $\mathcal{I}$;

(2) a standard Borel output space $\mathcal{O}$;

(3) a finite state set $\mathcal{S}$;

(4) a deterministic transition map

$$\tau : \mathcal{S} \times \mathcal{I} \to \mathcal{S};$$

(5) emission laws

$$P_s \in \mathcal{P}(\mathcal{O})$$

for each $s \in \mathcal{S}$;

(6) a stationary ergodic input process $(X_t)_{t \in \mathbb{Z}}$;

(7) a stationary state process $(S_t)_{t\in\mathbb{Z}}$ satisfying

$$S_t = \tau(S_{t-1}, X_t) \quad \text{a.s.};$$

(8) a stationary distribution

$$\pi_s = \mathbb{P}(S_t = s);$$

(9) a predictive future variable Z_t, by default the next output Y_t, such that S_t is sufficient:

$$Z_t \perp\!\!\!\perp \{S_u, X_u, Y_u : u < t\} \mid S_t.$$

The predictive distribution of state s is

$$P_s = \mathcal{L}(Z_t \mid S_t = s).$$

The stationary support is

$$\mathcal{S}^+ = \{s \in \mathcal{S} : \pi_s > 0\}.$$

Definition 3.15 (Stationary Controlled Unifilar Causal Machine)**.** A **stationary controlled unifilar causal machine** consists of a finite input alphabet $\mathcal{I}$, a standard Borel output space $\mathcal{O}$, a finite state set $\mathcal{S}$, emission kernels

$$P_s^x = \mathcal{L}\big(Y_t \mid S_t = s,\ X_t = x\big),$$

a measurable *unifilar* update

$$\tau : \mathcal{S} \times \mathcal{I} \times \mathcal{O} \to \mathcal{S}, \qquad S_{t+1} = \tau(S_t, X_t, Y_t) \quad \text{a.s.},$$

defined on jointly feasible triples, together with a stationary ergodic input process, a stationary distribution π, and a declared predictive target Z_t.

An update with $\tau(s, x, y) = \tau_0(s, x)$ for every feasible y is called *output-independent*; in that case the recursion collapses to $S_{t+1} = \tau_0(S_t, X_t)$ and, after an index shift, the machine is exactly a stationary controlled causal machine in the sense of Definition 3.14.

Remark 3.16 (The Input-Driven Model Is a Proper Subclass)**.** The containment is strict, and the witness is standard. Take $|\mathcal{I}| = 1$, $\mathcal{O} = \{0, 1\}$, two states A, B, with A emitting 0 or 1 with equal probability and B emitting 1 almost surely, and

$$\tau(A, \cdot, 0) = A, \qquad \tau(A, \cdot, 1) = B, \qquad \tau(B, \cdot, 1) = A.$$

Since the input alphabet is a singleton, the input history of length t is the unique word of that length, yet the state after two steps is A on the output history 00 and B on 01. The state is therefore not a function of the input history, and no map $\tau_0 : \mathcal{S} \times \mathcal{I} \to \mathcal{S}$ reproduces this machine.

Consequently Definition 3.14 describes an *input-driven* transducer, and the descriptions of it as a lumped ϵ-machine refer to this output-independent subclass rather than to general controlled ϵ-machines, whose causal states are functions of joint input–output histories (Remark 3.29). The retention theory is developed for the full unifilar class in Section 5.4, with the input-driven model as the output-independent specialization; the complexity constructions of Section 5.5 are stated for the input-driven model and transfer to the unifilar class by Remark 5.53.

Definition 3.17 (Z-predictive equivalence)**.** Fix the predictive target Z_t. On the stationary support define

$$s \sim_Z s' \iff \mathcal{L}(Z_t \mid S_t = s) = \mathcal{L}(Z_t \mid S_t = s').$$

If Z_t is the entire controlled future, $\sim_Z$ is causal-state equivalence. If $Z_t = Y_t$ is only the next output, $\sim_Z$ is one-step predictive equivalence, which is strictly coarser in general and need not identify causal states.

Throughout, "causal equivalence" means $\sim_Z$ for the declared target Z, and coincides with causal-state equivalence exactly when Z is the full future. All retention statements, including the zero-retention threshold, are to be read with respect to distinct Z-predictive states.

For unifilar machines with input-dependent emissions (Definition 3.15) the predictive object attached to a state is the whole kernel $x \mapsto P_s^x$, and $\sim_Z$ compares kernels on the input support: $s \sim_Z s'$ if and only if $P_s^x = P_{s'}^x$ for every input x of positive mass.

Finite output alphabets are the special case in which all P_s are supported on a finite set. Gaussian quadratic retention is obtained by taking $\mathcal{O} = \mathbb{R}^D$ and P_s Gaussian.

Definition 3.18 (Synchronized Finite-History Realization)**.** A stationary controlled causal machine has **synchronization depth** $L < \infty$ if there exists a surjection

$$\sigma_L : \mathcal{I}^L \to \mathcal{S}^+$$

such that, almost surely,

$$S_t = \sigma_L(X_{t-L+1}, \ldots, X_t).$$

For histories u with $|u| \geq L$, define

$$\sigma(u) = \sigma_L(\mathrm{suffix}_L(u)).$$

Then for all sufficiently long u,

$$\sigma(ux) = \tau(\sigma(u), x).$$

The finite-history trajectory is

$$\delta(u) = P_{\sigma(u)}.$$

Thus the original finite-history/right-congruence syntax is recovered for definite and synchronizing machines, including all NP-hardness constructions.

Definition 3.19 (Causal Equivalence)**.** On the stationary support, causal equivalence is

$$s \sim_\delta t \iff P_s = P_t.$$

In a synchronized machine, the corresponding history equivalence is

$$u \sim_\delta v \iff \sigma(u) = \sigma(v).$$

If the predictive distributions P_s are pairwise distinct on $\mathcal{S}^+$, then causal equivalence coincides with state identity on $\mathcal{S}^+$.

Definition 3.20 (Lumpable Quotient)**.** A map $\phi : \mathcal{S}^+ \to \mathcal{K}$ is **lumpable** if there exists a deterministic quotient transition

$$\tau_{\mathcal{K}} : \mathcal{K} \times \mathcal{I} \to \mathcal{K}$$

such that

$$\phi(\tau(s, x)) = \tau_{\mathcal{K}}(\phi(s), x)$$

for all $s \in \mathcal{S}^+$ and all inputs x in the support of the input process. For worst-case aggregation, require this for all $x \in \mathcal{I}$. The quotient process is

$$K_t = \phi(S_t).$$

Definition 3.21 (Unifilar Lumpability)**.** A quotient $\phi : \mathcal{S} \to \mathcal{K}$ of a unifilar machine is **unifilar-lumpable** if there is $\tau_{\mathcal{K}} : \mathcal{K} \times \mathcal{I} \times \mathcal{O} \to \mathcal{K}$ with

$$\phi\big(\tau(s, x, y)\big) = \tau_{\mathcal{K}}\big(\phi(s), x, y\big)$$

for every *jointly feasible* triple (s, x, y), that is, every triple with $P_s^x(\{y\}) > 0$ in the discrete case and, in general, every triple in the support of the joint law. The singleton quotient is always unifilar-lumpable.

The quotient has **block-uniform support** if

$$\phi(s) = \phi(s') \implies \operatorname{supp} P_s^x = \operatorname{supp} P_{s'}^x \qquad \text{for every } x$$

in the input support. More generally, say that ϕ has **connected support** if for every block C and every input x the graph on C joining s to s' when $\operatorname{supp} P_s^x \cap \operatorname{supp} P_{s'}^x \neq \varnothing$ is connected; block-uniform support is the special case in which that graph is complete.

Remark 3.22 (Unifilar Lumpability Does Not Reduce to Definition 3.20)**.** For an output-independent update Definition 3.21 is strictly weaker than Definition 3.20, and the two do not collapse onto each other. Writing $\tau(s, x, y) = \tau_0(s, x)$, unifilar lumpability requires $\phi(\tau_0(s, x)) = \phi(\tau_0(s', x))$ only for s, s' in a common block that share a feasible output y at input x; ordinary lumpability requires it for *all* s, s' in the block. When within-block emission supports are disjoint the requirement is vacuous and the unifilar notion is strictly weaker.

An explicit witness has $\mathcal{I} = \{0, 1\}$, $\mathcal{O} = \{0, 1\}$, three states, and

$$\tau_0(s_0, 0) = s_0, \quad \tau_0(s_0, 1) = s_1, \qquad \tau_0(s_1, 0) = s_0, \quad \tau_0(s_1, 1) = s_2,$$

$$\tau_0(s_2, 0) = s_0, \quad \tau_0(s_2, 1) = s_0,$$

with s_0, s_1 emitting 0 almost surely and s_2 emitting 1 almost surely. The chain is irreducible with strictly positive stationary distribution. The partition $\{\{s_0, s_2\}, \{s_1\}\}$ is unifilar-lumpable — s_0 and s_2 never share a feasible output, so no intertwining is demanded of them — yet it is not lumpable, since $\tau_0(s_0, 1) = s_1$ and $\tau_0(s_2, 1) = s_0$ lie in different blocks.

Consequently the two feasible sets differ, and the associated retention values can differ as well: on a three-state machine with $\tau_0(s_0, \cdot) = (s_1, s_2)$, $\tau_0(s_1, \cdot) = (s_2, s_0)$, $\tau_0(s_2, \cdot) = (s_0, s_0)$, predictive laws $P_{s_0} = \delta_2$, $P_{s_2} = \delta_1$, and P_{s_1} interior, the minimum of the retention cost over unifilar-lumpable quotients with two blocks is strictly below the minimum over lumpable quotients with two blocks. The reduction is therefore genuinely conditional, and the correct statement is Proposition 3.28: the two notions agree under connected support, in particular under block-uniform support, and in particular whenever all emission laws have full support.

Proposition 3.23 (Right-Congruence Correspondence in Synchronized Machines)**.** *Let the machine have synchronization depth L. A lumpable quotient $\phi : \mathcal{S}^+ \to \mathcal{K}$ induces a right congruence on sufficiently long histories by*

$$u \sim_\phi v \iff \phi(\sigma(u)) = \phi(\sigma(v)).$$

Conversely, assume that Z-predictive equivalence $\sim_Z$ of Definition 3.17 is itself a right congruence on $\mathcal{S}^+$, i.e. $s \sim_Z s'$ implies $\tau(s, x) \sim_Z \tau(s', x)$ for every feasible input x. This holds in particular when Z is the full controlled future, so that $\sim_Z$ is causal-state equivalence. Then any finite-index right congruence coarser than $\sim_Z$ induces a lumpable quotient of $\mathcal{S}^+$.

The hypothesis cannot be dropped: by Example 3.24, one-step predictive equivalence need not be a right congruence, and the induced partition need not be lumpable.

Proof. If ϕ is lumpable and $u \sim_\phi v$, then for sufficiently long u, v,

$$\phi(\sigma(ux)) = \phi(\tau(\sigma(u), x)) = \tau_{\mathcal{K}}(\phi(\sigma(u)), x) = \tau_{\mathcal{K}}(\phi(\sigma(v)), x) = \phi(\sigma(vx)),$$

so $ux \sim_\phi vx$. Hence $\sim_\phi$ is a right congruence on long histories.

Conversely, let $\sim$ be a finite-index right congruence coarser than $\sim_Z$, and assume $\sim_Z$ is a right congruence. Each $\sim$-class is then a union of $\sim_Z$-classes. Define $\phi(s)$ to be the $\sim$-class of any sufficiently long history u with $\sigma(u) = s$. This is well defined: $\sigma(u) = \sigma(v) = s$ makes u and v $\sim_Z$-equivalent, hence $\sim$-equivalent because $\sim$ is coarser, so the assigned $\sim$-class does not depend on the chosen history. Define the quotient transition *by* the intertwining,

$$\tau_{\mathcal{K}}(k, x) := \phi\big(\tau(s, x)\big) \qquad \text{for any } s \in \mathcal{S}^+ \text{ with } \phi(s) = k,$$

whose well-definedness is exactly the content of the right-congruence property: if $\phi(s) = \phi(s') = k$ and u, u' are long histories with $\sigma(u) = s$ and $\sigma(u') = s'$, then $u \sim u'$, hence $ux \sim u'x$ for every input x in the support, so $\phi(\tau(s, x)) = \phi(\tau(s', x))$ and $\tau_{\mathcal{K}}(k, x)$ is independent of the representative s. The pair $(\phi, \tau_{\mathcal{K}})$ is thus a lumpable quotient in the sense of Definition 3.20. $\square$

Example 3.24 (One-Step Predictive Equivalence Need Not Be a Congruence)**.** Let $\mathcal{S} = \{s_0, s_1, s_2, s_3\}$ with a single input letter a, $\mathcal{O} = \{0, 1\}$, and next-output laws

$$P_{s_0} = P_{s_1} = \text{Bernoulli}(\tfrac{1}{2}), \qquad P_{s_2} = \delta_0, \qquad P_{s_3} = \delta_1.$$

Let the transitions be

$$\tau(s_0, a) = s_2, \quad \tau(s_1, a) = s_3, \quad \tau(s_2, a) = s_2, \quad \tau(s_3, a) = s_3.$$

Taking $Z = Y$, the next output, gives $s_0 \sim_Y s_1$, since both are fair coins. But

$$\tau(s_0, a) = s_2 \not\sim_Y s_3 = \tau(s_1, a),$$

so $\sim_Y$ is not a right congruence, and the partition $\{\{s_0, s_1\}, \{s_2\}, \{s_3\}\}$ is not lumpable: the block $\{s_0, s_1\}$ has no well-defined successor block.

Taking instead Z to be the full future separates s_0 from s_1, because their continuations are the constant sequences 0^∞ and 1^∞. With that choice $\sim_Z$ is causal-state equivalence and the converse of Proposition 3.23 applies.

Since input-driven machines are unifilar (Remark 3.16), the same example shows that one-step predictive equivalence need not be a support-relative right congruence on joint histories in the general model.

A scope note: with a single input letter and this transition structure, every stationary distribution is supported on the absorbing pair $\{s_2, s_3\}$, while s_0 and s_1 — the states carrying the one-step equivalence — are transient. The example therefore concerns the transition structure and the reachability of the equivalence, not an instance posed on the stationary support in the sense of Definition 3.15; the structural point is unaffected by making all four states recurrent with a second input letter.

Remark 3.25 (Necessity of the Feasibility Restriction)**.** The restriction to feasible triples in Definition 3.21 is not dispensable. If (s, x, y) has emission probability zero the update $\tau(s, x, y)$ is unconstrained by the process, so requiring the intertwining there would reject quotients that are lumpable in every realizable sense. Over randomly generated unifilar machines with $|\mathcal{S}| \le 4$, $|\mathcal{I}| \le 2$ and $|\mathcal{O}| \le 3$ together with random partitions, approximately fifteen per cent of the quotients satisfying Definition 3.21 fail the corresponding condition stated without the feasibility restriction, so the two conditions are genuinely distinct.

Proposition 3.26 (Right Congruences on Joint Histories)**.** *Let a unifilar machine have finite synchronization depth L in joint histories, so that S_t is a function σ of the last L pairs (X_u, Y_u), and let F be the pruned history system of feasible joint histories of length at least L (Definition 2.6). Then:*

(i) if ϕ is unifilar-lumpable with block-uniform support, the relation $u \sim_\phi v$ defined by $\phi(\sigma(u)) = \phi(\sigma(v))$ is a support-relative right congruence on F in the sense of Definition 2.7. Without a support hypothesis it still satisfies clause (ii) on commonly feasible events, but clause (i) may fail, and then the quotient carries no well-defined feasible-event set (Remark 2.8);

(ii) conversely, if Z-predictive equivalence $\sim_Z$ on F is itself a support-relative right congruence, then every finite-index support-relative right congruence $\sim$ on F that is coarser *than $\sim_Z$ induces a well-defined unifilar-lumpable quotient of $\mathcal{S}^+$. The hypothesis on $\sim_Z$ holds whenever Z is the full controlled future.*

Proof. For (i), let $u \sim_\phi v$, and write k for the common value $\phi(\sigma(u)) = \phi(\sigma(v))$. Block-uniform support gives $\operatorname{supp} P^x_{\sigma(u)} = \operatorname{supp} P^x_{\sigma(v)}$ for every input x, so an event (x, y) is feasible from u exactly when it is feasible from v; this is clause (i) of Definition 2.7. For such an event, unifilar lumpability applied to the pair $\sigma(u), \sigma(v)$, which have equal ϕ-image, gives $\phi(\tau(\sigma(u), x, y)) = \tau_{\mathcal{K}}(k, x, y) = \phi(\tau(\sigma(v), x, y))$, and by unifilarity $\sigma(u\cdot(x, y)) = \tau(\sigma(u), x, y)$; hence $u \cdot (x, y) \sim_\phi v \cdot (x, y)$, which is clause (ii). Iterating gives closure under feasible words. Without block-uniform support the displayed computation still applies to events feasible from both histories, which is the weaker assertion recorded in the statement.

For (ii), the only issue is that $\sim$ descend to states, that is, that $\sigma(u) = \sigma(v)$ imply $u \sim v$; otherwise the induced map on $\mathcal{S}$ is not well defined. This is where the hypothesis enters. If $\sigma(u) = \sigma(v)$ then the two histories reach the same state and therefore carry the same Z-predictive law, so they are Z-predictive equivalent; since $\sim$ is coarser than Z-predictive equivalence, $u \sim v$. Define $\bar{\phi}(\sigma(u))$ to be the $\sim$-class of u, which is now unambiguous, and let $\mathcal{K}$ be the set of classes, finite by hypothesis. Right-congruence of $\sim$ under a feasible event (x, y) is exactly the intertwining $\bar{\phi}(\tau(s, x, y)) = \tau_{\mathcal{K}}(\bar{\phi}(s), x, y)$ required by Definition 3.21.

The calculation is the one in Proposition 3.23 with events (x, y) in place of letters, and support-relative feasibility handled as in the feasible-input clause of that proof.

It remains to check the final assertion, that $\sim_Z$ is a support-relative right congruence when Z is the full controlled future. Clause (ii) is the usual argument: conditioning an equality of conditional future laws on a commonly realized next event preserves the equality. Clause (i) is not an extra hypothesis but a consequence. If $u \sim_Z v$ then the two conditional laws of the controlled future agree; taking the one-step marginal under the input program that begins with x shows $P^x_{\sigma(u)} = P^x_{\sigma(v)}$, so the two emission laws have the same support and u, v have the same feasible events. Thus equality of full-future laws already forces equality of feasible sets, and no separate common-feasibility assumption is needed. ☐

Remark 3.27 (Which Hypothesis Does the Work in (ii))**.** Coarseness, not finite index, is what makes the converse true. Dropping it breaks descent: take a single-state machine with two feasible outputs, so every joint history reaches the same state, and let $u \sim v$ when $|u|$ and $|v|$ have the same parity. That relation is a right congruence of index two, yet $\sigma(u) = \sigma(v)$ for all u, v while $u \not\sim v$ whenever the lengths differ in parity, so no induced map on $\mathcal{S}$ exists. Here Z-predictive equivalence is the all-relation and the parity congruence is strictly finer, so the hypothesis of Proposition 3.26(ii) correctly excludes it.

Proposition 3.28 (Input-Driven Specialization)**.** *Let the machine be input-driven, $\tau(s, x, y) = \tau_0(s, x)$, and input-synchronized in the sense of Definition 3.18.*

(i) Lumpability in the sense of Definition 3.20 implies unifilar lumpability. The converse holds if ϕ has connected support, in particular if it has block-uniform support, and in

particular if every P_s^x has full support; it fails in general (Remark 3.22).

(ii) The relation $u \sim_\phi v \iff \phi(\sigma(u)) = \phi(\sigma(v))$ on feasible joint histories depends only on the input projections of u and v, and its projection to input histories is exactly the right congruence of Proposition 3.23.

Proof. (i) Given a quotient map $\tau_\mathcal{K}^0$ witnessing Definition 3.20, set $\tau_\mathcal{K}(k, x, y) = \tau_\mathcal{K}^0(k, x)$; the left-hand side of the unifilar condition is independent of y, so the identity holds at every feasible triple.

For the converse, fix a block C and an input x, and suppose $s, s' \in C$ share a feasible output $y \in \operatorname{supp} P_s^x \cap \operatorname{supp} P_{s'}^x$. Unifilar lumpability at the two triples (s, x, y) and (s', x, y) gives

$$\phi(\tau_0(s, x)) = \tau_\mathcal{K}(k, x, y) = \phi(\tau_0(s', x)),$$

so $\phi \circ \tau_0(\cdot, x)$ is constant along every edge of the support-overlap graph on C. If that graph is connected, the function is constant on C, which is Definition 3.20. Block-uniform support makes the graph complete, hence connected, and full supports make it complete for the same reason. Connectedness cannot be dropped: in the witness of Remark 3.22 the graph on the block $\{s_0, s_2\}$ has no edges, and the implication fails.

(ii) By Definition 3.18, $\sigma(u) = \sigma_L(\mathrm{suffix}_L(u_\mathcal{I}))$ depends only on the input projection $u_\mathcal{I}$, hence so does $\phi(\sigma(u))$. For an input extension x,

$$\phi(\sigma(u_\mathcal{I} x)) = \phi(\tau_0(\sigma(u_\mathcal{I}), x)) = \tau_\mathcal{K}^0\big(\phi(\sigma(u_\mathcal{I})), x\big),$$

which is the right-congruence computation of Proposition 3.23. Passing to joint extensions appends output letters that change neither the state nor the input projection. □

Remark 3.29 (Relation to ϵ-Machines)**.** An ϵ-machine in the sense of computational mechanics [32] is a unifilar predictive representation whose states are the causal states: classes of pasts carrying identical conditional laws of the future, minimal among such representations. A unifilar controlled machine (Definition 3.15) in which Z is the full controlled future, and whose states are the $\sim_Z$-classes of feasible joint histories, is a controlled ϵ-machine in that sense. The input-driven machines of Definition 3.14 form the subclass with output-independent transitions; in the synchronized case their causal states factor through input histories (Proposition 3.28(ii)). References elsewhere in this article to lumped ϵ-machines are to be read in this sense.

3.5 Full-KL Retention Gap

Definition 3.30 (Full-KL Retention Gap)**.** For a stationary controlled causal machine, define

$$\Delta_{\mathrm{ret}}^{\mathrm{KL}}(M) = \min_{\substack{\phi:\mathcal{S}^+\to\mathcal{K} \text{ lumpable} \\ |\mathcal{K}|\le M}} \sum_{k\in\mathcal{K}} \sum_{s:\phi(s)=k} \pi_s D_{\mathrm{KL}}(P_s \| \bar{P}_k),$$

where

$$\bar{P}_k = \frac{\sum_{s:\phi(s)=k} \pi_s P_s}{\sum_{s:\phi(s)=k} \pi_s}.$$

Blocks of zero stationary mass are ignored. Because $\mathcal{S}^+$ is finite, the minimum exists.

For unifilar machines with input-dependent emissions the operative object is the controlled gap of Definition 5.31, of which the present definition is the output-independent specialization (Corollary 5.47).

In a synchronized machine, lumpable quotients of $\mathcal{S}^+$ correspond to finite-index right congruences on sufficiently long histories by Proposition 3.23. The full-KL centroid $\bar{P}_k$ is a mixture distribution. It is not, in general, an arithmetic mean of parameters. The arithmetic-mean centroid appears only in restricted parametric families, such as the Gaussian quadratic restricted regime.

3.6 Gaussian Quadratic Restricted Retention Gap

Definition 3.31 (Gaussian Quadratic Restricted Retention)**.** A stationary controlled causal machine is in the **Gaussian quadratic restricted regime** if:

(1) the output space is the standard Borel space $\mathcal{O} = \mathbb{R}^D$;

(2) each positive-mass state $s \in \mathcal{S}^+$ has predictive distribution

$$P_s = \mathcal{N}(m_s, \tfrac{1}{2} I_D);$$

(3) admissible representatives are restricted to the same fixed-covariance Gaussian family:

$$Q_c = \mathcal{N}(c, \tfrac{1}{2} I_D);$$

(4) the cost is the KL divergence restricted to this family.

For this family,

$$D_{\mathrm{KL}}(P_s \| Q_c) = \|m_s - c\|^2 .$$

Equivalently, define Fisher-scaled features

$$y_s = \sqrt{2}\, m_s, \qquad c_y = \sqrt{2}\, c.$$

Then

$$D_{\mathrm{KL}}(P_s \| Q_c) = \frac{1}{2} \|y_s - c_y\|^2 .$$

The associated quadratic gap is

$$\Delta_{\mathrm{ret}}^{\mathrm{quad}}(M) = \min_{\substack{\phi : \mathcal{S}^+ \to \mathcal{K} \text{ lumpable} \\ |\mathcal{K}| \le M}} \sum_{k \in \mathcal{K}} \sum_{s : \phi(s) = k} \pi_s \|m_s - \bar{m}_k\|^2 ,$$

where

$$\bar{m}_k = \frac{\sum_{s:\phi(s)=k} \pi_s m_s}{\sum_{s:\phi(s)=k} \pi_s}.$$

Equivalently, in Fisher-scaled coordinates,

$$\Delta_{\mathrm{ret}}^{\mathrm{quad}}(M) = \min_{\substack{\phi:\mathcal{S}^+\to\mathcal{K}\ \text{lumpable}\\ |\mathcal{K}|\le M}} \frac{1}{2}\sum_{k\in\mathcal{K}}\sum_{s:\phi(s)=k} \pi_s \left\|y_s - \bar{y}_k\right\|^2,$$

where

$$\bar{y}_k = \frac{\sum_{s:\phi(s)=k} \pi_s y_s}{\sum_{s:\phi(s)=k} \pi_s}.$$

Blocks of zero stationary mass are ignored.

The restricted Gaussian centroid is the arithmetic mean, either in mean coordinates m_s or in Fisher-scaled coordinates y_s. The unrestricted full-KL centroid is the mixture distribution $\bar{P}_k$. These are not the same.

The quadratic gap is a weighted k-means objective with a lumpability constraint. It is the object used for the spectral retention converse and for the Gaussian quadratic NP-hardness theorem. It is not the full-KL retention gap unless the full-KL problem has been explicitly restricted to the fixed-covariance Gaussian family.

4 Structural Meta-Theorems

The schema supports several structural meta-theorems. These results are governed by finite-index right congruences, quotient factorization, and separated cost functionals. Their proofs remain regime-specific: Myhill–Nerode theory, operator approximation, and information processing supply the analytical engines.

4.1 Master Scope: A Typed Variational Picture

The results of this manuscript, across commitment, retention, and grounding, instantiate one typed variational picture along four pillars. Stating them together fixes what is unified and what is not; this subsection is a summary of scope, not a new derivation.

(I) **Typed variational schema.** Every regime gap has the shape

$$\Delta_{\mathbb{T}}(M) = \inf_{\text{budget-}M\text{ feasible }B} \mathcal{L}(\delta, B),$$

but the feasible class carries a regime-specific type: a finite right congruence on histories for commitment (Section 6.1), a stationary lumpable quotient for retention (Section 1.3), a deterministic Mealy machine for the symbolic grounding gap (Definition 7.27), and a finite-rank bounded operator for the unrestricted linear grounding relaxation (Theorem 7.3). The budget M is a right-congruence index in the first three cases and an operator rank in the fourth; these are not asserted equivalent (Remark 1.1).

(II) **Structural monotonicity and zero thresholds.** Each regime gap is non-increasing in M and has an exact zero threshold on its own feasible class: the Myhill–Nerode index for commitment (Theorem 6.2), the stationary support size of distinct predictive laws for full-KL retention (Theorem 5.2), and the operator rank of H_ν for the linear grounding relaxations (Proposition 7.8). These thresholds are proved separately in each regime; no single argument produces all three.

(III) **Conditional response-operator converse.** Whenever a regime's approximants admit a response operator of bounded rank dominating the task cost (Definition 8.9), the unified Schatten converse of Theorem 8.11 supplies a spectral-tail lower bound. This bridge is conditional on the domination hypothesis and on the domination modulus, which may itself depend on the budget (Remark 8.8); it does not by itself make the three regimes' numerical gaps comparable (Theorem 8.22).

(IV) **Regime-specific analytical engines.** The zero thresholds and quantitative rates are proved by different classical theories: Myhill–Nerode automata theory and minimax game values for commitment, information-bottleneck and Fisher-information asymptotics for retention, and Eckart–Young–Mirsky/Adamjan–Arov–Krein operator approximation for grounding. The schema organizes these engines under a common typed vocabulary; it does not replace them.

Sections 6–7 develop each regime in this order, and Section 12 states precisely which joint and comparative statements the typed schema does and does not support across regimes.

4.2 Monotonicity and Profinite Convergence

Meta-Theorem 4.1 (Monotonicity and Profinite Convergence)**.** *Let $\mathbb{T}$ be a task theory.*

(i) $\Delta_{\mathbb{T}}(M)$ *is non-increasing in* M.

(ii) If $\sim_\delta$ *has finite index, then, since* $\mathcal{E}$ *is reflexive by Definition 2.2,*

$$\Delta_{\mathbb{T}}(M) = 0 \quad \forall M \geq \operatorname{index}(\sim_\delta).$$

(iii) If $\mathcal{E}$ *is separated and* $\{0, \infty\}$*-valued, then:*

(a) for worst-case aggregation,

$$\Delta^{\sup}_{\mathbb{T}}(M) = 0 \textit{ for some } M \iff \operatorname{index}(\sim_\delta) < \infty;$$

(b) for μ*-average aggregation,*

$$\Delta^{\mu}_{\mathbb{T}}(M) = 0 \textit{ for some } M \iff \kappa_{\mathrm{obs}}(\delta, \mu) < \infty,$$

with κ_{obs} *as in Definition 4.3. If* μ *has full support this reduces to (a).*

(iv) Suppose $\mathbf{R}$ *is compact metrizable or compact uniform,* $\mathcal{E}$ *is uniformly continuous and separated, the gap is evaluated in the worst-case/uniform sense, and* $\mathcal{E}$ *is* uniformly compatible *with the uniformity of* $\mathbf{R}$*:*

$$\forall \textit{ entourages } U \ \exists \eta > 0 : \quad \max\{\mathcal{E}(X,Z), \mathcal{E}(Y,Z)\} < \eta \implies (X,Y) \in U.$$

Conversely, closeness in $\mathbf{R}$ *implies small cost, by uniform continuity of* $\mathcal{E}$*. On a compact metric space with* $\mathcal{E}(X,Z)=0 \iff X=Z$*, joint continuity already implies this compatibility; stating it explicitly is what makes the passage from "small cost to a common representative" to "close residual objects" valid without a hidden asymmetry. Then the following are equivalent:*

(a) δ *is uniformly continuous for the profinite uniformity;*

(b) δ *extends continuously to* $\widehat{\mathbf{H}}$*;*

(c)
$$\lim_{M\to\infty} \Delta_{\mathbb{T}}(M) = 0.$$

Proof. Parts (i)–(iii) follow from the definitions. Monotonicity follows because the feasible set of right congruences of index at most M is nested:

$$\{\mathrm{index}(\sim) \le M\} \subseteq \{\mathrm{index}(\sim) \le M+1\}.$$

If $\sim_\delta$ has finite index, quotient by $\sim_\delta$ and assign each class its common residual $\Psi([u]) = \delta(u)$, which is well defined since $u \sim_\delta v$ implies $\delta(u) = \delta(v)$. Every history then incurs $\mathcal{E}(\delta(u), \Psi([u])) = \mathcal{E}(\delta(u), \delta(u)) = 0$ by reflexivity, so the aggregated cost vanishes for any of the three aggregation schemes. Reflexivity is exactly what this step consumes; without it the quotient reproduces the trajectory but need not reproduce it at zero cost.

For the converse in (iii), suppose $\mathcal{E}$ is separated and $\{0,\infty\}$-valued and the worst-case gap is zero at budget M. Then $\mathcal{E}(\delta(u), \Psi([u])) = 0$ for every u, and separatedness upgrades this to $\delta(u) = \Psi([u])$. Hence histories in a common class have equal residuals after every continuation, so the state relation refines $\sim_\delta$ and $\mathrm{index}(\sim_\delta) \le M$. Separatedness is what this direction consumes, and it is assumed in (iii) but not in (ii).

For (iv), assume first that δ is uniformly continuous for the profinite uniformity. By the universal property of completion, δ extends continuously to a map

$$\widehat{\delta} : \widehat{\mathbf{H}} \to \mathbf{R}.$$

Since $\widehat{\mathbf{H}}$ is compact and $\mathbf{R}$ is compact uniform, $\widehat{\delta}$ is uniformly continuous. Hence for every cost tolerance $\varepsilon > 0$ there is a finite clopen partition of $\widehat{\mathbf{H}}$, equivalently a finite-index right congruence $\sim$, such that

$$u \sim v \implies \mathcal{E}(\delta(u), \delta(v)) < \varepsilon$$

in the uniform sense. Choosing representatives for each $\sim$-class gives an M-state approximation with worst-case gap less than ε.

Conversely, suppose $\Delta_{\mathbb{T}}(M) \to 0$ in the worst-case sense. For every $\varepsilon > 0$, choose a finite quotient $\sim$ and representative map Ψ such that

$$\sup_{u\in\mathcal{I}^*} \mathcal{E}(\delta(u), \Psi([u])) < \eta,$$

where $\eta > 0$ is chosen small enough, using compactness and separatedness of $\mathcal{E}$, to ensure that whenever two histories u, v are assigned the same representative, their residual objects are ε-close in the uniform structure of $\mathbf{R}$. If $u \sim v$, then $\Psi([u]) = \Psi([v])$, so $\delta(u)$ and $\delta(v)$ are ε-close. Thus δ is uniformly continuous for the profinite uniformity and therefore extends continuously to $\widehat{\mathbf{H}}$. $\square$

Remark 4.2 (Average-case aggregation)**.** Uniform worst-case approximation implies stationary average-case approximation. The converse direction, from average-case gap convergence to profinite uniform continuity, requires additional observability or positivity assumptions on the stationary process. The profinite equivalence is therefore stated for the worst-case/uniform gap unless such assumptions are supplied.

4.3 Boolean Zero-Error Dichotomy

Definition 4.3 (Observable Support Index)**.** Let μ be an average-case history law on $\mathcal{I}^*$ and let $S = \operatorname{supp} \mu$. Such a μ exists for discounted-prefix aggregation (Definition 2.4), where $\mu(u) = (1-\gamma)\gamma^{|u|} \Pr[\text{prefix } u]$ is a probability measure on finite histories. Stationary Cesàro aggregation does *not* in general supply one: Definition 2.3 averages a cost process along a stationary trajectory and, when the process has infinite support, the Cesàro averages need not converge to the expectation under any single probability measure on $\mathcal{I}^*$. The Cesàro case is treated separately in Remark 4.8. The **observable support index** $\kappa_{\text{obs}}(\delta, \mu)$ is the minimum index of a right congruence $\sim$ on $\mathcal{I}^*$ for which there exists a representative map Ψ with

$$\mathcal{E}\big(\delta(u), \Psi([u])\big) = 0 \qquad \text{for } \mu\text{-almost every } u.$$

If S is right-closed and $\mu(u) > 0$ for every $u \in S$, define in addition the *support-relative Myhill–Nerode relation*

$$u \sim_{\delta,S} v \iff \forall w \in \mathcal{I}^* \text{ with } uw, vw \in S : \quad \mathbf{R}\big(\delta(uw), \delta(vw)\big) = \mathbf{R}\big(\delta(vw), \delta(uw)\big) = 0.$$

Remark 4.4 (Restriction to the Support Is Not a Congruence)**.** The restriction $\sim_\delta\!\restriction_S$ is not in general a right congruence. If S is not closed under right extension, a pair $u, v \in S$ may have continuations $uw \notin S$ witnessing $u \not\sim_\delta v$ that the average cost never sees; and the restricted relation need not be compatible with the transition action. The operative object is therefore the index of a global congruence that is observationally exact on S, which is what Definition 4.3 names. For right-closed S the two indices are related by the sandwich of Meta-Theorem 4.7(ii); they coincide precisely when the completion of Lemma 4.5 is index-preserving, and Remark 4.6 exhibits a right-closed support where they differ.

Lemma 4.5 (Index-Preserving Extension off the Support)**.** *Let $S \subseteq \mathcal{I}^*$ be right-closed, i.e. $u \in S$ implies $uw \in S$ for every $w \in \mathcal{I}^*$. Let $\approx$ be an equivalence relation on S that is* stable*: $u \approx v$ and $x \in \mathcal{I}$ imply $ux \approx vx$. Then there is a right congruence $\sim$ on all of $\mathcal{I}^*$ whose restriction to S is $\approx$, and*

$$\operatorname{index}(\sim) = \operatorname{index}(\approx) + c_S,$$

where c_S is the number of classes the construction requires off S. One has $c_S = 0$ when $S = \mathcal{I}^*$, and $c_S \le 1$ whenever no history outside S has an extension inside S. In general c_S may be infinite, in which case the construction yields a right congruence of infinite index *and the upper bound is vacuous; the sandwich is informative exactly when the completion has finite index.*

Proof. Right-closure gives $u \in S \Rightarrow ux \in S$, so S is forward invariant. The complement is *not* forward invariant: a history outside S may have an extension inside S. For $S = \{u : u \text{ contains at least one } \mathtt{b}\}$, which is right-closed, one has $\mathtt{aaa} \notin S$ and $\mathtt{aaab} \in S$. The set $\{u : u \text{ ends in } \mathtt{b}\}$ would *not* serve here, since appending $\mathtt{a}$ leaves it and it is therefore not right-closed. The classes off S therefore cannot be merged freely, and the construction below is stated explicitly.

Define $\sim$ on $\mathcal{I}^*$ by declaring $u \sim v$ when either

(a) $u, v \in S$ and $u \approx v$; or

(b) $u, v \notin S$ and for every $y \in \mathcal{I}^*$ one has $uy \in S \iff vy \in S$, and $uy \in S$ implies $uy \approx vy$.

This is an equivalence relation, and no class meets both S and its complement.

Right congruence. Let $u \sim v$ and $x \in \mathcal{I}$. If $u, v \in S$ then $ux, vx \in S$ by right-closure and $ux \approx vx$ by stability, so $ux \sim vx$. Suppose $u, v \notin S$. For any $y \in \mathcal{I}^*$, applying (b) to the continuation xy gives $uxy \in S \iff vxy \in S$ and, when these hold, $uxy \approx vxy$. If $ux \notin S$ then $vx \notin S$ by (b) with $y = x$, and the displayed conditions are exactly (b) for the pair (ux, vx), so $ux \sim vx$. If $ux \in S$ then (b) with $y = x$ gives $vx \in S$ and $ux \approx vx$, so $ux \sim vx$ by (a).

Index. The classes inside S are exactly the $\approx$-classes, and those outside are the classes of (b); writing c_S for their number gives the stated count. If no history outside S has an extension inside S, then condition (b) is vacuous and all such histories form a single class, so $c_S \leq 1$. ☐

Remark 4.6 (The Extension Can Strictly Increase the Index)**.** The term c_S in Lemma 4.5 is not an artifact of the construction: it can be strictly positive even for right-closed S, so the equality $\kappa_{\text{obs}}(\delta, \mu) = \text{index}(\sim_{\delta,S})$ is false in general.

Let $\mathcal{I} = \{\mathtt{a}, \mathtt{b}\}$ and $S = \mathcal{I}^+$, the set of nonempty words, which is right-closed. Let the residual take two values $A \neq B$ and set

$$\delta(u) = \begin{cases} A, & u \text{ begins with } \mathtt{a}, \\ B, & u \text{ begins with } \mathtt{b}, \end{cases} \qquad u \in S,$$

with $\delta(\varepsilon)$ unconstrained since $\varepsilon \notin S$. For $u, v \in S$ every continuation uw, vw again lies in S and has the residual determined by the first letter, so $u \sim_{\delta,S} v$ exactly when u and v begin with the same letter:

$$\text{index}(\sim_{\delta,S}) = 2.$$

Now let $\sim$ be any right congruence on $\mathcal{I}^*$ admitting a representative map exact on S. The class of ε must send $\mathtt{a}$ into the A-class and $\mathtt{b}$ into the B-class. But the A-class sends *both* letters into the A-class, and the B-class sends both into the B-class, since $u\mathtt{a}$ and $u\mathtt{b}$ begin with the same letter as u. So ε has a transition pattern matching neither class and requires a third. Hence

$$\kappa_{\mathrm{obs}}(\delta, \mu) = 3 = \mathrm{index}(\sim_{\delta,S}) + 1,$$

attaining $c_S = 1$. A three-state witness is q_ε with $\tau(q_\varepsilon, \mathtt{a}) = q_A$, $\tau(q_\varepsilon, \mathtt{b}) = q_B$ and q_A, q_B absorbing.

Two consequences follow. First, the sandwich of Meta-Theorem 4.7(ii) is sharp at both ends. Second, the obstruction requires $\mu(\varepsilon) = 0$: if μ is a discounted-prefix law with $\mu(\varepsilon) > 0$, then $\varepsilon \in S$, and right-closure forces $S = \mathcal{I}^*$, whence $c_S = 0$ and the equality does hold.

Meta-Theorem 4.7 (Boolean Zero-Error Dichotomy)**.** *Let $\mathbb{T}$ be separated with $\mathcal{E}$ $\{0, \infty\}$-valued.*

(i) Worst-case aggregation. *If*

$$\Delta_{\mathbb{T}}^{\sup}(M) = \inf_{\mathrm{index}(\sim) \le M} \sup_{u \in \mathcal{I}^*} \mathcal{E}\big(\delta(u), \Psi([u])\big),$$

then

$$\Delta_{\mathbb{T}}^{\sup}(M) = 0 \iff \mathrm{index}(\sim_\delta) \le M.$$

(ii) Average-case aggregation. *If the aggregation is discounted-prefix with history law μ on $\mathcal{I}^*$, then*

$$\Delta_{\mathbb{T}}^{\mu}(M) = 0 \iff \kappa_{\mathrm{obs}}(\delta, \mu) \le M,$$

with κ_{obs} the observable support index of Definition 4.3. If μ has full support on $\mathcal{I}^*$ then $\kappa_{\mathrm{obs}}(\delta, \mu) = \mathrm{index}(\sim_\delta)$ and (ii) reduces to (i). If $S = \operatorname{supp} \mu$ is right-closed with positive mass, then

$$\mathrm{index}(\sim_{\delta,S}) \;\le\; \kappa_{\mathrm{obs}}(\delta, \mu) \;\le\; \mathrm{index}(\sim_{\delta,S}) + c_S,$$

where c_S is the number of additional classes required off S by Lemma 4.5; in particular $c_S = 0$ when $S = \mathcal{I}^*$, and equality $\kappa_{\mathrm{obs}}(\delta, \mu) = \mathrm{index}(\sim_{\delta,S})$ holds whenever the extension of Lemma 4.5 is index-preserving. For discounted-prefix laws the right-closed case never bites: since $\mu(\varepsilon) = 1 - \gamma > 0$ one has $\varepsilon \in S$, and right-closure then forces $S = \mathcal{I}^*$, whence $c_S = 0$ and the sandwich collapses to the equality (Remark 4.6). The sandwich is therefore informative only for average-case history laws whose support is not right-closed. In general the average-case index may be strictly smaller than the worst-case index, since a machine may err on μ-null histories at no cost.

Remark 4.8 (The Cesàro Case)**.** Clause (ii) is stated for discounted-prefix aggregation because that scheme carries a genuine probability measure on finite histories. Stationary Cesàro aggregation does not, so the phrase "μ-almost every history" has no direct meaning there and the clause must be reformulated.

Two formulations are available. On the *stationary state space*, when the process is a finite-state stationary ergodic chain with positive-mass state set $\mathcal{S}^+$, the natural statement replaces histories by states: zero Cesàro cost holds at budget M exactly when some right congruence of index at most M is exact on $\mathcal{S}^+$, which is Theorem 5.2 in the $\{0, \infty\}$-valued case. On *histories*, the correct surrogate for "μ-almost every" is *zero asymptotic frequency of error*: the Cesàro cost vanishes exactly when

$$\lim_{T\to\infty} \frac{1}{T} \left|\{t < T : \ \mathcal{E}(\delta(U_t), \Psi([U_t])) > 0\}\right| = 0$$

almost surely. For a finite-state ergodic chain the two agree, since a state of positive stationary mass is visited with positive asymptotic frequency. Outside that setting the second is the operative condition, and no formula in terms of a support-relative Myhill–Nerode relation is claimed.

Proof. (i) If $\mathrm{index}(\sim_\delta) \le M$, quotienting by $\sim_\delta$ and assigning each class its common residual gives an M-state approximation with zero cost at every history. Conversely, suppose an M-state approximation has zero worst-case cost. Then for *every* history u we have $\mathcal{E}(\delta(u), \Psi([u])) = 0$, so by separatedness $\delta(u) = \Psi([u])$. Hence any two histories in the same machine state have equal residuals, and, the state relation being a right congruence, equal residuals after every continuation; that is, the state relation refines $\sim_\delta$. Therefore $\mathrm{index}(\sim_\delta) \le \mathrm{index}(\sim) \le M$.

(ii) Suppose $\Delta^\mu_{\mathbb{T}}(M) = 0$. Then there are a right congruence $\sim$ of index at most M and a map Ψ with $\mathcal{E}(\delta(u), \Psi([u])) = 0$ for μ-almost every u. That is precisely the defining property in Definition 4.3, so $\kappa_{\mathrm{obs}}(\delta, \mu) \le M$.

Conversely, if $\kappa_{\mathrm{obs}}(\delta, \mu) \le M$, take a witnessing congruence and representative map; the induced M-state machine has zero μ-average cost by construction.

If μ has full support, "μ-almost every" is "every", and the argument of (i) applies verbatim, giving $\kappa_{\mathrm{obs}}(\delta, \mu) = \mathrm{index}(\sim_\delta)$.

If S is right-closed with positive mass, then for $u, v \in S$ the constraint is exactly that all continuations *within* S agree, which is $\sim_{\delta,S}$. Right-closure makes $\sim_{\delta,S}$ stable under the transition action in the sense of Lemma 4.5: if $u \sim_{\delta,S} v$ and $x \in \mathcal{I}$, then for every w with $uxw, vxw \in S$ the defining condition applied to the continuation xw gives $\delta(uxw) = \delta(vxw)$, so $ux \sim_{\delta,S} vx$.

For the lower bound, any global right congruence $\sim$ witnessing κ_{obs} must, restricted to S, refine $\sim_{\delta,S}$: two $\sim$-equivalent histories in S receive the same representative and hence the same residual after every continuation remaining in S. Therefore $\mathrm{index}(\sim_{\delta,S}) \le \kappa_{\mathrm{obs}}(\delta, \mu)$.

For the upper bound, apply Lemma 4.5 to $\approx = \sim_{\delta,S}$ to obtain a global right congruence restricting to it on S, of index $\mathrm{index}(\sim_{\delta,S}) + c_S$; assigning each class inside S its common residual gives zero cost μ-almost everywhere. This is the stated sandwich.

The extension step is not vacuous: a history outside S may have an extension inside S, so the classes off S cannot be merged arbitrarily without destroying the right-congruence property. Lemma 4.5 supplies the correct construction, and $c_S = 0$ exactly when it is index-preserving. $\square$

Remark 4.9 (Why the Qualifier Is Necessary)**.** The two clauses differ. Let δ be a residual trajectory requiring $\mathrm{index}(\sim_\delta) = k$ on all of $\mathcal{I}^*$, but suppose the input process never visits a set of histories whose removal collapses $\sim_\delta$ to $k' < k$ classes. Then a k'-state machine has zero average cost while failing to realize δ exactly, so the unqualified biconditional would be false. Statements elsewhere in this manuscript that invoke "zero error" should be read with clause (i) unless a stationary support is explicitly named, as in Theorem 5.2, which is a clause (ii) statement with $\operatorname{supp}\mu = \mathcal{S}^+$, so the relevant quantity there is κ_{obs} rather than the restriction of $\sim_\delta$ to the support.

Corollary 4.10 (Normalized $\{0, 1\}$ Commitment Cost)**.** *If*

$$\mathcal{E}(f, g) = \begin{cases} 0, & f = g, \\ 1, & f \neq g, \end{cases}$$

and the aggregation is worst-case, then

$$\Delta_{\mathrm{com}}(M) = \begin{cases} 0, & M \geq \operatorname{index}(\sim_\delta), \\ 1, & M < \operatorname{index}(\sim_\delta). \end{cases}$$

This corollary separates the structural Boolean valuation $\{0, \infty\}$ from the normalized operational valuation $\{0, 1\}$. The threshold is the same; only the positive mismatch value changes.

4.4 Exact Spectral Converse for Linear Hankel Gaps

Meta-Theorem 4.11 (Eckart–Young–Mirsky Spectral Converse)**.** *Let $\mathbb{T}$ be a Hilbert-module task theory with compact Hankel operator H_δ. Define the unrestricted linear rank-M gap by*

$$\Delta_{\mathbb{T}}^{\mathrm{unres}}(M) = \inf_{\operatorname{rank} B \leq M} \|H_\delta - B\|_{\mathrm{op}}.$$

Then

$$\Delta_{\mathbb{T}}^{\mathrm{unres}}(M) = \sigma_{M+1}(H_\delta).$$

Define the Hankel-structured finite-rank gap by

$$\Delta_{\mathbb{T}}^{\mathrm{Hank,str}}(M) = \inf_{\substack{\operatorname{rank} B \leq M \\ B \text{ Hankel}}} \|H_\delta - B\|_{\mathrm{op}}.$$

Then

$$\Delta_{\mathbb{T}}^{\mathrm{Hank,str}}(M) \geq \sigma_{M+1}(H_\delta),$$

with equality if an optimal rank-M truncation can be chosen Hankel, or under a valid Hardy-space Adamjan–Arov–Krein embedding satisfying its hypotheses.

Proof. The unrestricted equality is the Eckart–Young–Mirsky theorem for compact operators on Hilbert space:

$$\inf_{\operatorname{rank} B \leq M} \|H_\delta - B\|_{\mathrm{op}} = \sigma_{M+1}(H_\delta).$$

The Hankel-structured feasible set is a subset of the unrestricted feasible set, so restricting the infimum cannot decrease the value. Hence

$$\Delta_{\mathbb{T}}^{\mathrm{Hank,str}}(M) \geq \Delta_{\mathbb{T}}^{\mathrm{unres}}(M) = \sigma_{M+1}(H_\delta).$$

Equality requires that an unrestricted optimal rank-M approximant can be chosen Hankel, or that a separate Hankel approximation theorem applies. ☐

Remark 4.12 (AAK versus EYM)**.** The theorem above uses Eckart–Young–Mirsky [15, 16]. As already noted in Remark 3.12 for the Hilbert-module instantiation, the Adamjan–Arov–Krein theorem [12] applies only after a Hardy-space embedding has been separately established, and the model does not assume one by default.

Corollary 4.13 (Stable Residual-Norm Spectral Bound)**.** *Suppose a residual-cost functional satisfies*

$$\left\|h_\delta - h_{\widehat{\delta}}\right\|_{\infty,\rho} \leq \mathcal{L}_{\infty,\rho}(\delta, \widehat{\delta})$$

with $0 < \rho < 1/|\Sigma|$*. Then for every linear* M*-rank approximation* $\widehat{\delta}$ *with* $\operatorname{rank} H_{\widehat{\delta}} \leq M$*,*

$$\mathcal{L}_{\infty,\rho}(\delta, \widehat{\delta}) \geq (1 - |\Sigma|\rho)\sigma_{M+1}(H_\delta).$$

Proof. By Proposition 3.11,

$$\left\|H_\delta - H_{\widehat{\delta}}\right\|_{\mathrm{op}} \leq \frac{1}{1 - |\Sigma|\rho} \left\|h_\delta - h_{\widehat{\delta}}\right\|_{\infty,\rho}.$$

By Eckart–Young–Mirsky, since $\operatorname{rank} H_{\widehat{\delta}} \leq M$,

$$\left\|H_\delta - H_{\widehat{\delta}}\right\|_{\mathrm{op}} \geq \sigma_{M+1}(H_\delta).$$

Combining the two inequalities gives

$$\sigma_{M+1}(H_\delta) \leq \frac{1}{1 - |\Sigma|\rho} \left\|h_\delta - h_{\widehat{\delta}}\right\|_{\infty,\rho}.$$

Using the assumed domination of the weighted residual norm by the cost gives

$$\sigma_{M+1}(H_\delta) \leq \frac{1}{1 - |\Sigma|\rho} \mathcal{L}_{\infty,\rho}(\delta, \widehat{\delta}),$$

which rearranges to the stated bound. ☐

This corollary is a domination-bridge result. It does not assert that the residual-cost linear finite-rank gap equals the Hankel spectral tail unless the task cost is exactly the relevant operator or weighted residual norm and the feasible set is unrestricted.

4.5 Information-Bottleneck Meta-Theorem for Full KL

Meta-Theorem 4.14 (Finite-State Information Bottleneck)**.** *Let $S = S_t$, let $K = K_t = \phi(S_t)$ be a lumpable quotient, and let $Z = Z_t$ be the predictive future, by default the next output. For every lumpable quotient ϕ,*

$$\Delta_{\mathrm{ret}}^{\mathrm{KL}}(\phi) = I(S; Z \mid K).$$

Consequently,

$$\Delta_{\mathrm{ret}}^{\mathrm{KL}}(\phi) = I(S; Z) - I(K; Z).$$

Therefore,

$$\Delta_{\mathrm{ret}}^{\mathrm{KL}}(M) = I(S; Z) - \max_{\substack{\phi \ \textit{lumpable} \\ |\mathcal{K}| \le M}} I(K; Z).$$

Proof. Since $K = \phi(S)$, K is a function of S. Since S is sufficient for Z,

$$Z \perp\!\!\!\perp K \mid S.$$

Therefore,

$$I(K; Z \mid S) = 0.$$

For a fixed quotient K, the optimal representative in each block k is the conditional law $P_{Z|K=k}$, which equals the mixture centroid $\bar{P}_k$. Hence

$$\Delta_{\mathrm{ret}}^{\mathrm{KL}}(\phi) = \mathbb{E} D_{\mathrm{KL}}\big(P_{Z|S} \| P_{Z|K}\big) = I(S; Z \mid K).$$

By the chain rule,

$$I(S, K; Z) = I(K; Z) + I(S; Z \mid K).$$

But K is a function of S, so

$$I(S, K; Z) = I(S; Z).$$

The first identity $\Delta_{\mathrm{ret}}^{\mathrm{KL}}(\phi) = I(S; Z \mid K)$ holds in $[0, \infty]$ without restriction, regular conditional probabilities existing because Z is standard Borel. The subtracted form

$$I(S; Z \mid K) = I(S; Z) - I(K; Z)$$

requires $I(S; Z) < \infty$, since otherwise the right-hand side is the undefined difference $\infty - \infty$. In the present finite-state setting that condition is automatic: S takes finitely many values, so

$$I(S; Z) \le H(S) < \infty,$$

and $I(K; Z) \le I(S; Z) < \infty$ by data processing, so the rearrangement is always legitimate here. The statement carries the hypothesis explicitly because the identity is applied in the sequel to state sets that are finite by assumption rather than by construction. All variational statements below are to be read in the conditional-mutual-information form, which is unconditionally defined. Minimizing over lumpable quotients of size at most M is equivalent to maximizing $I(K; Z)$ over the same feasible set. $\square$

The identity is stated for the input-marginal target; its controlled analogue, conditional on the input and valid for general unifilar machines, is Theorem 5.32.

Corollary 4.15 (Lumpable Partition Form)**.** *If $\phi : \mathcal{S}^+ \to \mathcal{K}$ is a lumpable quotient with blocks $C_k = \{s \in \mathcal{S}^+ : \phi(s) = k\}$, then*

$$\Delta^{\mathrm{KL}}_{\mathrm{ret}}(M) = \min_{\substack{\phi:\mathcal{S}^+\to\mathcal{K} \text{ lumpable} \\ |\mathcal{K}|\le M}} \sum_{k\in\mathcal{K}} \sum_{s\in C_k} \pi_s D_{\mathrm{KL}}(P_s \| \bar{P}_k),$$

where

$$\bar{P}_k = \frac{\sum_{s\in C_k} \pi_s P_s}{\sum_{s\in C_k} \pi_s}.$$

Remark 4.16 (Role of the Right-Congruence Constraint)**.** The identity in Meta-Theorem 4.14 is the information bottleneck [24, 25] in native form. The schema constrains the optimizer: the maximum runs over finite-index right congruences in synchronized machines, or equivalently over lumpable deterministic quotients of the stationary support, rather than over arbitrary Markov encoders.

Remark 4.17 (Full KL versus Quadratic Surrogate)**.** The information-bottleneck identity is exact for the full-KL retention gap. The spectral covariance-tail bound proved later applies to the Gaussian quadratic restricted gap $\Delta^{\mathrm{quad}}_{\mathrm{ret}}(M)$, and locally to full KL under additional regularity assumptions. The full-KL centroid is the mixture distribution $\bar{P}_k$; the arithmetic mean centroid is a restricted Gaussian/Fisher phenomenon.

4.6 The Typed Principle

Meta-Theorem 4.18 (Typed Finite-State Rate–Distortion Principle)**.** *Fix a regime r specified by a rooted history right action H_{r} over an event alphabet $\mathcal{A}_{\mathsf{r}}$, a residual map $\delta_{\mathsf{r}} : H_{\mathsf{r}} \to \mathbf{R}_{\mathsf{r}}$, a reflexive cost $\mathcal{E}_{\mathsf{r}}$, a declared aggregation and support, and a class $\mathfrak{Q}_{\mathsf{r}}(M)$ of admissible finite-state factors of index at most M, with*

$$\Delta_{\mathsf{r}}(M) = \inf_{(Q,\Psi)\in\mathfrak{Q}_{\mathsf{r}}(M)} \mathcal{L}_{\mathsf{r}}\big(\delta_{\mathsf{r}}, \Psi \circ Q\big).$$

Then:

(a) $\Delta_{\mathsf{r}}(M)$ is nonincreasing in M (Meta-Theorem 4.1(i));

(b) exact zero distortion is governed by the least admissible factor through which δ_{r} factors on the declared support: for separated $\{0, \infty\}$-valued costs this is Meta-Theorem 4.1(iii), with the index taken relative to the support in the average-aggregation case;

(c) every quantitative converse beyond (b) requires a regime-specific analytical bridge; no such bound follows from the factorization syntax alone;

(d) if the regime admits a Schatten-p response representation with response operator $A_{\delta_{\mathsf{r}}}$, effective rank budget $r_{\mathsf{r}}(M)$ and modulus c_{r}, then

$$\Delta_{\mathsf{r}}(M) \;\ge\; c_{\mathsf{r}}^{-1}\, \Phi^{(p_{\mathsf{r}})}_{r_{\mathsf{r}}(M)}\big(A_{\delta_{\mathsf{r}}}\big)$$

(Theorem 8.11);

(e) *no comparison between two regimes follows without an explicit map identifying their history actions, admissible factors, costs and response operators; in the three principal regimes no such map exists (Theorem 8.22).*

In the three principal regimes the instantiation is:

regime	*admissible factor*	*exact invariant*	*analytical bridge*
commitment	*deterministic right congruence*	*Myhill–Nerode index*	*Boolean realizability*
retention	*lumpable quotient*	*Z-predictive support index*	*bottleneck / covariance*
grounding	*linear rank factor*	*Hankel rank (unrestricted gap)*	*Eckart–Young–Mirsky*

The grounding entry is stated for the unrestricted *gap* $\Delta_{\mathrm{grd}}^{\mathrm{unres}}$, *where* $\mathrm{rank}(H_\nu) = r < \infty$ *gives* $\Delta_{\mathrm{grd}}^{\mathrm{unres}}(M) = 0$ *for* $M \geq r$ *by Corollary 7.6. For the Hankel-structured gap* $\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}$ *only the inequality* $\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) \geq \sigma_{M+1}(H_\nu)$ *holds in general, and the corresponding exact statement needs the additional hypotheses of Theorems 7.9 and 7.11.*

Remark 4.19 (What the Principle Does and Does Not Unify)**.** The principle is a statement about shared *syntax* together with an explicit refusal to share *analysis*. Clauses (a) and (b) are genuinely regime-independent: monotonicity and the zero-distortion threshold follow from the factorization structure alone. Clause (d) is conditional on a representation being supplied, and the three regimes supply different operators — H_ν, $\mathbf{\Sigma}_\pi$, N_Λ — with different effective rank budgets M, $M-1$, M and unrelated moduli, which is precisely the content of clause (e).

Two things the principle deliberately does not assert. It does not claim a common history alphabet: commitment acts on $\mathcal{I}^*$ while retention in the unifilar generality of Definition 3.15 and grounding act on $(\mathcal{I} \times \mathcal{O})^*$, and the input-driven retention model of Definition 3.14 is the output-independent case in which the action factors through the input projection (Remark 3.16). And it does not assert a per-symbol rate: the resource is the quotient index, whose description length is $R(M) = \log M$ by Definition 9.2.

5 Retention

The retention regime approximates a stochastic predictive process by a finite-state stochastic predictor. The machine type is not a deterministic Mealy BSG. In general it is a stationary controlled unifilar predictor, and in the input-driven synchronized subclass a deterministic-transition, stochastic-emission lumped ϵ-machine. The cost is KL divergence between predictive distributions, and the structural constraint is lumpability of the stationary causal-state process, in the output-aware sense of Definition 3.21 for unifilar machines.

5.1 Retention Instantiation

In the retention instantiation, residual objects are probability distributions on standard Borel output spaces. In the synchronized finite-history subclass, the true trajectory is

$$\delta(u) = P_{\sigma(u)}$$

for sufficiently long histories u, where $\sigma(u)$ is the causal state associated with u. In the general stationary formulation, the residual at time t is the predictive distribution

$$P_{S_t} = \mathcal{L}(Z_t \mid S_t).$$

The cost profunctor is KL divergence:

$$\mathcal{E}(P, Q) = D_{\mathrm{KL}}(P \| Q).$$

The default aggregation is stationary average cost, either Cesàro or discounted; that finite prefixes do not define a probability measure over all lengths, and that this is what forces the two average-case schemes, is recorded once in Remark 2.5.

An M-state retention approximant is a stationary lumpable stochastic finite-state predictor. The machine type is a stationary controlled unifilar predictor (Definition 3.15); the input-driven synchronized subclass is the deterministic-transition case, and in it the approximant's states are lumped causal states. The transition structure is deterministic and must be lumpable with respect to the causal-state transition map τ. The emission associated with each machine state is a predictive distribution chosen to minimize stationary KL cost.

The full-KL retention gap is

$$\Delta_{\mathrm{ret}}^{\mathrm{KL}}(M) = \min_{\substack{\phi: \mathcal{S}^+ \to \mathcal{K} \text{ lumpable} \\ |\mathcal{K}| \le M}} \sum_{k \in \mathcal{K}} \sum_{s: \phi(s) = k} \pi_s D_{\mathrm{KL}}(P_s \| \bar{P}_k),$$

where

$$\bar{P}_k = \frac{\sum_{s:\phi(s)=k} \pi_s P_s}{\sum_{s:\phi(s)=k} \pi_s}.$$

When the quotient is synchronized, lumpable quotients of $\mathcal{S}^+$ correspond to finite-index right congruences on sufficiently long histories. The gap is then a constrained information-bottleneck objective over lumpable partitions of causal states.

5.2 Full-KL Retention

Lemma 5.1 (Mixture-Centroid Optimality)**.** *Let $P_1, \ldots, P_m$ be probability measures on a standard Borel space, and let $w_1, \ldots, w_m > 0$ with $\sum_i w_i = 1$. Define the mixture*

$$\bar{P} = \sum_{i=1}^{m} w_i P_i.$$

For any probability measure Q such that $\bar{P} \ll Q$,

$$\sum_{i=1}^{m} w_i D_{\mathrm{KL}}(P_i \| Q) = \sum_{i=1}^{m} w_i D_{\mathrm{KL}}(P_i \| \bar{P}) + D_{\mathrm{KL}}(\bar{P} \| Q).$$

Consequently,

$$\arg\min_{Q} \sum_{i=1}^{m} w_i D_{\mathrm{KL}}(P_i \| Q) = \bar{P},$$

and the minimum value is

$$\sum_{i=1}^{m} w_i D_{\mathrm{KL}}(P_i \| \bar{P}).$$

Proof. For each i,

$$D_{\mathrm{KL}}(P_i \| Q) = D_{\mathrm{KL}}(P_i \| \bar{P}) + \int \log \frac{d\bar{P}}{dQ}\, dP_i.$$

Multiplying by w_i and summing gives

$$\sum_i w_i D_{\mathrm{KL}}(P_i \| Q) = \sum_i w_i D_{\mathrm{KL}}(P_i \| \bar{P}) + \int \log \frac{d\bar{P}}{dQ}\, d\left(\sum_i w_i P_i \right).$$

The last integral is $D_{\mathrm{KL}}(\bar{P} \| Q)$. Since KL divergence is nonnegative and equals zero iff $\bar{P} = Q$, the unique minimizer is $Q = \bar{P}$. □

Theorem 5.2 (Zero Retention on Stationary Support)**.** *Assume the predictive distributions $\{P_s : s \in \mathcal{S}^+\}$ are pairwise distinct on the stationary support. Then*

$$\Delta_{\mathrm{ret}}^{\mathrm{KL}}(M) = 0 \iff M \geq |\mathcal{S}^+|.$$

For worst-case aggregation over all reachable states, replace $|\mathcal{S}^+|$ by the number of reachable states with pairwise distinct predictive distributions.

Proof. Zero KL cost requires that, for every block k,

$$P_s = \bar{P}_k$$

for every positive-mass state s in that block. If $P_s \neq P_t$ and $\pi_s, \pi_t > 0$, then merging s and t gives strictly positive KL cost by strict convexity of KL divergence in its second argument. Therefore zero cost requires every positive-mass predictive state to occupy its own block. The singleton partition is lumpable, so $M = |\mathcal{S}^+|$ suffices. Conversely, if $M < |\mathcal{S}^+|$, at least two distinct positive-mass predictive states are merged, giving positive cost. □

Example 5.3 (Parity)**.** Let $\mathcal{I} = \{0, 1\}$ with fair i.i.d. inputs, and let the output emit running input parity. There are two causal states with predictive distributions

$$P_0 = \delta_0, \qquad P_1 = \delta_1.$$

Then

$$\Delta^{\mathrm{KL}}_{\mathrm{ret}}(2) = 0.$$

The best one-state stochastic-emission approximation emits Bernoulli(q). Its stationary KL cost is

$$\frac{1}{2}D_{\mathrm{KL}}(\delta_0\|\mathrm{Bernoulli}(q)) + \frac{1}{2}D_{\mathrm{KL}}(\delta_1\|\mathrm{Bernoulli}(q)) = \frac{1}{2}[-\log q - \log(1-q)].$$

The minimum occurs at $q = 1/2$, giving

$$\Delta^{\mathrm{KL}}_{\mathrm{ret}}(1) = \log 2.$$

If one restricts the model to deterministic point-mass emissions, the one-state cost is infinite. That deterministic-emission model is not the stochastic-emission retention model studied here.

Lemma 5.4 (Coarsening Lemma)**.** *Let $\phi : \mathcal{S}^+ \to \mathcal{K}$ be a lumpable quotient with blocks*

$$C_k = \{s \in \mathcal{S}^+ : \phi(s) = k\}.$$

Then

$$\Delta^{\mathrm{KL}}_{\mathrm{ret}}(\phi) = \sum_{k\in\mathcal{K}}\sum_{s\in C_k} \pi_s D_{\mathrm{KL}}(P_s\|\bar{P}_k),$$

where

$$\bar{P}_k = \frac{\sum_{s\in C_k}\pi_s P_s}{\sum_{s\in C_k}\pi_s}.$$

Blocks of zero stationary mass are ignored.

Proof. For a fixed quotient, the stationary cost is

$$\sum_{k\in\mathcal{K}}\sum_{s\in C_k} \pi_s D_{\mathrm{KL}}(P_s\|Q_k),$$

where Q_k is the emission distribution assigned to block C_k. By Lemma 5.1, the minimizer over Q_k is the weighted mixture centroid $\bar{P}_k$. Substituting $\bar{P}_k$ gives the stated formula. □

Theorem 5.5 (Predictive-Information Decomposition)**.** *For every lumpable quotient ϕ,*

$$\Delta^{\mathrm{KL}}_{\mathrm{ret}}(\phi) = I(S; Z \mid K_\phi) = I(S;Z) - I(K_\phi; Z).$$

Consequently,

$$\Delta^{\mathrm{KL}}_{\mathrm{ret}}(M) = I(S;Z) - \max_{\substack{\phi \text{ lumpable} \\ |\mathcal{K}|\le M}} I(K_\phi; Z).$$

Proof. This is Meta-Theorem 4.14 specialized to the stationary controlled causal machine of Definition 3.14. □

Corollary 5.6 (Range, One-State Value, and Refinement Monotonicity)**.** *For every $M \geq 1$,*

$$0 \leq \Delta^{\mathrm{KL}}_{\mathrm{ret}}(M) \leq I(S; Z).$$

Moreover,

$$\Delta^{\mathrm{KL}}_{\mathrm{ret}}(1) = I(S; Z).$$

If ϕ and ψ are lumpable quotients and ϕ refines ψ, then

$$\Delta^{\mathrm{KL}}_{\mathrm{ret}}(\phi) \leq \Delta^{\mathrm{KL}}_{\mathrm{ret}}(\psi).$$

Consequently, the optimized gap $\Delta^{\mathrm{KL}}_{\mathrm{ret}}(M)$ is nonincreasing in M.

Proof. Since K_ϕ is a function of S,

$$0 \leq I(K_\phi; Z) \leq I(S; Z),$$

and the range follows from Theorem 5.5. For $M = 1$, the only lumpable quotient is constant, so $I(K_\phi; Z) = 0$, giving $\Delta^{\mathrm{KL}}_{\mathrm{ret}}(1) = I(S; Z)$. If ϕ refines ψ, then K_ψ is a function of K_ϕ, so the data-processing inequality gives

$$I(K_\psi; Z) \leq I(K_\phi; Z),$$

and therefore $\Delta^{\mathrm{KL}}_{\mathrm{ret}}(\phi) \leq \Delta^{\mathrm{KL}}_{\mathrm{ret}}(\psi)$. Monotonicity in M follows because the feasible set of quotients with at most M blocks is nested in M. □

Remark 5.7 (Role of the Right-Congruence Constraint in Retention)**.** Theorem 5.5 is the information bottleneck [24, 25] in native form, and what the schema contributes is the constraint on the optimizer, exactly as recorded in Remark 4.16: the maximum runs over finite-index right congruences in synchronized machines, equivalently over lumpable deterministic quotients of the stationary support, rather than over arbitrary Markov encoders.

5.3 Quadratic Restricted Spectral Converse

The spectral retention converse applies to the Gaussian quadratic restricted regime of Definition 3.31. It is not a global theorem for unrestricted full KL.

Let y_s denote Fisher-scaled predictive features for positive-mass states $s \in \mathcal{S}^+$. In the common-covariance Gaussian family with covariance $\frac{1}{2}I$, one may take

$$y_s = \sqrt{2}\, m_s,$$

so that

$$D_{\mathrm{KL}}(P_s \| Q_c) = \frac{1}{2} \left\| y_s - c_y \right\|^2.$$

Define the stationary Fisher covariance

$$\mathbf{\Sigma}_\pi = \sum_{s \in \mathcal{S}^+} \pi_s (y_s - \bar{y})(y_s - \bar{y})^\top, \qquad \bar{y} = \sum_{s \in \mathcal{S}^+} \pi_s y_s.$$

In mean coordinates m_s, the same covariance is scaled by a factor of 2. The quadratic gap in mean coordinates is

$$\sum_k \sum_{s:\phi(s)=k} \pi_s \left\| m_s - \bar{m}_k \right\|^2,$$

while in Fisher coordinates it is

$$\frac{1}{2} \sum_k \sum_{s:\phi(s)=k} \pi_s \left\| y_s - \bar{y}_k \right\|^2.$$

The spectral statement below uses the Fisher-coordinate covariance $\mathbf{\Sigma}_\pi$.

Theorem 5.8 (Spectral Converse for Quadratic Restricted Retention)**.** *In the Gaussian quadratic restricted regime, for every $M \geq 1$,*

$$\Delta_{\text{ret}}^{\text{quad}}(M) \geq \frac{1}{2} \sum_{i \geq M} \lambda_i(\mathbf{\Sigma}_\pi),$$

where the eigenvalues are ordered nonincreasingly. Equivalently,

$$\Delta_{\text{ret}}^{\text{quad}}(M) \geq \frac{1}{2} \Phi_{M-1}^{(1)}(\mathbf{\Sigma}_\pi).$$

Proof. In Fisher coordinates, the quadratic gap is weighted k-means:

$$\Delta_{\text{ret}}^{\text{quad}}(M) = \min_\phi \frac{1}{2} \sum_k \sum_{s:\phi(s)=k} \pi_s \left\| y_s - c_k \right\|^2.$$

For any lumpable quotient ϕ with $K \leq M$ cells, let

$$\Pi_k = \sum_{s:\phi(s)=k} \pi_s, \qquad m_k = \Pi_k^{-1} \sum_{s:\phi(s)=k} \pi_s y_s.$$

Completing the square inside each cluster gives

$$\sum_{s:\phi(s)=k} \pi_s \left\| y_s - c_k \right\|^2 = \sum_{s:\phi(s)=k} \pi_s \left\| y_s - m_k \right\|^2 + \Pi_k \left\| m_k - c_k \right\|^2.$$

Therefore

$$\frac{1}{2} \sum_k \sum_{s:\phi(s)=k} \pi_s \left\| y_s - c_k \right\|^2 \geq \frac{1}{2} \sum_k \sum_{s:\phi(s)=k} \pi_s \left\| y_s - m_k \right\|^2.$$

Define the within-cluster covariance

$$W = \sum_k \sum_{s:\phi(s)=k} \pi_s (y_s - m_k)(y_s - m_k)^\top$$

and the between-cluster covariance

$$B = \sum_k \Pi_k (m_k - \bar{y})(m_k - \bar{y})^\top.$$

The total covariance decomposes as

$$\mathbf{\Sigma}_\pi = W + B.$$

Hence

$$\frac{1}{2}\sum_k \sum_{s:\phi(s)=k} \pi_s \|y_s - m_k\|^2 = \frac{1}{2}\operatorname{tr}(W) = \frac{1}{2}\operatorname{tr}(\mathbf{\Sigma}_\pi - B).$$

Because

$$\sum_k \Pi_k(m_k - \bar{y}) = 0,$$

the centered cluster-mean vectors satisfy one linear relation, so

$$\operatorname{rank}(B) \le K - 1 \le M - 1.$$

By Ky Fan's maximum principle [17],

$$\operatorname{tr}(B) \le \sum_{i=1}^{M-1} \lambda_i(\mathbf{\Sigma}_\pi).$$

Therefore

$$\Delta_{\text{ret}}^{\text{quad}}(M) \ge \frac{1}{2}\operatorname{tr}(\mathbf{\Sigma}_\pi) - \frac{1}{2}\sum_{i=1}^{M-1} \lambda_i(\mathbf{\Sigma}_\pi) = \frac{1}{2}\sum_{i\ge M} \lambda_i(\mathbf{\Sigma}_\pi).$$

□

Corollary 5.9 (Trace Form and Equality Condition)**.** *For any lumpable quotient ϕ,*

$$\Delta_{\text{ret}}^{\text{quad}}(\phi) = \frac{1}{2}\operatorname{tr}(\mathbf{\Sigma}_\pi) - \frac{1}{2}\operatorname{tr}(B_\phi),$$

where

$$B_\phi = \sum_{k\in\mathcal{K}} \Pi_k(m_k - \bar{y})(m_k - \bar{y})^\top,$$

with

$$\Pi_k = \sum_{s:\phi(s)=k} \pi_s, \qquad m_k = \Pi_k^{-1} \sum_{s:\phi(s)=k} \pi_s y_s.$$

Consequently,

$$\Delta_{\text{ret}}^{\text{quad}}(M) = \frac{1}{2}\operatorname{tr}(\mathbf{\Sigma}_\pi) - \frac{1}{2}\max_{\substack{\phi \text{ lumpable} \\ |\mathcal{K}|\le M}} \operatorname{tr}(B_\phi).$$

The lower bound in Theorem 5.8 is tight iff there exists a lumpable quotient ϕ with $|\mathcal{K}| \le M$ such that

$$\operatorname{tr}(B_\phi) = \sum_{i=1}^{M-1} \lambda_i(\mathbf{\Sigma}_\pi).$$

Equivalently, the lumpable quotient realizes the constrained Ky Fan optimum capturing the top $M-1$ Fisher covariance directions.

Proof. The trace identity is the ANOVA decomposition used in the proof of Theorem 5.8. Maximizing $\mathrm{tr}(B_\phi)$ over lumpable quotients gives the stated formula for $\Delta_{\mathrm{ret}}^{\mathrm{quad}}(M)$. The equality condition is exactly the condition that the constrained maximizer attains the Ky Fan upper bound. $\square$

Remark 5.10 (Scope of the Spectral Theorem)**.** The theorem is a statement about $\Delta_{\mathrm{ret}}^{\mathrm{quad}}(M)$, not about $\Delta_{\mathrm{ret}}^{\mathrm{KL}}(M)$. For unrestricted full KL, the centroid is the mixture distribution $\bar{P}_k$, and the quadratic spectral bound can be loose or can exceed the total predictive information if interpreted as a full-KL bound.

Remark 5.11 (Tightness)**.** The bound is tight for the quadratic restricted gap whenever an optimal M-means solution splits along the top $M-1$ eigenvectors of $\mathbf{\Sigma}_\pi$. No tightness claim is made for full KL.

Lemma 5.12 (Uniform Second-Order Expansion of Mixture KL)**.** *Let $\{P_s^{(r)}\}_{s\in\mathcal{S}}$ be a finite family of distributions on a common measurable space, indexed by $r>0$. Suppose there exist a base probability measure P_0, functions $f_s \in L^2(P_0)$, and constants $C, C_3 < \infty$ such that, uniformly in s,*

$$\frac{dP_s^{(r)}}{dP_0} = 1 + r f_s + O(r^2),$$

with

$$\int f_s \, dP_0 = 0, \qquad \|f_s\|_{L^3(P_0)} \le C,$$

and third-order remainders bounded by $C_3 r^3$ in KL expansion. For any block $C \subseteq \mathcal{S}$ and weights

$$w_s = \frac{\pi_s}{\sum_{t\in C}\pi_t},$$

define the block mixture

$$\bar{P}_C^{(r)} = \sum_{s\in C} w_s P_s^{(r)}.$$

Then uniformly over all blocks,

$$\sum_{s\in C} w_s D_{\mathrm{KL}}(P_s^{(r)} \| \bar{P}_C^{(r)}) = \frac{r^2}{2}\sum_{s\in C} w_s \|f_s - \bar{f}_C\|^2_{L^2(P_0)} + O(r^3),$$

where

$$\bar{f}_C = \sum_{s\in C} w_s f_s.$$

If the family is a smooth exponential family with natural-parameter perturbations

$$\eta_s^{(r)} = \eta_0 + r\theta_s + o(r),$$

then

$$\|f_s - f_t\|^2_{L^2(P_0)} = \|\theta_s - \theta_t\|^2_{I_0} + o(1),$$

where I_0 is the Fisher information matrix at η_0.

Proof. Write densities

$$q_s^{(r)} = 1 + r f_s + O(r^2), \qquad q_C^{(r)} = 1 + r \bar{f}_C + O(r^2).$$

Using

$$\log(1+x) = x - \frac{x^2}{2} + O(x^3),$$

we obtain

$$\log \frac{q_s^{(r)}}{q_C^{(r)}} = r(f_s - \bar{f}_C) - \frac{r^2}{2}(f_s^2 - \bar{f}_C^2) + O(r^3).$$

Multiplying by $q_s^{(r)} = 1 + r f_s + O(r^2)$ gives

$$q_s^{(r)} \log \frac{q_s^{(r)}}{q_C^{(r)}} = r(f_s - \bar{f}_C) + r^2 \left[f_s(f_s - \bar{f}_C) - \frac{1}{2}(f_s^2 - \bar{f}_C^2) \right] + O(r^3).$$

Integrating against P_0, the first-order term vanishes because

$$\int (f_s - \bar{f}_C)\, dP_0 = 0.$$

The second-order term simplifies:

$$\int \left[f_s(f_s - \bar{f}_C) - \frac{1}{2}(f_s^2 - \bar{f}_C^2) \right] dP_0 = \frac{1}{2} \int (f_s - \bar{f}_C)^2\, dP_0.$$

Therefore

$$D_{\mathrm{KL}}(P_s^{(r)} \| \bar{P}_C^{(r)}) = \frac{r^2}{2} \|f_s - \bar{f}_C\|_{L^2(P_0)}^2 + O(r^3).$$

Multiplying by w_s and summing over $s \in C$ gives the block expansion.

Because $\mathcal{S}$ is finite and M is fixed, the number of lumpable partitions is finite. Hence the $O(r^3)$ remainder is uniform over all lumpable partitions of size at most M.

For a smooth exponential family, the score directions associated with natural-parameter perturbations θ_s have $L^2(P_0)$ inner product

$$\langle \theta_s, \theta_t \rangle_{I_0} = \theta_s^\top I_0 \theta_t.$$

Thus the $L^2(P_0)$ variance equals the Fisher-coordinate variance. $\square$

Corollary 5.13 (Uniform Fisher Remainder)**.** *Under the hypotheses of Theorem 5.14, stated immediately below, there exists a constant $C < \infty$ such that, for all sufficiently small $r > 0$ and all lumpable quotients ϕ with $|\mathcal{K}| \le M$,*

$$\left| \Delta_{\mathrm{ret}}^{\mathrm{KL},(r)}(\phi) - \frac{r^2}{2} \operatorname{tr}(\boldsymbol{\Sigma}_\pi - B_\phi) \right| \le C r^3,$$

where $\Delta_{\mathrm{ret}}^{\mathrm{KL},(r)}(\phi)$ denotes the full-KL cost of quotient ϕ in the perturbed family. Consequently,

$$\Delta_{\mathrm{ret}}^{\mathrm{KL},(r)}(M) = \frac{r^2}{2} \min_{\substack{\phi \text{ lumpable} \\ |\mathcal{K}| \le M}} \operatorname{tr}(\boldsymbol{\Sigma}_\pi - B_\phi) + O(r^3).$$

Proof. This follows from Lemma 5.12 and the finiteness of the set of lumpable quotients with at most M blocks. Uniformity over that finite set converts the blockwise $O(r^3)$ remainders into a uniform $O(r^3)$ remainder for the optimized gap. $\square$

Theorem 5.14 (Local Fisher Asymptotic Equality)**.** *Let $\{P_s^{(r)}\}_{s\in\mathcal{S}}$ be a family of distributions in a smooth finite-dimensional exponential family, indexed by $r > 0$. Suppose:*

(1) the natural parameters satisfy

$$\eta_s^{(r)} = \eta_0 + r\theta_s + o(r)$$

uniformly in s;

(2) the Fisher information matrix

$$I_0 = I(\eta_0)$$

is positive definite;

(3) the stationary weights satisfy

$$\pi_s > 0, \qquad \sum_s \pi_s = 1;$$

(4) third derivatives of the log-partition function are uniformly bounded in a neighborhood of η_0;

(5) the state set $\mathcal{S}$ and budget M are fixed.

Define Fisher-scaled local features

$$y_s = I_0^{1/2}\theta_s,$$

and the stationary local covariance

$$\mathbf{\Sigma}_\pi = \sum_s \pi_s(y_s - \bar{y})(y_s - \bar{y})^\top, \qquad \bar{y} = \sum_s \pi_s y_s.$$

Then, as $r \downarrow 0$,

$$\Delta_{\mathrm{ret}}^{\mathrm{KL},(r)}(M) = \frac{r^2}{2} \min_{\substack{\phi \ \textit{lumpable} \\ |\mathcal{K}|\le M}} \operatorname{tr}(\mathbf{\Sigma}_\pi - B_\phi) + o(r^2),$$

where

$$B_\phi = \sum_{k\in\mathcal{K}} \Pi_k (m_k - \bar{y})(m_k - \bar{y})^\top,$$

with

$$\Pi_k = \sum_{s:\phi(s)=k} \pi_s, \qquad m_k = \Pi_k^{-1} \sum_{s:\phi(s)=k} \pi_s y_s.$$

Consequently,

$$\Delta_{\mathrm{ret}}^{\mathrm{KL},(r)}(M) \ge \frac{r^2}{2} \sum_{i\ge M} \lambda_i(\mathbf{\Sigma}_\pi) + o(r^2).$$

If a lumpable partition attains the constrained PCA optimum capturing the top $M-1$ eigenvalues, then equality holds in the leading term:

$$\Delta_{\mathrm{ret}}^{\mathrm{KL},(r)}(M) = \frac{r^2}{2} \sum_{i \geq M} \lambda_i(\mathbf{\Sigma}_\pi) + o(r^2).$$

Proof. KL divergence agrees with the Fisher quadratic form to second order:

$$D_{\mathrm{KL}}(P_s^{(r)} \| P_t^{(r)}) = \frac{1}{2} \left\| \eta_s - \eta_t \right\|_{I_0}^2 + O(r^3),$$

where η_s are local Fisher coordinates and I_0 is the Fisher information at the base point. The assumptions give bounded third derivatives in a neighborhood of the base parameter.

Because the state set and the budget M are fixed, there are only finitely many lumpable quotients with at most M blocks. The $O(r^3)$ remainder is therefore uniform over all such quotients. Hence

$$\Delta_{\mathrm{ret}}^{\mathrm{KL},(r)}(M) = \frac{r^2}{2} \min_{\phi} \mathrm{tr}(\mathbf{\Sigma}_\pi - B_\phi) + o(r^2).$$

Since $\mathrm{rank}(B_\phi) \leq M-1$, Ky Fan's maximum principle [17] gives

$$\mathrm{tr}(B_\phi) \leq \sum_{i=1}^{M-1} \lambda_i(\mathbf{\Sigma}_\pi),$$

and therefore

$$\mathrm{tr}(\mathbf{\Sigma}_\pi - B_\phi) \geq \sum_{i \geq M} \lambda_i(\mathbf{\Sigma}_\pi).$$

This yields the spectral-tail lower bound. If a lumpable quotient attains the constrained PCA optimum, equality holds in the leading term. □

Remark 5.15 (Scope of the Local Fisher Theorem)**.** Theorem 5.14 is a small-radius asymptotic statement about the family $\{P_s^{(r)}\}$ as $r \downarrow 0$. It is not a global spectral equality for an arbitrary fixed full-KL retention instance.

Theorem 5.16 (Global Full-KL Spectral Converse in Probability Coordinates)**.** *Let $\mathcal{O}$ be finite and let each state $s \in \mathcal{S}^+$ carry the predictive probability vector $p_s \in \Delta^{|\mathcal{O}|-1}$. Put*

$$\bar{p} = \sum_{s \in \mathcal{S}^+} \pi_s p_s, \qquad \mathbf{\Sigma}_p = \sum_{s \in \mathcal{S}^+} \pi_s (p_s - \bar{p})(p_s - \bar{p})^\top.$$

Then for every lumpable quotient ϕ with at most M blocks,

$$\Delta_{\mathrm{ret}}^{\mathrm{KL}}(\phi) \ \geq \ \sum_{i \geq M} \lambda_i(\mathbf{\Sigma}_p),$$

and consequently

$$\Delta_{\mathrm{ret}}^{\mathrm{KL}}(M) \ \geq \ \sum_{i \geq M} \lambda_i(\mathbf{\Sigma}_p).$$

The bound is global: no small-radius, local-asymptotic, or regularity hypothesis is used, and $\mathbf{\Sigma}_p$ *is the covariance of the predictive* probability vectors*, not the Fisher covariance. The constant* 1 *is the largest possible: by Proposition 5.17 there are instances on which the ratio of the two sides tends to* 1*.*

Proof. Step 1: a quadratic minorant for KL, with the sharp constant. Pinsker's inequality gives $D_{\mathrm{KL}}(p\|q) \geq \frac{1}{2}\|p-q\|_1^2$. The crude bound $\|\cdot\|_2 \leq \|\cdot\|_1$ would lose a factor of two here; on the simplex a strictly better comparison is available, because the difference of two probability vectors is a *centred* vector.

Claim. If $p, q \in \Delta^{|\mathcal{O}|-1}$ and $\delta = p - q$, then $\frac{1}{2}\|\delta\|_1^2 \geq \|\delta\|_2^2$.

Indeed $\sum_i \delta_i = 0$, so the positive and negative parts of δ carry equal mass: writing $a = \sum_i (\delta_i)_+ = \sum_i (\delta_i)_-$ one has $\|\delta\|_1 = 2a$. Moreover $\|\delta\|_\infty \leq a$, since a single coordinate cannot exceed the total mass of its own sign class. Hence

$$\|\delta\|_2^2 \;\leq\; \|\delta\|_\infty \|\delta\|_1 \;\leq\; a \cdot 2a \;=\; 2a^2 \;=\; \tfrac{1}{2}\|\delta\|_1^2,$$

which is the claim. Combining it with Pinsker,

$$D_{\mathrm{KL}}(p\|q) \;\geq\; \tfrac{1}{2}\|p-q\|_1^2 \;\geq\; \|p-q\|_2^2 \tag{$*$}$$

for all $p, q \in \Delta^{|\mathcal{O}|-1}$, including boundary points, where both sides are interpreted in $[0, \infty]$.

Only $(*)$ is used below. Two remarks on why it is stated in this form. First, the centring step is what produces the constant 1 rather than $\frac{1}{2}$, and by Proposition 5.17 the constant 1 cannot be improved. Second, on the *interior* of the simplex the weaker inequality $D_{\mathrm{KL}}(p\|q) \geq \frac{1}{2}\|p-q\|_2^2$ is the statement that the negative entropy $\Phi(p) = \sum_i p_i \log p_i$, whose Bregman divergence is D_{KL}, has Hessian $\operatorname{diag}(1/p_i) \succeq I$ and is therefore 1-strongly convex in $\|\cdot\|_2$. That formulation is not available on the boundary, where the Hessian is undefined, and it is in any case lossy by the factor just recovered; this is why the proof is routed through Pinsker together with the centring claim rather than through strong convexity.

Step 2: mixture centroids. By Lemma 5.1 the block representative minimizing the π-weighted KL cost of a block C is the mixture centroid $\bar{p}_C = \big(\sum_{s\in C} \pi_s p_s\big)/\pi(C)$. Hence, writing ϕ for the partition into blocks $C_1, \ldots, C_M$,

$$\Delta_{\mathrm{ret}}^{\mathrm{KL}}(\phi) = \sum_k \sum_{s \in C_k} \pi_s\, D_{\mathrm{KL}}(p_s \| \bar{p}_{C_k}) \overset{(*)}{\geq} \sum_k \sum_{s\in C_k} \pi_s \|p_s - \bar{p}_{C_k}\|_2^2 = \operatorname{tr}(W_\phi),$$

where $W_\phi = \sum_k \sum_{s \in C_k} \pi_s (p_s - \bar{p}_{C_k})(p_s - \bar{p}_{C_k})^\top$ is the within-block covariance.

Step 3: ANOVA and Ky Fan. The usual variance decomposition gives $\mathbf{\Sigma}_p = W_\phi + B_\phi$ with $B_\phi = \sum_k \pi(C_k)(\bar{p}_{C_k} - \bar{p})(\bar{p}_{C_k} - \bar{p})^\top$ the between-block covariance. The M centroids satisfy the single affine relation $\sum_k \pi(C_k)(\bar{p}_{C_k} - \bar{p}) = 0$, so $\operatorname{rank}(B_\phi) \leq M - 1$. Both $\mathbf{\Sigma}_p$ and B_ϕ are positive semidefinite and symmetric, so Ky Fan's maximum principle [17] applies to eigenvalues and gives

$$\operatorname{tr}(B_\phi) \;\leq\; \sum_{i=1}^{M-1} \lambda_i(\mathbf{\Sigma}_p).$$

Therefore

$$\operatorname{tr}(W_\phi) = \operatorname{tr}(\boldsymbol{\Sigma}_p) - \operatorname{tr}(B_\phi) \;\ge\; \sum_{i \ge M} \lambda_i(\boldsymbol{\Sigma}_p),$$

and combining with Step 2 yields the claim. Taking the infimum over quotients with at most M blocks gives the statement for $\Delta_{\mathrm{ret}}^{\mathrm{KL}}(M)$. □

Proposition 5.17 (Optimality of the Constant)**.** *The constant* 1 *in Theorem 5.16 is the largest universal constant. Explicitly, let $\mathcal{O} = \{0,1\}$, let two predictive states carry weight $\frac{1}{2}$ each with*

$$p_\pm = \Big(\tfrac{1}{2} \pm \varepsilon,\ \tfrac{1}{2} \mp \varepsilon\Big), \qquad 0 < \varepsilon < \tfrac{1}{2},$$

and take $M = 1$. Then

$$\sum_{i \ge 1} \lambda_i(\boldsymbol{\Sigma}_p) = \operatorname{tr}(\boldsymbol{\Sigma}_p) = 2\varepsilon^2, \qquad \Delta_{\mathrm{ret}}^{\mathrm{KL}}(1) = 2\varepsilon^2 + O(\varepsilon^4),$$

so

$$\frac{\Delta_{\mathrm{ret}}^{\mathrm{KL}}(1)}{\sum_{i \ge 1} \lambda_i(\boldsymbol{\Sigma}_p)} \;\longrightarrow\; 1 \qquad (\varepsilon \downarrow 0).$$

Consequently no constant $c > 1$ satisfies $\Delta_{\mathrm{ret}}^{\mathrm{KL}}(M) \ge c \sum_{i \ge M} \lambda_i(\boldsymbol{\Sigma}_p)$ for all instances.

Proof. The stationary mixture is $\bar{p} = (\frac{1}{2}, \frac{1}{2})$, and $p_\pm - \bar{p} = \pm(\varepsilon, -\varepsilon)$, so

$$\boldsymbol{\Sigma}_p = \tfrac{1}{2}\begin{pmatrix} \varepsilon \\ -\varepsilon \end{pmatrix}\begin{pmatrix} \varepsilon & -\varepsilon \end{pmatrix} + \tfrac{1}{2}\begin{pmatrix} -\varepsilon \\ \varepsilon \end{pmatrix}\begin{pmatrix} -\varepsilon & \varepsilon \end{pmatrix} = \begin{pmatrix} \varepsilon^2 & -\varepsilon^2 \\ -\varepsilon^2 & \varepsilon^2 \end{pmatrix},$$

whose trace is $2\varepsilon^2$. With $M = 1$ the only quotient is the trivial one, whose representative is the mixture centroid $\bar{p}$ (Lemma 5.1), so

$$\Delta_{\mathrm{ret}}^{\mathrm{KL}}(1) = \tfrac{1}{2} D_{\mathrm{KL}}(p_+ \| \bar{p}) + \tfrac{1}{2} D_{\mathrm{KL}}(p_- \| \bar{p}) = \log 2 - h\big(\tfrac{1}{2} + \varepsilon\big),$$

h denoting the binary entropy in nats. Expanding $h(\frac{1}{2} + \varepsilon) = \log 2 - 2\varepsilon^2 - \frac{4}{3}\varepsilon^4 + O(\varepsilon^6)$ gives $\Delta_{\mathrm{ret}}^{\mathrm{KL}}(1) = 2\varepsilon^2 + \frac{4}{3}\varepsilon^4 + O(\varepsilon^6)$, and the stated ratio follows. □

Remark 5.18 (Where the Factor of Two Was Lost)**.** The constant is governed entirely by the quadratic minorant $(*)$. Bounding $\|\delta\|_2 \le \|\delta\|_1$ is valid on all of $\mathbb{R}^{|\mathcal{O}|}$ but ignores that differences of probability vectors are centred; the centring argument in Step 1 of Theorem 5.16 recovers the missing factor, and Proposition 5.17 shows that nothing further is available. The extremal configuration is the one that makes both inequalities in $(*)$ asymptotically tight at once: two antipodal binary perturbations of the uniform law, for which Pinsker is tight to leading order and δ has exactly two nonzero coordinates of equal magnitude.

Remark 5.19 (Why This Does Not Contradict the Fisher No-Go)**.** Theorem 5.16 is stated in the covariance $\boldsymbol{\Sigma}_p$ of the predictive *probability vectors*. The Fisher covariance $\boldsymbol{\Sigma}_\pi$ of Theorem 5.14 is a different object, computed in natural parameters, and the two are comparable only away from the simplex boundary.

A bi-Lipschitz reparameterization does not by itself give eigenvalue-by- eigenvalue comparability of the two covariance matrices, so the transfer is not immediate. What does hold is a converse stated directly in a fixed natural-parameter chart, by strong convexity of the log-partition function rather than by comparing covariances; this is Theorem 5.20.

Note also that "the natural-parameter Fisher covariance" is not canonical for a global family: it depends on a choice of minimal chart and, if one wishes a Fisher-weighted form, on a reference point. Fixing the natural parameter of the stationary mixture $\bar{p}$ and setting $\mathbf{\Sigma}_F = I(\eta_{\bar{p}})^{1/2}\mathbf{\Sigma}_\eta I(\eta_{\bar{p}})^{1/2}$ is one such choice. Theorem 5.23 is stated against a fixed chart in this sense, and rules out a universal constant for either form.

Theorem 5.20 (Global Interior Fisher-Coordinate Converse)**.** *Let $\mathcal{O}$ be finite and write the strictly positive laws on $\mathcal{O}$ in a fixed minimal natural-parameter chart*

$$p_\eta(y) = \exp\{\langle \eta, T(y)\rangle - A(\eta)\}.$$

Let K be a compact convex set of natural parameters containing the parameters η_s of all predictive laws and the parameters of all their mixture centroids*, and suppose*

$$\nabla^2 A(\eta) \ \succeq \ m_K I \qquad \textit{for all } \eta \in K,$$

for some $m_K > 0$. Put $\bar{\eta} = \sum_s \pi_s \eta_s$ and

$$\mathbf{\Sigma}_\eta = \sum_s \pi_s(\eta_s - \bar{\eta})(\eta_s - \bar{\eta})^\top.$$

Then

$$\Delta^{\mathrm{KL}}_{\mathrm{ret}}(M) \ \geq \ \frac{m_K}{2} \sum_{i \geq M} \lambda_i(\mathbf{\Sigma}_\eta).$$

Proof. Fix a block C with weights $w_s = \pi_s/\pi(C)$, mixture centroid $\bar{p}_C$, and let $\zeta_C \in K$ be its natural parameter. For an exponential family the Kullback–Leibler divergence is the Bregman divergence of A,

$$D_{\mathrm{KL}}(p_{\eta_s} \| p_{\zeta_C}) = B_A(\zeta_C, \eta_s),$$

so m_K-strong convexity of A on K gives

$$D_{\mathrm{KL}}(p_{\eta_s} \| p_{\zeta_C}) \ \geq \ \frac{m_K}{2}\|\eta_s - \zeta_C\|_2^2.$$

The hypothesis that K contains the centroid parameters is what places the segment $[\eta_s, \zeta_C]$ inside K; it is not implied by $K \supseteq \{\eta_s\}$, since $\eta \mapsto p_\eta$ is nonlinear and a mixture centroid need not have its parameter in the convex hull of the η_s.

Writing $\bar{\eta}_C = \sum_{s \in C} w_s \eta_s$ for the *parameter* mean and using that $\zeta \mapsto \sum_{s\in C} w_s \|\eta_s - \zeta\|_2^2$ is minimized at $\zeta = \bar{\eta}_C$,

$$\sum_{s \in C} w_s D_{\mathrm{KL}}(p_{\eta_s} \| \bar{p}_C) \ \geq \ \frac{m_K}{2} \sum_{s \in C} w_s \|\eta_s - \zeta_C\|_2^2 \ \geq \ \frac{m_K}{2} \sum_{s \in C} w_s \|\eta_s - \bar{\eta}_C\|_2^2.$$

Multiplying by $\pi(C)$ and summing over blocks gives $\Delta_{\text{ret}}^{\text{KL}}(\phi) \geq \frac{m_K}{2}\operatorname{tr}(W_\phi)$ with W_ϕ the within-block covariance of the η_s. The ANOVA decomposition $\mathbf{\Sigma}_\eta = W_\phi + B_\phi$ with $\operatorname{rank}(B_\phi) \leq M-1$ and Ky Fan's principle [17] give $\operatorname{tr}(W_\phi) \geq \sum_{i\geq M} \lambda_i(\mathbf{\Sigma}_\eta)$, exactly as in Theorem 5.16. Taking the infimum over quotients with at most M blocks completes the proof. □

Remark 5.21 (Degeneration at the Boundary)**.** The constant m_K is the operative quantity and it degrades toward the boundary: for the binary family, $\nabla^2 A(\eta) = p(1-p) \to 0$ as $p \to \{0,1\}$. This is the same mechanism recorded in Proposition 5.22, and it is why the theorem is stated on a compact interior set rather than globally.

Proposition 5.22 (The Two-Point Bernoulli Family Does Not Separate the Scales)**.** *Let $\pi_\pm = \frac{1}{2}$, let the output be binary, and set*

$$P_+ = \text{Bernoulli}(1-\varepsilon), \qquad P_- = \text{Bernoulli}(1-2\varepsilon), \qquad \varepsilon \downarrow 0.$$

Then, with $\mathbf{\Sigma}_\pi$ the Fisher covariance of Theorem 5.14 computed in the natural parameter,

$$\Delta_{\text{ret}}^{\text{KL}}(1) = \Theta(\varepsilon), \qquad \operatorname{tr}(\mathbf{\Sigma}_\pi) = \Theta(\varepsilon), \qquad \frac{\Delta_{\text{ret}}^{\text{KL}}(1)}{\operatorname{tr}(\mathbf{\Sigma}_\pi)} \longrightarrow \frac{\log 2 - \frac{3}{2}\log\frac{3}{2}}{\sqrt{2}\,(\frac{1}{2}\log 2)^2} = 0.50009\ldots$$

In particular the ratio is bounded away from 0, and this family does not *witness failure of a global Fisher-covariance converse.*

Proof. The information term. With $M = 1$ the only quotient is trivial, so $\Delta_{\text{ret}}^{\text{KL}}(1) = I(S;Y) = H(\bar{P}) - \frac{1}{2}H(P_+) - \frac{1}{2}H(P_-)$ with $\bar{P} = \text{Bernoulli}(1-\frac{3}{2}\varepsilon)$. Using

$$H\big(\text{Bernoulli}(1-a\varepsilon)\big) = a\varepsilon\big(-\log\varepsilon + 1 - \log a\big) + o(\varepsilon)$$

for fixed $a > 0$, the $-\log\varepsilon$ terms cancel because $\frac{3}{2} = \frac{1}{2}\cdot 1 + \frac{1}{2}\cdot 2$, leaving

$$\Delta_{\text{ret}}^{\text{KL}}(1) = \Big(\log 2 - \tfrac{3}{2}\log\tfrac{3}{2}\Big)\varepsilon + o(\varepsilon), \qquad \log 2 - \tfrac{3}{2}\log\tfrac{3}{2} = 0.08494\ldots > 0.$$

The Fisher term. Write the family in natural-parameter form $P_\eta(y) = \exp(\eta y - A(\eta))$ with $A(\eta) = \log(1+e^\eta)$ and $\eta(p) = \log\frac{p}{1-p}$. The Fisher information in the natural parameter is the second derivative of the log-partition function,

$$I(\eta) = A''(\eta) = p(1-p),$$

not $[p(1-p)]^{-1}$; the latter is the Fisher information in the mean parameter p, and the two are reciprocal because $dp/d\eta = p(1-p)$. Setting $\eta_\pm = \eta(p_\pm)$ and $\eta_0 = \frac{1}{2}(\eta_+ + \eta_-)$, equal weights give

$$\operatorname{tr}(\mathbf{\Sigma}_\pi) = I(\eta_0)\left(\frac{\eta_+ - \eta_-}{2}\right)^2.$$

As $\varepsilon \downarrow 0$ one has $\eta_+ - \eta_- \to \log 2$ and $p_0 = 1 - \sqrt{2}\,\varepsilon + o(\varepsilon)$, hence $I(\eta_0) = p_0(1-p_0) \sim \sqrt{2}\,\varepsilon$ and

$$\operatorname{tr}(\mathbf{\Sigma}_\pi) = \sqrt{2}\,\varepsilon\left(\tfrac{1}{2}\log 2\right)^2 + o(\varepsilon).$$

Conclusion. Both quantities are $\Theta(\varepsilon)$ and the ratio converges to the stated positive constant. $\square$

Theorem 5.23 (No Global Fisher-Coordinate Converse)**.** *Fix a minimal natural-parameter chart for the strictly positive laws on a finite $\mathcal{O}$, write η_s for the parameters of the predictive laws, and let $\boldsymbol{\Sigma}_\eta$ be their π-covariance in that chart; for the Fisher-weighted form fix in addition the reference point $\eta_{\bar{p}}$ of the stationary mixture and set $\boldsymbol{\Sigma}_F = I(\eta_{\bar{p}})^{1/2}\boldsymbol{\Sigma}_\eta I(\eta_{\bar{p}})^{1/2}$.*

There is no *universal constant $c > 0$, independent of the instance, with*

$$\Delta_{\mathrm{ret}}^{\mathrm{KL}}(M) \;\geq\; c\sum_{i\geq M}\lambda_i(\boldsymbol{\Sigma}_\eta) \qquad \text{or} \qquad \Delta_{\mathrm{ret}}^{\mathrm{KL}}(M) \;\geq\; c\sum_{i\geq M}\lambda_i(\boldsymbol{\Sigma}_F)$$

for all instances. The failure occurs already for two binary predictive states and $M = 1$.

Proof. Let $\mathcal{O} = \{0, 1\}$, let two predictive states carry weight $\frac{1}{2}$ each, and for $\varepsilon \in (0, \frac{1}{2})$ set

$$P_- = \mathrm{Bernoulli}(\varepsilon), \qquad P_+ = \mathrm{Bernoulli}(1-\varepsilon).$$

The divergence is bounded. The stationary mixture is $\bar{p} = \frac{1}{2}(P_- + P_+) = \mathrm{Bernoulli}(\frac{1}{2})$. With $M = 1$ the only quotient is trivial and its representative is the mixture centroid (Lemma 5.1), so

$$\Delta_{\mathrm{ret}}^{\mathrm{KL}}(1) = \tfrac{1}{2}D_{\mathrm{KL}}(P_+\|\bar{p}) + \tfrac{1}{2}D_{\mathrm{KL}}(P_-\|\bar{p}) = \log 2 - h(\varepsilon) \;\leq\; \log 2,$$

where h is the binary entropy in nats. In particular $\Delta_{\mathrm{ret}}^{\mathrm{KL}}(1)$ stays bounded, and indeed increases to $\log 2$, as $\varepsilon \downarrow 0$.

The parameter covariance diverges. In the minimal natural parameter $\eta(p) = \log\frac{p}{1-p}$ the two states sit at

$$\eta_\pm = \pm L_\varepsilon, \qquad L_\varepsilon = \log\frac{1-\varepsilon}{\varepsilon}.$$

Equal weights give $\bar{\eta} = 0$ and hence the scalar covariance

$$\boldsymbol{\Sigma}_\eta = L_\varepsilon^2 \;\longrightarrow\; \infty \qquad (\varepsilon \downarrow 0).$$

Conclusion for the unweighted form. Therefore

$$\frac{\Delta_{\mathrm{ret}}^{\mathrm{KL}}(1)}{\lambda_1(\boldsymbol{\Sigma}_\eta)} \;\leq\; \frac{\log 2}{L_\varepsilon^2} \;\longrightarrow\; 0,$$

so no positive c can work.

Conclusion for the Fisher-weighted form. The reference point is $\bar{p} = \frac{1}{2}$, at which the natural-parameter Fisher information is $I(\eta_{\bar{p}}) = A''(\eta_{\bar{p}}) = \bar{p}(1-\bar{p}) = \frac{1}{4}$ (Proposition 5.22 records this normalization). Hence $\boldsymbol{\Sigma}_F = \frac{1}{4}L_\varepsilon^2 \to \infty$ as well, and $\Delta_{\mathrm{ret}}^{\mathrm{KL}}(1)/\lambda_1(\boldsymbol{\Sigma}_F) \leq 4\log 2/L_\varepsilon^2 \to 0$. Weighting by the Fisher information at the mixture therefore does not repair the bound: the reference point stays in the interior while the states escape to the boundary. $\square$

Remark 5.24 (Reading the Two Fisher Statements Together)**.** Theorem 5.23 and Theorem 5.20 are complementary rather than competing, and together they settle the natural-parameter question completely.

The interior theorem gives $\Delta^{\mathrm{KL}}_{\mathrm{ret}}(M) \geq \frac{m_K}{2} \sum_{i \geq M} \lambda_i(\mathbf{\Sigma}_\eta)$ with $m_K = \inf_{\eta \in K} \lambda_{\min}(\nabla^2 A(\eta))$ over a compact K containing the state parameters and their mixture centroids. The no-go theorem shows that the dependence of the constant on K is not an artefact of the proof: $m_K \downarrow 0$ as K is enlarged toward the boundary, and no boundary-independent constant survives. The witness is not the family of Proposition 5.22, in which the two states approach the boundary at comparable rates and the ratio tends to $0.50009\ldots$; it is the family in which the two states approach *opposite* faces of the simplex, so that the parameter separation L_ε diverges while the divergence saturates at $\log 2$.

The contrast with probability coordinates is the substance of the matter. In those coordinates Theorem 5.16 holds globally with the sharp constant 1, because the simplex is bounded and $\|p_s - \bar{p}\|_2$ cannot exceed $\sqrt{2}$. The natural-parameter chart is unbounded, and it is exactly this unboundedness that the counterexample exploits.

Remark 5.25 (What Survives, and Why)**.** The delicate regime is the boundary of the simplex. In the natural parameter the Fisher information of a Bernoulli law is $p(1-p)$, which *vanishes* at the boundary, while $\Delta^{\mathrm{KL}}_{\mathrm{ret}}$ vanishes there as well; the two scales must therefore be compared rather than assumed to separate, and Proposition 5.22 shows that on the natural two-point family they do not. The local theorem (Theorem 5.14) is unaffected in any case, since there the family is fixed and only the perturbation scale r tends to zero.

The hypotheses under which a global Fisher-covariance converse is known are those that make the parameter-to-probability map bi-Lipschitz: uniform interiority of the predictive laws, or equivalently a fixed compact exponential-family parameter set bounded away from the boundary. Under any of them Theorem 5.16 transfers, by Remark 5.19. Without them no such converse exists at all: Theorem 5.23 exhibits binary families on which $\Delta^{\mathrm{KL}}_{\mathrm{ret}}$ stays bounded while the parameter covariance diverges. The interiority hypothesis is therefore necessary, not merely convenient.

Remark 5.26 (Machine-Checked Fragments)**.** Several of the inequalities on which the results below depend have been formalized in Lean 4 against Mathlib and are machine-checked: the centring step of Theorem 5.16, namely that $\sum_i d_i = 0$ implies $\|d\|_2^2 \leq \frac{1}{2}\|d\|_1^2$, together with its two supporting lemmas; the halving step $c_2 \leq c_1$ and $c_1 + c_2 \leq n$ imply $2c_2 \leq n$ of Proposition 13.19, in a form making its independence of $|\mathcal{O}|$ explicit; the comparison $\min\{a(M), b(M)\} \leq a(N) + b(N)$ underlying Lemma 9.28 and the mediant inequality underlying Lemma 9.26; the parallel-axis identity behind the ANOVA split and the minimality of the weighted mean; and the counting core of Proposition 13.37, that a bounded strictly increasing block count admits at most $M-1$ increases. Seventeen statements are checked in total, across seven modules, with no appeal to `sorry` and no axioms beyond Lean's standard three, namely propositional extensionality, the axiom of choice, and quotient soundness. The sources, a build script, and the axiom audit accompany the manuscript as supplementary material.

The scope of this is deliberately narrow and should not be overstated. What is verified is the arithmetic skeleton: the inequalities, identities and counting arguments. Pinsker's inequality is taken as a hypothesis rather than imported; Ky Fan's maximum principle, the Kullback–Leibler functional itself, and all automata-theoretic constructions — Mealy machines, version spaces, the adaptive games — are not formalized, so the surrounding theorems are not verified end to end.

Remark 5.27 (Sharpness of the Quadratic Minorant)**.** The constant in the minorant $(*)$ of Theorem 5.16 is exactly 1, and both inequalities composing it are individually sharp.

For the centring step, $\sup\{\|\delta\|_2^2/(\frac{1}{2}\|\delta\|_1^2)\}$ over nonzero centred $\delta \in \mathbb{R}^k$ equals 1 for every $k \geq 2$, attained precisely when δ has exactly two nonzero coordinates of equal magnitude and opposite sign; direct maximization over $k = 2, \ldots, 12$ returns 1.000000000000. For the composite, $\inf\{D_{\mathrm{KL}}(p\|q)/\|p-q\|_2^2\}$ over distinct $p, q \in \Delta^{k-1}$ equals 1 and is not attained: over 58,186 deliberately near-degenerate pairs evaluated in 60-digit arithmetic the minimum observed ratio is 1.0000000053. Along the extremal family $p = (\frac{1}{2}+t, \frac{1}{2}-t)$, $q = (\frac{1}{2}-t, \frac{1}{2}+t)$ one has $D_{\mathrm{KL}}(p\|q)/\|p-q\|_2^2 = 1 + \frac{4}{3}t^2 + O(t^4)$, decreasing to 1.

A caution on floating-point evaluation: at separations $\|p-q\|_2^2 \sim 10^{-12}$ the ratio computed in double precision can read below 1 purely through cancellation in the logarithms. Re-evaluating the same pairs in extended precision restores the inequality.

Remark 5.28 (Numerical Verification)**.** The predictive-information identity of Theorem 5.5 is an exact identity and can be checked numerically on finite instances: over stationary chains with $|\mathcal{S}^+| \leq 5$ and $|\mathcal{O}| \leq 4$, enumerating *all* set partitions, the three quantities $\Delta_{\mathrm{ret}}^{\mathrm{KL}}(\phi)$, $I(S; Z \mid K_\phi)$ and $I(S; Z) - I(K_\phi; Z)$ agree to 7.6×10^{-16}.

Theorem 5.16 admits an end-to-end check of the same kind, and each step of its proof can be isolated. Over 2,434 (instance, partition) pairs with $|\mathcal{S}^+| \leq 5$, $|\mathcal{O}| \leq 4$, again enumerating all set partitions, the maximum violation is 7.5×10^{-17} for the minorant $\Delta_{\mathrm{ret}}^{\mathrm{KL}}(\phi) \geq \mathrm{tr}(W_\phi)$, 1.1×10^{-16} for the ANOVA decomposition $\mathbf{\Sigma}_p = W_\phi + B_\phi$, 1.7×10^{-16} for the Ky Fan bound $\mathrm{tr}(B_\phi) \leq \sum_{i \leq M-1} \lambda_i$, and 7.5×10^{-17} for the composite conclusion $\Delta_{\mathrm{ret}}^{\mathrm{KL}}(\phi) \geq \sum_{i \geq M} \lambda_i(\mathbf{\Sigma}_p)$. The smallest ratio of the two sides over those pairs is 1.0015, so the bound is not vacuous on generic instances.

Theorem 5.20 requires more care to test, since m_K is the least eigenvalue of $\nabla^2 A$ over a convex set containing the state parameters *and* the centroid parameters, not a coarser proxy such as the least predictive probability. Computing m_K by minimizing $\lambda_{\min}(\nabla^2 A)$ over the convex hull gives 0 violations on 1,664 (instance, partition) pairs, with least ratio 1.0011; substituting a proxy for m_K produces spurious violations and tests a different statement.

The quadratic-surrogate spectral bound can be checked directly on finite weighted point clouds. The local full-KL Fisher expansion can be checked on small-radius exponential-family instances by verifying the predicted r^2 leading term and an $O(r^3)$ remainder. These computations illustrate the theorems; they do not establish their hypotheses.

Remark 5.29 (Computational Conventions)**.** The computational observations reported throughout this manuscript are held to the following conventions, stated once here and cross-referenced wherever an enumeration, search, or numerical evaluation is quoted. *Distinct machines* are labelled transition/output table pairs counted up to renaming of the state set, so a count of N machines means N renaming classes, and *minimality* is decided by Moore partition refinement before counting. *Tie-breaking* is lexicographic on a canonical encoding of the tables, so extremal machines are identified up to renaming. *Structured subclasses* are named at each site where they are searched (for example, minimal machines whose first input acts as a permutation and whose output function has a single probe state). *Arithmetic* is exact rational where stated, 60-digit floating point where stated, and never double precision where a near-degenerate comparison is at issue. Search is either exhaustive over the stated class or hill-climbing with the restart schedule reported at the site. These conventions apply to the retention checks of Remark 5.28 and to the machine-table searches of Section 13.

An independent verification suite reproduces the recomputable subset of these observations exactly — the counter-family values, both non-convexity instances, the Csiszár identity, the adaptive-depth and cyclic-shift witnesses, and the random-instance minorant checks — and re-implements the exhaustive searches of Section 13 under the conventions above; the suite, the machine tables of the extremal witnesses, and its exact outputs are distributed as part of the supplementary package.

Proposition 5.30 (Relation between Full-KL and Quadratic Restricted Gaps)**.** *In the Gaussian quadratic restricted regime, for every $M \geq 1$,*

$$\Delta_{\mathrm{ret}}^{\mathrm{KL}}(M) \leq \Delta_{\mathrm{ret}}^{\mathrm{quad}}(M).$$

Proof. Fix a lumpable quotient ϕ. In the full-KL problem, the block representative is unrestricted, and the optimal representative is the mixture centroid $\bar{P}_k$. In the quadratic restricted problem, the representative is constrained to the fixed-covariance Gaussian family $Q_c = \mathcal{N}(c, \frac{1}{2}I)$. A constrained minimization problem has value no smaller than the corresponding unconstrained problem. Hence, for each ϕ,

$$\Delta_{\mathrm{ret}}^{\mathrm{KL}}(\phi) \leq \Delta_{\mathrm{ret}}^{\mathrm{quad}}(\phi).$$

Minimizing over lumpable quotients gives $\Delta_{\mathrm{ret}}^{\mathrm{KL}}(M) \leq \Delta_{\mathrm{ret}}^{\mathrm{quad}}(M)$. □

5.4 Unifilar Retention and the Controlled Information Bottleneck

The retention theory extends from the input-driven model to general unifilar machines without new analytical mechanism: the information-theoretic identities depend only on the sufficiency structure $K = \phi(S)$, and the spectral converses apply fiberwise in the input. What does change is the feasible set, which is now the unifilar-lumpable quotients of Definition 3.21, and with it the zero-retention threshold.

Definition 5.31 (Controlled Full-KL Retention Gap)**.** Let the input process be i.i.d. with law p and independent of the state process. For a unifilar-lumpable quotient $\phi : \mathcal{S}^+ \to \mathcal{K}$ with blocks C_k and representatives $\{Q_k^x\}$, set

$$\Delta_{\mathrm{ret}}^{\mathrm{KL,ctrl}}(\phi) = \sum_x p(x) \sum_k \sum_{s \in C_k} \pi_s \, D_{\mathrm{KL}}\big(P_s^x \,\big\|\, Q_k^x\big),$$

the representatives being optimized for each ϕ, and

$$\Delta_{\text{ret}}^{\text{KL,ctrl}}(M) = \min_{\substack{\phi \text{ unifilar-lumpable} \\ |\mathcal{K}| \le M}} \Delta_{\text{ret}}^{\text{KL,ctrl}}(\phi).$$

Blocks of zero stationary mass are ignored.

The independence of the input from the state is a genuine hypothesis and not a normalization: it is what makes $\mathbb{P}(S_t = s \mid K_t = k, X_t = x)$ equal to $\pi_s/\pi(C_k)$, and hence what identifies the block centroid with a conditional law. Remark 5.33 records what fails without it.

Theorem 5.32 (Controlled Information-Bottleneck Identity)**.** *Under the hypotheses of Definition 5.31, for every unifilar-lumpable ϕ,*

$$\Delta_{\text{ret}}^{\text{KL,ctrl}}(\phi) = I(S; Y \mid K_\phi, X) = I(S; Y \mid X) - I(K_\phi; Y \mid X),$$

and consequently

$$\Delta_{\text{ret}}^{\text{KL,ctrl}}(M) = I(S; Y \mid X) - \max_{\substack{\phi \text{ unifilar-lumpable} \\ |\mathcal{K}| \le M}} I(K_\phi; Y \mid X).$$

The optimal representative is the controlled mixture centroid

$$\bar{P}_k^x = \frac{\sum_{s \in C_k} \pi_s P_s^x}{\sum_{s \in C_k} \pi_s}.$$

All the mutual informations are finite, since S is finite and $I(S; Y \mid X) \le H(S)$.

Proof. Fix an input x. Conditionally on $X_t = x$, the block cost $\sum_{s \in C_k} \pi_s D_{\text{KL}}(P_s^x \| Q_k^x)$ is minimized over Q_k^x at the mixture centroid $\bar{P}_k^x$, by Lemma 5.1 applied to the family $\{P_s^x : s \in C_k\}$ with weights proportional to π_s. Substituting the centroids and summing against $p(x)$,

$$\Delta_{\text{ret}}^{\text{KL,ctrl}}(\phi) = \mathbb{E}\, D_{\text{KL}}\big(P_{Y|S,X} \,\big\|\, P_{Y|K,X}\big),$$

where the identification $P_{Y|K,X}(\cdot \mid k, x) = \bar{P}_k^x$ uses independence of X_t from S_t: it gives $\mathbb{P}(S_t = s \mid K_t = k, X_t = x) = \pi_s/\pi(C_k)$ for $s \in C_k$, so the conditional law of Y_t given $(K_t, X_t) = (k, x)$ is precisely the π-weighted mixture $\bar{P}_k^x$. Since $K = \phi(S)$ is a function of S, we have $P_{Y|S,K,X} = P_{Y|S,X}$, and the displayed expectation is by definition $I(S; Y \mid K, X)$.

The chain rule for conditional mutual information gives

$$I(S; Y \mid X) = I(K; Y \mid X) + I(S; Y \mid K, X),$$

valid because K is a function of S, whence the subtracted form. The rearrangement is legitimate because $I(S; Y \mid X) \le H(S) < \infty$, so no difference of infinities arises. Minimizing over unifilar-lumpable quotients with at most M blocks turns the identity into the stated variational form. $\square$

Remark 5.33 (Role of the Independence Hypothesis)**.** Definition 3.15 requires only that the input process be stationary and ergodic, which permits X_t to be correlated with S_t. Under such correlation the two sides of Theorem 5.32 differ: the conditional weight of s within its block given $X_t = x$ is not $\pi_s/\pi(C_k)$, so the centroid $\bar{P}_k^x$ is not a conditional law. Already with two states, binary inputs and a ternary output alphabet, a non-product stationary law of (S, X) makes the quantity of Definition 5.31 exceed $I(S; Y \mid X) - I(K; Y \mid X)$ by more than 0.13 nats on the one-block quotient.

The hypothesis is a property of the cost functional rather than of the machine, and it can be removed by measuring the cost against the fiberwise conditional weights. This is carried out in Theorem 5.35 below, of which Theorem 5.32 is the independent case.

Definition 5.34 (Fiberwise Controlled Retention Gap)**.** Let (S_t, X_t) have an arbitrary stationary joint law, with input marginal $p(x) = \mathbb{P}(X_t = x)$ and fiberwise conditional state weights

$$\pi_s^x = \mathbb{P}(S_t = s \mid X_t = x), \qquad x \in \operatorname{supp} p.$$

For a unifilar-lumpable quotient ϕ with blocks C_k and representatives $\{Q_k^x\}$, define

$$\Delta_{\mathrm{ret}}^{\mathrm{KL,gen}}(\phi) = \sum_x p(x) \sum_k \sum_{s \in C_k} \pi_s^x \, D_{\mathrm{KL}}\big(P_s^x \,\big\|\, Q_k^x\big),$$

the representatives being optimized for each ϕ, and

$$\Delta_{\mathrm{ret}}^{\mathrm{KL,gen}}(M) = \min_{\substack{\phi \text{ unifilar-lumpable} \\ |\mathcal{K}| \le M}} \Delta_{\mathrm{ret}}^{\mathrm{KL,gen}}(\phi).$$

Fibers and blocks of zero mass are ignored. When X_t is independent of S_t one has $\pi_s^x = \pi_s$ and Definition 5.31 is recovered.

Theorem 5.35 (Controlled Information-Bottleneck Identity, General Inputs)**.** *Let (S_t, X_t) be stationary with arbitrary dependence, and let ϕ be unifilar-lumpable. Then*

$$\Delta_{\mathrm{ret}}^{\mathrm{KL,gen}}(\phi) = I(S; Y \mid K_\phi, X) = I(S; Y \mid X) - I(K_\phi; Y \mid X),$$

and consequently

$$\Delta_{\mathrm{ret}}^{\mathrm{KL,gen}}(M) = I(S; Y \mid X) - \max_{\substack{\phi \text{ unifilar-lumpable} \\ |\mathcal{K}| \le M}} I(K_\phi; Y \mid X).$$

The optimal representative is the fiberwise conditional centroid

$$\bar{P}_k^x = \frac{\sum_{s \in C_k} \pi_s^x P_s^x}{\sum_{s \in C_k} \pi_s^x} = \mathcal{L}\big(Y_t \mid K_t = k,\ X_t = x\big).$$

Proof. Fix $x \in \operatorname{supp} p$ and a block C_k of positive conditional mass. Conditionally on $X_t = x$, the inner sum $\sum_{s \in C_k} \pi_s^x D_{\mathrm{KL}}(P_s^x \| Q_k^x)$ is minimized over Q_k^x at the π^x-weighted mixture of the P_s^x, by Lemma 5.1 applied with weights proportional to π_s^x. That mixture is exactly $\mathcal{L}(Y_t \mid K_t = k, X_t = x)$: conditioning on $X_t = x$ first and on the block second,

$$\mathbb{P}(S_t = s \mid K_t = k, X_t = x) = \frac{\pi_s^x}{\sum_{s' \in C_k} \pi_{s'}^x} \qquad (s \in C_k),$$

which holds for *any* joint law of (S_t, X_t), no independence being used; independence was needed in Theorem 5.32 only to replace π_s^x by π_s. Substituting the centroids,

$$\Delta_{\text{ret}}^{\text{KL,gen}}(\phi) = \mathbb{E}\, D_{\text{KL}}\big(P_{Y|S,X} \,\big\|\, P_{Y|K,X}\big) = I(S; Y \mid K, X),$$

the last equality being the definition of conditional mutual information, together with $P_{Y|S,K,X} = P_{Y|S,X}$ because $K = \phi(S)$. The chain rule $I(S;Y \mid X) = I(K;Y \mid X) + I(S;Y \mid K, X)$ gives the subtracted form, and is legitimate since $I(S;Y \mid X) \leq H(S) < \infty$. Minimizing over unifilar-lumpable quotients with at most M blocks yields the variational statement. □

Corollary 5.36 (Elementary Properties, General Inputs)**.** $0 \leq \Delta_{\text{ret}}^{\text{KL,gen}}(M) \leq I(S;Y \mid X)$ *with* $\Delta_{\text{ret}}^{\text{KL,gen}}(1) = I(S;Y \mid X)$*; the gap is nonincreasing in* M*; and* $\Delta_{\text{ret}}^{\text{KL,gen}}(\phi) \leq \Delta_{\text{ret}}^{\text{KL,gen}}(\psi)$ *whenever the unifilar-lumpable* ϕ *refines* ψ*.*

Proof. As in Corollary 5.39, the independent-input special case stated below, every step being conditional on X and using Theorem 5.35 in place of Theorem 5.32. □

Corollary 5.37 (Fiberwise Spectral Converse, General Inputs)**.** *For finite* $\mathcal{O}$*, let* $\mathbf{\Sigma}_p^x$ *denote the covariance of the predictive probability vectors* $\{p_s^x\}$ *under the* conditional *weights* $\{\pi_s^x\}$*. Then*

$$\Delta_{\text{ret}}^{\text{KL,gen}}(M) \;\geq\; \sum_x p(x) \sum_{i \geq M} \lambda_i\big(\mathbf{\Sigma}_p^x\big).$$

Proof. On each fiber the family $\{p_s^x\}$ carries the probability weights $\{\pi_s^x\}$ and the quotient is the same for every fiber, the lumpability constraint being x-independent. Theorem 5.16 applies verbatim on the fiber with those weights; multiplying by $p(x)$ and summing gives the bound. □

Remark 5.38 (The Reweighting Is Not Cosmetic)**.** In Corollary 5.37 the fiber covariance must be formed with the conditional weights π_s^x. Substituting the unconditional weights π_s, which is the form that Corollary 5.50 inherits from the independent case, produces a quantity that is not a lower bound: among randomly generated instances with correlated (S, X) and all budgets M, the unconditional-weight expression exceeds $\Delta_{\text{ret}}^{\text{KL,gen}}(M)$ in several per cent of cases, while the conditional-weight bound of Corollary 5.37 is never violated. Dependence between input and state therefore changes the operative covariance, not only the block weights.

Corollary 5.39 (Range, One-State Value, and Refinement Monotonicity, Controlled)**.** $0 \leq \Delta_{\text{ret}}^{\text{KL,ctrl}}(M) \leq I(S;Y \mid X)$*, with* $\Delta_{\text{ret}}^{\text{KL,ctrl}}(1) = I(S;Y \mid X)$*; if* ϕ *refines* ψ *and both are unifilar-lumpable then* $\Delta_{\text{ret}}^{\text{KL,ctrl}}(\phi) \leq \Delta_{\text{ret}}^{\text{KL,ctrl}}(\psi)$*; and* $\Delta_{\text{ret}}^{\text{KL,ctrl}}(M)$ *is nonincreasing in* M*.*

Proof. This is Corollary 5.6 with every quantity conditioned on X. A single block makes K constant, so $I(K;Y \mid X) = 0$; refinement is data processing conditional on X, since K_ψ is then a function of K_ϕ; and the feasible sets are nested in M. Feasibility of the one-block quotient is Definition 3.21. □

The zero-retention threshold does *not* transfer verbatim. In the input-driven theory the singleton quotient is lumpable and the predictive partition can always be realized, so the threshold is the number of distinct predictive laws (Theorem 5.2). For unifilar machines the partition by predictive kernel need not be unifilar-lumpable, and the correct threshold is the index of its coarsest unifilar-lumpable refinement.

Corollary 5.39 is the controlled counterpart of the elementary facts recorded in Corollary 5.6, and the boundary behaviour of the natural-parameter converses is unchanged under control, as Remark 5.52 records.

Definition 5.40 (Kernel Partition and Its Stable Refinement)**.** Let ϕ_{ker} be the partition of $\mathcal{S}^+$ by predictive kernel, so $\phi_{\text{ker}}(s) = \phi_{\text{ker}}(s')$ if and only if $P_s^x = P_{s'}^x$ for every x in the input support. Define ϕ_{ker}^* to be the coarsest unifilar-lumpable quotient refining ϕ_{ker}, and write $N^* = |\phi_{\text{ker}}^*(\mathcal{S}^+)|$ for its index.

Proposition 5.41 (Existence of the Stable Refinement)**.** *ϕ_{ker}^* exists and is unique, and is computed by the refinement recursion*

$$\phi^{(0)} = \phi_{\text{ker}}, \qquad \phi^{(m+1)}(s) = \Big(\phi^{(m)}(s),\ \big(\phi^{(m)}(\tau(s,x,y))\big)_{(x,y)\ \textit{feasible from } s}\Big),$$

which stabilizes after at most $|\mathcal{S}^+|$ steps.

Proof. Call a partition *admissible* if it is unifilar-lumpable and refines ϕ_{ker}. The singleton partition is admissible, so the family is nonempty. The recursion produces a decreasing chain of partitions, each refining ϕ_{ker}, in a set of size $|\mathcal{S}^+|$; the number of blocks is nondecreasing and bounded by $|\mathcal{S}^+|$, so the chain stabilizes after at most $|\mathcal{S}^+|$ steps, at a partition ϕ^∞. At the fixed point, $\phi^\infty(s) = \phi^\infty(s')$ implies $\phi^\infty(\tau(s,x,y)) = \phi^\infty(\tau(s',x,y))$ for every commonly feasible (x,y), which is unifilar lumpability with $\tau_{\mathcal{K}}(\phi^\infty(s),x,y) = \phi^\infty(\tau(s,x,y))$; and ϕ^∞ refines $\phi_{\text{ker}} = \phi^{(0)}$ by construction. So ϕ^∞ is admissible.

For maximality, let ψ be any admissible partition. We show by induction that ψ refines $\phi^{(m)}$ for every m. For $m = 0$ this is the hypothesis that ψ refines ϕ_{ker}. If ψ refines $\phi^{(m)}$ and $\psi(s) = \psi(s')$, then $\phi^{(m)}(s) = \phi^{(m)}(s')$, and unifilar lumpability of ψ gives $\psi(\tau(s,x,y)) = \psi(\tau(s',x,y))$ at every commonly feasible event, hence $\phi^{(m)}(\tau(s,x,y)) = \phi^{(m)}(\tau(s',x,y))$ by the inductive hypothesis; the two defining data of $\phi^{(m+1)}$ therefore agree, so $\phi^{(m+1)}(s) = \phi^{(m+1)}(s')$. Taking m at the stabilization index shows ψ refines ϕ^∞. Thus ϕ^∞ is the coarsest admissible partition, and uniqueness follows. □

Theorem 5.42 (Zero Controlled Retention)**.** $\Delta_{\text{ret}}^{\text{KL,ctrl}}(M) = 0$ *if and only if $M \geq N^*$, with N^* the index of Definition 5.40.*

Proof. Suppose $\Delta_{\mathrm{ret}}^{\mathrm{KL,ctrl}}(\phi) = 0$ for a unifilar-lumpable ϕ. The cost is a sum of nonnegative terms $p(x)\pi_s D_{\mathrm{KL}}(P_s^x \| \bar{P}_k^x)$, so every term vanishes; by the equality condition for KL divergence, $P_s^x = \bar{P}_{\phi(s)}^x$ for every positive-mass s and every x in the input support. Two states in a common block therefore have equal kernels, so ϕ refines ϕ_{ker}, and being unifilar-lumpable it is admissible in the sense of Proposition 5.41. Hence ϕ refines ϕ_{ker}^* and has at least N^* blocks.

Conversely ϕ_{ker}^* is itself unifilar-lumpable and refines ϕ_{ker}, so each of its blocks carries a single kernel, every block centroid equals that kernel, and the cost is zero. It has N^* blocks, so $M \geq N^*$ suffices. □

Remark 5.43 (The Threshold Is Not the Number of Distinct Kernels)**.** In the input-driven setting of Theorem 5.2 the threshold is the number of distinct predictive laws, because the predictive partition is automatically lumpable there. In the unifilar class the two quantities separate, and the gap $N^* - |\phi_{\mathrm{ker}}(\mathcal{S}^+)|$ can be positive. Take $\mathcal{I}$ a singleton, $\mathcal{O} = \{0, 1\}$, three states A, B, C, kernels

$$P_A = P_B = \mathrm{Bernoulli}(\tfrac{1}{2}), \qquad P_C = \mathrm{Bernoulli}(\tfrac{1}{10}),$$

and the unifilar update

$$\tau(A,0) = A, \quad \tau(A,1) = C, \qquad \tau(B,0) = C, \quad \tau(B,1) = B, \qquad \tau(C,0) = A, \quad \tau(C,1) = B.$$

The stationary distribution is strictly positive. There are two distinct kernels, but the partition $\{\{A, B\}, \{C\}\}$ is not unifilar-lumpable: A and B are in one block and the output 0 is feasible from both, yet $\tau(A, 0) = A$ and $\tau(B, 0) = C$ lie in different blocks. The refinement of Proposition 5.41 separates A from B, so $N^* = 3$ while $|\phi_{\mathrm{ker}}(\mathcal{S}^+)| = 2$, and indeed $\Delta_{\mathrm{ret}}^{\mathrm{KL,ctrl}}(2) > 0$ whereas $\Delta_{\mathrm{ret}}^{\mathrm{KL,ctrl}}(3) = 0$. The separation is not exceptional: over randomly generated unifilar machines with at most five states and at least one coincidence of predictive kernels, the kernel partition fails to be unifilar-lumpable in approximately fifty-five per cent of instances.

This is exactly the distinction between predictive equivalence and causal-state equivalence recorded in Definition 3.17: the refinement ϕ_{ker}^* is the causal-state partition for the full controlled future, and ϕ_{ker} is its one-step coarsening.

The separation of Remark 5.43 is as large as the state set permits, and the refinement recursion is essentially as slow as Proposition 5.41 allows. Both are witnessed by one family.

Definition 5.44 (Counter Family $\mathcal{C}_M$)**.** For $M \geq 3$ and $\beta, \gamma \in (0, 1)$ with $\beta \neq \gamma$, let $\mathcal{C}_M$ be the unifilar machine with a single input letter, output alphabet $\mathcal{O} = \{0, 1\}$, state set $\{0, 1, \ldots, M-1\}$, update

$$\tau(s, 0) = \min\{s + 1,\, M - 1\}, \qquad \tau(s, 1) = 0,$$

and emission laws

$$P_s = \mathrm{Bernoulli}(\beta) \;\; (0 \leq s \leq M-2), \qquad P_{M-1} = \mathrm{Bernoulli}(\gamma).$$

Output 0 advances the counter and saturates at $M-1$; output 1 resets it. The restriction to the open interval is required for Proposition 5.45(i): at $\gamma = 0$ the state $M-1$ emits 0 almost surely and, being saturated, is absorbing, while at $\beta = 1$ every state below $M-1$ resets almost surely and $M-1$ is never reached. In either case the chain fails to be irreducible and some state leaves the stationary support. The input alphabet being a singleton, X_t is deterministic and hence independent of S_t, so Theorem 5.32 applies and $\Delta_{\mathrm{ret}}^{\mathrm{KL,gen}}$ coincides with $\Delta_{\mathrm{ret}}^{\mathrm{KL,ctrl}}$ on this family.

Proposition 5.45 (The Refinement Gap Is Maximal and the Bound Is Tight)**.** *Let $\beta, \gamma \in (0,1)$ with $\beta \neq \gamma$. For $\mathcal{C}_M$:*

(i) the chain is irreducible with strictly positive stationary distribution, so $\mathcal{S}^+$ is the whole state set;

(ii) the kernel partition has exactly two blocks, $|\phi_{\mathrm{ker}}(\mathcal{S}^+)| = 2$;

(iii) the stable refinement is the discrete partition, $N^* = M$, so the gap $N^* - |\phi_{\mathrm{ker}}(\mathcal{S}^+)| = M-2$ is the largest possible on M states;

(iv) the refinement recursion of Proposition 5.41 stabilizes after exactly $M-1$ steps, so its bound of $|\mathcal{S}^+|$ steps is tight to within one;

(v) consequently $\Delta_{\mathrm{ret}}^{\mathrm{KL,ctrl}}(M-1) > 0$ while $\Delta_{\mathrm{ret}}^{\mathrm{KL,ctrl}}(M) = 0$.

Proof. (i) Every state emits both outputs with positive probability: states $0, \ldots, M-2$ because $\beta \in (0,1)$, and state $M-1$ because $\gamma \in (0,1)$. Consequently output 1 carries any state to 0, and a run of $M-1$ outputs 0 carries 0 to $M-1$, so every state reaches every other. The chain is therefore irreducible on a finite set, and its stationary distribution is strictly positive, so $\mathcal{S}^+ = \{0, \ldots, M-1\}$.

(ii) States $0, \ldots, M-2$ carry the law Bernoulli(β) and state $M-1$ carries Bernoulli$(\gamma) \neq$ Bernoulli(β), giving exactly two kernel classes.

(iii) and *(iv)* Write $\phi^{(m)}$ for the m-th iterate, with $\phi^{(0)} = \phi_{\mathrm{ker}}$. We claim, by induction on $m \geq 0$, that $\phi^{(m)}$ separates exactly the top $m+1$ states, that is

$$\phi^{(m)}(s) = \phi^{(m)}(s') \iff s = s' \quad \text{or} \quad s, s' \leq M-2-m.$$

For $m = 0$ this is (ii). Assume it for m and let $s < s' \leq M-2-m$. The recursion refines by the pair of successor classes $\big(\phi^{(m)}(\tau(s,0)), \phi^{(m)}(\tau(s,1))\big)$. Since $s, s' \leq M-2-m \leq M-2$, neither is saturated, so $\tau(s,0) = s+1$ and $\tau(s',0) = s'+1$, while $\tau(s,1) = \tau(s',1) = 0$. By the inductive hypothesis $s+1$ and $s'+1$ are separated precisely when at least one of them exceeds $M-2-m$, that is precisely when $s' = M-2-m$. Hence $\phi^{(m+1)}$ splits $M-2-m$ off from the states below it and leaves $0, \ldots, M-3-m$ together, which is the claim for $m+1$.

At $m = M-2$ the block $\{s : s \leq M-2-m\} = \{0\}$ is a singleton, so $\phi^{(M-2)}$ is the discrete partition and $N^* = M$, giving (iii). The recursion detects stabilization one step later, at $m = M-1$, since $\phi^{(M-2)}$ and $\phi^{(M-1)}$ have the same number of blocks while $\phi^{(M-3)}$ has fewer; the number of iterations is therefore exactly $M-1$, which is (iv). The bound $N^* \leq |\mathcal{S}^+|$ of

Proposition 5.41 is attained, and no partition of an M-element set has more than M blocks, so the gap $M-2$ is maximal.

(v) Immediate from Theorem 5.42 and $N^* = M$. $\square$

Remark 5.46 (Scope of the Extremal Family)**.** Proposition 5.45 settles the worst case, not the typical one. It shows that no bound of the form $N^* \le f(|\phi_{\mathrm{ker}}(\mathcal{S}^+)|)$ can hold, since $\mathcal{C}_M$ has two kernel classes for every M, and that the round count of Proposition 5.41 cannot be improved below $|\mathcal{S}^+|-1$. It uses a single input letter, so the phenomenon is driven entirely by the output-dependence of the update and is invisible to any input-driven analysis. For $M = 3, \dots, 7$ the values $\Delta_{\mathrm{ret}}^{\mathrm{KL,ctrl}}(M-1)$ are 0.0481, 0.0321, 0.0192, 0.0107 and 0.0057 nats at $\beta = \frac{1}{2}$, $\gamma = \frac{1}{10}$, decreasing in M as the extra resolved state carries less stationary mass.

Corollary 5.47 (Recovery of the Input-Driven Theory)**.** *Let the emissions be output-independent, $P_s^x = P_s$ for every x, as in every construction of Section 5.5. Then $\Delta_{\mathrm{ret}}^{\mathrm{KL,ctrl}}(\phi) = \Delta_{\mathrm{ret}}^{\mathrm{KL}}(\phi)$ for every quotient ϕ feasible in both theories. If moreover the machine is input-driven and every quotient under consideration has connected support — in particular if every P_s has full support — the two feasible sets coincide by Proposition 3.28(i), and therefore $\Delta_{\mathrm{ret}}^{\mathrm{KL,ctrl}}(M) = \Delta_{\mathrm{ret}}^{\mathrm{KL}}(M)$ for every M, the controlled identity reducing to Meta-Theorem 4.14.*

Proof. The first claim is immediate: with $P_s^x = P_s$ the centroids $\bar{P}_k^x$ are independent of x and $\sum_x p(x) = 1$. The second follows because equality of the two minima requires equality of the feasible sets over which they are taken; Remark 5.48 shows that this hypothesis cannot be dropped. $\square$

Remark 5.48 (Why the Support Hypothesis Cannot Be Dropped Here)**.** Without a support hypothesis the two optimized gaps genuinely differ, even for output-independent emissions, because the unifilar feasible set is strictly larger. The three-state witness of Remark 3.22 has $\Delta_{\mathrm{ret}}^{\mathrm{KL,ctrl}}(2) < \Delta_{\mathrm{ret}}^{\mathrm{KL}}(2)$, the two values differing by more than 0.15 nats. Corollary 5.47 is therefore stated with the hypothesis, and the complexity transfer of Remark 5.53 is argued from full supports rather than from the general reduction.

Corollary 5.49 (Controlled Quadratic Spectral Converse)**.** *In the controlled Gaussian quadratic regime $P_s^x = \mathcal{N}(m_s^x, \frac{1}{2} I_D)$, with Fisher features $y_s^x = \sqrt{2}\, m_s^x$ and per-input covariances*

$$\boldsymbol{\Sigma}_\pi^x = \sum_s \pi_s (y_s^x - \bar{y}^x)(y_s^x - \bar{y}^x)^\top, \qquad \bar{y}^x = \sum_s \pi_s y_s^x,$$

the controlled quadratic gap satisfies

$$\Delta_{\mathrm{ret}}^{\mathrm{quad,ctrl}}(M) \;\ge\; \frac{1}{2} \sum_x p(x) \sum_{i \ge M} \lambda_i(\boldsymbol{\Sigma}_\pi^x).$$

Proof. The lumpability constraint does not depend on x, so a single quotient is used on every fiber. Fix x: the inner cost is the Gaussian quadratic restricted gap of the family $\{y_s^x\}$ under that quotient, and the ANOVA-plus-Ky-Fan argument of Theorem 5.8 bounds it below by $\frac{1}{2} \sum_{i \ge M} \lambda_i(\boldsymbol{\Sigma}_\pi^x)$, the effective rank of the between-block covariance being $M-1$ on each fiber.

Multiplying by $p(x)$ and summing gives the bound, and taking the minimum over quotients preserves it since the bound is quotient-free. $\square$

Corollary 5.50 (Controlled Probability-Coordinate Spectral Converse)**.** *For finite $\mathcal{O}$, let p_s^x be the predictive probability vectors and $\mathbf{\Sigma}_p^x$ their π-covariance on fiber x. Then, globally and with no regularity hypothesis,*

$$\Delta_{\mathrm{ret}}^{\mathrm{KL,ctrl}}(M) \;\geq\; \sum_x p(x) \sum_{i \geq M} \lambda_i(\mathbf{\Sigma}_p^x),$$

and the constant 1 is sharp on any fiber of positive mass.

Proof. Apply Theorem 5.16 fiberwise, exactly as in Corollary 5.49; the inequality $(*)$ of that proof is valid at boundary points, so no interiority is needed. Sharpness follows by placing the two-point family of Proposition 5.17 on a single fiber with $p(x) > 0$ and making the emissions constant in x elsewhere. $\square$

Corollary 5.51 (Controlled Interior Fisher Converse)**.** *Suppose that for each x the strictly positive laws $\{p_s^x\}$ lie in a fixed minimal natural-parameter chart with a compact convex parameter set K^x containing the state parameters* and *their mixture-centroid parameters, and that $\nabla^2 A(\eta) \succeq m_K^x I$ on K^x. Then*

$$\Delta_{\mathrm{ret}}^{\mathrm{KL,ctrl}}(M) \;\geq\; \frac{1}{2}\Big(\min_x m_K^x\Big) \sum_x p(x) \sum_{i \geq M} \lambda_i(\mathbf{\Sigma}_\eta^x),$$

where $\mathbf{\Sigma}_\eta^x$ is the π-covariance of the natural parameters on fiber x.

Proof. Theorem 5.20 applied on each fiber, with the worst fiber constant taken out of the p-weighted sum. As there, the hypothesis that K^x contains the centroid parameters is what places the segments $[\eta_s, \zeta_C]$ inside K^x. $\square$

Remark 5.52 (The Fisher No-Go Persists Under Control)**.** The counterexample of Theorem 5.23 lives on a single fiber, so it is a controlled instance with constant emissions. No universal constant exists for a natural-parameter converse in the controlled theory either: interiority is necessary exactly as in the unconditional case, and Corollary 5.50 is the boundary-safe statement.

Remark 5.53 (Complexity Results Transfer to the Unifilar Class)**.** Every machine appearing in Theorems 5.56, 5.57, 5.58 and Corollary 5.59 is input-driven with output-independent emissions, hence a valid instance of the unifilar class. Transfer of hardness requires an argument, since enlarging the feasible set of quotients can only lower the optimum and could in principle convert a NO instance into a YES instance. It is valid here because in each construction the two feasible sets coincide.

In the reset construction of Theorem 5.56, and in the reset machine reused in Theorem 5.58, *every* partition of the state set is already lumpable, since states in a common block have identical successors $\tau(s_i, \ell) = s_\ell$; the unifilar feasible set, being a subset of the set of all partitions, therefore coincides with the lumpable one. In the depth-two construction of Theorem 5.57 not every partition is lumpable, but the NO-case bound is proved against an arbitrary machine-state variable K with $|K| \le k$ through the fractional k-means Lemma 5.55; a fractional assignment dominates every partition, lumpable or not, so the lower bound already covers the unifilar feasible set. In Theorem 5.58 the predictive laws $p_i = u + \delta z_i$ have all coordinates strictly positive, so supports are full and block-uniform for every partition, and Proposition 3.28(i) applies directly.

The NP-completeness, promise NP-hardness, and APX-hardness statements therefore hold verbatim when the retention decision problem ranges over unifilar machines; certificates and membership are unchanged.

5.5 Computational Complexity of Retention

The complexity results below are theorems of the Gaussian quadratic restricted retention regime. They do not establish NP-hardness for unrestricted full-KL retention.

Lemma 5.54 (Polynomial Gap Amplification for Rational k-Means)**.** *Given a rational Euclidean k-means instance*

$$v_1, \dots, v_n \in \mathbb{Q}^d,$$

integer k, and rational threshold τ, there is a polynomial-time transformation to integer points

$$a_1, \dots, a_n \in \mathbb{Z}^D$$

and a rational threshold T such that exactly one of the following holds:

$$\mathrm{OPT}_a \le T,$$

or

$$\mathrm{OPT}_a \ge T + 2n.$$

The bit complexity of the transformed instance is polynomial in the original bit complexity.

Proof. Clear denominators to obtain integer points b_i and rational threshold τ'. Let

$$D_0 = \mathrm{lcm}(1, 2, \dots, n) \cdot \mathrm{denom}(\tau').$$

For integer points, the ordinary k-means objective of a cluster C of size m is

$$W(C) = \sum_{i \in C} \|b_i\|^2 - \frac{1}{m} \left\| \sum_{i \in C} b_i \right\|^2 .$$

The first term is an integer. The second term has denominator dividing m. Therefore $mW(C) \in \mathbb{Z}$, and since D_0 is a multiple of every $m \le n$,

$$D_0 W(C) \in \mathbb{Z}.$$

Summing over clusters shows that every k-means objective value lies in

$$\frac{1}{D_0}\mathbb{Z}.$$

Let

$$q = \lfloor D_0 \tau' \rfloor.$$

Because all objective values lie on the grid $\frac{1}{D_0}\mathbb{Z}$,

$$\mathrm{OPT}_b \le \tau' \implies \mathrm{OPT}_b \le \frac{q}{D_0},$$

while

$$\mathrm{OPT}_b > \tau' \implies \mathrm{OPT}_b \ge \frac{q+1}{D_0}.$$

Now scale all coordinates by an integer

$$L = \left\lceil \sqrt{2nD_0} \right\rceil,$$

and set

$$a_i = Lb_i.$$

Every k-means objective scales by L^2. Define

$$T = \frac{L^2 q}{D_0}.$$

Then in the YES case,

$$\mathrm{OPT}_a \le \frac{L^2 q}{D_0} = T,$$

while in the NO case,

$$\mathrm{OPT}_a \ge \frac{L^2(q+1)}{D_0} = T + \frac{L^2}{D_0} \ge T + 2n.$$

The bit length of D_0 is polynomial because

$$\log D_0 = O(\log \mathrm{denom}(\tau')) + O(\log \mathrm{lcm}(1, \ldots, n)),$$

and $\log \mathrm{lcm}(1, \ldots, n) = O(n)$. Therefore $\log L$ is polynomial, and the scaled instance has polynomial bit complexity. □

Lemma 5.55 (Fractional Assignment Cannot Beat Hard k-Means)**.** *Let $a_1, \ldots, a_n$ be points in Euclidean space. For any fractional assignment weights*

$$w_{jr} \ge 0, \qquad \sum_{r=1}^{k} w_{jr} = 1,$$

and any centers $c_1, \dots, c_k$, define the fractional objective

$$F = \frac{1}{n} \sum_{j=1}^{n} \sum_{r=1}^{k} w_{jr} \|a_j - c_r\|^2.$$

Let OPT *be the optimal hard k-means objective. Then*

$$F \geq \mathrm{OPT}.$$

Proof. For fixed centers $c_1, \dots, c_k$, define the hard nearest-center assignment

$$r(j) \in \arg\min_r \|a_j - c_r\|^2.$$

Then

$$\frac{1}{n} \sum_{j=1}^{n} \|a_j - c_{r(j)}\|^2 \leq F.$$

Recomputing the centers as the means of the assigned clusters cannot increase the hard k-means objective. Therefore

$$\mathrm{OPT} \leq \frac{1}{n} \sum_{j=1}^{n} \|a_j - c_{r(j)}\|^2 \leq F.$$

□

Theorem 5.56 (Reset-Vacuity NP-Completeness)**.** *In the Gaussian quadratic restricted retention regime with common covariance $\frac{1}{2}I$, the decision problem*

$$\Delta_{\mathrm{ret}}^{\mathrm{quad}}(M) < \theta$$

is NP-complete for rational data. NP-hardness holds even for minimal definite machines of synchronization depth one.

Proof. Reduce from Euclidean k-means [31]. By Lemma 5.54, assume we are given integer points

$$a_1, \dots, a_n \in \mathbb{Z}^d,$$

integer k, and rational threshold T such that either

$$\mathrm{OPT}_a \leq T$$

or

$$\mathrm{OPT}_a \geq T + 2n.$$

Construct a retention instance as follows. Let

$$\mathcal{S} = \{s_1, \dots, s_n\}, \qquad \mathcal{I} = \{1, \dots, n\}.$$

Use uniform i.i.d. inputs. Define reset transitions

$$\tau(s_i, \ell) = s_\ell.$$

After one input, the state equals the input symbol. Hence the synchronization depth is one.

Let e_i be the i-th standard basis vector in $\mathbb{R}^n$, and set

$$\eta = \frac{1}{8}.$$

Define Gaussian means

$$m_i = (a_i, \eta e_i) \in \mathbb{R}^{d+n}.$$

Let emissions be

$$P_i = \mathcal{N}(m_i, \tfrac{1}{2}I).$$

The stationary distribution is uniform on $\mathcal{S}$.

Every partition of $\mathcal{S}$ is lumpable. Indeed, if s_i, s_j lie in the same block, then for every input ℓ,

$$\tau(s_i, \ell) = s_\ell = \tau(s_j, \ell),$$

so the successors are identical and therefore lie in the same block.

For a partition $\mathcal{C} = \{C_1, \dots, C_K\}$ with $K \le k$, the quadratic retention cost is

$$\Delta_{\mathrm{ret}}^{\mathrm{quad}}(\mathcal{C}) = \frac{1}{n} \sum_{r=1}^{K} \sum_{i \in C_r} \|m_i - \bar{m}_r\|^2.$$

Decompose into main and tag coordinates.

The main-coordinate cost is

$$\frac{1}{n} \sum_{r=1}^{K} \sum_{i \in C_r} \|a_i - \bar{a}_r\|^2,$$

which is exactly the ordinary k-means objective divided by n.

For a block of size m, the tag-coordinate cost is

$$\frac{1}{n} \sum_{i \in C_r} \left\| \eta e_i - \frac{\eta}{m} \sum_{j \in C_r} e_j \right\|^2 = \frac{\eta^2}{n}(m-1).$$

Summing over blocks gives

$$\text{tag cost} = \eta^2 \frac{n-K}{n} \le \eta^2 = \frac{1}{64}.$$

If $\mathrm{OPT}_a \le T$, choose an optimal k-clustering. Then

$$\Delta_{\mathrm{ret}}^{\mathrm{quad}}(k) \le \frac{T}{n} + \frac{1}{64}.$$

If $\mathrm{OPT}_a \geq T + 2n$, then for every partition into at most k blocks, the main-coordinate cost is at least

$$\frac{\mathrm{OPT}_a}{n} \geq \frac{T + 2n}{n} = \frac{T}{n} + 2.$$

The tag cost is nonnegative, so

$$\Delta_{\mathrm{ret}}^{\mathrm{quad}}(k) \geq \frac{T}{n} + 2.$$

Set

$$\theta = \frac{T}{n} + 1.$$

Then

$$\mathrm{OPT}_a \leq T \implies \Delta_{\mathrm{ret}}^{\mathrm{quad}}(k) < \theta,$$

while

$$\mathrm{OPT}_a \geq T + 2n \implies \Delta_{\mathrm{ret}}^{\mathrm{quad}}(k) > \theta.$$

Thus the strict decision problem is NP-hard.

Membership in NP follows because a certificate is a partition of the n states. For each block, the centroid is a rational vector whose coordinates have polynomial bit length. The quadratic cost is a sum of squared Euclidean distances with rational coefficients. The tag contribution is

$$\eta^2 \frac{n - K}{n}, \qquad \eta = \frac{1}{8},$$

which is rational with constant denominator. The total cost can therefore be computed exactly by rational arithmetic in time polynomial in n, d, and the input bit length, and compared with the rational threshold θ. Hence the problem is NP-complete. $\square$

Theorem 5.57 (NP-Hardness for Minimal Definite Depth-2 ϵ-Machines)**.** *In the Gaussian quadratic restricted retention regime with common covariance $\frac{1}{2}I$, the decision problem*

$$\Delta_{\mathrm{ret}}^{\mathrm{quad}}(M) < \theta$$

is NP-hard even when restricted to minimal definite ϵ-machines of synchronization depth 2*. With rational means and rational threshold, it is NP-complete.*

Proof. Use the tagged depth-2 construction. By Lemma 5.54, assume we are given integer points $a_1, \ldots, a_n$, integer k, and rational threshold T with either

$$\mathrm{OPT}_a \leq T$$

or

$$\mathrm{OPT}_a \geq T + 2n.$$

Let the input alphabet be

$$\mathcal{I} = \{1, \ldots, n\},$$

with uniform i.i.d. inputs. Create states

$$s_{ij}, \qquad i, j \in \{1, \dots, n\}.$$

Define the transition

$$\tau(s_{ij}, \ell) = s_{j\ell}.$$

After two inputs, the state is determined by the last two inputs, independently of the initial state. Hence the synchronization depth is 2, and the machine is definite of depth 2.

Let e_{ij} be the standard basis vector indexed by (i, j) in $\mathbb{R}^{n^2}$, and set

$$\eta = \frac{1}{8}.$$

Define means

$$m_{ij} = (a_j, \eta e_{ij}) \in \mathbb{R}^{d+n^2}.$$

Let the emission be

$$P_{ij} = \mathcal{N}(m_{ij}, \tfrac{1}{2}I).$$

Because the covariance is $\frac{1}{2}I$,

$$D_{\mathrm{KL}}(P_{ij} \| P_c) = \|m_{ij} - c\|^2$$

for Gaussian representatives $P_c = \mathcal{N}(c, \frac{1}{2}I)$.

The means m_{ij} are all distinct because the tag coordinates ηe_{ij} are distinct. Therefore the one-step predictive distributions are all distinct, and the minimal ϵ-machine has n^2 causal states.

Suppose first that $\mathrm{OPT}_a \leq T$. Let $C_1, \dots, C_k$ be a k-clustering of $\{1, \dots, n\}$ achieving objective at most T. Define

$$D_r = \{s_{ij} : j \in C_r\}.$$

This partition is lumpable: if $s_{ij}, s_{i'j'} \in D_r$, then $j, j' \in C_r$, and for every input ℓ,

$$\tau(s_{ij}, \ell) = s_{j\ell}, \qquad \tau(s_{i'j'}, \ell) = s_{j'\ell},$$

both of which have second coordinate ℓ, hence lie in the same block $D_{c(\ell)}$, where $c(\ell)$ is the cluster containing ℓ.

The stationary distribution on pairs (i, j) is uniform. The main-coordinate cost is therefore

$$\frac{1}{n} \sum_{r=1}^{k} \sum_{j \in C_r} \|a_j - \bar{a}_r\|^2 \leq \frac{T}{n}.$$

The tag-coordinate cost is at most $1/64$. Hence

$$\Delta_{\mathrm{ret}}^{\mathrm{quad}}(k) \leq \frac{T}{n} + \frac{1}{64}.$$

Now suppose that $\mathrm{OPT}_a \geq T + 2n$. Consider any k-state approximation. It induces a machine-state variable K with $|K| \leq k$. Decompose the cost into main and tag coordinates. The tag cost is nonnegative.

For the main coordinate, let J be the second state index. Under the uniform stationary distribution, J is uniform on $\{1, \dots, n\}$. Let

$$w_{jr} = \mathbb{P}(K = r \mid J = j).$$

The main cost is

$$\frac{1}{n} \sum_{j=1}^{n} \sum_{r=1}^{k} w_{jr} \left\| a_j - c_r^{\mathrm{main}} \right\|^2 .$$

This is a fractional k-means objective on the points a_j, multiplied by $1/n$. By Lemma 5.55, fractional assignment cannot beat the optimal hard k-means objective. Therefore the main cost is at least

$$\frac{\mathrm{OPT}_a}{n} \geq \frac{T + 2n}{n} = \frac{T}{n} + 2.$$

Set

$$\theta = \frac{T}{n} + 1.$$

Then YES implies

$$\Delta_{\mathrm{ret}}^{\mathrm{quad}}(k) < \theta,$$

while NO implies

$$\Delta_{\mathrm{ret}}^{\mathrm{quad}}(k) > \theta.$$

Thus the retention decision problem is NP-hard. Rational parameters give NP-completeness by the same certificate argument as in Theorem 5.56. $\square$

Theorem 5.58 (Promise NP-Hardness of Finite-Alphabet Full-KL Retention)**.** *Finite-alphabet full-KL retention is NP-hard under a promise, already for synchronization-depth-one reset machines in which every state partition is lumpable. Precisely, there are polynomial-time computable rationals $\theta_{\mathrm{yes}} < \theta_{\mathrm{no}}$ such that deciding*

$$\Delta_{\mathrm{ret}}^{\mathrm{KL}}(k) \leq \theta_{\mathrm{yes}} \quad \textit{versus} \quad \Delta_{\mathrm{ret}}^{\mathrm{KL}}(k) \geq \theta_{\mathrm{no}}$$

is NP-hard, the promise being that one of the two holds.

Proof. Reduce from Euclidean k-means. By Lemma 5.54 we may assume integer points $a_1, \dots, a_n \in \mathbb{Z}^d$, an integer k, and a rational T with either $\mathrm{OPT}_a \leq T$ or $\mathrm{OPT}_a \geq T + 2n$, all of polynomial bit complexity.

The machine. Take $\mathcal{S} = \{s_1, \dots, s_n\}$ and $\mathcal{I} = \{1, \dots, n\}$ with reset transitions $\tau(s_i, \ell) = s_\ell$ and uniform i.i.d. inputs, exactly as in Theorem 5.56. The synchronization depth is one, the stationary distribution is uniform, and every partition is lumpable, since s_i, s_j in a common block have identical successors $\tau(s_i, \ell) = s_\ell = \tau(s_j, \ell)$.

The output alphabet and the tangent embedding. Let $\mathcal{O} = \{1, \ldots, 2d\}$, so predictive laws are points of Δ^{2d-1}. Embed each a_i in the tangent space of the simplex at the uniform law by doubling coordinates with opposite signs:

$$z_i = \big(a_{i1}, -a_{i1},\ a_{i2}, -a_{i2},\ \ldots,\ a_{id}, -a_{id}\big) \in \mathbb{Z}^{2d}, \qquad \sum_j (z_i)_j = 0.$$

The doubling is what makes the embedding *centred*, hence tangent to the simplex, and it preserves the Euclidean geometry up to a factor of two:

$$\|z_i - z_{i'}\|_2^2 = 2\|a_i - a_{i'}\|_2^2.$$

Let $u = (1/2d, \ldots, 1/2d)$ and set

$$p_i = u + \delta z_i, \qquad \delta = 2^{-\Lambda},$$

with Λ a positive integer fixed below. Each p_i is a rational probability vector, and for $\delta < \big(2d \max_{i,j} |(z_i)_j|\big)^{-1}$ all coordinates are strictly positive, so every p_i lies in a fixed compact subset of the interior. Assign $P_i = p_i$ as the predictive law of state s_i.

The objective is a Jensen–Shannon cost. For a block C with uniform weights $w_i = 1/|C|$, Lemma 5.1 makes the optimal representative the mixture centroid $\bar{p}_C$, so the block cost is

$$J_C = \sum_{i \in C} w_i\, D_{\mathrm{KL}}(p_i \| \bar{p}_C),$$

the Jensen–Shannon divergence of the block, and $\Delta_{\mathrm{ret}}^{\mathrm{KL}}(k) = \min_{|\mathcal{C}| \le k} \sum_C \frac{|C|}{n} J_C$.

Uniform quadratic expansion. Write $\bar{z}_C = \sum_{i \in C} w_i z_i$, so $\bar{p}_C = u + \delta \bar{z}_C$. Since all points lie in a compact interior set, a third-order Taylor expansion of D_{KL} about u, with the linear terms vanishing because $\sum_j (z_i - \bar{z}_C)_j = 0$, gives

$$J_C = d\,\delta^2 \sum_{i \in C} w_i \|z_i - \bar{z}_C\|_2^2 + \rho_C, \qquad |\rho_C| \ \le\ C_0\, d^2 \delta^3 Z^3,$$

where $Z = \max_i \|z_i\|_1$ and C_0 is absolute; the Hessian of $p \mapsto \sum_j p_j \log p_j$ at u is $2d \cdot I$ on the tangent space, which supplies the factor d after the $\frac{1}{2}$ of the quadratic term. The remainder bound is uniform over blocks and partitions because it depends only on d, δ and Z. By the geometry of the embedding,

$$\sum_{i \in C} w_i \|z_i - \bar{z}_C\|_2^2 = 2 \sum_{i \in C} w_i \|a_i - \bar{a}_C\|_2^2,$$

so the leading term is $2d\delta^2$ times the k-means cost of C. Summing over blocks with weights $|C|/n$,

$$\left|\ \Delta_{\mathrm{ret}}^{\mathrm{KL}}(k) - \frac{2d\delta^2}{n}\,\mathrm{OPT}_a\ \right| \ \le\ C_0\, d^2 \delta^3 Z^3,$$

the same bound applying to the value of every individual partition, so that the optimizing partitions of the two objectives need not coincide for the estimate to hold.

Choice of δ and the thresholds. The promise gap in the k-means objective is $2n$, so the gap in the leading term is $2d\delta^2 \cdot 2n/n = 4d\delta^2$. Choose Λ polynomial in the input bit length so that

$$C_0\, d^2\delta^3 Z^3 \;<\; d\,\delta^2,$$

which holds as soon as $\delta < \big(C_0 d Z^3\big)^{-1}$; since Z has polynomially many bits, such a Λ is polynomial and $\delta = 2^{-\Lambda}$ is computable in polynomial time. Set

$$\theta_{\text{yes}} = \frac{2d\delta^2}{n}T + d\delta^2, \qquad \theta_{\text{no}} = \frac{2d\delta^2}{n}(T+2n) - d\delta^2 = \theta_{\text{yes}} + 2d\delta^2,$$

both rational with polynomially many bits and $\theta_{\text{yes}} < \theta_{\text{no}}$. In the YES case $\mathrm{OPT}_a \le T$ gives $\Delta^{\mathrm{KL}}_{\mathrm{ret}}(k) \le \theta_{\text{yes}}$; in the NO case $\mathrm{OPT}_a \ge T + 2n$ gives $\Delta^{\mathrm{KL}}_{\mathrm{ret}}(k) \ge \theta_{\text{no}}$. The reduction is polynomial, and the promise is met in both cases. □

Corollary 5.59 (APX-Hardness of Finite-Alphabet Full-KL Retention)**.** *There is a constant $\varepsilon_0 > 0$ such that approximating $\Delta^{\mathrm{KL}}_{\mathrm{ret}}(k)$ to within a factor $1+\varepsilon_0$ is NP-hard, for finite output alphabets and already for synchronization-depth-one reset machines in which every partition is lumpable. In particular the problem admits no PTAS unless* $\mathsf{P} = \mathsf{NP}$*.*

Proof. Euclidean k-means is APX-hard: there is $\varepsilon_1 > 0$ such that approximating it within $1+\varepsilon_1$ is NP-hard [29], with the constant subsequently improved [30]. We transfer this through the embedding of Theorem 5.58, for which it suffices to control the objective *relatively* rather than only across a promise gap.

Fix an instance $a_1, \dots, a_n \in \mathbb{Z}^d$ and let Ξ denote its k-means objective, so that for a partition $\mathcal{C}$ with at most k blocks the embedded cost obeys, uniformly in $\mathcal{C}$,

$$\Delta^{\mathrm{KL}}_{\mathrm{ret}}(\mathcal{C}) = \frac{2d\delta^2}{n}\,\Xi(\mathcal{C}) \;+\; \rho(\mathcal{C}), \qquad |\rho(\mathcal{C})| \;\le\; C_0 d^2\delta^3 Z^3,$$

by the uniform third-order expansion established in the proof of Theorem 5.58.

Two normalizations make the transfer relative. First, we may assume $\Xi(\mathcal{C}) \ge 1$ for every partition that is not cost-zero: the points are integral, so by the granularity argument of Lemma 5.54 every nonzero value of Ξ is a rational with denominator dividing $\mathrm{lcm}(1,\dots,n)$, and rescaling the input by that integer — a polynomial-time operation multiplying Ξ by its square — makes all nonzero values at least 1. Second, choose $\delta = 2^{-\Lambda}$ with Λ polynomial in the input size so that

$$C_0 d^2\delta^3 Z^3 \;\le\; \frac{\varepsilon_1}{4}\cdot\frac{2d\delta^2}{n},$$

which holds as soon as $\delta \le \varepsilon_1/(2C_0\, d\, n\, Z^3)$ and therefore costs only polynomially many bits.

Write $\beta = 2d\delta^2/n$. For every partition,

$$\beta\,\Xi(\mathcal{C}) - \tfrac{\varepsilon_1}{4}\beta \;\le\; \Delta^{\mathrm{KL}}_{\mathrm{ret}}(\mathcal{C}) \;\le\; \beta\,\Xi(\mathcal{C}) + \tfrac{\varepsilon_1}{4}\beta,$$

so, using $\Xi \ge 1$ on the relevant partitions,

$$\big(1 - \tfrac{\varepsilon_1}{4}\big)\beta\,\Xi(\mathcal{C}) \;\le\; \Delta^{\mathrm{KL}}_{\mathrm{ret}}(\mathcal{C}) \;\le\; \big(1 + \tfrac{\varepsilon_1}{4}\big)\beta\,\Xi(\mathcal{C}).$$

Hence $\Delta^{\mathrm{KL}}_{\mathrm{ret}}$ and $\beta\Xi$ agree to within the factor $(1+\frac{\varepsilon_1}{4})/(1-\frac{\varepsilon_1}{4})$ *simultaneously for all partitions*, and the same therefore holds for their minima.

The uniformity is the operative point and is visible numerically. Enumerating *all* partitions with at most k blocks for $d \in \{2,3,4\}$ and $n \in \{5,6\}$, the ratio $\Delta^{\mathrm{KL}}_{\mathrm{ret}}(\mathcal{C})/(\beta\,\Xi(\mathcal{C}))$ has spread at most 9.0×10^{-5} at $\delta = 10^{-3}$, 9.0×10^{-9} at $\delta = 10^{-5}$, and 9.0×10^{-13} at $\delta = 10^{-7}$: the spread falls as δ^2, so a polynomial number of bits in δ makes the two-sided bound hold for every partition at once. In particular the minimizing partitions of the two objectives coincide on the instances tested.

Suppose an algorithm returned a partition $\widehat{\mathcal{C}}$ with $\Delta^{\mathrm{KL}}_{\mathrm{ret}}(\widehat{\mathcal{C}}) \le (1+\varepsilon_0)\min_{\mathcal{C}} \Delta^{\mathrm{KL}}_{\mathrm{ret}}(\mathcal{C})$. Then

$$\Xi(\widehat{\mathcal{C}}) \;\le\; \frac{1+\frac{\varepsilon_1}{4}}{1-\frac{\varepsilon_1}{4}}\,(1+\varepsilon_0)\,\min_{\mathcal{C}}\Xi(\mathcal{C}),$$

and choosing $\varepsilon_0 > 0$ small enough that this prefactor is at most $1+\varepsilon_1$ — for instance $\varepsilon_0 = \varepsilon_1/4$, for $\varepsilon_1 \le 1$ — yields a $(1+\varepsilon_1)$-approximation to k-means, which is NP-hard. The machine construction, lumpability of all partitions, and depth-one synchronization are exactly as in Theorem 5.58. □

Remark 5.60 (Why Only a Promise, and Only Hardness)**.** Two limitations of Theorem 5.58 are intrinsic to the construction and should not be read as slack.

First, the statement is a promise problem. The reduction controls the objective only up to the cubic remainder ρ_C, so it separates instances whose values differ by at least $2d\delta^2$ but says nothing about instances whose values straddle a threshold more closely. Exact comparison of $\Delta^{\mathrm{KL}}_{\mathrm{ret}}(k)$ with a rational threshold is a different question, and the promise formulation is the honest one.

Second, no NP-*completeness* is claimed. Membership in NP would require verifying, in polynomial time, an inequality between a rational threshold and a sum of terms of the form $p\log(p/q)$ with rational p, q. Deciding such inequalities is not known to be in P, and the general problem of comparing sums of logarithms of rationals to a rational is a well-known obstacle to exact membership. Proposition 5.61 below reduces that sum to a single comparison $R \ge e^s$ between a factored rational and an exponential, which is the sharp form of the obstruction and is discussed in Remarks 5.62 and 5.63. The Gaussian quadratic regime of Theorem 5.56 avoids this entirely, since its objective is a rational function of the data; that is precisely why completeness is available there and only hardness here.

The obstruction just described can be located exactly. The sum of logarithms collapses to a single comparison between a rational and an exponential, so neither the number of logarithmic terms nor their coefficients is the difficulty.

Proposition 5.61 (Algebraic Form of the Full-KL Verification Problem)**.** *Let a full-KL retention instance have rational stationary weights π_s and rational, strictly positive predictive laws P_s on a finite alphabet, and fix a partition ϕ. Then there are a positive integer D and a positive rational R with*

$$\Delta^{\mathrm{KL}}_{\mathrm{ret}}(\phi) = \frac{1}{D}\log R,$$

where D and a factored *representation $R = \prod_j b_j^{m_j}$, with the b_j rational and the m_j integers, are computable in time polynomial in the instance size. Consequently, comparing $\Delta^{\mathrm{KL}}_{\mathrm{ret}}(\phi)$ with a rational threshold τ reduces in polynomial time to the problem*

$$\begin{aligned}&\textsc{RationalExpCompare}:\\ &\quad \textit{given a factored positive rational } R \textit{ and a rational } s,\\ &\quad \textit{decide whether } R \geq e^s.\end{aligned}$$

Proof. For the fixed partition, each block centroid $\bar{P}_k = \big(\sum_{s\in C_k} \pi_s P_s\big)/\big(\sum_{s\in C_k} \pi_s\big)$ is a rational probability vector computable by rational arithmetic. Expanding the definition,

$$\Delta^{\mathrm{KL}}_{\mathrm{ret}}(\phi) = \sum_k \sum_{s\in C_k} \sum_y \pi_s P_s(y) \log \frac{P_s(y)}{\bar{P}_k(y)} = \sum_j a_j \log b_j,$$

a sum of at most $|\mathcal{S}^+||\mathcal{O}|$ terms with all a_j and b_j rational and $b_j > 0$. Let D be a common denominator of the a_j and write $a_j = m_j/D$ with $m_j \in \mathbb{Z}$. Then

$$\Delta^{\mathrm{KL}}_{\mathrm{ret}}(\phi) = \frac{1}{D}\sum_j m_j \log b_j = \frac{1}{D}\log\Big(\prod_j b_j^{m_j}\Big),$$

so $R = \prod_j b_j^{m_j}$, negative exponents contributing reciprocals. Both D and the list of pairs (b_j, m_j) have polynomial bit size. Finally $\Delta^{\mathrm{KL}}_{\mathrm{ret}}(\phi) \leq \tau$ if and only if $R \leq e^{D\tau}$, and $s = D\tau$ is rational of polynomial bit size. □

Remark 5.62 (The Factored Representation Is Necessary)**.** Proposition 5.61 asserts polynomial computability of R only in factored form, and the qualification is not a formality. The exponents $m_j = a_j D$ are integers of polynomial bit *length* but exponential *magnitude*, so the expanded numerator and denominator of $R = \prod_j b_j^{m_j}$ have bit size $\sum_j |m_j| \log b_j$, which grows exponentially in the instance size: on rational instances with $|\mathcal{S}^+| \leq 6$ and denominators below 32, inputs of a few hundred bits already produce expanded representations of billions of bits, while the factored representation stays in the low thousands. RationalExpCompare must therefore be posed for R given as a product of powers, that is by a straight-line program, and it is in that form that it is a PosSLP-type question.

Remark 5.63 (Exact Status of Membership)**.** Proposition 5.61 isolates the obstruction to membership in NP: it is RationalExpCompare, and not the number of logarithmic terms, which collapse into one. That problem is decidable. For $s = 0$ one compares the rational R with 1, which is exact arithmetic on the factored form. For $s \neq 0$ the Lindemann–Weierstrass theorem makes e^s transcendental while R is rational, hence $R \neq e^s$, so the comparison is strict and is resolved by evaluating e^s to sufficient precision. What is missing is a polynomial bound on the precision required, equivalently a polynomial lower bound on $|R - e^s|$ in terms of the bit sizes of the factored R and of s.

Accordingly, finite-alphabet full-KL retention lies in NP whenever RationalExpCompare admits polynomial-time verifiable certificates. This is the precise sense in which the Gaussian quadratic regime of Theorem 5.56, whose objective is a rational function of the data and so never meets a transcendental comparison, admits NP-completeness, while the full-KL regime of Theorem 5.58 admits promise NP-hardness only.

Remark 5.64 (The Output Alphabet of the Embedding)**.** The tangent-space embedding of Theorem 5.58 needs a map $T : \mathbb{R}^d \to \mathbb{R}^{|\mathcal{O}|}$ subject to three constraints: its image must lie in the zero-sum subspace, so that $u + \delta T z$ is a probability vector; it must be a similarity, $T^\top T = cI$, so that the leading Jensen–Shannon term is a constant multiple of the k-means objective; and it must be rational, so that integer inputs give rational laws and the promise structure survives. The doubling map $a \mapsto (a_1, -a_1, \ldots, a_d, -a_d)$ meets all three with $|\mathcal{O}| = 2d$ and $c = 2$.

The zero-sum subspace of $\mathbb{R}^{|\mathcal{O}|}$ has dimension $|\mathcal{O}| - 1$, so $|\mathcal{O}| \geq d + 1$ is forced. Whether the minimum is attained is a rational-equivalence question, and the answer depends on d rather than on a universal factor.

For $d = 2$ the minimum is *not* attained. In the basis $(1, -1, 0)$, $(1, 1, -2)$ the zero-sum subspace of $\mathbb{Q}^3$ carries the Gram matrix $\operatorname{diag}(2, 6)$, of discriminant 12, whose class modulo rational squares is 3; an orthogonal rational basis of equal norms would give $\operatorname{diag}(c, c)$, of discriminant c^2 and class 1. Since $3 \not\equiv 1$ modulo squares, no rational centred similarity $\mathbb{R}^2 \to \mathbb{R}^3$ exists, and $|\mathcal{O}| = 4 = 2d$ is optimal.

For $d = 3$, by contrast, the minimum *is* attained. The three non-constant rows of a Hadamard matrix of order 4,

$$(-1, -1, 1, 1), \qquad (-1, 1, -1, 1), \qquad (-1, 1, 1, -1),$$

are zero-sum, pairwise orthogonal and of common norm 4, and therefore define a rational centred similarity $\mathbb{R}^3 \to \mathbb{R}^4$ with $c = 4$. The alphabet may thus be taken of size $d + 1 = 4$ rather than $2d = 6$. The same construction applies whenever a Hadamard matrix of order $d + 1$ exists, so $|\mathcal{O}| = d + 1$ suffices for $d = 3, 7, 15, \ldots$ — the Sylvester orders $2^k - 1$ — and for every other Hadamard order as well: $d = 11$, from the order-12 Hadamard matrix, is the smallest qualifier outside the Sylvester pattern. Wherever an order-$(d + 1)$ Hadamard matrix exists, the factor two is slack.

The operative statement is therefore that $|\mathcal{O}|$ may be taken as small as the least $n \geq d+1$ for which the zero-sum subspace of $\mathbb{Q}^n$ contains d pairwise orthogonal rational vectors of equal norm, a condition of Hadamard type; $n = 2d$ always works and $n = d+1$ works exactly when the corresponding quadratic form is rationally equivalent to a scalar matrix. In every case $|\mathcal{O}|$ grows with d, since the zero-sum subspace must have dimension at least d. Hardness of full-KL retention for a *fixed* output alphabet therefore remains open and would require a reduction from a source problem hard in bounded dimension, Euclidean k-means in fixed dimension not being known to be so.

Remark 5.65 (Scope of the Complexity Theorems)**.** Theorems 5.56 and 5.57 are theorems of the Gaussian quadratic restricted retention regime, where the objective is rational in the data and NP-completeness is available. The reset construction shows that the right-congruence constraint can be made vacuous while preserving reachability and minimality. The depth-2 construction uses tagged means to make all n^2 predictive distributions distinct.

For unrestricted full-KL retention over a finite output alphabet, Theorem 5.58 gives NP-hardness under a promise, by embedding k-means into a small interior neighborhood of the uniform law so that the Jensen–Shannon block cost is a controlled perturbation of the Euclidean one. What is not established for that regime is exact NP-completeness, for the arithmetic reason recorded in Remark 5.60.

6 Commitment

The commitment regime concerns deterministic finite-state realization and enforcement. Its exact zero-error theory is governed by Myhill–Nerode residual equivalence. Its quantitative theory is governed by dynamic programming and, in simultaneous-move settings, by strategic-spread lower bounds. The exponent correspondence draws an *analogy* between commitment and the Booleanized symmetrized $\alpha = 0$ support vertex: both are zero-set / support-type objects rather than magnitude-type objects. This is a structural analogy, not an identification; commitment is not raw one-sided Rényi order 0, and it is not ordinary Schatten rank. Section 6.2 below supplies a genuinely quantitative, non-threshold commitment gap for readers who want a magnitude-type commitment object.

6.1 Deterministic Specifications

Definition 6.1 (Causal Length-Preserving Specification)**.** A **deterministic specification** is a map

$$F : \mathcal{I}^* \to \mathcal{O}^*$$

that is **length-preserving**, $|F(u)| = |u|$ for every $u \in \mathcal{I}^*$, and **causal**: $F(u)$ is a prefix of $F(uw)$ for every $u, w \in \mathcal{I}^*$. Causality and length-preservation make F a well-typed finite-word transduction, and by the standard inverse-limit argument for prefix-consistent finite-word maps, F extends uniquely to a continuous causal map $F^\infty : \mathcal{I}^{\mathbb{N}} \to \mathcal{O}^{\mathbb{N}}$ on infinite input streams, recovering the BSG map type of Section 1.1. All commitment quantities below are stated for finite $u, w \in \mathcal{I}^*$, for which $uw \in \mathcal{I}^*$ and every displayed expression is well typed.

For each finite input history u, causality lets us define the residual

$$F_u(w) = F(uw)_{|u|+1:|u|+|w|}, \qquad w \in \mathcal{I}^*,$$

the unique word with $F(uw) = F(u)F_u(w)$. Both u and w range over finite words, so $uw \in \mathcal{I}^*$ remains in the domain of F, and F_u is the continuation behavior required after the history u.

Define the Myhill–Nerode equivalence

$$u \sim_F v \iff F_u = F_v,$$

and let

$$\kappa_{\det}(F) = \operatorname{index}(\sim_F).$$

Theorem 6.2 (Exact Commitment Bound)**.** *A deterministic BSG with M states implements F iff*

$$M \geq \kappa_{\det}(F).$$

Under normalized worst-case mismatch cost,

$$\Delta_{\mathrm{com}}(M) = \begin{cases} 0, & M \geq \kappa_{\det}(F), \\ 1, & M < \kappa_{\det}(F). \end{cases}$$

Proof. Necessity is the Myhill–Nerode residual lower bound: if two histories induce distinct residuals, no deterministic machine can assign them the same state and still implement F on all continuations. Sufficiency is realized by the residual automaton whose states are the equivalence classes of $\sim_F$, with transition

$$[u] \cdot x = [ux]$$

and output determined by the residual class. □

The structural Boolean version uses the $\{0, \infty\}$ valuation:

$$\mathcal{E}(f, g) = \begin{cases} 0, & f = g, \\ \infty, & f \neq g. \end{cases}$$

The normalized operational version uses the $\{0, 1\}$ valuation. The threshold is the same; only the positive mismatch value changes; the schema-layer statement of this separation for arbitrary separated task theories is Corollary 4.10, of which the present pair of valuations is the commitment instance.

6.2 Quantitative Commitment: A Distributional Rate–Distortion Gap

The exact commitment gap $\Delta_{\mathrm{com}}(M)$ of Theorem 6.2 is a threshold: it is 0 once $M \geq \kappa_{\det}(F)$ and constant and positive otherwise, and it carries no information about how a sub-threshold budget should be spent. We now supply a genuinely quantitative, rate-distortion-typed commitment gap over right congruences, parallel in shape to the retention information-bottleneck gap $\Delta_{\mathrm{ret}}^{\mathrm{KL}}(M)$ and to the grounding tracking deficit (Definition 7.35), with its own exact fixed-budget formula and its own support-sensitive zero threshold.

Fix a discounted-prefix history law μ on $\mathcal{I}^*$,

$$\mu(u) = (1-\gamma)\gamma^{|u|} \Pr[\text{prefix } u], \qquad \gamma \in (0,1),$$

as in Definition 4.3; this is a genuine probability measure on all of $\mathcal{I}^*$, since finite-length prefix masses are mixed geometrically across lengths. Let F be a causal length-preserving specification (Definition 6.1).

Definition 6.3 (Implemented Machine and One-Step Mismatch)**.** For a right congruence $\sim$ of index at most M (Definition 2.11) and an output rule

$$\rho : \big(\mathcal{I}^*/\!\sim\big) \times \mathcal{I} \to \mathcal{O},$$

the pair $(\sim, \rho)$ realizes the Mealy machine that, having read a nonempty history u of length t, outputs $\rho\big([u_{1:t-1}]_\sim,\, u_t\big)$ at position t. This is a BSG output map of type $\lambda : (\mathcal{I}^*/\!\sim) \times \mathcal{I} \to \mathcal{O}$ (Section 1.1), so $(\sim, \rho)$ is implementable with $\mathrm{index}(\sim)$ states. Its one-step mismatch indicator at u is

$$\mathrm{e}_{\sim,\rho}(u) = \mathbf{1}\Big\{\rho\big([u_{1:t-1}]_\sim,\, u_t\big) \neq F(u)_t\Big\}, \qquad t = |u| \geq 1.$$

Definition 6.4 (Distributional Commitment Rate–Distortion Gap)**.**

$$\Delta^{\mathrm{RD}}_{\mathrm{com}}(M) = \min_{\substack{\sim \text{ right congruence} \\ \mathrm{index}(\sim) \leq M}} \min_{\rho} \sum_{u \in \mathcal{I}^* \setminus \{\varepsilon\}} \mu(u)\, \mathrm{e}_{\sim,\rho}(u).$$

The minimum is attained: up to renaming of the quotient states there are only finitely many Mealy machines with at most M states over the finite alphabets $\mathcal{I}$ and $\mathcal{O}$, and the objective depends on $(\sim, \rho)$ only through the induced machine's transition structure and one-step output rule.

For a class k of $\sim$ and a symbol $a \in \mathcal{I}$, let

$$\mu(k, a) = \sum_{\substack{u \neq \varepsilon \\ [u_{1:|u|-1}]_\sim = k,\ u_{|u|} = a}} \mu(u), \qquad q_{k,a}(o) = \frac{1}{\mu(k, a)} \sum_{\substack{u \neq \varepsilon \\ [u_{1:|u|-1}]_\sim = k,\ u_{|u|} = a \\ F(u)_{|u|} = o}} \mu(u) \quad (\mu(k, a) > 0),$$

the μ-conditional law, given class k and next symbol a, of the true next output demanded by F.

Definition 6.5 (One-Step Determination Index)**.** The **one-step determination index** $\kappa_{\mathrm{pair}}(F, \mu)$ is the minimum index of a right congruence $\sim$ on $\mathcal{I}^*$ for which $F(u)_{|u|}$ is μ-almost-surely determined by the pair $\big([u_{1:|u|-1}]_\sim,\, u_{|u|}\big)$: explicitly, there exists a map $\rho : \big(\mathcal{I}^*/\sim\big) \times \mathcal{I} \to \mathcal{O}$ with $\rho\big([u_{1:|u|-1}]_\sim, u_{|u|}\big) = F(u)_{|u|}$ for μ-almost every nonempty u. Only nonempty u carry a one-step output, so the empty word — which has positive mass, $\mu(\varepsilon) = 1 - \gamma$, under a discounted law (Definition 2.4) — does not enter the condition; the index depends on μ only through its support.

Theorem 6.6 (Exact Fixed-Budget Commitment Rate–Distortion Formula)**.** *For every right congruence $\sim$,*

$$\min_{\rho} \sum_{u \neq \varepsilon} \mu(u)\, \mathrm{e}_{\sim,\rho}(u) = \sum_{\substack{k,a \\ \mu(k,a) > 0}} \mu(k, a)\Big(1 - \max_{o \in \mathcal{O}} q_{k,a}(o)\Big),$$

attained by the greedy rule $\rho(k, a) \in \arg\max_o q_{k,a}(o)$. Consequently

$$\Delta^{\mathrm{RD}}_{\mathrm{com}}(M) = \min_{\mathrm{index}(\sim) \leq M} \sum_{\substack{k,a \\ \mu(k,a) > 0}} \mu(k, a)\Big(1 - \max_{o} q_{k,a}(o)\Big).$$

Moreover $\Delta^{\mathrm{RD}}_{\mathrm{com}}(M) = 0$ if and only if $M \geq \kappa_{\mathrm{pair}}(F, \mu)$, the one-step determination index of Definition 6.5. In particular $\kappa_{\mathrm{pair}}(F, \mu) \leq \kappa_{\det}(F)$, the threshold of Theorem 6.2: the two indices coincide when μ has full support, and the inequality is strict under restricted supports (Remark 6.7).

Proof. Fix $\sim$. Group the sum defining $\Delta^{\mathrm{RD}}_{\mathrm{com}}$ by the pair $(k, a) = ([u_{1:|u|-1}]_\sim, u_{|u|})$: for fixed (k, a) with $\mu(k, a) > 0$, the rule ρ assigns a single value $\rho(k, a) \in \mathcal{O}$ to every u in that group, and

$$\sum_{\substack{u \neq \varepsilon \\ [u_{1:|u|-1}]_\sim = k,\, u_{|u|} = a}} \mu(u)\, \mathbf{1}\{\rho(k, a) \neq F(u)_{|u|}\} = \mu(k, a)\big(1 - q_{k,a}(\rho(k, a))\big) \geq \mu(k, a)\big(1 - \max_o q_{k,a}(o)\big),$$

with equality exactly when $\rho(k, a)$ is a mode of $q_{k,a}$. Summing over (k, a) and choosing the mode at every pair proves the displayed formula; groups with $\mu(k, a) = 0$ contribute 0 under any choice of ρ and may be assigned arbitrarily. Minimizing further over $\sim$ of index at most M gives $\Delta^{\mathrm{RD}}_{\mathrm{com}}(M)$.

For the zero threshold: $\Delta^{\mathrm{RD}}_{\mathrm{com}}(M) = 0$ for a given $\sim$ if and only if every visited pair (k, a) has $q_{k,a}$ a point mass, i.e. $F(u)_{|u|}$ is μ-almost surely determined by $([u_{1:|u|-1}]_\sim, u_{|u|})$, which is precisely the determination condition of Definition 6.5; hence $\Delta^{\mathrm{RD}}_{\mathrm{com}}(M) = 0$ exactly for $M \geq \kappa_{\mathrm{pair}}(F, \mu)$. The inequality $\kappa_{\mathrm{pair}}(F, \mu) \leq \kappa_{\det}(F)$ holds because the full Myhill–Nerode congruence $\sim_F$ determines the one-step output through the pair: $[u_{1:|u|-1}]_{\sim_F}$ determines the residual $F_{u_{1:|u|-1}}$, whose value on the one-letter continuation $u_{|u|}$ is $F(u)_{|u|}$, so the residual automaton of Theorem 6.2 is one-step exact on all of $\mathcal{I}^*$. When μ has full support the reverse inequality holds as well: a pair-determining right congruence then forces $u \sim v$ to imply $F_u = F_v$, because each continuation output $F(uw)_{|u|+j}$ is a function of $([uw_{1:j-1}]_\sim, w_j)$ and right invariance identifies the two arguments along u and v. The two indices then coincide, and strictness of the inequality is a support phenomenon, as in Remark 6.7. $\square$

Remark 6.7 (Why the Pair, Not the Class)**.** Determination by the pair $([u_{1:|u|-1}]_\sim, u_{|u|})$ is strictly weaker than determination by the class $[u]_\sim$ alone: the pair refines the class, since $[u]_\sim = [u_{1:|u|-1}]_\sim \cdot u_{|u|}$, but a single class is reached by many distinct pairs, so a class-level representative is a stronger requirement than a pair-level output rule. For the identity transduction $F(u) = u$ over a binary alphabet under a full-support μ, every residual coincides, so $\kappa_{\det}(F) = 1$: the one-state rule $\rho(*, x) = x$ is one-step exact, $\Delta^{\mathrm{RD}}_{\mathrm{com}}(1) = 0$, and $\kappa_{\mathrm{pair}}(F, \mu) = 1$, whereas determining $u_{|u|}$ from the class $[u]_\sim$ alone requires two classes. The gap between κ_{pair} and $\kappa_{\det}$, by contrast, is a support phenomenon: for the transduction $F(u)_t = u_t \oplus u_{t-1}$ (with $u_0 = 0$) the two indices coincide at 2 under a full-support law, but if the input process is supported on strictly alternating words, then $u_{t-1} = 1 - u_t$ on the support, the one-step output is constant there, and $\kappa_{\mathrm{pair}}(F, \mu) = 1$.

Remark 6.8 (Relation to the Three Commitment Notions)**.** Three distinct commitment quantities are now in play, kept apart by notation. $\Delta_{\mathrm{com}}(M)$ (Theorem 6.2) is the exact worst-case Boolean threshold, testing equality of full residual functions on every history. $\Delta^{\mathrm{RD}}_{\mathrm{com}}(M)$ (Definition 6.4) is a distributional, single-step quantitative loss averaged over a fixed history law μ, whose zero set can be strictly larger than that of Δ_{com} because it only demands agreement on the very next symbol and only on the μ-visited part of history space. $\Delta_{\mathrm{com}}^{\mathrm{game}}(M)$ (Section 6.5) is a simultaneous-move strategic gap against an adversarial input sequence chosen online, and is not an averaging or a threshold quantity at all. No general inequality between $\Delta_{\mathrm{com}}^{\mathrm{RD}}$ and $\Delta_{\mathrm{com}}^{\mathrm{game}}$ is asserted; they are posed on different protocols.

6.3 General Safety Properties

For a safety property

$$\Lambda \subseteq (\mathcal{I} \times \mathcal{O})^*,$$

there may be many safe output choices. A deterministic specification F is **admissible** for Λ if every finite input-output prefix generated by F lies in Λ. We write this as

$$F \subseteq \Lambda.$$

A BSG enforces Λ if its induced specification is admissible for every input sequence.

The complexity is the minimal Myhill–Nerode index of a deterministic admissible strategy.

Definition 6.9 (Controller Complexity of a Safety Property)**.**

$$\kappa_{\mathrm{ctrl}}(\Lambda) = \min_{F \subseteq \Lambda} \kappa_{\mathrm{det}}(F),$$

where the minimum is over deterministic specifications whose graphs lie inside Λ. If no finite-state admissible specification exists, set $\kappa_{\mathrm{ctrl}}(\Lambda) = \infty$.

Theorem 6.10 (Exact Safety Enforcement)**.** *A deterministic BSG with M states can enforce Λ iff*

$$M \geq \kappa_{\mathrm{ctrl}}(\Lambda).$$

If Λ admits a unique admissible residual specification F_Λ, then

$$\kappa_{\mathrm{ctrl}}(\Lambda) = \kappa_{\mathrm{det}}(F_\Lambda).$$

Proof. If a BSG enforces Λ, it implements some admissible deterministic specification $F \subseteq \Lambda$. Its number of states is at least $\kappa_{\mathrm{det}}(F)$, hence at least $\kappa_{\mathrm{ctrl}}(\Lambda)$. Conversely, if $M \geq \kappa_{\mathrm{ctrl}}(\Lambda)$, choose an admissible specification $F \subseteq \Lambda$ attaining the minimum and implement it by its residual automaton.

If Λ admits a unique admissible residual specification F_Λ, then the minimum is attained uniquely up to residual equivalence, so

$$\kappa_{\mathrm{ctrl}}(\Lambda) = \kappa_{\mathrm{det}}(F_\Lambda).$$

□

The distinction between $\kappa_{\mathrm{det}}(F)$ and $\kappa_{\mathrm{ctrl}}(\Lambda)$ is that a safety property may admit many strategies, and the minimal controller may be smaller than the complexity of an arbitrary admissible specification.

Theorem 6.11 (Finite-Monitor Winning Region for Regular Safety)**.** *Suppose $\Lambda \subseteq (\mathcal{I} \times \mathcal{O})^*$ is a regular safety property, recognized by a deterministic finite automaton $\mathcal{M} = (\mathcal{Q}, q_0, \delta_{\mathcal{M}}, \mathrm{Acc})$ over the joint alphabet $\mathcal{I} \times \mathcal{O}$, in the sense that $w \in \Lambda$ iff every prefix of w has its $\delta_{\mathcal{M}}$-run staying inside* Acc*. Define the* ***safe winning region*** *$W \subseteq \mathcal{Q}$ by the greatest-fixpoint recursion*

$$W_0 = \mathrm{Acc}, \qquad W_{n+1} = \Big\{ q \in W_n : \forall a \in \mathcal{I}\ \exists b \in \mathcal{O} \ \ \delta_{\mathcal{M}}(q,(a,b)) \in W_n \Big\}, \qquad W = \bigcap_{n \geq 0} W_n.$$

Since $\mathcal{Q}$ is finite the recursion stabilizes after at most $|\mathcal{Q}|$ steps. Then:

(i) A finite-state admissible strategy enforcing Λ from q_0 exists if and only if $q_0 \in W$.

(ii) When $q_0 \in W$, the residual automaton on state space W with output rule $\beta(q,a) \in \{b \in \mathcal{O} : \delta_{\mathcal{M}}(q,(a,b)) \in W\}$ (any fixed selection) enforces Λ and has at most $|W| \leq |\mathcal{Q}|$ states; consequently

$$\kappa_{\mathrm{ctrl}}(\Lambda) \leq |W| \leq |\mathcal{Q}|.$$

(iii) W is computable in time polynomial in $|\mathcal{Q}| \cdot |\mathcal{I}| \cdot |\mathcal{O}|$ by the stated fixpoint iteration.

Proof. This is the classical attractor / safety-game construction, specialized to a Mealy controller choosing b in response to an adversarial a. Monotonicity of the recursion ($W_{n+1} \subseteq W_n$) and finiteness of $\mathcal{Q}$ give stabilization in at most $|\mathcal{Q}|$ steps, since each strict decrease removes at least one state. If $q_0 \in W$, then from every $q \in W$ and every $a \in \mathcal{I}$ there is $b \in \mathcal{O}$ with $\delta_{\mathcal{M}}(q,(a,b)) \in W$ by construction, so the selection β keeps every play inside $W \subseteq \mathrm{Acc}$ forever, which is exactly admissibility for Λ; this proves the "if" direction of (i) and (ii). Conversely, suppose a finite-state admissible strategy enforces Λ from q_0. An induction on n shows every state reachable by that strategy lies in W_n for every n: a violation of membership in W_{n+1} at a reachable state would exhibit an adversary input a after which every output b leaves W_n, and admissibility of the strategy would then force the next reached state to already violate W_n, contradicting the inductive hypothesis at the previous step. Hence every reachable state, in particular q_0, lies in $\bigcap_n W_n = W$, proving the "only if" direction of (i). Claim (iii) is immediate from the fixpoint iteration, which performs at most $|\mathcal{Q}|$ rounds each costing $O(|\mathcal{Q}| \cdot |\mathcal{I}| \cdot |\mathcal{O}|)$ to test the two quantifiers over $\mathcal{Q} \times \mathcal{I} \times \mathcal{O}$. □

Remark 6.12 (Relation to κ_{ctrl})**.** Theorem 6.11 gives a constructive upper bound $\kappa_{\mathrm{ctrl}}(\Lambda) \leq |W|$ computable directly from a recognizing automaton for Λ, complementing the abstract characterization of Definition 6.9 and Theorem 6.10. The bound $|W|$ need not equal $\kappa_{\mathrm{ctrl}}(\Lambda)$: further residual minimization of the W-automaton, exactly as in Theorem 6.2, can strictly reduce the state count while preserving admissibility.

6.4 Quantitative Commitment: Alternating Moves

Let a deterministic reward automaton be

$$\mathcal{R} = (\mathcal{Q}, q_0, \delta_{\mathcal{R}}, r),$$

with bounded reward

$$r : \mathcal{Q} \times \mathcal{I} \times \mathcal{O} \to \mathbb{R}$$

and discount $\gamma \in (0, 1)$. In the alternating-move interpretation, the adversary chooses an input $a \in \mathcal{I}$, the agent observes a, and then chooses an output $b \in \mathcal{O}$.

Two value functions occur below and must be kept apart, since they belong to different protocols.

The **alternating-move** value is the state function

$$V_{\text{alt}}(q) = \min_{a \in \mathcal{I}} \max_{b \in \mathcal{O}} \left[r(q, a, b) + \gamma \, V_{\text{alt}}(\delta_{\mathcal{R}}(q, a, b)) \right],$$

in which the adversary moves first and minimizes and the agent, having observed the current input, maximizes. Write $V_{\text{alt}}(\infty) = V_{\text{alt}}(q_0)$ for its value at the initial reward state; the agent here is unrestricted in memory, since V_{alt} is computed on the reward automaton itself.

The simultaneous protocol cannot be realized by a Mealy BSG. A Mealy output map has the type $\lambda : \mathcal{S} \times \mathcal{I} \to \mathcal{O}$, so it reads the current input a_t; a policy that must commit to b_t before seeing a_t is a different object and requires a different machine model.

Definition 6.13 (Pre-Input-Output Controller)**.** A **pre-input-output controller** of budget M is a tuple

$$\pi = (\mathcal{M}, m_0, \beta, \eta), \qquad |\mathcal{M}| \leq M,$$

with output map $\beta : \mathcal{M} \to \mathcal{O}$ and update map $\eta : \mathcal{M} \times \mathcal{I} \times \mathcal{O} \to \mathcal{M}$, acting by

$$b_t = \beta(m_t), \qquad m_{t+1} = \eta(m_t, a_t, b_t).$$

The output at time t depends on the memory state alone, so it is committed before a_t is revealed; the memory may then absorb a_t. Write Π_M for the set of such controllers. A Mealy BSG is *not* a pre-input-output controller, since its output map is a function of $(\mathcal{S}, \mathcal{I})$.

The **simultaneous-move** value at budget M is

$$V_{\text{sim}}(M) = \sup_{\pi \in \Pi_M} \inf_{a_{0:\infty} \in \mathcal{I}^{\mathbb{N}}} \sum_{t \geq 0} \gamma^t \, r(q_t, a_t, b_t),$$

the trajectory (q_t) being generated on the reward automaton by the realized pairs (a_t, b_t).

The M-state commitment gap is the difference between the two protocols,

$$\Delta_{\text{com}}^{\text{game}}(M) = V_{\text{alt}}(\infty) - V_{\text{sim}}(M).$$

This is the **simultaneous-move strategic gap**, kept notationally distinct from the exact Boolean gap $\Delta_{\text{com}}(M)$ of Theorem 6.2 and from the distributional rate-distortion gap $\Delta_{\text{com}}^{\text{RD}}(M)$ of Section 6.2: it compares two action protocols on the same reward automaton, not two feasible classes of the same protocol. It is an **observation-and-memory gap**: a positive value combines the finite-memory restriction with the loss of access to the current input, and the two contributions are not separated by this definition. Even at $M = \infty$ the gap need not vanish, since the alternating protocol grants the agent information that no pre-input-output controller possesses.

6.5 Quantitative Commitment: Simultaneous Moves

In simultaneous-move games, the adversary chooses a_t and the agent chooses b_t without observing each other's current choice. The agent may use finite memory and past observations, but not the current adversary action.

Define the strategic spread

$$\alpha(q) = V_{\text{alt}}(q) - \max_{b \in \mathcal{O}} \min_{a \in \mathcal{I}} \left[r(q, a, b) + \gamma\, V_{\text{alt}}(\delta_{\mathcal{R}}(q, a, b)) \right].$$

The first term is the alternating-move value, in which the agent observes the current input before responding. The second term is the best one-step simultaneous-move guarantee using a pure output b. By the elementary minimax inequality,

$$\max_b \min_a \leq \min_a \max_b,$$

so $\alpha(q) \geq 0$.

Theorem 6.14 (Strategic Spread Lower Bound, Precise Version)**.** *Let V_{alt} be the alternating-move value function of Section 6.5. For a pre-input-output controller $\pi \in \Pi_M$ of Definition 6.13, let $b_t = \beta(m_t)$ be its output at time t, committed before a_t is revealed. Define the myopic worst-case adversary by*

$$a_t \in \arg\min_{a \in \mathcal{I}} \left[r(q_t, a, b_t) + \gamma V_{\text{alt}}(\delta_{\mathcal{R}}(q_t, a, b_t)) \right].$$

Let q_t^π be the reward-automaton trajectory generated by π and this adversary. Define

$$\alpha_\gamma(\mathcal{R}, M) = \inf_{\pi \in \Pi_M} \sum_{t=0}^{\infty} \gamma^t \alpha(q_t^\pi),$$

where the infimum is over pre-input-output controllers of budget M (Definition 6.13), these being the policies admissible in the simultaneous protocol.

Then the finite-memory simultaneous-move commitment gap satisfies

$$\Delta_{\text{com}}^{\text{game}}(M) \geq \alpha_\gamma(\mathcal{R}, M).$$

Proof. For each t,

$$V_{\text{alt}}(q_t) \geq \min_a \left[r(q_t, a, b_t) + \gamma V_{\text{alt}}(q_{t+1}) \right],$$

because $V_{\text{alt}}(q_t) = \min_a \max_b [\cdots]$ and the maximum over b is at least the value at the particular output b_t.

By the choice of a_t,

$$r(q_t, a_t, b_t) + \gamma V_{\text{alt}}(q_{t+1}) = \min_a \left[r(q_t, a, b_t) + \gamma V_{\text{alt}}(\delta_{\mathcal{R}}(q_t, a, b_t)) \right].$$

Therefore,

$$V_{\text{alt}}(q_t) - r(q_t, a_t, b_t) - \gamma V_{\text{alt}}(q_{t+1}) \geq \alpha(q_t).$$

Multiply by γ^t and sum from $t = 0$ to $T - 1$:

$$V_{\text{alt}}(q_0) - \sum_{t=0}^{T-1} \gamma^t r(q_t, a_t, b_t) - \gamma^T V_{\text{alt}}(q_T) \geq \sum_{t=0}^{T-1} \gamma^t \alpha(q_t).$$

Since rewards are bounded and $\gamma < 1$,

$$\gamma^T V_{\text{alt}}(q_T) \to 0.$$

Thus,

$$V_{\text{alt}}(q_0) - \sum_{t=0}^{\infty} \gamma^t r(q_t, a_t, b_t) \geq \sum_{t=0}^{\infty} \gamma^t \alpha(q_t).$$

For each deterministic finite-memory policy π, the adversary constructed above is admissible in the simultaneous-move value because π is deterministic and therefore its current output is predictable from the history. Hence

$$\inf_{\text{adv}} \text{Payoff}(\pi, \text{adv}) \leq \text{Payoff}(\pi, \text{myopic adversary}).$$

For every π,

$$\text{Payoff}(\pi, \text{myopic adversary}) \leq V_{\text{alt}}(q_0) - \sum_{t=0}^{\infty} \gamma^t \alpha(q_t^\pi).$$

Therefore

$$\max_{\pi} \inf_{\text{adv}} \text{Payoff}(\pi, \text{adv}) \leq V_{\text{alt}}(q_0) - \inf_{\pi} \sum_{t=0}^{\infty} \gamma^t \alpha(q_t^\pi).$$

The left-hand side is $V_{\text{sim}}(M)$. Hence

$$V_{\text{sim}}(M) \leq V_{\text{alt}}(q_0) - \alpha_\gamma(\mathcal{R}, M),$$

and therefore

$$\Delta_{\text{com}}^{\text{game}}(M) = V_{\text{alt}}(q_0) - V_{\text{sim}}(M) \geq \alpha_\gamma(\mathcal{R}, M).$$

$\square$

Remark 6.15 (Memory Tracking and Adversary Specification)**.** The proof tracks the machine memory state m_t through the augmented state (q_t, m_t). The lower bound applies to every finite-memory deterministic policy, and therefore to every finite-state BSG. The adversary is explicitly specified as the myopic worst-case response to the deterministic current output b_t.

Corollary 6.16 (Stateless Games)**.** *For a stateless game,*

$$\Delta_{\text{com}}^{\text{game}}(M) = \frac{\alpha}{1 - \gamma}$$

for all $M \geq 1$, the budget being irrelevant here because a pre-input-output controller in a stateless game emits an output depending only on its memory state, against which the *adversary best-responds; additional memory therefore cannot improve the guarantee. Here α is the* one-stage *strategic spread*

$$\alpha = \min_{a\in\mathcal{I}} \max_{b\in\mathcal{O}} r(a,b) \;-\; \max_{b\in\mathcal{O}} \min_{a\in\mathcal{I}} r(a,b).$$

Both terms are one-stage quantities in the immediate reward r. Neither is the discounted alternating-move value $V_{\rm alt}$, which in the stateless case satisfies $V_{\rm alt} = \bigl(\min_a \max_b r(a,b)\bigr)/(1-\gamma)$.

Proof. Write $m_1 = \min_a \max_b r(a,b)$ and $m_2 = \max_b \min_a r(a,b)$ for the two one-stage values, so that $\alpha = m_1 - m_2 \geq 0$ by the minimax inequality.

In a stateless game the reward-automaton state is constant, so the Bellman equation of Section 6.5 reads $V_{\rm alt} = m_1 + \gamma V_{\rm alt}$, whence

$$V_{\rm alt} = \frac{m_1}{1-\gamma}.$$

The strategic spread at the unique state then evaluates to

$$\alpha(q) = V_{\rm alt} - \max_b \min_a \bigl[r(a,b) + \gamma V_{\rm alt}\bigr] = V_{\rm alt} - \gamma V_{\rm alt} - \max_b \min_a r(a,b) = (1-\gamma)V_{\rm alt} - m_2 = m_1 - m_2,$$

which is α and is independent of t. The lower bound of Theorem 6.14 gives

$$\Delta^{\rm game}_{\rm com}(M) \geq \sum_{t=0}^{\infty} \gamma^t \alpha = \frac{\alpha}{1-\gamma}.$$

For the reverse inequality, choose a constant output

$$b^\star \in \arg\max_b \min_a r(a,b).$$

A single-state pre-input-output controller emitting $b^\star$ at every time guarantees discounted payoff

$$\frac{1}{1-\gamma} \max_b \min_a r(a,b).$$

Since the alternating-move value is $V_{\rm alt} = m_1/(1-\gamma)$, the gap is at most

$$\frac{m_1 - \max_b \min_a r(a,b)}{1-\gamma} = \frac{m_1 - m_2}{1-\gamma} = \frac{\alpha}{1-\gamma}.$$

Thus equality holds. Finite memory does not improve the guarantee, because the deterministic policy's current output is predictable from the history, and the adversary can choose the myopic worst-case response in each period. $\square$

6.6 Commitment as the Booleanized $\alpha = 0$ Vertex

The exact zero-error commitment cost is not raw Rényi order-0 divergence. Raw D_0 tests one-sided support inclusion, not equality, and is not Boolean-valued. The construction uses symmetrization followed by Boolean valuation.

For probability vectors p, q, recall the classical order-0 Rényi divergence

$$D_0(p\|q) = -\log q(\operatorname{supp} p),$$

with the usual extended-value conventions. Define the symmetrized support mismatch

$$S_0(p, q) = D_0(p\|q) + D_0(q\|p),$$

and a Boolean valuation

$$v(t) = \begin{cases} 0, & t = 0, \\ 1, & t > 0, \end{cases}$$

or, in the extended convention, $v(t) = \infty$ for $t > 0$. The Booleanized support cost is

$$B_0(p, q) = v(S_0(p, q)).$$

For probability vectors,

$$D_0(p\|q) = 0 \iff q(\operatorname{supp} p) = 1 \iff \operatorname{supp} q \subseteq \operatorname{supp} p$$

up to null sets. Therefore

$$S_0(p, q) = 0 \iff \operatorname{supp} p = \operatorname{supp} q,$$

and hence

$$B_0(p, q) = 0 \iff \operatorname{supp} p = \operatorname{supp} q.$$

Commitment residuals are usually Boolean or symbolic functions rather than probability vectors. Let a residual row be a Boolean function

$$r_u : \Sigma^* \to \{0, 1\}.$$

Its support is

$$\operatorname{supp} r_u = \{w \in \Sigma^* : r_u(w) = 1\}.$$

Equality of Boolean functions is equality of supports:

$$r_u = r_v \iff \operatorname{supp} r_u = \operatorname{supp} r_v.$$

Thus the Booleanized support cost recovers the commitment zero-error test:

$$B_0(r_u, r_v) = 0 \iff r_u = r_v.$$

For general deterministic residuals with non-Boolean output alphabet, one may encode the residual as the Boolean support of its graph:

$$\operatorname{supp} F_u = \{(w, o) : F_u(w) = o\}.$$

Equality of residuals is again equality of supports. The primitive commitment cost remains equality; the Booleanized support construction is an encoding, not a literal probabilistic divergence on arbitrary residual objects.

Consequently, if κ_{MN} is the Myhill–Nerode index, the commitment gap satisfies

$$\Delta_{\mathrm{com}}(M) = 0 \iff M \geq \kappa_{\mathrm{MN}},$$

and under the normalized 0/1 convention,

$$\Delta_{\mathrm{com}}(M) = 1 \quad \text{if } M < \kappa_{\mathrm{MN}}.$$

Remark 6.17 (Scope of the Boolean Vertex)**.** Raw one-sided D_0 tests support inclusion, not equality, and is not Boolean-valued. The construction uses symmetrization and Boolean valuation. The relevant object is Boolean residual equality / Myhill–Nerode index, not ordinary real or complex rank.

Remark 6.18 (Not Ordinary Schatten Rank)**.** The commitment vertex is a Boolean-rank / Myhill–Nerode threshold. Ordinary Schatten $p \to 0$ limits count ordinary singular-value rank, not deterministic Boolean realization rank. Boolean Hankel matrices need not define compact Hilbert-space operators, and no ordinary Schatten-norm limit is used to derive the commitment threshold.

Remark 6.19 (The $p \to 0$ Limit and Why It Is Not the Boolean Vertex)**.** It is natural to ask whether commitment is recovered from the Schatten family by letting $p \downarrow 0$. It is not, and the obstruction is worth isolating, because it explains why the Boolean vertex is a separate construction rather than a limiting case of Definition 8.9.

For a finite-rank operator A the *unpowered* tail sum does converge to a rank count:

$$\lim_{p \downarrow 0} \sum_{i>r} \sigma_i(A)^p = \#\{i > r : \sigma_i(A) > 0\},$$

since $\sigma^p \to 1$ for every $\sigma > 0$. Composing with the threshold $t \mapsto \mathbf{1}_{\{t>0\}}$ then yields a Boolean valuation. That composition is an additional operation, not part of the Schatten family.

The tail functional actually used in the template, $\Phi_r^{(p)}(A) = \left(\sum_{i>r} \sigma_i(A)^p\right)^{1/p}$, does *not* converge to a $\{0, \infty\}$-valued map. If the tail has $k \geq 2$ nonzero singular values then $\sum_{i>r} \sigma_i^p \to k > 1$, so $\Phi_r^{(p)}(A) = \left(\sum_{i>r} \sigma_i^p\right)^{1/p} \to +\infty$ irrespective of the magnitudes: for $\sigma = (3, \frac{1}{2}, \frac{1}{4})$ and $r = 1$ one has $\Phi_1^{(1)} = 0.75$ but $\Phi_1^{(0.01)} \approx 4.5 \times 10^{29}$. If the tail has exactly one nonzero entry the limit is that entry, and if the tail is zero the limit is 0. The limit is therefore not two-valued, and the divergence is driven by the tail *count* rather than by any norm.

Accordingly, commitment replaces ordinary rank by the deterministic Boolean rank $\operatorname{rank}_{\mathbb{B}}^{\det} N_\Lambda$ and the real-valued norm by a Boolean realizability valuation. It is the Boolean-rank vertex of the rank-control principle, not a corollary of Mirsky's theorem, and it does not require N_Λ to define a compact Hilbert-space operator.

Remark 6.20 (Commitment Threshold Behavior)**.** Exact Boolean commitment does not produce a smooth spectral-tail budget curve. The gap is a threshold governed by κ_{MN}. Smooth bias–variance behavior appears only in quantitative commitment variants, such as the discounted reward models above, or in approximate-commitment losses explicitly defined elsewhere.

7 Grounding

The grounding regime concerns approximation of stochastic channels or residual kernels. The spectral converse is a theorem about linear finite-rank realizations, not about deterministic Mealy machines. Exact symbolic deterministic grounding is a different problem. The regime also contains intrinsic lower bounds for deterministic approximation: a superpolynomial determinism gap and a stochasticity floor.

7.1 Grounding Instantiation

In the grounding instantiation, residual objects are bounded operators, stochastic kernels, or elements of a Banach or C^*-algebraic residual space. When a Banach representation B is chosen, the cost is typically an operator-norm or Banach-space distance:

$$\mathcal{E}(\nu, \widehat{\nu}) = \|\nu - \widehat{\nu}\|_{\mathrm{op}}.$$

The scalar impulse response used by the Hankel theory is obtained through a bounded linear functional

$$\phi : B \to \mathbb{C}, \qquad h(u) = \phi(\nu_u).$$

The spectral converse concerns linear finite-rank realizations of the induced Hankel operator. It does not assert that a deterministic Mealy machine with the same state count realizes or approximates the channel. Deterministic symbolic grounding and linear finite-rank grounding are distinct approximation problems.

Remark 7.1 (Residual Norm versus Hankel Norm)**.** As in Remark 3.6, stated here for the grounding instantiation: the norm used for residual-operator comparisons is the norm of the chosen Banach representation B, whereas the Hankel gap uses the ℓ^2 operator norm of the Hankel operator H_ν. No domination between these norms is assumed unless explicitly stated.

7.2 Linear Finite-Rank Spectral Converse

Remark 7.2 (Algebraic Hankel Rank Versus Bounded-Operator Rank)**.** A finite-dimensional weighted automaton factors its formal Hankel matrix as

$$h(uv) = \alpha^\top T_u T_v \eta,$$

which bounds the *algebraic* matrix rank by the state dimension. This does not by itself imply that the infinite matrix defines a bounded, let alone compact, operator on ℓ^2: the factor rows and columns need not be square summable. The spectral statements below are therefore conditional on compactness of H_ν, and algebraic-rank hypotheses do not substitute for it.

Theorem 7.3 (Spectral Bound for Linear Grounding)**.** *Let ν be a stochastic channel whose impulse response defines a compact Hankel operator H_ν on $\ell^2((\mathcal{I} \times \mathcal{O})^*)$. Define the unrestricted linear rank-M gap by*

$$\Delta_{\mathrm{grd}}^{\mathrm{unres}}(M) = \inf_{\operatorname{rank} B \le M} \|H_\nu - B\|_{\mathrm{op}}.$$

Then

$$\Delta_{\mathrm{grd}}^{\mathrm{unres}}(M) = \sigma_{M+1}(H_\nu).$$

Define the Hankel-structured finite-rank realization gap by

$$\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) = \inf_{\substack{\operatorname{rank} B \le M \\ B \ \ \textit{Hankel}}} \|H_\nu - B\|_{\mathrm{op}}.$$

Then

$$\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) \ge \sigma_{M+1}(H_\nu).$$

Equality holds if an optimal rank-M truncation can be chosen Hankel. Theorem 7.9 below identifies a checkable hypothesis under which equality is automatic.

Proof. The unrestricted equality is Meta-Theorem 4.11, applied to the Hilbert-module task theory induced by ν. By definition,

$$\Delta_{\mathrm{grd}}^{\mathrm{unres}}(M) = \inf_{\operatorname{rank} B \le M} \|H_\nu - B\|_{\mathrm{op}}.$$

Eckart–Young–Mirsky gives

$$\inf_{\operatorname{rank} B \le M} \|H_\nu - B\|_{\mathrm{op}} = \sigma_{M+1}(H_\nu).$$

The Hankel-structured feasible set is a subset of the unrestricted feasible set, so restricting the infimum cannot decrease the value. Hence

$$\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) \ge \Delta_{\mathrm{grd}}^{\mathrm{unres}}(M) = \sigma_{M+1}(H_\nu).$$

□

Corollary 7.4 (Spectral Converse under Operator-Norm Domination)**.** *Let $\mathcal{L}_\infty$ be a grounding cost and let $\Delta_{\mathrm{grd}}^{\mathrm{lin}}(M)$ denote the linear finite-rank residual-cost gap. If*

$$\mathcal{L}_\infty(\nu, \widehat{\nu}) \ \ge \ \|H_\nu - H_{\widehat{\nu}}\|_{\mathrm{op}}$$

for every finite-rank residual approximant $\widehat{\nu}$, then

$$\Delta_{\mathrm{grd}}^{\mathrm{lin}}(M) \ \ge \ \sigma_{M+1}(H_\nu).$$

Proof. Every budget-M approximant induces a residual operator $H_{\widehat{\nu}}$ of rank at most M. By hypothesis the grounding cost dominates the operator-norm distance, so the infimum of the cost over such approximants is at least the infimum of $\|H_\nu - H_{\widehat{\nu}}\|_{\mathrm{op}}$ over rank-M operators, which is $\sigma_{M+1}(H_\nu)$ by Eckart–Young–Mirsky [15, 16], as in Theorem 7.3. □

Remark 7.5 (Domination Is the Only Input)**.** Corollary 7.4 is the grounding instance of the domination pattern used throughout: a converse in a spectral quantity follows from a single inequality between the task cost and an operator norm, with no further structure on $\mathcal{L}_\infty$. It supplies a lower bound only; the reverse inequality requires the achievability hypotheses discussed in Section 9.

Corollary 7.6 (Finite-Rank and Compact Limits)**.** *Under the compactness hypothesis of Theorem 7.3,*

$$\Delta_{\mathrm{grd}}^{\mathrm{unres}}(M) \longrightarrow 0 \qquad \textit{as } M \to \infty.$$

If moreover $\mathrm{rank}(H_\nu) = r < \infty$*, then* $\Delta_{\mathrm{grd}}^{\mathrm{unres}}(M) = 0$ *for all* $M \geq r$*. The corresponding statements for* $\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M)$ *require the additional Hankel-approximation hypotheses of Theorems 7.9 and 7.11.*

Proof. Compactness of H_ν gives $\sigma_{M+1}(H_\nu) \to 0$, and Theorem 7.3 identifies $\Delta_{\mathrm{grd}}^{\mathrm{unres}}(M)$ with $\sigma_{M+1}(H_\nu)$. If $\mathrm{rank}(H_\nu) = r$ then $\sigma_{M+1}(H_\nu) = 0$ for every $M \geq r$. □

Proposition 7.7 (Finite-Section Certificates for Linear Grounding)**.** *Let H be a compact Hankel operator on $\ell^2(\Sigma^*)$, and let P_N denote the orthogonal projection onto the span of the words of length at most N. Put $H_N = P_N H P_N$. Then, for every $M \geq 0$,*

$$\sigma_{M+1}(H_N) \leq \sigma_{M+1}(H) = \Delta_{\mathrm{grd}}^{\mathrm{unres}}(M) \leq \sigma_{M+1}(H_N) + \|H - H_N\|_{\mathrm{op}}.$$

Consequently $\sigma_{M+1}(H_N) \to \sigma_{M+1}(H)$, with the displayed norm error, and the same lower certificate applies to the structured gap:

$$\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) \geq \sigma_{M+1}(H_N).$$

If the finite matrix entries of H_N are computable (respectively rational), these are computable (respectively algebraic) finite certificates. Any independent bound on $\|H - H_N\|_{\mathrm{op}}$ turns the certificate into a two-sided error interval.

Proof. For every rank-$\leq M$ operator B,

$$\|H - B\|_{\mathrm{op}} \geq \|P_N(H - B)P_N\|_{\mathrm{op}} \geq \inf_{\mathrm{rank}\, C \leq M} \|H_N - C\|_{\mathrm{op}} = \sigma_{M+1}(H_N),$$

where the last equality is finite-dimensional Eckart–Young–Mirsky. Taking the infimum over B proves the first inequality. The reverse estimate follows from the Lipschitz bound $|\sigma_j(A) - \sigma_j(B)| \leq \|A - B\|_{\mathrm{op}}$. Since $P_N \to I$ strongly and H is compact, $\|H - P_N H P_N\|_{\mathrm{op}} \to 0$. Finally, the structured feasible set is a subset of the unrestricted one, so restricting the infimum in $\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M)$ cannot decrease it below $\sigma_{M+1}(H_N)$ by the same argument applied to $\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}$ in place of $\Delta_{\mathrm{grd}}^{\mathrm{unres}}$. □

Proposition 7.8 (Exact Zero Threshold for the Structured Linear Problem)**.** *Let H be a bounded Hankel operator. Then*

$$\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) = 0 \quad \Longleftrightarrow \quad \mathrm{rank}(H) \le M.$$

Thus, whenever H has finite rank r, both the unrestricted and the Hankel-structured linear gaps vanish exactly for $M \ge r$.

Proof. If $\mathrm{rank}(H) \le M$, then H itself is an admissible structured approximant, so $\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) = 0$. Conversely, suppose $\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) = 0$; choose rank-$\le M$ Hankel operators B_n with $\|H - B_n\|_{\mathrm{op}} \to 0$. The set of bounded operators of rank at most M is norm closed: if $\mathrm{rank}(H) > M$, choose $M+1$ vectors whose images under H are linearly independent; their Gram determinant is positive and remains positive under sufficiently small operator-norm perturbations, so no operator of rank at most M can lie arbitrarily close to H in operator norm. Hence $\mathrm{rank}(H) \le M$. □

Theorem 7.9 (Hankel-Structured Equality via Adamjan–Arov–Krein)**.** *We treat the scalar case first. Suppose $|\Sigma| = 1$, so that $\ell^2(\Sigma^*) \cong \ell^2(\mathbb{N})$ carries a single unilateral shift S of multiplicity one, and an operator B is Hankel exactly when $S^*B = BS$. Let S_+ be the unilateral shift on the Hardy space H^2 of the disc. (For $|\Sigma| > 1$ see Theorem 7.11 and Remark 7.12: the free monoid Σ^* does* not *carry a natural shift of multiplicity one, and the scalar theory below does not apply directly. In the present one-letter case, by contrast, the canonical unitary $e_{a^n} \mapsto z^n$ from $\ell^2(\Sigma^*) \cong \ell^2(\mathbb{N})$ to H^2 satisfies hypothesis (a) below automatically, so that hypothesis is a checkable structural fact here rather than an additional datum; it is nonetheless stated as a hypothesis because the theorem is quoted in the generality in which U is arbitrary.) Assume there is a unitary $U : \ell^2(\Sigma^*) \to H^2$ such that*

*(a) U **intertwines the shifts**: $US = S_+U$; consequently B is Hankel in the sense above if and only if UBU^* satisfies $S_+^*(UBU^*) = (UBU^*)S_+$, i.e. if and only if UBU^* is a classical Hardy-space Hankel operator, so U carries the feasible set*

$$\{B : \mathrm{rank}\, B \le M,\ B \text{ Hankel}\}$$

bijectively *onto the classical rank-$\le M$ Hankel feasible set;*

(b) $UH_\nu U^* = H_\varphi$ for some symbol $\varphi \in L^\infty(\mathbb{T})$ with H_φ compact, equivalently $\varphi \in H^\infty + C(\mathbb{T})$.

Then for every $M \ge 0$

$$\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) = \Delta_{\mathrm{grd}}^{\mathrm{unres}}(M) = \sigma_{M+1}(H_\nu).$$

Moreover an optimal approximant may be taken of the form H_ψ with $\mathbb{P}_-\psi$ rational of degree at most M, and the optimal error is attained. Uniqueness of the optimal rank-$\le M$ Hankel approximant is not *asserted in general; it holds under the classical simplicity hypothesis $\sigma_{M+1}(H_\varphi) > \sigma_{M+2}(H_\varphi)$, equivalently that the Schmidt pair at level $M+1$ is unique up to a unimodular scalar [3, Ch. 4]. Absent that hypothesis, existence and the value σ_{M+1} are the safe AAK claims made here. The two descriptions of the feasible set agree by Kronecker's* *theorem: a Hankel operator has finite rank exactly when the antianalytic part of its symbol is rational, and then*

$$\operatorname{rank} H_\psi = \deg \mathbb{P}_-\psi,$$

where for a rational $r = p/q$ in lowest terms $\deg r = \max(\deg p, \deg q)$, equivalently the sum of the multiplicities of the poles of r including a possible pole at infinity [3, Sec. 4].

Indexing convention. *Singular values are indexed here from $\sigma_1 = \|H_\nu\|_{\mathrm{op}}$, so the rank-at-most-$M$ error is σ_{M+1}. The operator-theory literature indexes from $s_0(T) = \|T\|$, defining $s_m(T) = \inf\{\|T-R\| : \operatorname{rank} R \le m\}$; the two conventions are related by $\sigma_{m+1} = s_m$, and the statement above is the case $m = M$. Rank budget and symbol degree carry the* same *index M, because Kronecker's theorem is an equality rather than an inequality; there is no off-by-one between them.*

Proof. Under the stated hypotheses the Adamjan–Arov–Krein theorem [12] states exactly that the distance from H_φ to the set of Hankel operators of rank at most M equals $\sigma_{M+1}(H_\varphi)$, and that the infimum is attained by a Hankel operator Γ_M of rank at most M, equivalently one whose symbol has rational antianalytic part of degree at most M. In the source convention this reads $\|\Gamma - \Gamma_m\| = s_m(\Gamma)$ with $s_m = \sigma_{m+1}$. We do not reproduce the proof, which is classical; see also [3] for a modern treatment.

Transporting by U preserves singular values and the operator norm, as for any unitary. Preservation of *Hankel structure*, by contrast, is not automatic for a general unitary and is supplied here by hypothesis (a): the intertwining relation $US = S_+U$ gives, for any bounded B,

$$S^*B = BS \quad \Longleftrightarrow \quad S_+^*(UBU^*) = (UBU^*)S_+,$$

so U maps the Hankel feasible set onto the classical Hankel feasible set and U^* maps it back. Since U also preserves rank, the two constrained infima defining $\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M)$ and its Hardy-space counterpart are over corresponding feasible sets and have equal values. Hence the identity descends to H_ν. Combining with the unrestricted equality of Theorem 7.3 gives the chain

$$\sigma_{M+1}(H_\nu) = \Delta_{\mathrm{grd}}^{\mathrm{unres}}(M) \le \Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) = \sigma_{M+1}(H_\nu),$$

so all three coincide. $\square$

Remark 7.10 (Two Rank Conventions, and Why They Differ)**.** Two spectral-tail conventions appear in this paper. They are both correct and must not be conflated; the difference is structural, not notational.

(G) Operator-approximation form. In the grounding regime the budget M constrains an operator directly, $\operatorname{rank} B \le M$, and Eckart–Young–Mirsky or Adamjan–Arov–Krein give the error σ_{M+1}. Budget M, rank M, tail starting at index $M+1$.

(R) Quotient form. In the retention regime the budget M constrains a *partition* into M blocks, and the induced between-block covariance B_ϕ has $\operatorname{rank} B_\phi \le M-1$: the M centroids satisfy the single affine relation $\sum_k \pi(C_k)(\bar{p}_{C_k} - \bar{p}) = 0$, so centring removes one degree of freedom. Ky Fan then bounds $\operatorname{tr}(B_\phi)$ by $\sum_{i=1}^{M-1} \lambda_i$ and the tail is $\sum_{i \ge M} \lambda_i$. Budget M, effective rank $M-1$, tail starting at index M.

The one-step offset between the two is therefore the codimension of the centring constraint, and it is a property of the feasible set rather than of the indexing. Writing the retention tail as $\sum_{i>M} \lambda_i$ would silently import convention (G) into a regime where it is false, and would not recover the converse. Conversely, a grounding tail written $\sum_{i>M} \sigma_i$ is the same as σ_{M+1} onward and is correct. These are the "effective rank budgets" M and $M-1$ recorded in Theorem 8.22, and they are one of the reasons no unconditional cross-regime inequality holds.

Theorem 7.11 (Multi-Letter Case, Conditional Form)**.** *Let $|\Sigma| > 1$. The free monoid Σ^* carries $|\Sigma|$ prefix shifts S_a, $a \in \Sigma$, jointly an isometry of multiplicity $|\Sigma|$, rather than a single unilateral shift of multiplicity one; consequently the classical scalar Adamjan–Arov–Krein theorem does not apply to H_ν as stated. Suppose nevertheless that there exist a Hilbert space $\mathcal{K}$ carrying a multi-shift or free-semigroup Hankel structure, a unitary $U : \ell^2(\Sigma^*) \to \mathcal{K}$ intertwining the respective shift systems and carrying*

$$\{B : \operatorname{rank} B \le M,\ B \text{ Hankel on } \ell^2(\Sigma^*)\}$$

bijectively onto the corresponding rank-$\le M$ Hankel class on $\mathcal{K}$, and an Adamjan–Arov–Krein/Nehari-type finite-rank approximation theorem valid on $\mathcal{K}$ for the transported operator $UH_\nu U^*$ *— that is, $UH_\nu U^*$ belongs to the class to which the theorem applies (in particular it is compact, consistently with the standing spectral admissibility of Definition 3.5), and the theorem supplies*

$$\inf_{\substack{C \text{ Hankel on } \mathcal{K} \\ \operatorname{rank} C \le M}} \|UH_\nu U^* - C\|_{\mathrm{op}} = \sigma_{M+1}(UH_\nu U^*).$$

Then

$$\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) = \Delta_{\mathrm{grd}}^{\mathrm{unres}}(M) = \sigma_{M+1}(H_\nu).$$

Absent such a structure-preserving embedding, only the Eckart–Young–Mirsky inequality

$$\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) \ \ge \ \sigma_{M+1}(H_\nu)$$

is asserted.

Proof. The transport is the same as in the scalar case, and only transport of structure is used. Unitary conjugation by U preserves rank, the operator norm, and singular values, and by hypothesis it carries the Hankel feasible set of $\ell^2(\Sigma^*)$ bijectively onto the Hankel class of $\mathcal{K}$, so

$$\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) = \inf_{\substack{C \text{ Hankel on } \mathcal{K} \\ \operatorname{rank} C \le M}} \|UH_\nu U^* - C\|_{\mathrm{op}}.$$

The hypothesized approximation theorem on $\mathcal{K}$ supplies the value $\sigma_{M+1}(UH_\nu U^*) = \sigma_{M+1}(H_\nu)$ for this constrained infimum, and Eckart–Young–Mirsky identifies $\Delta_{\mathrm{grd}}^{\mathrm{unres}}(M) = \sigma_{M+1}(H_\nu)$ unconditionally, so the three quantities coincide. The intertwining clause of the hypothesis is not separately consumed by this argument — the bijectivity of the Hankel classes is what is used — but it is retained because it is the property any natural candidate for U would have to satisfy, and because it excludes transporting the feasible sets by a unitary unrelated to the shift systems. $\square$

Remark 7.12 (Multi-Alphabet Scope)**.** The hypothesis of Theorem 7.11 is strong and is *not* known to be checkable in general. Since this manuscript routinely uses the joint alphabet $\Sigma = \mathcal{I} \times \mathcal{O}$, for which $|\Sigma| > 1$ in every nondegenerate case, the scalar AAK theorem should not be presented as automatically applicable. Identifying the correct free-semigroup or vector-valued Hardy model, and determining whether finite-rank AAK approximation survives the equivalence, is an operator-theoretic problem; it is recorded as Open Problem 7.13.

Open Problem 7.13 (Intrinsic Multi-Alphabet Hankel Equality)**.** For $|\Sigma| > 1$, characterize intrinsically – in terms of the channel ν – when the Hankel operator on $\ell^2(\Sigma^*)$ admits a structure-preserving embedding into a vector-valued Hardy or full Fock space Hankel class carrying an Adamjan–Arov–Krein/Nehari finite-rank approximation theorem, and hence when $\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) = \sigma_{M+1}(H_\nu)$. *Established baseline:* Theorem 7.9 settles the scalar case $|\Sigma| = 1$ unconditionally, and Corollary 7.14 converts a strict gap into an obstruction, but only in that scalar model. *Falsifiable success criterion:* an intrinsic (i.e. stated purely in terms of ν, with no auxiliary embedding datum) necessary and sufficient condition for $\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) = \sigma_{M+1}(H_\nu)$ to hold for all M simultaneously, or a proof that no such ν-intrinsic condition exists (e.g. by exhibiting two channels with identical ν-intrinsic invariants of every fixed finite type but differing equality status).

Corollary 7.14 (When the Structured Gap Is Strict)**.** *If $\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) > \sigma_{M+1}(H_\nu)$ for some M, then no* shift-intertwining *Hardy-space symbol as in Theorem 7.9 exists; that is, either no unitary intertwines the two shifts, or no compact Hardy symbol is available. Consequently a strict structured gap is an obstruction to Hardy-space representability, not merely a failure of the approximation argument.*

Proof. Contrapositive of Theorem 7.9. □

Remark 7.15 (Scope: Scalar Shift Model Only)**.** Corollary 7.14 is stated for, and its proof uses, the scalar shift model of Theorem 7.9, i.e. $|\Sigma| = 1$. In the multi-letter setting $|\Sigma| > 1$, Theorem 7.11 requires an explicit multi-shift/free-semigroup embedding hypothesis that is not established in general; without it the contrapositive has no content, since the hypothesis of Theorem 7.9 does not apply and a strict structured gap is then not automatically an obstruction to any representability notion beyond the one stated in Theorem 7.11. Corollary 7.14 should not be read as an intrinsic representability obstruction outside the scalar model absent the extra structure assumed there.

Remark 7.16 (Unrestricted versus Hankel-Restricted Feasible Sets)**.** The two gaps stand in the relation recorded in Theorem 7.3: the unrestricted equality $\Delta_{\mathrm{grd}}^{\mathrm{unres}}(M) = \sigma_{M+1}(H_\nu)$ is Eckart–Young–Mirsky for the rank-M operator-norm problem; the Hankel-structured gap obeys $\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) \geq \sigma_{M+1}(H_\nu)$ because restricting the feasible set cannot decrease the infimum; and equality is not guaranteed without the additional structure of Theorems 7.9 and 7.11.

Remark 7.17 (Infinite Support Does Not Imply Unboundedness)**.** A deterministic Mealy machine may induce an impulse response with infinite support. Infinite support alone does not imply that the associated Hankel operator is unbounded: this is the grounding-side instance of Remark 3.13, whose content — boundedness requires decay or summability conditions, decay suffices, and non-decaying bounded symbolic responses are decided case by case — transfers verbatim from the abstract impulse-response setting of Section 3.3. For example,

$$h(u) = \gamma^{|u|}$$

with $0 < \gamma < 1/|\Sigma|$ defines a bounded Hankel-type operator by the Schur-test domination of Proposition 3.11, the same criterion as there. Bounded symbolic impulse responses that fail to decay may or may not define bounded Hankel operators; boundedness is checked case by case.

Remark 7.18 (AAK Scope)**.** The default spectral converse is Eckart–Young–Mirsky; the precise role of the Hardy-space embedding in making Adamjan–Arov–Krein applicable is stated once, in Remarks 3.12 and 4.12 and Theorems 7.9 and 7.11, and no such embedding is assumed by default anywhere in the grounding theory.

Open Problem 7.19 (Exact Symbolic Grounding Gap)**.** Determine the exact symbolic grounding gap

$$\Delta_{\mathrm{grd}}^{\mathrm{symb}}(M) = \Delta_{\mathrm{grd}}(M; \gamma)$$

of Definition 7.27, for deterministic Mealy machines. The spectral tail $\sigma_{M+1}(H_\nu)$ is an exact theorem for the unrestricted linear finite-rank relaxation, not for exact symbolic deterministic approximation. Relating the linear relaxation gap to the symbolic deterministic gap requires additional structure, such as Boolean realization theorems, stochasticity floors, or superpolynomial determinism gaps. *Established baseline:* Theorem 7.29 and Proposition 7.31 give the $M \to \infty$ limit and certified finite-horizon intervals; see the discussion in Section 17.4. *Falsifiable success criterion:* either an exact closed form for $\Delta_{\mathrm{grd}}^{\mathrm{symb}}(M)$ at finite M, or a proof that no such closed form exists in a stated class of formulas (e.g. rational functions of the transition/emission data and γ).

7.3 Max-Divergence and Operator-Norm Distance

The exponent correspondence organizes grounding by the $\alpha = \infty$ supremum vertex. The $\alpha = \infty$ Rényi vertex is max-divergence. The exact grounding cost is operator-norm distance. These are distinct functionals, but they are locally equivalent under bounded-overlap hypotheses.

Definition 7.20 (Symmetrized Max-Divergence)**.** For finite-dimensional positive definite states ρ, σ, define

$$D_{\max}(\rho \| \sigma) = \log \left\| \sigma^{-1/2} \rho \sigma^{-1/2} \right\|_\infty ,$$

with $+\infty$ if $\operatorname{supp} \rho \nsubseteq \operatorname{supp} \sigma$. Define the symmetrized version

$$D_{\max}^{\mathrm{sym}}(\rho \| \sigma) = \max\{D_{\max}(\rho \| \sigma), D_{\max}(\sigma \| \rho)\}.$$

Proposition 7.21 (Local Equivalence with Operator-Norm Distance)**.** *Let ρ, σ be finite-dimensional states.*

(a) Suppose $\rho, \sigma \geq mI$ *for some* $m > 0$*, and*

$$\|\rho - \sigma\|_\infty \leq \varepsilon.$$

Then

$$D_{\max}^{\mathrm{sym}}(\rho\|\sigma) \leq \log\left(1 + \frac{\varepsilon}{m}\right).$$

(b) Conversely, if

$$D_{\max}^{\mathrm{sym}}(\rho\|\sigma) \leq \delta,$$

then

$$e^{-\delta}\sigma \leq \rho \leq e^{\delta}\sigma,$$

and therefore

$$\|\rho - \sigma\|_\infty \leq (e^{\delta} - 1)\,\|\sigma\|_\infty.$$

For density operators, $\|\sigma\|_\infty \leq 1$*, so*

$$\|\rho - \sigma\|_\infty \leq e^{\delta} - 1.$$

Proof. For (a), if $\|\rho - \sigma\|_\infty \leq \varepsilon$, then

$$\rho \leq \sigma + \varepsilon I.$$

Since $\sigma \geq mI$, we have $I \leq \sigma/m$, hence

$$\rho \leq \sigma + \frac{\varepsilon}{m}\sigma = \left(1 + \frac{\varepsilon}{m}\right)\sigma.$$

Therefore

$$D_{\max}(\rho\|\sigma) \leq \log\left(1 + \frac{\varepsilon}{m}\right).$$

Exchanging ρ and σ gives the symmetrized bound.

For (b), $D_{\max}^{\mathrm{sym}}(\rho\|\sigma) \leq \delta$ means both

$$\rho \leq e^{\delta}\sigma$$

and

$$\sigma \leq e^{\delta}\rho.$$

The second inequality is equivalent to

$$e^{-\delta}\sigma \leq \rho.$$

Thus

$$-(1 - e^{-\delta})\sigma \leq \rho - \sigma \leq (e^{\delta} - 1)\sigma.$$

Since $1 - e^{-\delta} \leq e^{\delta} - 1$, we obtain

$$-(e^{\delta} - 1)\sigma \leq \rho - \sigma \leq (e^{\delta} - 1)\sigma.$$

Taking operator norms gives

$$\|\rho - \sigma\|_\infty \leq (e^{\delta} - 1)\,\|\sigma\|_\infty.$$

For density operators, $\|\sigma\|_\infty \leq 1$. $\square$

Theorem 7.22 (Grounding Vertex). *The grounding cost is the C^*-norm / Schatten-∞ distance*

$$\textit{grounding cost} = \|a - b\|_\infty$$

between residual operators or kernels. The $\alpha = \infty$ Rényi vertex is the max-divergence $D_{\max}$, or its symmetrization $D_{\max}^{\mathrm{sym}}$. These are distinct functionals.

On commuting diagonal states,

$$D_{\max}(p\|q) = \log \max_i \frac{p_i}{q_i}, \qquad \|\mathrm{diag}(p) - \mathrm{diag}(q)\|_\infty = \max_i |p_i - q_i|.$$

They are related by Proposition 7.21 under bounded-overlap hypotheses, but they are not identical.

Remark 7.23 (Scope of the Grounding-Vertex Identification). Theorem 7.22 identifies the grounding cost with the C^*-norm / Schatten-∞ distance only in the special operator-norm grounding model of Section 7.1; the preceding general instantiation allows an arbitrary Banach residual norm, for which no such identification is asserted absent the domination hypothesis of Corollary 7.4. Moreover, Proposition 7.21 compares *finite-dimensional density operators* locally in $D_{\max}^{\mathrm{sym}}$ versus $\|\cdot\|_\infty$; it does *not* compare $D_{\max}$ to the infinite-dimensional Hankel-operator distance $\|H_\nu - H_{\widehat{\nu}}\|_{\mathrm{op}}$, and no such comparison is established in this manuscript.

Remark 7.24 (Supremum Vertex as Organization). The max-divergence vertex is the divergence-face analogue of worst-case or supremum aggregation. It organizes the grounding regime conceptually, but the exact spectral converse remains the Schatten-∞ / operator-norm tail of Theorem 7.3 for the unrestricted linear finite-rank Hankel relaxation, with the Hankel-structured gap obeying only the lower bound recorded there and equality requiring the additional structural hypotheses of Theorems 7.9 and 7.11.

7.4 Determinism Gap for Formal Hankel Rank

The linear finite-rank spectral converse can be small even when deterministic symbolic approximation is hard. The following theorem gives a superpolynomial separation between stochastic linear rank and deterministic state complexity; see Remark 7.26 for why "exponential" would overstate the proved rate.

Theorem 7.25 (Determinism Gap for Formal Hankel Rank). *Let $\mathcal{S}(k)$ denote any lower bound on the number of states of a deterministic finite automaton recognizing a language that some k-state probabilistic finite automaton recognizes with a fixed isolated cutpoint. Then there exists a family of causal channels $\{\nu_k\}$ whose formal Hankel matrices have* algebraic *rank at most k, such that any deterministic Mealy machine approximating ν_k with worst-case per-step total-variation error $< 1/4$ requires at least $\mathcal{S}(k)/|\mathcal{O}|$ states.*

The best unconditional value of $\mathcal{S}$ available in the literature is that of Ambainis (Theorem 1 of [5]): a k-state probabilistic automaton with isolated cutpoint whose smallest equivalent deterministic automaton has

$$\mathcal{S}(k) = \Omega\left(2^{k \log\log k / \log k}\right),$$

states, a bound which is superpolynomial but sub-*exponential, since the exponent is $o(k)$. A genuinely exponential $\mathcal{S}(k) = 2^{\Omega(k)}$ is not established unconditionally by the classical references and is not assumed here: the later non-constructive constructions of Freivalds [7] reach the logarithmic state regime, but their strongest form rests on Artin's conjecture on primitive roots, and no unconditional exponential value is quoted here.*

Proof. In the isolated-cutpoint framework of [4, 6], for every k there exists a regular language L_k recognized by a k-state probabilistic finite automaton with bounded error at least $1/4$:

$$x \in L_k \implies p_{\mathrm{acc}}(x) > 3/4,$$

and

$$x \notin L_k \implies p_{\mathrm{acc}}(x) < 1/4,$$

while the minimal deterministic finite automaton for L_k has at least $\mathcal{S}(k)$ states.

Construct a causal binary-output channel ν_k that, after reading prefix x, emits a Bernoulli random variable with parameter $p_{\mathrm{acc}}(x)$. This channel is realizable by a k-state stochastic automaton. Its scalar impulse response has a k-dimensional linear representation

$$h_k(uv) = \alpha^\top T_u T_v \eta,$$

so the formal Hankel matrix $\mathsf{H}_k(u,v) = h_k(uv)$ has *algebraic* rank at most k. This factorization does *not* assert that H_k defines a bounded, let alone compact, operator on unweighted ℓ^2; see Remark 7.2.

A deterministic Mealy machine outputs a bit $b(x) \in \{0,1\}$ at each prefix x. Its per-step total-variation error for a Bernoulli output is

$$|p_{\mathrm{acc}}(x) - b(x)|.$$

If this error is $< 1/4$ for all prefixes x, then:

- if $x \in L_k$, then $p_{\mathrm{acc}}(x) > 3/4$, so $b(x) = 1$;
- if $x \notin L_k$, then $p_{\mathrm{acc}}(x) < 1/4$, so $b(x) = 0$.

Thus the deterministic machine decides L_k on all prefixes.

To conclude a state lower bound we pass from the Mealy machine to an acceptor. A Mealy machine assigns outputs to transitions rather than to states, so the conversion is not immediate. Given a Mealy machine $(\mathcal{S}, s_0, \tau, \lambda)$ whose final emitted symbol decides membership, form the acceptor with state set $\mathcal{S} \times \mathcal{O}$, initial state $(s_0, 0)$, transitions $\big((s,y),x\big) \mapsto \big(\tau(s,x), \lambda(s,x)\big)$, and accepting set $\mathcal{S} \times \{1\}$. This records the last emitted symbol in the state, recognizes exactly the language decided by the Mealy machine, and multiplies the state count by $|\mathcal{O}|$, a constant.

Hence an acceptor for L_k with $|\mathcal{O}| \cdot |\mathcal{S}|$ states exists, so $|\mathcal{O}| \cdot |\mathcal{S}| \geq \mathcal{S}(k)$ and $|\mathcal{S}| \geq \mathcal{S}(k)/|\mathcal{O}|$, which for fixed $\mathcal{O}$ is $\Omega(\mathcal{S}(k))$. □

Remark 7.26 (Scope of the Superpolynomial Gap)**.** The theorem concerns deterministic Mealy approximation of a stochastic channel under worst-case per-step total-variation error. It does not contradict the linear finite-rank spectral converse: the channel has low *algebraic* Hankel rank as a formal linear object, but deterministic symbolic realization may require exponentially many states.

Two scope restrictions are essential. First, the separation is between *formal linear realization dimension* and deterministic state complexity; it is not an instance of the compact-operator spectral theorem, because boundedness of H_k on unweighted ℓ^2 is not established. Second, the separation is quantitative rather than exponential. The classical references [4, 6] supply the isolated-cutpoint framework but no rate; the best proved rate is $\mathcal{S}(k) = \Omega(2^{k \log\log k/\log k})$ [5], whose exponent is $o(k)$. The gap is therefore superpolynomial and sub-exponential, and describing it as "exponential" would overstate what is known. Whether a truly exponential family exists is open.

7.5 Stochasticity Floor

Even when state complexity is not the obstruction, deterministic machines cannot eliminate intrinsic stochasticity.

For a fixed input sequence x, let

$$h_t = (x_{1:t}, Y_{1:t-1})$$

be the history available before emitting Y_t, and define the local stochastic spread

$$s(h_t) = 1 - \max_{b \in \mathcal{O}} p_{h_t}(b),$$

where

$$p_{h_t}(b) = \mathbb{P}(Y_t = b \mid h_t).$$

The stochasticity coefficient is

$$\sigma_\gamma(\nu) = \sup_{x \in \mathcal{I}^{\mathbb{N}}} \sum_{t=1}^{\infty} \gamma^{t-1} \mathbb{E}[s(h_t) \mid X = x].$$

Define the discounted Kantorovich–Rubinstein / Hamming distance between the channel output law $\nu \mid x$ and a deterministic output sequence $y = A(x)$ by

$$d_{\mathrm{KR},\gamma}\big(\nu \mid x, y\big) = \mathbb{E}_{Y \sim \nu|x} \sum_{t=1}^{\infty} \gamma^{t-1} \mathbf{1}\{Y_t \neq y_t\}.$$

This is the Kantorovich–Rubinstein distance for the discounted Hamming ground metric.

Definition 7.27 (Discounted Symbolic Grounding Gap)**.** The **discounted symbolic grounding gap** over deterministic Mealy machines with at most M states is

$$\Delta_{\mathrm{grd}}(M;\gamma) = \inf_{\substack{\mathcal{A} \text{ deterministic Mealy} \\ |\mathcal{S}_\mathcal{A}| \le M}} \sup_{x \in \mathcal{I}^{\mathbb{N}}} d_{\mathrm{KR},\gamma}\big(\nu \mid x, \mathcal{A}(x)\big).$$

This is the symbolic counterpart of the linear relaxation gaps $\Delta^{\mathrm{unres}}_{\mathrm{grd}}(M)$ and $\Delta^{\mathrm{Hank,str}}_{\mathrm{grd}}(M)$: the feasible set consists of deterministic machines rather than finite-rank operators, and the cost is a discounted Hamming distance rather than an operator norm. Determining it exactly is Open Problem 7.19, where it is written $\Delta^{\mathrm{symb}}_{\mathrm{grd}}(M)$ when the discount is suppressed.

Theorem 7.28 (Stochasticity Lower Bound)**.** *Assume* $0 < \gamma < 1$*. For every deterministic Mealy machine* $\mathcal{A}$*,*

$$\sup_{x \in \mathcal{I}^{\mathbb{N}}} d_{\mathrm{KR},\gamma}\big(\nu \mid x, \mathcal{A}(x)\big) \geq \sigma_\gamma(\nu).$$

Consequently, with $\Delta_{\mathrm{grd}}(M;\gamma)$ *as in Definition 7.27,*

$$\Delta_{\mathrm{grd}}(M;\gamma) \geq \sigma_\gamma(\nu)$$

for every M*. Moreover* $\sigma_\gamma(\nu) > 0$ *precisely when there exist an input sequence* x *and a time* t *such that, with positive probability under* $\nu \mid x$*, the conditional output law at time* t *is not a point mass. Nondegeneracy on unreachable or null histories is insufficient.*

Proof. Let $b^{\mathcal{A}}_t$ be the deterministic output emitted by $\mathcal{A}$ at time t under input sequence x. Conditional on the history h_t,

$$\mathbb{E}[\mathbf{1}_{\{Y_t \neq b^{\mathcal{A}}_t\}} \mid h_t] = 1 - p_{h_t}(b^{\mathcal{A}}_t) \geq 1 - \max_{b \in \mathcal{O}} p_{h_t}(b) = s(h_t).$$

Therefore

$$\mathbb{E}[\mathbf{1}_{\{Y_t \neq b^{\mathcal{A}}_t\}} \mid X = x] \geq \mathbb{E}[s(h_t) \mid X = x].$$

Multiplying by γ^{t-1}, summing over t, and taking the supremum over x gives

$$\sup_{x} d_{\mathrm{KR},\gamma}\big(\nu \mid x, \mathcal{A}(x)\big) \geq \sigma_\gamma(\nu).$$

Taking the infimum over M-state deterministic machines gives the stated lower bound on $\Delta_{\mathrm{grd}}(M;\gamma)$.

For the stated equivalence, suppose first that there exist x and t such that, with positive probability under $\nu \mid x$, the conditional law $p_{h_t}(\cdot)$ is not a point mass, i.e. $s(h_t) > 0$ on an event of positive $\nu \mid x$-probability. Then $\mathbb{E}[s(h_t) \mid X = x] > 0$, and since every summand of $\sigma_\gamma(\nu)$ is non-negative, the t-th term alone gives $\sigma_\gamma(\nu) \geq \gamma^{t-1}\mathbb{E}[s(h_t) \mid X = x] > 0$. Conversely, if for every x and every t the conditional law $p_{h_t}(\cdot)$ is a point mass whenever h_t occurs with positive $\nu \mid x$-probability, then $s(h_t) = 0$ on every event of positive probability, so $\mathbb{E}[s(h_t) \mid X = x] = 0$ for every x, t and $\sigma_\gamma(\nu) = 0$. This proves the claimed criterion directly, in terms of positive probability on reachable histories, without invoking any further undefined non-degeneracy notion. □

Theorem 7.29 (Observable Deterministic Floor and Its Finite-State Limit)**.** *Assume* $0 < \gamma < 1$ *and a finite input alphabet. For an input word* $u \in \mathcal{I}^t$*, let*

$$q_u(b) = \mathbb{P}(Y_t = b \mid X_{1:t} = u),$$

where all earlier outputs have been integrated out, and define

$$F_\gamma(\nu) = \sup_{x \in \mathcal{I}^{\mathbb{N}}} \sum_{t \geq 1} \gamma^{t-1}\bigl(1 - \max_{b \in \mathcal{O}} q_{x_{1:t}}(b)\bigr).$$

Then every deterministic causal input–output map A satisfies

$$\sup_x d_{\mathrm{KR},\gamma}(\nu \mid x, A(x)) \geq F_\gamma(\nu).$$

Equality is attained among unrestricted deterministic causal maps by choosing, at each input history u, a symbol in $\arg\max_b q_u(b)$. Moreover,

$$\lim_{M \to \infty} \Delta_{\mathrm{grd}}(M; \gamma) = F_\gamma(\nu).$$

In particular, $\sigma_\gamma(\nu) \leq F_\gamma(\nu)$, and the inequality can be strict, because $\sigma_\gamma(\nu)$ conditions its local spread on past random outputs $Y_{<t}$ unavailable to a Mealy machine acting on the input alone, whereas q_u has those outputs integrated out.

Proof. For a deterministic map A, its time-t symbol is a function $a(u)$ of the observable input history $u = x_{1:t}$. Hence

$$\mathbb{P}(Y_t \neq a(u) \mid X_{1:t} = u) = 1 - q_u(a(u)) \geq 1 - \max_b q_u(b).$$

Summation over t and the supremum over x prove the lower bound. Pointwise modal choices attain equality because the choice at one input history changes no other summand. To obtain finite-state approximants, store the input prefix through depth N, use those modal choices through time N, and thereafter enter one sink state with arbitrary fixed output. This is a finite Mealy machine with at most $1 + |\mathcal{I}| + \cdots + |\mathcal{I}|^N$ states, and its excess discounted loss over $F_\gamma(\nu)$ is at most $\gamma^N/(1-\gamma)$, since every post-depth-N summand is bounded by $\gamma^{t-1} \cdot 1$ and $\sum_{t>N} \gamma^{t-1} = \gamma^N/(1-\gamma)$. Letting $N \to \infty$ proves the limit, using also the lower bound $\Delta_{\mathrm{grd}}(M;\gamma) \geq F_\gamma(\nu)$ for every M from the first part. Finally, conditional Jensen applied to the map $y \mapsto \max_b p_{h_t}(b \mid Y_{<t} = y)$ gives

$$\mathbb{E}\bigl[\max_b \mathbb{P}(Y_t = b \mid X_{1:t}, Y_{<t}) \bigm| X_{1:t}\bigr] \geq \max_b \mathbb{P}(Y_t = b \mid X_{1:t}) = \max_b q_{X_{1:t}}(b),$$

so summing $\gamma^{t-1}\bigl(1 - \mathbb{E}[\max_b p_{h_t}(b) \mid X_{1:t}]\bigr) \leq \gamma^{t-1}(1 - \max_b q_{X_{1:t}}(b))$ and comparing to the definitions of σ_γ and F_γ gives $\sigma_\gamma(\nu) \leq F_\gamma(\nu)$. □

Remark 7.30 (Scope: Input-Only Deterministic Class)**.** Theorem 7.29 is stated for the manuscript's input-only deterministic Mealy class of Definition 7.27. If the intended deterministic approximant instead observes feedback of past outputs, both the definition of q_u and the formula for F_γ must be changed consistently; no such extension is claimed here.

Proposition 7.31 (Finite-Horizon Symbolic Certificates)**.** *Assume* $0 < \gamma < 1$. *Define*

$$\Delta^{(T)}_{\mathrm{grd}}(M;\gamma) = \min_{\substack{A \text{ deterministic Mealy} \\ |A| \leq M}} \max_{x \in \mathcal{I}^T} \sum_{t=1}^{T} \gamma^{t-1} \mathbb{P}(Y_t \neq A(x)_t \mid X_{1:t} = x_{1:t}).$$

Then

$$\Delta^{(T)}_{\mathrm{grd}}(M;\gamma) \le \Delta_{\mathrm{grd}}(M;\gamma) \le \Delta^{(T)}_{\mathrm{grd}}(M;\gamma) + \frac{\gamma^T}{1-\gamma}.$$

If the target is a finite-state stochastic channel with rational transition and emission probabilities and γ is rational, then $\Delta^{(T)}_{\mathrm{grd}}(M;\gamma)$ is a decidable rational number: one may enumerate the finitely many labeled M-state Mealy machines and the finitely many input words of length T, evaluating each displayed loss by finite forward recursion.

Proof. For each fixed machine and input word, the omitted tail $\sum_{t>T}\gamma^{t-1}\mathbb{P}(Y_t \neq A(x)_t \mid X_{1:t} = x_{1:t})$ lies in $[0, \gamma^T/(1-\gamma)]$, since each summand is a probability in $[0,1]$ scaled by γ^{t-1}. Taking first the maximum over input words of length T and then the minimum over M-state machines gives both inequalities: the machine and input word optimal for the truncated problem, extended arbitrarily beyond depth T, gives the upper bound on $\Delta_{\mathrm{grd}}(M;\gamma)$, while restricting any machine's supremum over infinite input sequences to length-T prefixes and truncating the sum gives the lower bound on $\Delta^{(T)}_{\mathrm{grd}}(M;\gamma)$. Under the stated rational finite-state assumptions, every forward recursion computing $\mathbb{P}(Y_t \neq A(x)_t \mid X_{1:t} = x_{1:t})$ from the rational transition and emission probabilities is a finite composition of rational operations, and every discounted sum over $t \le T$ of rational γ-powers and rational probabilities is again rational, while the candidate machine set (finitely many labeled M-state Mealy machines) and input set ($|\mathcal{I}|^T$ words) are finite, so the minimum and maximum are attained and computable. □

Corollary 7.32 (Exact Memoryless Stochasticity Floor)**.** *Let ν be a memoryless output channel emitting i.i.d. outputs with law p on $\mathcal{O}$, independent of history and input. Then for every $M \ge 1$,*

$$\Delta_{\mathrm{grd}}(M;\gamma) = \frac{1-\max_{b\in\mathcal{O}} p_b}{1-\gamma}.$$

In particular, for a Bernoulli(p) *channel,*

$$\Delta_{\mathrm{grd}}(M;\gamma) = \frac{\min\{p, 1-p\}}{1-\gamma}.$$

Proof. For a memoryless channel, the local stochastic spread is constant:

$$s(h_t) = 1 - \max_{b\in\mathcal{O}} p_b.$$

Therefore

$$\sigma_\gamma(\nu) = \sum_{t=1}^{\infty}\gamma^{t-1}\left(1-\max_b p_b\right) = \frac{1-\max_b p_b}{1-\gamma}.$$

The lower bound follows from Theorem 7.28.

For the upper bound, choose

$$b^\star \in \arg\max_{b\in\mathcal{O}} p_b$$

and use the one-state deterministic machine that emits $b^\star$ at every time. Its discounted Hamming error is exactly

$$\sum_{t=1}^{\infty} \gamma^{t-1}(1 - p_{b^\star}) = \frac{1 - \max_b p_b}{1 - \gamma}.$$

Thus the lower bound is attained. □

Remark 7.33 (Deterministic Machines Cannot Remove Intrinsic Randomness)**.** The stochasticity floor is independent of the state budget M. No deterministic Mealy machine, however large, can match a genuinely stochastic channel in discounted Hamming/Kantorovich–Rubinstein distance better than the channel's intrinsic stochastic spread. This obstruction persists even with unbounded deterministic memory.

Remark 7.34 (Relation to the Linear Spectral Converse)**.** The stochasticity floor concerns deterministic symbolic grounding. The linear spectral converse concerns finite-rank linear realizations. A channel may have small linear Hankel gap while having a positive stochasticity floor for deterministic machines. The two results measure different approximation regimes.

7.6 State Compression Above the Floor

Theorem 7.28 bounds the symbolic grounding gap below by a quantity independent of the budget M. It therefore says nothing about how the gap varies with M. The following decomposition separates the budget-independent floor from a budget-dependent term, and exhibits that term as a mode-prediction analogue of the retention gap.

Throughout this subsection the channel is a stationary finite-state channel with state set $\mathcal{S}$, stationary law π, and predictive laws $P_s = \mathcal{L}(Y_t \mid S_t = s)$ on a finite output alphabet, and ϕ ranges over maps $\mathcal{S}^+ \to \mathcal{K}$ with $|\mathcal{K}| \le M$. Write $\pi(C_k) = \sum_{s \in C_k} \pi_s$ for the mass of the block $C_k = \phi^{-1}(k)$.

Definition 7.35 (Block Symbol and Tracking Deficit)**.** For a block C_k of positive mass, the **block symbol** is

$$\hat{b}_k \in \arg\max_{b \in \mathcal{O}} \sum_{s \in C_k} \pi_s P_s(b),$$

ties broken by any fixed order, and the **tracking deficit** of ϕ is

$$D(\phi) = \sum_{s \in \mathcal{S}^+} \pi_s \Big[\max_b P_s(b) - P_s\big(\hat{b}_{\phi(s)}\big)\Big].$$

The one-step stationary floor is $\sigma_1 = \sum_{s \in \mathcal{S}^+} \pi_s \big(1 - \max_b P_s(b)\big)$.

Proposition 7.36 (Floor Plus Tracking Decomposition)**.** *The following is first an* oracle *state-compression identity: the emitted symbol may use the current latent state S_t through its block label $\phi(S_t)$, which is not in general available to a deterministic Mealy machine acting on the input alone. It becomes a realizable deterministic Mealy construction only under the implementability condition of Proposition 7.37 below. Let ϕ be any state compression and* *let $\mathcal{A}_\phi$ denote the (possibly non-implementable) rule that, given oracle access to S_t, occupies block $\phi(S_t)$ and emits $\hat{b}_{\phi(S_t)}$. Then the stationary one-step error of $\mathcal{A}_\phi$ decomposes as*

$$\sum_{s \in \mathcal{S}^+} \pi_s \Big(1 - P_s\big(\hat{b}_{\phi(s)}\big)\Big) = \sigma_1 + D(\phi),$$

and

(i) $D(\phi) \geq 0$, with equality if and only if $\hat{b}_{\phi(s)}$ attains $\max_b P_s(b)$ for every $s \in \mathcal{S}^+$;

(ii) D vanishes on the singleton compression, so $\min_{|\mathcal{K}| \leq M} D = 0$ whenever $M \geq |\mathcal{S}^+|$;

(iii) D is monotone under refinement: if ϕ refines ψ, then

$$D(\phi) \leq D(\psi).$$

Consequently $\min_{|\mathcal{K}| \leq M} D(\phi)$ is nonincreasing in M.

Proof. The identity is immediate from $1 - P_s(\hat{b}_{\phi(s)}) = \big(1 - \max_b P_s(b)\big) + \big(\max_b P_s(b) - P_s(\hat{b}_{\phi(s)})\big)$, weighted by π_s and summed. For (i), each bracket is a maximum minus a coordinate of the same vector, hence nonnegative, and the sum of nonnegative terms vanishes exactly when each does. For (ii), the singleton compression has $\hat{b}_{\{s\}} = \arg\max_b P_s(b)$, so every bracket is zero.

For (iii), write $w_s(b) = \pi_s P_s(b)$. The statewise modal term $\sum_s \pi_s \max_b P_s(b) = \sum_s \max_b w_s(b) = 1 - \sigma_1$ in the decomposition is constant across partitions, whereas the blockwise accuracy is

$$\sum_{C \in \mathcal{P}} \max_b \sum_{s \in C} w_s(b),$$

so that

$$D(\phi) = \sum_{s \in \mathcal{S}^+} \max_b w_s(b) - \sum_{C \in \mathcal{P}_\phi} \max_b \sum_{s \in C} w_s(b),$$

the first term being the partition-independent total modal mass $1 - \sigma_1$, for the partition $\mathcal{P}_\phi$ induced by ϕ, and likewise for ψ. If a block $C \in \mathcal{P}_\psi$ is split by the refinement ϕ into $C_1, \ldots, C_r \in \mathcal{P}_\phi$, then, since a maximum of a sum is at most the sum of the maxima taken termwise over a partition of the summation range,

$$\sum_{j=1}^{r} \max_b \sum_{s \in C_j} w_s(b) \geq \max_b \sum_{j=1}^{r} \sum_{s \in C_j} w_s(b) = \max_b \sum_{s \in C} w_s(b).$$

Applying this to every block of $\mathcal{P}_\psi$ and summing gives

$$\sum_{C' \in \mathcal{P}_\phi} \max_b \sum_{s \in C'} w_s(b) \;\geq\; \sum_{C \in \mathcal{P}_\psi} \max_b \sum_{s \in C} w_s(b),$$

and subtracting from the partition-independent total $\sum_s \max_b w_s(b)$ reverses the inequality to $D(\phi) \leq D(\psi)$. The feasible sets are nested in the budget: every partition with at most M

blocks is feasible at every budget $M' \geq M$, and each budget-M' feasible partition that splits a budget-M feasible partition refines it. Combining the nesting of the feasible sets with the refinement monotonicity of D just established therefore gives the conclusion: enlarging the budget admits refinements of each budget-M feasible partition, and D cannot increase under refinement, so $\min_{|\mathcal{K}|\leq M} D(\phi)$ is nonincreasing in M. □

Proposition 7.37 (Implementability of a Block-Mode Policy)**.** *Suppose the channel is input-driven and synchronized: there is a map $s : \mathcal{I}^* \to \mathcal{S}$ with $S_t = s(X_{1:t})$. If ϕ is a right congruence for this update, namely there exists $\bar{\tau} : \mathcal{K} \times \mathcal{I} \to \mathcal{K}$ such that*

$$\phi(s(ux)) = \bar{\tau}(\phi(s(u)), x) \qquad (u \in \mathcal{I}^*,\ x \in \mathcal{I}),$$

then the block symbols $\hat{b}_k$ define a deterministic Mealy machine with at most $|\mathcal{K}|$ states, and its stationary one-step error is $\sigma_1 + D(\phi)$ as in Proposition 7.36. Without this condition the same formula is only the error of a genie that reveals $\phi(S_t)$ to the predictor.

Proof. The displayed identity makes the block label $\phi(S_t) = \phi(s(X_{1:t}))$ a deterministic finite-state update from the input history alone, via $\bar{\tau}$, so the machine with state set $\mathcal{K}$, transition $\bar{\tau}$, and output rule $k \mapsto \hat{b}_k$ is a well-typed Mealy machine of the form of Section 1.1, and outputting $\hat{b}_k$ at state k realizes exactly the oracle rule of Proposition 7.36. Conditioning on $S_t = s$ gives error $1 - P_s(\hat{b}_{\phi(s)})$; averaging over π and using $1 - P_s(\hat{b}_{\phi(s)}) = (1 - \max_b P_s(b)) + (\max_b P_s(b) - P_s(\hat{b}_{\phi(s)}))$ proves the formula. If the label $\phi(S_t)$ is not a function of the input history alone – for instance if S_t depends on the channel's own latent randomness beyond what $X_{1:t}$ determines – no input-only Mealy machine can in general implement $\hat{b}_{\phi(S_t)}$, and the decomposition remains only a genie bound. □

Remark 7.38 (What the Deficit Is, and What It Is Not)**.** Two points fix the scope of Proposition 7.36.

First, the deficit must be measured against the *block symbol* evaluated under the true state law, $P_s(\hat{b}_{\phi(s)})$, and not against the mode of the block mixture, $\max_b P_{\phi(s)}(b)$. The latter substitution yields a quantity that is not a deficit at all: for a two-state block with $P_{s_1} = (\frac{1}{2}, \frac{1}{2})$, $P_{s_2} = (\frac{9}{10}, \frac{1}{10})$ and equal weights, the mixture is $(\frac{7}{10}, \frac{3}{10})$ and $\max_b P_{s_1}(b) - \max_b P_{\phi(s_1)}(b) = -\frac{1}{5} < 0$. With the block symbol the corresponding bracket is $\frac{1}{2} - \frac{1}{2} = 0$.

Second, a positive deficit is not forced by $M < |\mathcal{S}^+|$. Distinct predictive laws may share a mode, and then merging them costs nothing in this currency: for $\mathcal{O} = \{1, 2, 3\}$, three states of equal weight with $P_{s_1} = (\frac{3}{5}, \frac{1}{5}, \frac{1}{5})$, $P_{s_2} = (\frac{7}{10}, \frac{1}{5}, \frac{1}{10})$ and $P_{s_3} = (\frac{4}{5}, \frac{1}{10}, \frac{1}{10})$, the one-block compression has $\hat{b} = 1$ and $D = 0$ although $M = 1 < 3 = |\mathcal{S}^+|$. The vanishing condition is mode preservation, which is strictly weaker than separating the predictive states; contrast Theorem 5.2, where the KL currency does force one block per distinct predictive law.

Remark 7.39 (Relation to Retention)**.** Proposition 7.36 gives the symbolic grounding problem above the floor the same shape as retention: a compression ϕ of the stationary state set, a representative per block, and a π-weighted cost of merging. The currencies differ. Retention charges $D_{\mathrm{KL}}(P_s \| \bar{P}_k)$ against the mixture centroid, which by Theorem 5.2 vanishes only when each block carries a single predictive law. Grounding charges $\max_b P_s(b) - P_s(\hat{b}_k)$

against a single symbol, which vanishes under the weaker mode-preservation condition of Remark 7.38. This comparison of zero sets is itself an oracle-state comparison unless the implementability condition of Proposition 7.37 holds for a shared family of partitions on both sides; under that condition the zero-set inclusion is valid (mode preservation is weaker than equality of predictive laws), because both problems then range over the same implementable feasible class. No stronger claim should be read into it: it does not by itself show that the grounding compression problem is "never harder" than retention in any global sense, only that its zero set is no smaller. No inequality between the two *gap values* follows, the costs being incomparable; this is the Independence Theorem 12.6 in the present instance.

8 The Conditional Schatten Converse Template

The analytic converses for grounding and quadratic retention have a common operator-approximation mechanism. This section isolates that mechanism in a conditional template. The Schatten tail form follows once a regime supplies a rank-controlled response operator and a Schatten domination inequality. Contextwise ℓ^p aggregation does not by itself imply Schatten-p domination.

Proposition 8.1 (Comparison of Schatten Tails of a Common Operator)**.** *Let A have finite rank κ and let $0 \le r < \kappa$. For $1 \le p \le q \le \infty$, set $n = \kappa - r$. Then*

$$\Phi_r^{(q)}(A) \le \Phi_r^{(p)}(A) \le n^{1/p-1/q}\, \Phi_r^{(q)}(A),$$

with the convention $1/\infty = 0$. In particular

$$\Phi_r^{(\infty)}(A) \le \Phi_r^{(2)}(A) \le \sqrt{n}\, \Phi_r^{(\infty)}(A), \qquad \Phi_r^{(\infty)}(A) \le \Phi_r^{(1)}(A) \le n\, \Phi_r^{(\infty)}(A).$$

Proof. Apply the monotonicity and equivalence inequalities for ℓ^p norms on $\mathbb{R}^n$, namely $\|x\|_q \le \|x\|_p \le n^{1/p-1/q} \|x\|_q$ for $1 \le p \le q \le \infty$ [1, Ch. 2], to the finite tail vector $(\sigma_{r+1}(A), \ldots, \sigma_\kappa(A))$, whose length is $n = \kappa - r$. The displayed special cases are $q = \infty$ with $p = 2$ and $p = 1$. □

Remark 8.2 (Scope of the Comparison)**.** Proposition 8.1 compares different Schatten exponents for *one* operator and one rank budget. It does not compare the different regimes, whose response operators, effective rank budgets, and constants differ; that absence is the content of Theorem 8.22.

Remark 8.3 (Relation to Aggregation and Domination)**.** The definition separates three levels:

(1) contextwise aggregation, for example an ℓ^p-type norm over contexts;

(2) operator approximation, for example a Schatten-p norm over singular values;

(3) finite-state rank control, for example rank $\le M$, rank $\le M-1$, or Boolean rank $\le M$.

Mirsky's theorem governs the passage from operator approximation and rank control to a spectral tail. The implication

$$\text{contextwise } \ell^p \text{ aggregation} \quad \Longrightarrow \quad \text{Schatten-}p \text{ domination}$$

is not automatic. The exponent p in the tail is the exponent of the operator norm appearing in the domination bridge, not automatically the exponent of contextwise aggregation.

A sufficient way to prove domination is to exhibit a linear measurement or embedding map from response-operator errors to context errors with a frame-type lower bound

$$\|\text{context error}\|_{\ell^p} \geq \frac{1}{c_p} \left\|A_\delta - A_{\widehat{\delta}}\right\|_{\mathcal{S}_p}.$$

Without such a bridge, the template does not yield a regime-specific converse.

Remark 8.4 (Nonempty Instantiation)**.** The schema is nonempty. Let the exact object be a Hilbert–Schmidt operator $A \in \mathcal{S}_2$, let budget-M approximants be arbitrary operators B with $\operatorname{rank}(B) \leq M$, and define the task cost by

$$\mathcal{L}(A, B) = \|A - B\|_{\mathcal{S}_2}.$$

Then

$$A_\delta = A, \qquad A_{\widehat{\delta}} = B, \qquad r(M) = M, \qquad c_2 = 1,$$

and

$$\Delta(M) = \Phi_M^{(2)}(A) = \left(\sum_{i>M} \sigma_i(A)^2\right)^{1/2}.$$

This is an operator-approximation instantiation. It does not assert the existence of a finite-state $\mathcal{L}^2$ regime satisfying Schatten-2 domination.

Remark 8.5 (Conditional Status of the Template)**.** The template is conditional. It yields a Schatten-tail lower bound only after a regime supplies:

(1) a response operator A_δ;

(2) an effective rank budget $r_{\mathbb{T}}(M)$;

(3) a domination inequality relating task loss to a Schatten norm.

Without these data, the template does not apply.

For a compact operator A on a Hilbert space, write its singular values in nonincreasing order:

$$\sigma_1(A) \geq \sigma_2(A) \geq \cdots \geq 0.$$

For $1 \leq p < \infty$, if A belongs to the Schatten ideal $\mathcal{S}_p$, the Schatten-p norm is

$$\|A\|_{\mathcal{S}_p} = \left(\sum_i \sigma_i(A)^p\right)^{1/p},$$

and for a compact operator outside $\mathcal{S}_p$ we use the extended value $\|A\|_{\mathcal{S}_p} = +\infty$, so that all displayed inequalities are read in $[0, \infty]$. Finally

$$\|A\|_{\mathcal{S}_\infty} = \sigma_1(A).$$

For an integer rank budget $r \geq 0$, define the Schatten-p tail

$$\Phi_r^{(p)}(A) = \begin{cases} \left(\sum_{i>r} \sigma_i(A)^p\right)^{1/p}, & 1 \leq p < \infty, \\ \sigma_{r+1}(A), & p = \infty, \end{cases}$$

with the convention $\Phi_r^{(p)}(A) = 0$ if $r \geq \operatorname{rank}(A)$ in the finite-rank case.

Lemma 8.6 (Linear Measurement Bridges)**.** *Suppose the residual difference $z = \delta - \widehat{\delta}$ enters the response operators through a* linear *map T, so that $A_\delta - A_{\widehat{\delta}} = T(z)$, and suppose the task cost is a contextwise weighted ℓ^p aggregation, $\mathcal{L}_{\mathbb{T}} = \|z\|_{\ell^p(w)}$. If T satisfies the frame-type bound*

$$\|T(z)\|_{\mathcal{S}_p} \;\leq\; C_p \;\|z\|_{\ell^p(w)} \qquad \textit{for all } z,$$

then the domination inequality of Definition 8.9 holds with modulus $c_p = C_p$. Conversely, any modulus c_p valid for all z yields such a bound with $C_p = c_p$, so for linear T the two conditions are the same.

Proof. Immediate: $\mathcal{L}_{\mathbb{T}} = \|z\|_{\ell^p(w)} \geq C_p^{-1} \|T(z)\|_{\mathcal{S}_p} = c_p^{-1} \left\|A_\delta - A_{\widehat{\delta}}\right\|_{\mathcal{S}_p}$, and the converse is the same inequality with the roles of the two sides exchanged. ☐

Proposition 8.7 (No Dimension-Free Hankel Bridge)**.** *Let T be the Hankel embedding sending a sequence $z = (z_0, \ldots, z_{2n-2})$ to the $n \times n$ matrix $H(z)_{ij} = z_{i+j}$, and let the aggregation be unweighted ℓ^p. Then the optimal constants satisfy*

$$C_2(n) = \sqrt{n}, \qquad C_1(n) \;\geq\; n, \qquad C_\infty(n) \;\geq\; n.$$

In particular no dimension-free frame bridge exists for the Hankel embedding at $p \in \{1, 2, \infty\}$, and any modulus c_p obtained this way degrades with the budget.

Proof. For $p = 2$, $\|H(z)\|_{\mathcal{S}_2}^2 = \sum_k m_k |z_k|^2$ with multiplicity $m_k = \#\{(i,j) : i + j = k\} = \min(k+1, n, 2n-1-k)$, so the ratio is maximized by the atom $z = e_{n-1}$ on the main anti-diagonal, where $m_{n-1} = n$; hence $C_2(n) = \sqrt{n}$ exactly.

For $p = 1$ take the same atom. Then $H(e_{n-1})$ is the anti-identity, whose n singular values all equal 1, so $\|H(e_{n-1})\|_{\mathcal{S}_1} = n$ while $\|e_{n-1}\|_{\ell^1} = 1$, giving $C_1(n) \geq n$.

For $p = \infty$ take $z \equiv 1$. Then $H(z)$ is the all-ones matrix, of rank one with operator norm n, while $\|z\|_{\ell^\infty} = 1$, giving $C_\infty(n) \geq n$. ☐

Remark 8.8 (Consequence for the Template)**.** Lemma 8.6 identifies exactly what a bridge must supply, and Proposition 8.7 shows that the most natural candidate does not supply a constant one. Two readings are available.

If one insists on a dimension-free (constant) modulus, the linear-measurement route fails for Hankel embeddings and a different mechanism is needed; this is the content of the open problem on domination bridges. Definition 8.9 and Theorem 8.11 are stated for the general budget-dependent modulus $c_p(M)$ precisely to accommodate this case: the template still applies with $c_p(M)$ in place of a constant c_p, but the converse it yields is correspondingly weaker, and the weakening must be carried explicitly into every downstream tail bound rather than absorbed into a constant.

Two scoping points prevent this from being read too broadly.

First, the diagonal embedding $z \mapsto \mathrm{diag}(z)$ is a Schatten isometry for every p, since the singular values of $\mathrm{diag}(z)$ are the $|z_i|$; the retention instance therefore admits a dimension-free bridge with $C_p = 1$. The obstruction is specific to the overlap structure of Hankel operators, in which one sequence entry occupies up to n matrix positions.

Second, the grounding instance of Corollary 8.14 is *not* exposed to the obstruction. There the cost is itself the operator-norm distance $\|H_\nu - H_{\widehat{\nu}}\|_{\mathrm{op}}$ and the response operator is the same object, so domination holds with equality and $c_\infty = 1$, with no frame constant to pay. Proposition 8.7 bears on a different situation: a regime whose cost aggregates residual *sequence entries* in ℓ^p, which is then to be compared with a Schatten norm of their Hankel embedding. That is precisely the implication "contextwise ℓ^p implies Schatten-p domination" which is not automatic, and the proposition quantifies its failure.

Definition 8.9 (Schatten-p Response Representation)**.** Fix $p \in [1, \infty]$. A task theory $\mathbb{T}$ has a **Schatten-p response representation** with response operator A_δ, effective rank budget

$$r_{\mathbb{T}} : \mathbb{N} \to \mathbb{N} \cup \{0\},$$

and modulus

$$c_p : \mathbb{N} \to (0, \infty)$$

if every budget-M approximant $\widehat{\delta}$ is assigned a response operator $A_{\widehat{\delta}}$ such that:

(a) $\mathrm{rank}(A_{\widehat{\delta}}) \le r_{\mathbb{T}}(M)$;

(b) the domination inequality holds:

$$\mathcal{L}_{\mathbb{T}}(\delta, \widehat{\delta}) \ge \frac{1}{c_p(M)} \left\|A_\delta - A_{\widehat{\delta}}\right\|_{\mathcal{S}_p}.$$

A **constant-modulus** representation is the special case $c_p(M) \equiv c_p$ for all M; this is the case used by Corollary 8.14, where $c_\infty \equiv 1$, and it is the only case in which the bracketed notation c_p (without an argument) is used below.

Theorem 8.10 (Eckart–Young–Mirsky)**.** *The following is classical and is quoted without proof; see [15, 16], and [3] for the operator setting. Let A be a compact operator on a Hilbert space, and let $|||\cdot|||$ be a unitarily invariant norm. For every integer $r \ge 0$,*

$$\min_{\mathrm{rank}\, B \le r} |||A - B||| = \left|\left|\left|\mathrm{diag}\big(\sigma_{r+1}(A), \sigma_{r+2}(A), \dots\big)\right|\right|\right|.$$

In particular, for every $1 \le p \le \infty$,

$$\min_{\operatorname{rank} B \le r} \|A - B\|_{\mathcal{S}_p} = \Phi_r^{(p)}(A).$$

Theorem 8.11 (Conditional Unified Spectral Converse)**.** *Suppose a task theory $\mathbb{T}$ has a Schatten-p response representation in the sense of Definition 8.9, with response operator A_δ, effective rank budget $r_{\mathbb{T}}(M)$, and modulus $c_p(\cdot)$. Then*

$$\Delta_{\mathbb{T}}(M) \ge \frac{1}{c_p(M)} \Phi^{(p)}_{r_{\mathbb{T}}(M)}(A_\delta).$$

Equivalently, for $1 \le p < \infty$,

$$\Delta_{\mathbb{T}}(M) \ge \frac{1}{c_p(M)} \left(\sum_{i > r_{\mathbb{T}}(M)} \sigma_i(A_\delta)^p \right)^{1/p},$$

and for $p = \infty$,

$$\Delta_{\mathbb{T}}(M) \ge \frac{1}{c_\infty(M)} \sigma_{r_{\mathbb{T}}(M)+1}(A_\delta).$$

When the modulus is constant, $c_p(M) \equiv c_p$, these recover the familiar budget-independent bounds, and we then write c_p for $c_p(M)$ as in Corollary 8.14.

Proof. For any budget-M approximant $\widehat{\delta}$,

$$\operatorname{rank}(A_{\widehat{\delta}}) \le r_{\mathbb{T}}(M).$$

Hence, by Theorem 8.10,

$$\left\|A_\delta - A_{\widehat{\delta}}\right\|_{\mathcal{S}_p} \ge \min_{\operatorname{rank} B \le r_{\mathbb{T}}(M)} \|A_\delta - B\|_{\mathcal{S}_p} = \Phi^{(p)}_{r_{\mathbb{T}}(M)}(A_\delta).$$

By domination at budget M,

$$\mathcal{L}_{\mathbb{T}}(\delta, \widehat{\delta}) \ge \frac{1}{c_p(M)} \Phi^{(p)}_{r_{\mathbb{T}}(M)}(A_\delta).$$

Taking the infimum over budget-M approximants gives the result. □

Remark 8.12 (Numerical Verification Scope)**.** Numerical checks are consistency checks, not proofs of the hypotheses.

- Mirsky/Eckart–Young identities at $p \in \{1, 2, \infty\}$ can be checked numerically on random finite matrices. Such checks verify the operator-approximation theorem, not the domination hypotheses.
- The ℓ^p tail comparisons of Proposition 8.1 can be checked on random finite spectra. Such checks verify only elementary norm monotonicity.

- The retention check uses the tail
$$\sum_{i\geq M} \lambda_i(\mathbf{\Sigma}_\pi),$$
not $\sum_{i>M} \lambda_i(\mathbf{\Sigma}_\pi)$. A check using the latter does not recover the retention converse.
- Numerical experiments cannot prove Hankel domination for grounding, cannot prove that a given KL or finite-state cost satisfies Schatten domination, cannot prove Myhill–Nerode rank identities, and cannot prove the Independence Theorem.

Such checks verify operator identities and indexing; they do not verify full applicability of the template.

Remark 8.13 (Equality versus Lower Bound)**.** If the task cost is exactly the scaled Schatten-p response error and the feasible response operators include the truncated singular-value approximants, then Theorem 8.11 becomes an equality. In general, domination gives only a lower bound.

8.1 Grounding as the Schatten-∞ Instance

Corollary 8.14 (Grounding as $p = \infty$)**.** *Let H_ν be the response Hankel operator of the target channel. For the unrestricted linear finite-rank Hankel relaxation gap,*

$$\Delta_{\mathrm{grd}}^{\mathrm{unres}}(M) = \inf_{\operatorname{rank} B \leq M} \|H_\nu - B\|_{\mathcal{S}_\infty}.$$

Theorem 8.11 with

$$A_\delta = H_\nu, \qquad r_{\mathrm{grd}}(M) = M, \qquad c_\infty = 1$$

gives

$$\Delta_{\mathrm{grd}}^{\mathrm{unres}}(M) = \Phi_M^{(\infty)}(H_\nu) = \sigma_{M+1}(H_\nu).$$

For the Hankel-structured finite-rank realization gap the feasible set is restricted to Hankel operators of rank at most M, so Theorem 7.3 gives $\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) \geq \sigma_{M+1}(H_\nu)$, with equality requiring the additional structure of Theorems 7.9 and 7.11.

Proof. The unrestricted gap is by definition the rank-M operator-norm approximation problem for H_ν, so Eckart–Young–Mirsky [15, 16] gives equality with $\sigma_{M+1}(H_\nu)$; equivalently, this is Theorem 8.11 applied with $A_\delta = H_\nu$, $r_{\mathrm{grd}}(M) = M$ and $c_\infty = 1$, for which the domination inequality holds with equality.

The Hankel-structured gap restricts the feasible set from all rank-M operators to rank-M *Hankel* operators. An infimum over a subset is at least the infimum over the ambient set, so $\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) \geq \Delta_{\mathrm{grd}}^{\mathrm{unres}}(M) = \sigma_{M+1}(H_\nu)$. Nothing forces equality: the optimal rank-M truncation supplied by Eckart–Young–Mirsky need not be Hankel. Adamjan–Arov–Krein is not invoked here, and is available only when an independent Hardy-space embedding is supplied (Theorem 7.9).

For the residual-cost gap $\Delta_{\mathrm{grd}}^{\mathrm{lin}}(M)$, apply Theorem 8.11 with the assumed domination inequality, as in Corollary 7.4. □

Remark 8.15 (Scope of the Grounding Instance)**.** This is a converse for the *unrestricted* linear finite-rank grounding relaxation. The Hankel-structured finite-rank realization gap satisfies the same singular-value lower bound, since restricting the feasible set to Hankel operators cannot decrease the infimum, but equality there requires the additional structural hypotheses of Theorems 7.9 and 7.11. Transfer to an exact symbolic deterministic gap requires a separate comparison theorem.

8.2 Retention as the Schatten-1 Instance with Effective Rank $M-1$

Corollary 8.16 (Retention as $p = 1$ with Rank Budget $M - 1$)**.** *In the Gaussian quadratic restricted retention regime, let $\mathbf{\Sigma}_\pi \succeq 0$ be the stationary Fisher covariance. For $M \geq 1$,*

$$\Delta_{\mathrm{ret}}^{\mathrm{quad}}(M) \geq \frac{1}{2}\Phi_{M-1}^{(1)}(\mathbf{\Sigma}_\pi) = \frac{1}{2}\sum_{i \geq M} \lambda_i(\mathbf{\Sigma}_\pi).$$

If the retention gap is exactly this quadratic surrogate gap, this recovers the retention converse. If the retention gap is an expected KL gap in a small-radius regime, the same formula is the second-order spectral lower bound under the regularity assumptions of Theorem 5.14.

Proof. Work in Fisher coordinates, with weighted predictive points $\{y_s\}$ and weights π_s summing to one. Write

$$\bar{y} = \sum_s \pi_s y_s, \qquad \mathbf{\Sigma}_\pi = \sum_s \pi_s (y_s - \bar{y}) \otimes (y_s - \bar{y}).$$

A lumpable quotient with at most M blocks induces a partition into $K \leq M$ nonempty cells $C_1, \ldots, C_K$ with weights $\Pi_k = \sum_{s \in C_k} \pi_s$ and centroids m_k. Define the between-cluster covariance

$$B = \sum_{k=1}^{K} \Pi_k (m_k - \bar{y}) \otimes (m_k - \bar{y}).$$

Since $\sum_k \Pi_k (m_k - \bar{y}) = 0$, the centred centroid vectors satisfy one nontrivial linear relation, so

$$\operatorname{rank}(B) \leq K - 1 \leq M - 1.$$

For the quadratic distortion with representatives c_k, completing the square inside each cell gives

$$E = \frac{1}{2}\sum_{k=1}^{K}\sum_{s \in C_k} \pi_s \left\|y_s - c_k\right\|^2 \;\geq\; \frac{1}{2}\sum_{k=1}^{K}\sum_{s \in C_k} \pi_s \left\|y_s - m_k\right\|^2 = \frac{1}{2}\operatorname{tr}(\mathbf{\Sigma}_\pi - B),$$

the last step being the ANOVA identity $\mathbf{\Sigma}_\pi = W + B$ with $W \succeq 0$ the within-cluster covariance.

Because $\mathbf{\Sigma}_\pi - B = W \succeq 0$, its trace coincides with its nuclear norm,

$$\operatorname{tr}(\mathbf{\Sigma}_\pi - B) = \left\|\mathbf{\Sigma}_\pi - B\right\|_{\mathcal{S}_1},$$

which is precisely what places this instance at the Schatten exponent $p = 1$. Since $\mathrm{rank}(B) \le M - 1$, Theorem 8.10 gives

$$\|\mathbf{\Sigma}_\pi - B\|_{\mathcal{S}_1} \;\ge\; \min_{\mathrm{rank}\, P \le M-1} \|\mathbf{\Sigma}_\pi - P\|_{\mathcal{S}_1} = \sum_{i > M-1} \lambda_i(\mathbf{\Sigma}_\pi) = \sum_{i \ge M} \lambda_i(\mathbf{\Sigma}_\pi).$$

Hence $E \ge \frac{1}{2} \sum_{i \ge M} \lambda_i(\mathbf{\Sigma}_\pi)$, and taking the infimum over feasible quotients gives the stated converse, with effective rank $M - 1$ and $\Phi^{(1)}_{M-1}(\mathbf{\Sigma}_\pi) = \sum_{i \ge M} \lambda_i(\mathbf{\Sigma}_\pi)$. □

Remark 8.17 (Why the Exponent Is 1)**.** The retention cost is not literally an ℓ^1 norm of residuals. In the quadratic regime it is a mean of squared errors. The Schatten-1 tail appears because the residual covariance is positive semidefinite, so its trace equals its nuclear norm. The exponent $p = 1$ is justified by the covariance-residual structure, and does not follow from the aggregation being a mean.

8.3 A Conditional Schatten-2 / Frobenius Regime

Corollary 8.18 (Conditional Frobenius Converse)**.** *If a task theory has a Schatten-2 response representation with response operator A_δ, effective rank budget $r_{\mathbb{T}}(M)$, and modulus c_2, then*

$$\Delta_{\mathbb{T}}(M) \ge \frac{1}{c_2} \Phi^{(2)}_{r_{\mathbb{T}}(M)}(A_\delta) = \frac{1}{c_2} \left(\sum_{i > r_{\mathbb{T}}(M)} \sigma_i(A_\delta)^2 \right)^{1/2}.$$

Proof. This is Theorem 8.11 with $p = 2$. □

8.4 Commitment as the Boolean-Rank Vertex

Commitment is not a Schatten-norm instance. It is the Boolean realizability vertex of the rank-control principle.

Definition 8.19 (Deterministic Boolean Hankel Rank)**.** Let $\Lambda \subseteq \Sigma^*$ be a language. Its Boolean Hankel matrix is

$$N_\Lambda(u, v) = \mathbf{1}_{\{uv \in \Lambda\}}.$$

Define

$$\mathrm{rank}^{\det}_{\mathbb{B}}\, N_\Lambda$$

to be the minimum number of states of a deterministic finite automaton recognizing Λ. By the Myhill–Nerode theorem,

$$\mathrm{rank}^{\det}_{\mathbb{B}}\, N_\Lambda = \kappa_{\mathrm{MN}}(\Lambda),$$

the number of Myhill–Nerode equivalence classes. If Λ is not regular, set $\kappa_{\mathrm{MN}}(\Lambda) = \infty$.

Remark 8.20 (Scope of the Boolean Realization Rank)**.** The quantity $\mathrm{rank}^{\det}_{\mathbb{B}}\, N_\Lambda$ is the *deterministic* Boolean realization rank, the invariant governing commitment. It is distinct from the real or complex rank of N_Λ, and from Boolean matrix ranks admitting nondeterministic factorizations, all three of which may differ from it.

Proposition 8.21 (Commitment Exact Threshold)**.** *Use the* 0/1 *commitment loss:*

$$\mathcal{E}_{\text{com}}(f,g) = \begin{cases} 0, & f = g, \\ 1, & f \neq g. \end{cases}$$

Then

$$\Delta_{\text{com}}(M) = \begin{cases} 0, & M \geq \kappa_{\text{MN}}(\Lambda), \\ 1, & M < \kappa_{\text{MN}}(\Lambda), \end{cases}$$

equivalently $\Delta_{\text{com}}(M) = v_0\big(\kappa_{\text{MN}}(\Lambda) - M\big)$ *with*

$$v_0(t) = \begin{cases} 0, & t \leq 0, \\ 1, & t > 0. \end{cases}$$

Under the extended-real Boolean valuation, in which the nonzero value is ∞ *rather than* 1*, replace* 1 *by* ∞ *throughout; the threshold* $\kappa_{\text{MN}}(\Lambda)$ *is unchanged.*

Proof. By the Myhill–Nerode theorem, a deterministic finite automaton recognizes Λ with M states iff $M \geq \kappa_{\text{MN}}(\Lambda)$. Therefore the zero-error commitment gap is zero exactly in that case, and positive otherwise. With 0/1 loss, the positive value is 1. □

8.5 Scope Boundary: No Unconditional Cross-Regime Ordering

Theorem 8.22 (No Universal Cross-Regime Inequality)**.** *The comparison inequalities of Proposition 8.1 require three things simultaneously: a common response operator, a common effective rank budget, and comparable domination moduli. The three regimes supply none of them. Their response operators are*

$$H_\nu, \qquad \mathbf{\Sigma}_\pi, \qquad N_\Lambda,$$

their effective rank budgets are

$$M, \qquad M-1, \qquad M,$$

and their domination moduli c_p *are unrelated.*

Therefore no unconditional inequality of the form

$$\Delta_{\text{grd}}(M) \leq \Delta_{\text{ret}}(M), \qquad \Delta_{\text{ret}}(M) \leq \Delta_{\text{com}}(M),$$

or any fixed monotone relation among the three gaps holds for all tasks. The unified Mirsky template unifies the functional form *of the converses under shared response representations; it does not unify the operators, the effective rank budgets, or the numerical gaps.*

Proof. This is a direct consequence of the Independence Theorem of Section 12, which constructs product tasks whose grounding, retention, and commitment obstructions vary independently. That construction is self-contained and does not invoke the present theorem, so the forward reference is one of presentation order only. □

9 The Two-Axis Oracle Inequality

The offline approximation theory and the online estimation theory are two distinct axes. The offline axis measures the deficit of the best budget-M predictor in a regime-specific hypothesis class. The online axis measures regret against that budget-M class. This section composes the two axes, within a fixed regime, into an adaptive structural-risk principle.

The bridge is type-correct: the online hypothesis class matches the offline approximation class. Retention uses stochastic finite-state predictors; grounding spectral relaxations use linear finite-rank predictors; commitment uses deterministic Mealy machines. Spectral tails enter online rates only through explicit spectral-rate assumptions.

9.1 Type-Correct Axes on One Clock

Fix a regime r. Let

$$\mathcal{H}_1^{\mathsf{r}} \subseteq \mathcal{H}_2^{\mathsf{r}} \subseteq \cdots$$

be the nested budget classes in that regime, and let

$$\mathcal{H}_\infty^{\mathsf{r}} = \bigcup_{M \geq 1} \mathcal{H}_M^{\mathsf{r}},$$

or its appropriate closure when stochastic or linear predictors are involved.

The regime-specific typing is as follows:

- **Commitment.** $\mathcal{H}_M^{\mathrm{com}}$ is the class of deterministic Mealy machines with at most M states. The loss is the 0/1 mismatch loss.
- **Retention.** $\mathcal{H}_M^{\mathrm{ret}}$ is a class of stochastic finite-state predictors, for example stationary controlled causal machines whose admissible quotients are lumpable quotients of the stationary support $\mathcal{S}^+$ with at most M blocks. The loss is log-loss or a bounded proper surrogate.
- **Grounding spectral relaxation.** $\mathcal{H}_M^{\mathrm{grd,lin}}$ is the class of linear finite-rank realizations of rank at most M, matching the linear finite-rank grounding converse. The exact symbolic deterministic Mealy grounding problem is a different object and is not substituted here.

Let

$$L_T(h) = \sum_{t=1}^{T} \ell_t(h)$$

denote cumulative loss. The estimation-axis envelopes invoked below are of the standard concentration type [26].

Definition 9.1 (Finite-State Benchmark)**.** For regime r, define

$$L_T^{\star,\mathsf{r}} = \inf_{h \in \mathcal{H}_\infty^{\mathsf{r}}} \mathbb{E}\, L_T(h).$$

This is the best loss achievable by the unrestricted finite-state model class in that regime. It equals the Bayes-optimal true-process loss only if the Bayes predictor lies in the closure of $\mathcal{H}_\infty^{\mathsf{r}}$.

Definition 9.2 (State Rate)**.** The *state rate* of a finite-state factor with at most M classes is

$$R(M) = \log M,$$

the description length of a state index, and the associated state-rate distortion function of a task theory $\mathbb{T}$ is $D_{\mathbb{T}}(R) = \Delta_{\mathbb{T}}(\lfloor e^R \rfloor)$.

Throughout, curves are parameterized by the budget M rather than by R, which is a change of variable and affects no theorem.

The object $D_{\mathbb{T}}$ is a lossy-compression trade-off in the informal sense: a source $\delta_{\mathbb{T}}$, a description budget $R = \log M$, and a fidelity criterion $\mathcal{E}_{\mathbb{T}}$. Three properties separate it from a Shannon rate–distortion function, and the third is the substantive one. The quantity $\log M$ is a one-shot description length for the memory, not a per-symbol transmission rate. The distortion axis is a regime-specific cost, not a single-letter expected distortion. And $D_{\mathbb{T}}$ need not be convex, so the operational apparatus that convexity underwrites — in particular the time-sharing argument that makes the Shannon curve the lower convex envelope of achievable pairs — is unavailable here (Proposition 9.3). The trade-off is therefore genuine but is not an instance of Shannon's theory, and the results below are stated for $\Delta_{\mathbb{T}}(M)$ directly.

Proposition 9.3 (The Finite-State Curve Need Not Be Convex)**.** *There is a retention instance, with every partition lumpable, for which $R \mapsto D_{\mathbb{T}}(R)$ is not convex on $\{\log M : 1 \leq M \leq |\mathcal{S}^+|\}$.*

Proof. Take $|\mathcal{S}^+| = 5$, $\mathcal{O} = \{1, 2, 3\}$, the reset transition $\tau(s_i, \ell) = s_\ell$ of Theorem 5.56, so that every partition is lumpable, stationary weights proportional to

$$(0.0344,\ 0.3506,\ 0.1906,\ 0.2176,\ 0.2068),$$

and predictive laws the row normalizations of

$$(0.4805, 0.4113, 0.1082),\ (0.2746, 0.4018, 0.3236),\ (0.2960, 0.5426, 0.1614),$$

$$(0.2334, 0.5019, 0.2648),\ (0.6498, 0.0548, 0.2954).$$

Minimizing $\Delta^{\mathrm{KL}}_{\mathrm{ret}}$ over all partitions with at most M blocks gives

$$D(0) = 0.0948616, \quad D(\log 2) = 0.0148089, \quad D(\log 3) = 0.0049099,$$

$$D(\log 4) = 0.0021747, \quad D(\log 5) = 0,$$

whose successive chord slopes are -0.115492, -0.024414, -0.009508, -0.009746. Convexity requires the slopes to be nondecreasing; the last decreases, so D is not convex. The computation is exact rational-plus-logarithm arithmetic, carried out at 60 significant digits, and the final slope decrement is -2.38×10^{-4}, far above any rounding effect. $\square$

Remark 9.4 (Why Convexity Fails)**.** The Shannon curve is convex because rates may be time-shared: given codes at rates R_1 and R_2 one randomizes between them and attains every point of the chord. The finite-state budget admits no such operation. The constraint

$|\mathcal{K}| \le M$ is a hard cardinality bound on a *deterministic* lumpable quotient, and a randomization between quotients of sizes M_1 and M_2 is not itself a quotient of any intermediate size. Convexity therefore has no mechanism to hold, and by Proposition 9.3 it indeed fails; over randomly generated retention instances with $|\mathcal{S}^+| \le 5$ it fails in roughly one case in eight.

The failure is not an artefact of measuring the budget logarithmically. Convexity is a property of the pair (rate axis, distortion axis), so one might hope that some other parameterization of the budget restores it. It does not. Take $|\mathcal{S}^+| = 4$, $\mathcal{O} = \{1, 2, 3\}$, stationary weights proportional to $(17, 18, 22, 21)$, and predictive laws the row normalizations of

$$(20, 25, 2), \quad (2, 34, 30), \quad (37, 1, 27), \quad (20, 9, 1).$$

The optimal costs are $D = 0.2887482,\ 0.1601480,\ 0.0145216,\ 0$ at $M = 1, 2, 3, 4$. Against the raw budget $R = M$ the chord slopes are -0.128600, -0.145626, -0.014522, and against $R = \log M$ they are -0.185531, -0.359159, -0.050478; both sequences decrease at the first step, so the curve is non-convex under either parameterization. Since the obstruction is the absence of a convex combination of feasible quotients, and that is a property of the feasible set rather than of the axis, no rescaling of the rate can supply a convexity guarantee.

Definition 9.5 (Finite-Horizon Approximation Deficit)**.** The budget-M approximation deficit is

$$A_T^{\mathsf{r}}(M) = \inf_{h \in \mathcal{H}_M^{\mathsf{r}}} \mathbb{E}\, L_T(h) - L_T^{\star;\mathsf{r}}.$$

The per-symbol deficit is

$$\Delta_T^{\mathsf{r}}(M) = \frac{1}{T} A_T^{\mathsf{r}}(M).$$

Quantification over processes. As written, $A_T^{\mathsf{r}}(M)$ depends on the process P generating the expectation, and the minimax statements below require an envelope that does not. Two readings make this precise, and the choice must be fixed once and used consistently. Either fix a class $\mathcal{P}$ of processes and take the worst case,

$$a_M(\mathcal{P}) = \sup_{P \in \mathcal{P}} \Big[\inf_{h \in \mathcal{H}_M^{\mathsf{r}}} \mathbb{E}_P L_T(h) - L_T^{\star;\mathsf{r}}(P) \Big],$$

or fix for each M an explicit hard subclass $\mathcal{P}_M$ on which the deficit is bounded below. Throughout the oracle section the first reading is in force, with $\mathcal{P}$ the process class of the regime r; displays such as $\sup_P \mathrm{Reg}_T \ge \sup_M \min\{A_T^{\mathsf{r}}(M), \mathsf{E}_M^{\mathsf{r}}(T)\}$ are to be read with $A_T^{\mathsf{r}}(M) = a_M(\mathcal{P})$, so that both sides quantify over the same class and the inequality is between deterministic envelopes. Where a single fixed P is intended, as in the fixed-budget decomposition, the process is named explicitly.

This definition is finite-horizon and does not presuppose stationarity. Under stationary ergodic conditions one has

$$A_T^{\mathsf{r}}(M) = T\Delta^{\mathsf{r}}(M) + o(T),$$

where $\Delta^{\mathsf{r}}(M)$ is the stationary per-symbol gap of the corresponding regime; the identification of $\Delta_T^{\mathsf{r}}(M)$ with a stationary gap requires that hypothesis and is not automatic.

Definition 9.6 (Online Estimation Rate)**.** An estimation rate for budget M is any function $\mathsf{E}^{\mathsf{r}}_M(T)$ for which there exists a base learner $\mathcal{A}_M$ satisfying

$$\mathbb{E}\,L_T(\mathcal{A}_M) - \inf_{h\in\mathcal{H}^{\mathsf{r}}_M} \mathbb{E}\,L_T(h) \le \mathsf{E}^{\mathsf{r}}_M(T).$$

9.2 Exact Fixed-Budget Decomposition

Proposition 9.7 (Fixed-Budget Bias–Variance Decomposition)**.** *For any learner $\mathcal{A}_M$ whose predictions lie in or are generated from $\mathcal{H}^{\mathsf{r}}_M$, if*

$$\mathbb{E}\,L_T(\mathcal{A}_M) - \inf_{h\in\mathcal{H}^{\mathsf{r}}_M} \mathbb{E}\,L_T(h) \le \mathsf{E}^{\mathsf{r}}_M(T),$$

then its total excess loss against the unrestricted finite-state benchmark satisfies

$$\mathrm{Reg}_T(\mathcal{A}_M) := \mathbb{E}\,L_T(\mathcal{A}_M) - L_T^{\star,\mathsf{r}} \le A^{\mathsf{r}}_T(M) + \mathsf{E}^{\mathsf{r}}_M(T).$$

Proof. The in-class infimum need not be attained, so fix $\varepsilon > 0$ and choose an ε-approximate in-class optimum $h^{\varepsilon}_M \in \mathcal{H}^{\mathsf{r}}_M$ with

$$\mathbb{E}\,L_T(h^{\varepsilon}_M) \le \inf_{h\in\mathcal{H}^{\mathsf{r}}_M} \mathbb{E}\,L_T(h) + \varepsilon.$$

Inserting it and splitting,

$$\mathrm{Reg}_T(\mathcal{A}_M) = \big(\mathbb{E}\,L_T(\mathcal{A}_M) - \mathbb{E}\,L_T(h^{\varepsilon}_M)\big) + \big(\mathbb{E}\,L_T(h^{\varepsilon}_M) - L_T^{\star,\mathsf{r}}\big).$$

The first bracket is at most $\mathsf{E}^{\mathsf{r}}_M(T) + \varepsilon$ by the estimation hypothesis, and the second is at most $A^{\mathsf{r}}_T(M) + \varepsilon$ by the choice of h^{ε}_M. Hence

$$\mathrm{Reg}_T(\mathcal{A}_M) \le A^{\mathsf{r}}_T(M) + \mathsf{E}^{\mathsf{r}}_M(T) + 2\varepsilon,$$

and letting $\varepsilon \downarrow 0$ gives the claim. When the infimum is attained one may take $\varepsilon = 0$ directly. $\square$

9.3 Adaptive Oracle Inequalities

Theorem 9.8 (Adaptive Agnostic Oracle Inequality)**.** *Assume losses are bounded in* $[0, 1]$*. For each $M \ge 1$, let $\mathcal{A}_M$ be a base learner satisfying*

$$\mathbb{E}\,L_T(\mathcal{A}_M) - \inf_{h\in\mathcal{H}^{\mathsf{r}}_M} \mathbb{E}\,L_T(h) \le \mathsf{E}^{\mathsf{r}}_M(T).$$

There exists an aggregated learner $\mathcal{A}^{\mathrm{agg}}$ such that for every $M \ge 1$,

$$\mathrm{Reg}_T(\mathcal{A}^{\mathrm{agg}}) \le A^{\mathsf{r}}_T(M) + \mathsf{E}^{\mathsf{r}}_M(T) + \Psi_M(T),$$

where

$$\Psi_M(T) \le C\left(\sqrt{T\ln(eM)} + \ln(eM)\right)$$

for a universal constant C, uniformly in T. This is sharper than the $\ln(eM)$ *form obtainable from a learning-rate grid: the second-order term is* $\ln(eM)$*, with no dependence on T. Consequently, if*

$$M_T^\star \in \arg\min_{M\ge1}\left[A_T^{\mathsf{r}}(M) + \mathsf{E}_M^{\mathsf{r}}(T)\right],$$

then

$$\mathrm{Reg}_T(\mathcal{A}^{\mathrm{agg}}) \le \min_{M\ge1}\left[A_T^{\mathsf{r}}(M) + \mathsf{E}_M^{\mathsf{r}}(T)\right] + \Psi_{M_T^\star}(T).$$

The learner does not know $M_T^\star$, $A_T^{\mathsf{r}}(\cdot)$, or the spectral decay.

Proof. Treat the base learners $\{\mathcal{A}_M\}_{M\ge1}$ as a countable expert set with prior

$$\pi_M = \frac{6}{\pi^2 M^2}, \qquad \ln(1/\pi_M) = 2\ln M + O(1).$$

Aggregate them with a parameter-free prior-weighted expert algorithm enjoying the standard second-order guarantee

$$\mathbb{E}\, L_T(\mathcal{A}^{\mathrm{agg}}) - \mathbb{E}\, L_T(\mathcal{A}_M) \le C\sqrt{T\ln(1/\pi_M)} + C\ln(1/\pi_M)$$

simultaneously for every M, with no tuning to M or T [27, 28]. Substituting $\ln(1/\pi_M) = 2\ln M + O(1)$ gives an aggregation penalty

$$\Psi_M(T) \le C\left(\sqrt{T\ln(eM)} + \ln(eM)\right).$$

Adding the fixed-budget decomposition of Proposition 9.7 and taking the infimum over M gives the result.

A single exponential-weights instance carries one learning rate, and a grid of rates η_j does not by itself yield a per-expert tuned bound; a parameter-free or adaptive-rate aggregator is therefore required. □

Theorem 9.9 (Realizable 0/1 Model Selection)**.** *Assume* $0/1$ *loss. Suppose each base learner $\mathcal{A}_M$ makes expected mistakes at most*

$$\mathsf{E}_M^{\mathrm{real}}(T)$$

more than the best predictor in $\mathcal{H}_M^{\mathsf{r}}$. There exists a weighted-majority aggregation over the base learners, with prior $w_M = 6/(\pi^2 M^2)$ on the budget index, such that for every M,

$$\mathrm{Reg}_T(\mathcal{A}^{\mathrm{wm}}) \le C\,A_T^{\mathsf{r}}(M) + C\,\mathsf{E}_M^{\mathrm{real}}(T) + C\ln(eM),$$

for a universal constant C.

Proof. The weighted-majority mistake bound is standard [9, 27] and is not reproved here. If expert M makes m_M mistakes and has prior weight w_M, weighted majority with a constant multiplicative update makes at most

$$C_1 m_M + C_2\ln(1/w_M)$$

mistakes in expectation, up to finite-alphabet constants. Since

$$m_M \le \inf_{h\in\mathcal{H}_M} \mathbb{E}L_T(h) + \mathsf{E}_M^{\mathrm{real}}(T),$$

subtracting $L_T^{\star,\mathsf{r}}$ gives the claim. □

Remark 9.10 (Realizable and Agnostic Laws Are Separate)**.** The realizable theorem does not use the bounded-loss $\sqrt{T}$ aggregation overhead. Realizable and agnostic budget laws are therefore stated separately.

Theorem 9.11 (Mixable Log-Loss Oracle Inequality)**.** *Let $\mathcal{H}_M^{\mathsf{r}}$ consist of stochastic predictors emitting probability distributions over a finite output alphabet. Assume either positivity or clipped log-loss so that the loss is mixable. If each base learner $\mathcal{A}_M$ satisfies*

$$\mathbb{E}\, L_T(\mathcal{A}_M) - \inf_{h\in\mathcal{H}_M^{\mathsf{r}}} \mathbb{E}\, L_T(h) \le \mathsf{E}_M^{\mathsf{r}}(T),$$

then there exists an aggregating learner $\mathcal{A}^{\mathrm{mix}}$ such that for every M,

$$\mathrm{Reg}_T(\mathcal{A}^{\mathrm{mix}}) \le A_T^{\mathsf{r}}(M) + \mathsf{E}_M^{\mathsf{r}}(T) + C\ln(eM).$$

Proof. For mixable losses, Vovk's aggregating algorithm [23] with prior weights w_M gives

$$L_T(\mathcal{A}^{\mathrm{mix}}) \le \min_M \left[L_T(\mathcal{A}_M) + \ln(1/w_M)\right]$$

up to mixability constants. Taking expectations and applying Proposition 9.7 yields the result. □

9.4 Spectral Input: Converses versus Achievability

The offline schema supplies spectral converses. These are lower bounds. They do not by themselves give online regret upper rates.

Definition 9.12 (Spectral-Rate Assumption)**.** We say that regime r satisfies a spectral-rate assumption with tail $S^{\mathsf{r}}(M)$ if there exist constants $c, C > 0$ such that

$$c\, S^{\mathsf{r}}(M) \le \Delta_T^{\mathsf{r}}(M) \le C\, S^{\mathsf{r}}(M) + o(1)$$

uniformly in the relevant asymptotic range. Equivalently,

$$A_T^{\mathsf{r}}(M) \le T\, C\, S^{\mathsf{r}}(M) + o(T).$$

The lower inequality is the converse. The upper inequality is an achievability bound.

For grounding, the unrestricted relaxation satisfies

$$S^{\mathrm{grd}}(M) = \sigma_{M+1}(H_\nu), \qquad \Delta_{\mathrm{grd}}^{\mathrm{unres}}(M) = \sigma_{M+1}(H_\nu),$$

while the Hankel-structured gap satisfies

$$\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) \ge \sigma_{M+1}(H_\nu).$$

For quadratic retention,

$$S^{\mathrm{ret}}(M) = \sum_{i\ge M} \lambda_i(\boldsymbol{\Sigma}_\pi) = \Phi_{M-1}^{(1)}(\boldsymbol{\Sigma}_\pi).$$

9.5 Budget Laws under Spectral-Rate Assumptions

Define the oracle objective

$$\mathsf{F}^{\mathsf{r}}_T(M) = A^{\mathsf{r}}_T(M) + \mathsf{E}^{\mathsf{r}}_M(T).$$

For the remainder of this subsection write

$$a^{\mathsf{r}}(M) = \frac{1}{T} A^{\mathsf{r}}_T(M)$$

for the per-round approximation deficit. Every displayed upper rate below assumes a corresponding *achievability* bound on $a^{\mathsf{r}}(M)$, not merely a spectral converse.

Corollary 9.13 (Agnostic Geometric Decay)**.** *Suppose the agnostic estimation rate is*

$$\mathsf{E}^{\mathsf{r}}_M(T) = O\Big(\sqrt{T\,M\log(eM)}\Big),$$

and suppose

$$a^{\mathsf{r}}(M) \le C\rho^M, \qquad 0 < \rho < 1.$$

Then

$$M^\star_T = \Theta(\log T),$$

and the adaptive aggregated learner satisfies

$$\mathrm{Reg}_T = \widetilde{O}(\sqrt{T}).$$

If, in addition, a minimax lower bound of the same order is available for the specified protocol and process class, then

$$\mathrm{Reg}_T = \widetilde{\Theta}(\sqrt{T}).$$

Proof. Balance

$$T\rho^M \asymp \sqrt{T\,M\log M}.$$

Squaring gives

$$T^2\rho^{2M} \asymp T\,M\log M,$$

so

$$\rho^{2M} \asymp \frac{M\log M}{T}.$$

Thus

$$M = \frac{1}{2\ln(1/\rho)}\ln T + O(\ln\ln T).$$

Substitution gives both terms of order $\sqrt{T}$ times polylogarithmic factors. The adaptive overhead from Theorem 9.8 is absorbed into the $\widetilde{O}$ notation. ∎

Corollary 9.14 (Agnostic Polynomial Decay)**.** *Suppose*

$$a^{\mathsf{r}}(M) \le CM^{-\alpha}, \qquad \alpha > 0.$$

Then

$$M_T^\star = \widetilde{\Theta}\big(T^{1/(2\alpha+1)}\big)\,,$$

and

$$\mathrm{Reg}_T = \widetilde{O}\big(T^{(\alpha+1)/(2\alpha+1)}\big)\,.$$

If, in addition, a minimax lower bound of the same order is available for the specified protocol and process class, then

$$\mathrm{Reg}_T = \widetilde{\Theta}\big(T^{(\alpha+1)/(2\alpha+1)}\big)\,.$$

Proof. Balance

$$TM^{-\alpha} \asymp \sqrt{T\,M\log M}.$$

Squaring gives

$$T^2M^{-2\alpha} \asymp T\,M\log M,$$

so

$$M^{2\alpha+1} \asymp \frac{T}{\log M}.$$

Hence

$$M_T^\star = \widetilde{\Theta}\big(T^{1/(2\alpha+1)}\big)\,.$$

The balanced value is

$$TM^{-\alpha} = \widetilde{\Theta}\big(T^{1-\alpha/(2\alpha+1)}\big) = \widetilde{\Theta}\big(T^{(\alpha+1)/(2\alpha+1)}\big)\,.$$

□

Corollary 9.15 (Realizable Geometric Decay)**.** *If the realizable or mixable estimation rate is*

$$\mathsf{E}^{\mathsf{r}}_M(T) = O(M\log(eM)),$$

and

$$a^{\mathsf{r}}(M) \le C\rho^M, \qquad 0<\rho<1,$$

then

$$M_T^\star = \Theta(\log T),$$

and

$$\mathrm{Reg}_T = \widetilde{O}(\log T).$$

If, in addition, a minimax lower bound of the same order is available for the specified protocol and process class, then

$$\mathrm{Reg}_T = \widetilde{\Theta}(\log T).$$

Proof. Balance

$$T\rho^M \asymp M\log M.$$

Thus

$$M = \frac{1}{\ln(1/\rho)}\ln T + O(\ln\ln T),$$

and the common objective value is

$$M \log M = \Theta(\log T \log\log T).$$

The logarithmic model-selection overhead is lower order. $\square$

Corollary 9.16 (Realizable Polynomial Decay)**.** *If*

$$a^{\mathsf{r}}(M) \le CM^{-\alpha}, \qquad \alpha > 0,$$

and the realizable or mixable estimation rate is

$$\mathsf{E}^{\mathsf{r}}_M(T) = O(M \log(eM)),$$

then

$$M^\star_T = \widetilde{\Theta}\big(T^{1/(\alpha+1)}\big)\,,$$

and

$$\mathrm{Reg}_T = \widetilde{O}\big(T^{1/(\alpha+1)}\big)\,.$$

If, in addition, a minimax lower bound of the same order is available for the specified protocol and process class, then

$$\mathrm{Reg}_T = \widetilde{\Theta}\big(T^{1/(\alpha+1)}\big)\,.$$

Proof. Balance

$$TM^{-\alpha} \asymp M \log M.$$

Thus

$$M^{\alpha+1} \asymp \frac{T}{\log M},$$

so

$$M^\star_T = \widetilde{\Theta}\big(T^{1/(\alpha+1)}\big)\,.$$

The balanced objective is

$$M \log M = \widetilde{\Theta}\big(T^{1/(\alpha+1)}\big)\,.$$

$\square$

Remark 9.17 (Different Exponents Are Structural)**.** The agnostic exponent

$$\frac{\alpha+1}{2\alpha+1}$$

and the realizable exponent

$$\frac{1}{\alpha+1}$$

are not interchangeable. They reflect the different costs of estimation:

$$\sqrt{TM\log M} \quad \text{versus} \quad M \log M.$$

9.6 Continuous Bias–Variance Sandwich and Lower-Bound Status

The oracle inequality proved above is an upper bound. A separate question is whether the resulting bias–variance envelope is minimax-optimal. This subsection establishes an envelope comparison valid under regularity hypotheses, and distinguishes it from the stronger minimax statement of Theorem 9.34.

Proposition 9.18 (Necessity of the Aggregation Penalty)**.** *Consider online aggregation over M base predictors with losses in $[0,1]$, in which the aggregator observes each base predictor's loss each round and competes with the best of them. There is a universal $c > 0$ such that for all $M \geq 2$ and all $T \geq \log_2 M$, every aggregation rule, deterministic or randomized, suffers*

$$\sup \Big[\mathbb{E}\, L_T(\textit{aggregator}) - \min_{m \leq M} L_T(m) \Big] \;\geq\; c\sqrt{T \ln M}$$

on some loss sequence. Consequently the $\sqrt{T\ln(eM)}$ term of $\Psi_M(T)$ cannot be removed at the level of generality at which the oracle inequality is stated.

Proof. This is the classical lower bound for prediction with expert advice; we indicate the standard argument and cite it rather than reproduce it in full. Draw each base predictor's loss on each round as an independent fair coin in $\{0,1\}$. Every aggregation rule then has expected cumulative loss exactly $T/2$, while the expected minimum over M independent symmetric random walks of length T is $T/2 - \Theta(\sqrt{T \ln M})$ for $M \leq 2^T$, by the standard maximal-inequality estimate for the maximum of M sub-Gaussian variables. The gap is therefore $\Omega(\sqrt{T \ln M})$ in expectation over the random sequence, hence for some fixed sequence. See [27, Thm. 3.7] and [26, Ch. 2] for the precise constants. □

Remark 9.19 (What the Lower Bound Does and Does Not Settle)**.** Proposition 9.18 closes the question of necessity at the generality of *arbitrary* base predictors treated as experts: no adaptive rule competing with M unrelated predictors avoids $\Omega(\sqrt{T \ln M})$. It does not settle whether the penalty is necessary for the *nested* families arising here, where the base predictors are the budget-M' learners for $M' \leq M$ and are therefore highly dependent: the best predictor in a nested family is monotone in the budget, and an aggregator might exploit that structure. Whether $\Psi_M(T)$ can be improved for nested finite-state classes remains open.

Hypothesis 9.20 (Regular Bias–Variance Envelope)**.** Let M be relaxed to a continuous budget variable $x \in [1, \infty)$. Define an approximation envelope $a_T(x)$ and an estimation envelope $b_T(x)$ as follows.

The approximation envelope is a continuous interpolation of the spectral approximation tail, with the index convention

$$a_T(x) = T\,\Delta(x), \qquad \Delta(x) \approx \frac{1}{2} \sum_{i \geq \lceil x \rceil} \lambda_i,$$

or the corresponding regime-specific spectral tail. In the quadratic retention regime, the discrete tail is

$$\Delta(M) = \frac{1}{2} \sum_{i \geq M} \lambda_i(\boldsymbol{\Sigma}_\pi),$$

with effective rank $M-1$.

The estimation envelope $b_T(x)$ is a continuous interpolation of the type-correct online estimation cost, including all logarithmic factors and, where relevant, the adaptive aggregation penalty

$$\Psi_M(T) = O\Big(\sqrt{T\ln(eM)} + \ln(eM)\Big).$$

Assume:

(i) $a_T(x) > 0$ and a_T is continuous and nonincreasing;

(ii) $b_T(x) > 0$ and b_T is continuous and nondecreasing;

(iii) there exists a crossing point x_0 such that

$$a_T(x_0) = b_T(x_0).$$

Proposition 9.21 (Continuous Bias–Variance Sandwich)**.** *Under Assumption 9.20, define*

$$B_T = \sup_{x\geq 1} \min\{a_T(x), b_T(x)\},$$

and

$$A_T = \inf_{x\geq 1}\big[a_T(x) + b_T(x)\big].$$

Then

$$B_T \leq A_T \leq 2B_T.$$

Proof. Let x_0 be a crossing point with

$$a_T(x_0) = b_T(x_0) = B_T.$$

Because a_T is nonincreasing and b_T is nondecreasing, for $x \leq x_0$ we have $a_T(x) \geq B_T$, while for $x \geq x_0$ we have $b_T(x) \geq B_T$. Hence for every x,

$$a_T(x) + b_T(x) \geq B_T.$$

Taking the infimum over x gives $A_T \geq B_T$.

Conversely, evaluating the sum at the crossing point gives

$$A_T \leq a_T(x_0) + b_T(x_0) = 2B_T.$$

Therefore

$$B_T \leq A_T \leq 2B_T.$$

□

Remark 9.22 (The Sandwich Is Not by Itself a Minimax Lower Bound)**.** Proposition 9.21 is an envelope comparison. It shows that, under regularity, the sum envelope A_T and the sup–min envelope B_T are within a factor of 2. It does not by itself prove

$$\inf_{\text{learner}} \sup_P \mathrm{Reg}_T \geq B_T.$$

Promoting the sandwich to a minimax lower bound requires explicit approximation floors, explicit estimation floors, and a reduction valid for arbitrary adaptive learners. Such floors are supplied, as hypotheses, by Assumption 9.24; they are not established unconditionally for the regimes considered here.

Remark 9.23 (Conditions Governing the Envelope Comparison)**.** Four conditions govern the use of the bias–variance sandwich, and each is required for the statements of this section to be read correctly.

(1) The continuous sandwich is valid under Assumption 9.20, that is, for envelopes that are positive, continuous, monotone in opposite directions, and crossing.

(2) The retention spectral tail uses the index convention

$$\sum_{i \geq M},$$

not $\sum_{i>M}$: the effective rank is $M-1$, because centring removes one degree of freedom (Theorem 5.8).

(3) The estimation envelope must retain its logarithmic factors and, where model selection is adaptive, the aggregation penalty $\Psi_M(T)$.

(4) Rounding a continuous optimal budget to an integer one is not universally harmless; it requires regularity near the optimum, for example bounded adjacent spectral ratios or the absence of large spectral gaps.

The envelope comparison is an inequality between deterministic envelopes. It becomes a minimax lower bound under the explicit floors of Assumption 9.24, which yield Theorem 9.34. The minimax optimality of the adaptive oracle envelope is therefore conditional on those floors.

Hypothesis 9.24 (Explicit Approximation and Estimation Floors)**.** Fix a regime r and horizon T. Assume:

(i) **Packing floor.** For each M there are finitely many processes $P_M^1, \ldots, P_M^m$ in the class, $m = m(M)$, and a quantity $\Delta_M > 0$, such that, writing

$$R_i(h) = \mathbb{E}_{P_M^i} \ell_T(h) - L_T^{\star,\mathsf{r}}(P_M^i),$$

every predictor h, adaptive or not, satisfies

$$\frac{1}{m} \sum_{i=1}^{m} R_i(h) \;\geq\; \Delta_M.$$

That is, no single predictor is simultaneously near-optimal across the family, the *average* shortfall being at least Δ_M.

(ii) **Envelope calibration.** The packing parameter dominates the envelope minimum,

$$\Delta_M \;\geq\; \min\bigl\{A^{\mathsf{r}}_T(M), \mathsf{E}^{\mathsf{r}}_M(T)\bigr\}.$$

The two clauses must be kept apart, and conflating them is a genuine hazard. Defining Δ_M to *equal* $\min\{A^{\mathsf{r}}_T(M), \mathsf{E}^{\mathsf{r}}_M(T)\}$ would make the assumption vacuous in every realizable regime. There the model class contains the truth, so $A^{\mathsf{r}}_T(M) = 0$ for all M at or above the true state count, the minimum is 0, and the floor asserts only $\frac{1}{m}\sum_i R_i(h) \geq 0$, which is automatic. The packing floor and the envelope calibration are therefore separate demands: the first is a statement about the family, proved by exhibiting mutually distinguishable processes; the second is a comparison between that family's information content and the envelope whose optimality is at issue. Only the conjunction yields a matching lower bound, and in a realizable regime the calibration clause is where the content resides, since the relevant comparison is then against $\mathsf{E}^{\mathsf{r}}_M(T)$ alone.

The averaging is over a family whose size may grow with M, and this is essential rather than cosmetic. A two-point version of this condition, with $R_0(h) + R_1(h) \geq \Delta_M$, is unsatisfiable once Δ_M exceeds $2\log 2$. Indeed under log loss $R_i(h) = D_{\mathrm{KL}}(P^i \| Q_h)$ for the transcript law Q_h induced by h, and

$$\min_Q \bigl[D_{\mathrm{KL}}(P^0\|Q) + D_{\mathrm{KL}}(P^1\|Q)\bigr] = 2\,\mathrm{JS}(P^0, P^1) \;\leq\; 2\log 2,$$

the minimum being attained at the mixture $\frac{1}{2}(P^0 + P^1)$. Since $\mathsf{E}^{\mathsf{r}}_M(T)$ diverges in T in every regime considered here, a two-point floor would fail for all large T. With m processes the same computation gives

$$\min_Q \frac{1}{m}\sum_{i=1}^{m} D_{\mathrm{KL}}(P^i\|Q) = I(V;Y), \qquad V \sim \mathrm{Unif}[m],$$

which is at most $\log m$ and equals $\log m$ when the transcript laws are mutually singular. Carrying $\Delta_M = \Theta(M\log M)$ therefore requires $m = \exp(\Theta(M\log M))$, of the order of $|\mathcal{H}_M|$.

(iii) **Monotonicity.** $A^{\mathsf{r}}_T(\cdot)$ is nonincreasing and $\mathsf{E}^{\mathsf{r}}_{(\cdot)}(T)$ is nondecreasing.

Remark 9.25 (What a Single Family Can and Cannot Deliver)**.** Clauses (i) and (ii) constrain a single packing. The consequence for regimes in which *both* envelopes are active is the following.

A packing internal to $\mathcal{H}_M$ certifies estimation difficulty: its members are all realizable at budget M, so it lower-bounds the price of not knowing which member is in force, and calibrates against $\mathsf{E}^{\mathsf{r}}_M(T)$. A packing certifying approximation difficulty must instead place its members *outside* the reach of $\mathcal{H}_M$, so that $\inf_{h\in\mathcal{H}^{\mathsf{r}}_M} R_P(h) \geq a_M > 0$. These are different families with different type signatures, and exhibiting each separately does not by itself certify that one adversary forces both burdens at once: a learner facing the approximation-hard family may pay a_M while facing no estimation burden, and conversely.

It does not follow, however, that a bound on the *sum* is out of reach without a direct-sum construction. The sum-form envelope $\inf_M[a_M + b_M]$ is the quantity of interest, and it is already reachable. The discrete sandwich of Lemma 9.28 already converts between the two forms: with $a(M) = A^{\mathsf{r}}_T(M)$, $b(M) = \mathsf{E}^{\mathsf{r}}_M(T)$, $B = \sup_M \min\{a(M), b(M)\}$ and $A = \inf_M[a(M) + b(M)]$, it gives $B \le A \le (1+\kappa)B$ under its nondegeneracy and crossing conditions. Consequently a floor calibrated against the *minimum* automatically yields a bound against the *sum*,

$$\sup_P \mathrm{Reg}_T \;\ge\; c\,B \;\ge\; \frac{c}{1+\kappa}\,\inf_M\bigl[A^{\mathsf{r}}_T(M) + \mathsf{E}^{\mathsf{r}}_M(T)\bigr],$$

which is the second display of Theorem 9.34. The loss is the jump ratio κ at the crossing index, and nothing more.

The role of a direct-sum construction is therefore narrower than it may appear. It is not needed to reach the sum-form envelope at all; it is needed only to remove the factor $(1+\kappa)$. That factor is itself harmless: by Lemma 9.26 the jump ratio of every estimation envelope occurring here is bounded by an absolute constant — at most $1+\log 2$ in the agnostic regime and $2(1+\log 2)$ in the realizable regime — independently of the horizon T and of where the crossing falls. The sum-form minimax lower bound therefore holds with an absolute constant, and *no* direct-sum construction is required for it.

What a both-axes construction would still buy is a bound in which the two components are certified by a single adversary, which is a statement about the *source* of the difficulty rather than about the size of the constant. Accordingly Assumption 9.24 is stated for the single quantity Δ_M calibrated against the minimum, which is what Lemma 9.28 consumes, and nothing is lost in passing to the sum.

Lemma 9.26 (The Jump Ratio Is an Absolute Constant)**.** *Let the estimation envelope have the regularly varying form*

$$b(M) = \gamma\, M^{\alpha}\bigl(\log eM\bigr)^{\beta}, \qquad \alpha, \beta \ge 0,\ (\alpha,\beta) \ne (0,0),\ \gamma > 0.$$

Then for every $M \ge 2$,

$$\frac{b(M)}{b(M-1)} \;\le\; \frac{b(2)}{b(1)} = 2^{\alpha}\bigl(1+\log 2\bigr)^{\beta},$$

and consequently the jump ratio of Lemma 9.28 satisfies

$$\kappa \;\le\; 2^{\alpha}(1+\log 2)^{\beta}$$

whatever the crossing index. The bound depends only on the exponents: it is independent of the horizon T, of the scale γ, and of the approximation envelope.

If b is a sum of finitely many such terms, $b = \sum_j b_j$ with exponents (α_j, β_j), then $\kappa \le \max_j 2^{\alpha_j}(1+\log 2)^{\beta_j}$.

Proof. Write $b(M)/b(M-1) = u(M)\, v(M)$ with

$$u(M) = \left(\tfrac{M}{M-1}\right)^{\alpha}, \qquad v(M) = \left(\tfrac{\log eM}{\log e(M-1)}\right)^{\beta}.$$

Both factors are nonincreasing on $M \geq 2$. For u this is immediate, since $M \mapsto M/(M-1) = 1 + 1/(M-1)$ is decreasing and $t \mapsto t^{\alpha}$ is increasing for $\alpha \geq 0$. For v it suffices to treat $\beta = 1$ and put $\ell(x) = \log ex = 1 + \log x$, which is positive, increasing and concave on $x \geq 1$. For a positive concave ℓ the ratio $\ell(x)/\ell(x-1)$ is nonincreasing: writing $x \mapsto \ell(x)/\ell(x-1)$ and differentiating, the sign of the derivative is that of $\ell'(x)\ell(x-1) - \ell(x)\ell'(x-1)$, and concavity gives $\ell'(x) \leq \ell'(x-1)$ while monotonicity gives $\ell(x-1) \leq \ell(x)$, so with $\ell' > 0$ the expression is at most $\ell'(x-1)\big[\ell(x-1) - \ell(x)\big] \leq 0$. Raising to the power $\beta \geq 0$ preserves the monotonicity.

Conclusion. The product of two nonincreasing positive functions is nonincreasing, so $b(M)/b(M-1)$ attains its supremum over $M \geq 2$ at $M = 2$, where its value is $2^{\alpha}(1+\log 2)^{\beta}$ because $\log e \cdot 1 = 1$ and $\log e \cdot 2 = 1 + \log 2$. Since κ is by definition $b(M^*)/b(M^*-1)$ for the crossing index $M^* \geq 2$, or 1 when $M^* = 1$, the stated bound holds in all cases.

Sums. For $b = \sum_j b_j$ the mediant inequality gives

$$\frac{\sum_j b_j(M)}{\sum_j b_j(M-1)} \;\leq\; \max_j \frac{b_j(M)}{b_j(M-1)} \;\leq\; \max_j 2^{\alpha_j}(1+\log 2)^{\beta_j},$$

since for positive x_j, y_j one has $(\sum_j x_j)/(\sum_j y_j) \leq \max_j (x_j/y_j)$. □

Remark 9.27 (The Jump Ratio for the Envelopes of This Paper)**.** Applying Lemma 9.26 to the estimation envelopes used here:

estimation envelope	(α, β)	$\kappa \leq$
$\sqrt{T \log eM} + \log eM$	$(0, \frac{1}{2}), (0, 1)$	$1 + \log 2 \approx 1.6931$
$\sqrt{TM \log eM}$	$(\frac{1}{2}, \frac{1}{2})$	$\sqrt{2(1+\log 2)} \approx 1.8402$
$M \log eM$	$(1, 1)$	$2(1 + \log 2) \approx 3.3863$

Each is an absolute constant. Consequently the second display of Theorem 9.34 holds with an absolute constant in front of the sum-form envelope: for instance $c/(1+\kappa) \geq c/2.70$ in the agnostic finite-class regime and $c/4.39$ in the realizable regime.

Two points of scope. First, the normalization $\log eM$ rather than $\log M$ is what makes b positive at $M = 1$, as condition (a) of Lemma 9.28 requires; the bare $M \log M$ vanishes there. If one insists on $M \log M$ the crossing index is at least 3 and the supremum of consecutive ratios over $M \geq 3$ is $3 \log 3/(2 \log 2) \approx 2.3774$, again absolute. Second, the lemma constrains only the estimation envelope: no regularity is imposed on the approximation envelope beyond the monotonicity already assumed.

Lemma 9.28 (Discrete Bias–Variance Sandwich)**.** *Let $a, b : \mathbb{N}_{\geq 1} \to (0, \infty)$ with a nonincreasing and b nondecreasing. Assume* *(a)* $b(1) \le a(1)$*;*

(b) $b(M) \ge a(M)$ *for some* M*.*

Let $M^* = \min\{M : b(M) \ge a(M)\}$*, which exists by (b), and set*

$$\kappa = \begin{cases} b(M^*)/b(M^*-1), & M^* \ge 2, \\ 1, & M^* = 1, \end{cases}$$

the jump ratio *of the estimation envelope at the crossing index; note* $\kappa \ge 1$ *since* b *is nondecreasing and positive. Put*

$$B = \sup_{M \ge 1} \min\{a(M), b(M)\}, \qquad A = \inf_{M \ge 1} \big[a(M) + b(M)\big].$$

Then

$$B \;\le\; A \;\le\; (1+\kappa)\, B.$$

No continuity and no exact crossing point are required.

Proof. Left inequality. Fix M and let N be arbitrary. If $N \ge M$ then $b(N) \ge b(M) \ge \min\{a(M), b(M)\}$ because b is nondecreasing; if $N < M$ then $a(N) \ge a(M) \ge \min\{a(M), b(M)\}$ because a is nonincreasing. Since both a and b are positive, in either case

$$a(N) + b(N) \;\ge\; \min\{a(M), b(M)\}.$$

Taking the infimum over N gives $A \ge \min\{a(M), b(M)\}$, and then the supremum over M gives $A \ge B$.

Right inequality. Suppose first $M^* \ge 2$. By minimality of M^* we have $b(M^*-1) < a(M^*-1)$, so

$$B \;\ge\; \min\{a(M^*-1), b(M^*-1)\} = b(M^*-1),$$

while $b(M^*) \ge a(M^*)$ gives

$$B \;\ge\; \min\{a(M^*), b(M^*)\} = a(M^*).$$

Hence, using $b(M^*) = \kappa\, b(M^*-1)$,

$$A \;\le\; a(M^*) + b(M^*) \;=\; a(M^*) + \kappa\, b(M^*-1) \;\le\; B + \kappa B \;=\; (1+\kappa)B.$$

If instead $M^* = 1$, then $b(1) \ge a(1)$, which with (a) forces $b(1) = a(1)$; thus $B \ge \min\{a(1), b(1)\} = a(1)$ and $A \le a(1) + b(1) = 2a(1) \le 2B = (1+\kappa)B$ since $\kappa = 1$. $\square$

Remark 9.29 (Role of the Jump Ratio)**.** The constant cannot be improved to a universal one. For $a = (10, 1)$ and $b = (1, 9)$ on $\{1, 2\}$ one has $B = 1$ and $A = 10$, so $A/B = 10 = 1 + \kappa$ with $\kappa = 9$. What the continuous statement (Proposition 9.21) buys by assuming continuity and an exact crossing is precisely $\kappa = 1$, giving the factor 2. In the discrete setting the estimation envelope typically satisfies $b(M)/b(M-1) = 1 + O(1/M)$ – this holds for $b(M) = \sqrt{T \ln(eM)} + \ln(eM)$ – so $\kappa \to 1$ and the factor tends to 2.

Lemma 9.30 (The Packing Floor Is an Information Condition)**.** *Under log loss, let $P^1, \dots, P^m$ be processes in the class and let V be uniform on $[m]$. Then for every predictor h, with Q_h the transcript law it induces,*

$$\frac{1}{m}\sum_{i=1}^{m} R_i(h) \;\geq\; \min_{Q} \frac{1}{m}\sum_{i=1}^{m} D_{\mathrm{KL}}(P^i \| Q) = I(V; Y_{1:T}),$$

the minimum being attained at the mixture $\bar{P} = \frac{1}{m}\sum_i P^i$. Consequently Assumption 9.24(i) holds with a given Δ_M if and only if some finite subfamily of the class satisfies

$$I(V; Y_{1:T}) \;\geq\; \Delta_M.$$

Proof. Under log loss $R_i(h) = D_{\mathrm{KL}}(P^i \| Q_h)$, so the average is $\frac{1}{m}\sum_i D_{\mathrm{KL}}(P^i \| Q_h)$. For any Q,

$$\frac{1}{m}\sum_{i} D_{\mathrm{KL}}(P^i \| Q) = \frac{1}{m}\sum_{i} D_{\mathrm{KL}}(P^i \| \bar{P}) + D_{\mathrm{KL}}(\bar{P} \| Q) = I(V; Y_{1:T}) + D_{\mathrm{KL}}(\bar{P} \| Q),$$

by the compensation identity, and $D_{\mathrm{KL}}(\bar{P} \| Q) \geq 0$ with equality iff $Q = \bar{P}$. The middle expression is the mutual information because V is uniform and $Y_{1:T} \mid V = i$ has law P^i. $\square$

Remark 9.31 (What Discharging the Floor Requires in Each Regime)**.** Lemma 9.30 reduces the floor to exhibiting a subfamily carrying Δ_M nats about its index by horizon T. The regimes differ sharply in how hard that is.

In the *deterministic* realizable regimes the transcripts of distinct members are eventually distinct, so the induced laws are mutually singular and $I(V; Y) = \log m$ exactly, as soon as the horizon exceeds the separating length. Proposition 9.32 carries this out.

In the *stochastic* regimes distinctness is statistical rather than logical, and $I(V; Y)$ grows only as fast as the family can be estimated. For the natural retention packing — M hidden Bernoulli biases spaced $\Theta(1/M)$ apart, indexed by $b \in [M]^M$, so $\log m = M \log M$ — each coordinate needs $\Omega(M^2)$ samples to resolve, hence $T = \Omega(M^3)$ before $I(V; Y)$ approaches $\log m$. At $T = \Theta(M \log M)$ the mutual information is a small fraction of $\log m$.

Consequently the floor in a stochastic regime should be read with an explicit horizon hypothesis, and Δ_M taken as the information actually available at that horizon rather than as $\log m$. Constructing packings that are efficient in T for the retention, grounding, and commitment regimes is open (Section 17.4).

Proposition 9.32 (The Packing Floor Is Satisfied in the Realizable Stream Regime)**.** *Let r be the realizable persistent-stream regime with $\mathsf{E}^{\mathsf{r}}_M(T) = \Theta(M \log M)$, and let $\mathcal{G}_M^{\mathrm{act}}$ be the gated family of Definition 13.44 on the forcing stream w of Theorem 13.5, of length $O(M \log M)$. Take $\{P_M^i\}$ to be the $m = M^M$ laws induced by the members of $\mathcal{G}_M^{\mathrm{act}}$. Then the transcript laws are mutually singular, so*

$$\min_{Q} \frac{1}{m}\sum_{i=1}^{m} D_{\mathrm{KL}}(P_M^i \| Q) = I(V; Y) = \log m = M \log M,$$

and the packing floor of Assumption 9.24(i) holds with $\Delta_M = M \log M$.

Moreover the envelope calibration of Assumption 9.24(ii) holds in this regime. The regime is realizable, so $A^{\mathsf{r}}_T(M) = 0$ and $\min\{A^{\mathsf{r}}_T(M), \mathsf{E}^{\mathsf{r}}_M(T)\} = 0 \leq \Delta_M$; the substantive comparison is with the estimation envelope, and $\Delta_M = M \log M = \Theta(\mathsf{E}^{\mathsf{r}}_M(T))$ matches it in order.

Proof. Two distinct maps $g \neq g'$ differ at some $v \in Q$ and some coordinate j. The stream w contains the block $w_v\, \mathtt{c}\, \mathtt{d}^L$, whose readout rounds emit the coordinates of $g(v)$ in order, and by the gating property of Definition 13.44 those rounds report g at the argument v and at no other. Hence the transcripts of g and g' differ in the j-th readout of that block, and the m transcripts are pairwise distinct. Being deterministic machines on a fixed input stream, each induces a point mass, so the m laws are mutually singular and $I(V;Y) = H(V) = \log m$ with V uniform. Finally $|\mathcal{G}^{\mathrm{act}}_M| = M^M$ gives $\log m = M \log M$.

The reduction to the floor is Lemma 9.30: every predictor satisfies $\frac{1}{m}\sum_i R_i(h) \geq I(V;Y) = M \log M$. The horizon suffices because the forcing stream has length $O(M \log M)$ and separates every pair of members within it. $\square$

Remark 9.33 (Status of the Floor Across Regimes)**.** Proposition 9.32 discharges Assumption 9.24(i) and (ii) for the realizable persistent-stream regime, so Theorem 9.34 is unconditional there. Four qualifications delimit what this does and does not settle.

Zeroth, and prior to the other three, the loss types differ and must not be silently identified. Lemma 9.30 and Proposition 9.32 are statements about logarithmic loss, with $R_i(h) = D_{\mathrm{KL}}(P^i \| Q_h)$ and the packing certified by mutual singularity of transcript laws; Theorem 13.5, by contrast, is a 0/1 mistake theorem. The mutual-information computation therefore does not by itself yield a 0/1 lower bound, and the two should be read as separate results about the same family: an unconditional $\Theta(M \log M)$ mistake bound in the realizable stream protocol, and a log-loss packing statement with $\Delta_M = M \log M$. Neither is derived from the other here.

First, the regime is realizable, so the approximation envelope vanishes and the floor is calibrated against the estimation envelope alone. The proposition therefore certifies the estimation axis; it says nothing about the approximation axis; by Remark 9.25 the sum-form envelope is nevertheless reached through the discrete sandwich, at the cost of the jump ratio $(1+\kappa)$, which Lemma 9.26 bounds by an absolute constant.

Second, the corresponding constructions for the retention, grounding, and commitment regimes are not carried out here and remain open (Section 17.4). Those regimes are stochastic, and by Remark 9.31 a packing there must additionally be efficient in the horizon.

Third, what the proposition establishes in general terms is that the hypothesis is satisfiable and has at least one verified instance, which the two-point form of the floor did not: that form is unsatisfiable for $\Delta_M > 2 \log 2$, as recorded in Assumption 9.24(i).

Theorem 9.34 (Minimax Lower Bound for the Oracle Envelope)**.** *Under Assumption 9.24 the bounds below hold with $c = 1$; they are written with a general constant $c > 0$ since only* *its positivity is used. For every learner, adaptive over M or not,*

$$\sup_P \mathrm{Reg}_T \;\geq\; c \sup_{M\geq 1} \min\bigl\{A^{\mathsf{r}}_T(M),\, \mathsf{E}^{\mathsf{r}}_M(T)\bigr\}.$$

If in addition the discrete nondegeneracy and crossing conditions (a), (b) of Lemma 9.28 hold for $a(M) = A^{\mathsf{r}}_T(M)$ and $b(M) = \mathsf{E}^{\mathsf{r}}_M(T)$, with jump ratio $\kappa = \kappa(T)$ at the crossing index, then

$$\sup_P \mathrm{Reg}_T \;\geq\; \frac{c}{1+\kappa} \inf_{M\geq 1} \left[A^{\mathsf{r}}_T(M) + \mathsf{E}^{\mathsf{r}}_M(T)\right],$$

which matches the adaptive upper bound of Theorem 9.8 up to the aggregation penalty $\Psi_M(T)$ and the factor $1+\kappa(T)$.

The matching is up to universal *constants precisely when $\sup_T \kappa(T) < \infty$; otherwise the constant degrades with $\kappa(T)$ and the two-sided statement is correspondingly weaker. For the estimation envelopes arising here the condition holds with room to spare: for $\mathsf{E}^{\mathsf{r}}_M(T) = \sqrt{T\ln(eM)} + \ln(eM)$ one has $\kappa(T) = 1 + O(1/M^*)$, so $\kappa(T) \to 1$ and the factor tends to 2, recovering the constant of the continuous sandwich (Proposition 9.21) without its continuity and exact-crossing hypotheses.*

Proof. Fix M and let $\mathcal{A}$ be any learner, deterministic or randomized, adaptive or not. A learner is a measurable map from transcripts to predictions, so it is in particular an admissible h in the packing floor. Applying that floor to $h = \mathcal{A}$ gives

$$\frac{1}{m}\sum_{i=1}^{m} R_i(\mathcal{A}) \;\geq\; \Delta_M,$$

and a maximum dominates an average, so

$$\max_{1\leq i\leq m} R_i(\mathcal{A}) \;\geq\; \Delta_M.$$

Since the adversary may select the process,

$$\sup_P \mathrm{Reg}_T(\mathcal{A}) \;\geq\; \Delta_M,$$

and as M was arbitrary,

$$\sup_P \mathrm{Reg}_T \;\geq\; \sup_{M\geq 1} \min\bigl\{A^{\mathsf{r}}_T(M), \mathsf{E}^{\mathsf{r}}_M(T)\bigr\},$$

which is the first display with $c = 1$. The averaged form is in fact stronger than the two-point form it replaces: the latter yielded only $\frac{1}{2}\Delta_M$ through $\max\{u, v\} \geq \frac{1}{2}(u+v)$, whereas here the maximum dominates the average outright.

For the second display, put $a(M) = A^{\mathsf{r}}_T(M)$ and $b(M) = \mathsf{E}^{\mathsf{r}}_M(T)$. These are positive, a nonincreasing and b nondecreasing, by the monotonicity clause of Assumption 9.24. Monotonicity alone is not enough: Lemma 9.28 additionally requires the nondegeneracy condition $b(1) \leq a(1)$ and the crossing condition $b(M) \geq a(M)$ for some M, and these are hypotheses of the present clause, not consequences of monotonicity. Under them, and with κ the jump ratio at the crossing index,

$$\sup_M \min\{a,b\} = B \;\geq\; \frac{1}{1+\kappa}A = \frac{1}{1+\kappa}\inf_M\big[a(M)+b(M)\big].$$

Combining with the first display, and recalling $c=1$, yields

$$\sup_P \mathrm{Reg}_T \;\geq\; \frac{1}{1+\kappa}\inf_{M\geq 1}\left[A^{\mathsf{r}}_T(M)+\mathsf{E}^{\mathsf{r}}_M(T)\right].$$

The continuous route through Assumption 9.20 and Proposition 9.21 gives the same conclusion with $1+\kappa$ replaced by 2, but at the price of assuming continuity of the interpolated envelopes and an exact crossing point; the discrete route assumes neither. $\square$

Remark 9.35 (Sum-Separation versus Two-Point Testing)**.** Two distinct hypotheses yield a minimax lower bound of this type, and they require different arguments. Under the packing floor of Assumption 9.24, which constrains *every* predictor, the bound follows directly and no indistinguishability argument enters; the floor is an averaged Fano-type condition, and Proposition 9.32 exhibits a family satisfying it. Under the weaker hypothesis that only the *optimal* predictors of the two processes are separated, a Le Cam or Hellinger two-point argument is required, together with a demonstration that any learner with small regret on both processes would distinguish the transcript laws; see [22, Ch. 2]. Combining a total-variation affinity condition with a packing condition is redundant, since the latter alone suffices. The formulation adopted here is the packing form.

Remark 9.36 (What Remains Genuinely Open)**.** Theorem 9.34 closes the two-sided characterization *up to the aggregation penalty* $\Psi_M(T)$ and universal constants, conditional on the explicit floors of Assumption 9.24. Two finer questions remain. First, whether the penalty $\Psi_M(T) = O(\sqrt{T\log(eM)})$ is itself necessary, i.e. whether adaptive aggregation over an unknown budget is strictly harder than the oracle problem; the second-order $\log(eM)$ term in the parameter-free bound suggests a separation, though a small one. Second, whether the floors of Assumption 9.24 hold unconditionally for the specific finite-state regimes of this manuscript, which is a statement about spectral decay of the associated operators rather than about the oracle mechanism.

9.7 Scope and Conditions

- **Type correctness.** The bridge is stated within a fixed regime and a fixed hypothesis-class type.
- **Benchmark.** The benchmark $L^{\star;\mathsf{r}}_T$ is the best finite-state predictor in the regime-specific union. It is called the true-process or Bayes benchmark only under an explicit richness assumption.
- **Converse versus achievability.** Spectral converses are lower bounds on $A^{\mathsf{r}}_T(M)$. They become regret upper rates only under the spectral-rate assumption of Definition 9.12. The oracle upper bounds of Theorems 9.8, 9.9 and 9.11 are stated in terms of the true

approximation deficit $A_T^{\mathsf{r}}(M)$; expressing the resulting regret as a spectral rate requires an *upper* bound on that deficit by a spectral tail, which a converse does not supply.

- **Active synchronization.** The active realizable estimation rate is

$$\mathsf{E}_M^{\text{active}}(T) = \Theta(M \log M)$$

unconditionally on the full class, by Corollary 13.47. The oracle inequality remains valid with this rate.

- **No cross-regime unification.** The bridge composes the offline and online axes within one regime. It does not unify the three regimes and does not produce a universal ordering of grounding, retention, and commitment gaps.

10 The Linear Second-Order Safety Surrogate

This section develops a linear second-order safety surrogate. The surrogate is exact for a linear block-local problem, but it is not an exact formula for the discrete right-congruence Price of Safety. The distinction is type-theoretic: pinching and majorization live in a linear operator surrogate, whereas the discrete Price of Safety is a constrained optimization over right congruences.

10.1 Type-Correct Objects

Let $\mathcal{S}$ be a finite state set with stationary weights $\pi_s > 0$, $\sum_s \pi_s = 1$. Let $y_s \in \mathbb{R}^d$ be the predictive or Fisher feature of state s, and define the centered features

$$x_s = y_s - \bar{y}, \qquad \bar{y} = \sum_s \pi_s y_s.$$

Let

$$D_\pi = \operatorname{diag}(\pi_s)$$

and let X be the matrix with rows $x_s^\top$. The feature covariance is

$$\boldsymbol{\Sigma}_\pi = X^\top D_\pi X.$$

The state-indexed centered Gram matrix is

$$G = D_\pi^{1/2} X X^\top D_\pi^{1/2},$$

with entries

$$G_{st} = \sqrt{\pi_s \pi_t}\, \langle x_s, x_t \rangle.$$

The matrix G is positive semidefinite and satisfies

$$G\sqrt{\pi} = 0, \qquad \sqrt{\pi} := (\sqrt{\pi_s})_{s\in\mathcal{S}}.$$

Let

$$\mathcal{B} = \{B_1, \ldots, B_r\}$$

be a safety partition of the state set. Let P_b be the coordinate projection onto B_b. The block-diagonal algebra is

$$\mathcal{A} = \left\{ X : \ X = \sum_b P_b X P_b \right\}.$$

The associated pinching map is

$$\mathrm{Pinch}_{\mathcal{A}}(X) = \sum_b P_b X P_b.$$

This map is unital, positive, trace-preserving, idempotent, and, in the complexified setting, completely positive.

For an arbitrary matrix X, let

$$s_1(X) \geq s_2(X) \geq \cdots \geq 0$$

denote its singular values, and define the Ky Fan r-norm by

$$\|X\|_{(r)} = \sum_{i=1}^{r} s_i(X), \qquad \|X\|_{(0)} = 0.$$

If $X \succeq 0$ this coincides with the eigenvalue form

$$\|X\|_{(r)} = \sum_{i=1}^{r} \lambda_i^{\downarrow}(X),$$

so the two conventions agree on G and $\mathrm{Pinch}_{\mathcal{A}} G$. The singular-value form is the one used for the indefinite difference $G - \mathrm{Pinch}_{\mathcal{A}} G$ in Corollary 10.9, where it is a genuine norm and in particular satisfies the triangle inequality.

Remark 10.1 (A More Partition-Faithful Linear Safe Relaxation)**.** If one wants a linear relaxation closer to safe partitions, require the safe linear projection to contain the safety-block label subspace. Let

$$g_b = P_b \sqrt{\pi}$$

be the weighted indicator of safety block B_b, and let

$$\mathcal{S}_{\mathcal{A}} = \mathrm{span}\{g_b : b = 1, \dots, r\}.$$

A partition-faithful safe linear relaxation is

$$\mathrm{Safe}_{\mathrm{lin}}^{\mathrm{part}}(M) = \frac{1}{2} \max_{\substack{P = P^\top = P^2,\ P \in \mathcal{A} \\ \mathrm{rank}\, P \leq M,\ \mathcal{S}_{\mathcal{A}} \subseteq \mathrm{range}\, P}} \mathrm{tr}(PG),$$

when the feasible set is nonempty, in particular when M is at least the number of safety blocks. If the feasible set is empty, the constrained maximum is infeasible.

This constrained PCA problem respects the fact that a safe partition must at least distinguish safety blocks. It does not reduce to the simple pinching formula $\|\mathrm{Pinch}_{\mathcal{A}} G\|_{(M-1)}$. The pinching law of this section is the block-local surrogate, not this partition-faithful relaxation.

Remark 10.2 (Compatibility with Causal States)**.** If the retention variables are causal states, the safety blocks should be unions of causal states, i.e. the causal-state equivalence should refine the safety equivalence. In the stationary formulation, the safety blocks should be compatible with lumpable quotients of $\mathcal{S}^+$.

If instead safety splits causal states, the construction below may still be applied to a history-indexed Gram matrix, but the resulting quantity is a history-level safety surrogate rather than the causal-state Price of Safety, and the two should not be identified.

10.2 The Linear Second-Order Block-Local Surrogate

Definition 10.3 (Free Linear Retained Information)**.** For an M-state budget, set $r = M-1$. The free linear second-order retained information is

$$\mathrm{Free}_{\mathrm{lin}}(M) = \frac{1}{2} \left\| G \right\|_{(M-1)} .$$

Definition 10.4 (Block-Local Safe Linear Retained Information)**.** A block-local linear summary of dimension r is an orthogonal projection P of rank at most r lying in the block algebra $\mathcal{A}$. Define

$$\mathrm{Safe}^{\mathrm{loc}}_{\mathrm{lin}}(M) = \frac{1}{2} \max_{\substack{P=P^\top=P^2 \\ \mathrm{rank}\, P \le M-1 \\ P \in \mathcal{A}}} \mathrm{tr}(PG).$$

Lemma 10.5 (Safe Block-Local Maximum as a Ky Fan Norm)**.** *For $r = M-1$,*

$$\max_{\substack{P=P^\top=P^2 \\ \mathrm{rank}\, P \le r \\ P \in \mathcal{A}}} \mathrm{tr}(PG) = \left\| \mathrm{Pinch}_{\mathcal{A}}\, G \right\|_{(r)} .$$

Consequently,

$$\mathrm{Safe}^{\mathrm{loc}}_{\mathrm{lin}}(M) = \frac{1}{2} \left\| \mathrm{Pinch}_{\mathcal{A}}\, G \right\|_{(M-1)} .$$

Proof. If $P \in \mathcal{A}$, then $P = \sum_b P_b P P_b$, and therefore

$$\mathrm{tr}(PG) = \mathrm{tr}(P\, \mathrm{Pinch}_{\mathcal{A}}\, G).$$

Since $\mathrm{Pinch}_{\mathcal{A}}\, G$ is block diagonal, its spectral projections can be chosen block diagonal, i.e. inside $\mathcal{A}$. Therefore the maximum over block-diagonal projections of rank at most r equals the unrestricted Ky Fan maximum for $\mathrm{Pinch}_{\mathcal{A}}\, G$, namely $\left\| \mathrm{Pinch}_{\mathcal{A}}\, G \right\|_{(r)}$. □

Remark 10.6 (What the Surrogate Compares)**.** The surrogate of Definition 10.7 compares unconstrained linear summaries with *block-local* linear summaries. This is not the safe right-congruence problem. A safe right congruence is a discrete partition whose classes lie inside safety blocks; such a partition may distinguish whole safety blocks from one another, and the corresponding centered contrast vectors are then not block-local. The block-local constraint is therefore strictly more restrictive than safety, and the surrogate is a different optimization object, related to the discrete problem only through the relaxation gaps of Proposition 10.11.

Definition 10.7 (Linear Block-Local Price of Safety Surrogate)**.** The linear surrogate is

$$\mathrm{PoS}^{\mathrm{loc}}_{\mathrm{lin}}(M) = \mathrm{Free}_{\mathrm{lin}}(M) - \mathrm{Safe}^{\mathrm{loc}}_{\mathrm{lin}}(M) = \frac{1}{2}\,\|G\|_{(M-1)} - \frac{1}{2}\,\|\mathrm{Pinch}_{\mathcal{A}}\,G\|_{(M-1)}\,.$$

10.3 Pinching Law, Majorization, and Coupling

Theorem 10.8 (Pinching Majorization and Surrogate Nonnegativity)**.** *For positive semidefinite G,*

$$\lambda(\mathrm{Pinch}_{\mathcal{A}}\,G) \prec \lambda(G).$$

Hence for every $r \geq 0$,

$$\|\mathrm{Pinch}_{\mathcal{A}}\,G\|_{(r)} \leq \|G\|_{(r)}\,,$$

and therefore

$$\mathrm{PoS}^{\mathrm{loc}}_{\mathrm{lin}}(M) \geq 0.$$

Proof. The pinching map is an average over block sign or block phase unitaries. In the complex case,

$$\mathrm{Pinch}_{\mathcal{A}}(X) = \int_{[0,2\pi]^q} U_\theta X U_\theta^*\, d\theta, \qquad U_\theta = \sum_{b=1}^{q} e^{i\theta_b} P_b,$$

with $d\theta$ normalized Haar measure on the torus; in the real case one averages instead over the 2^q block sign flips $U_\varepsilon = \sum_b \varepsilon_b P_b$, $\varepsilon_b \in \{\pm 1\}$. In either case the off-block entries average to zero and the diagonal blocks are preserved, which is the asserted identity for $\mathrm{Pinch}_{\mathcal{A}}$. Ky Fan norms are unitarily invariant and convex, so

$$\|\mathrm{Pinch}_{\mathcal{A}}\,G\|_{(r)} \leq \int \|U_\theta G U_\theta^*\|_{(r)}\, d\theta = \|G\|_{(r)}\,.$$

This is majorization of the pinched spectrum by the original spectrum. □

Corollary 10.9 (Coupling Upper Bound)**.** *For every $M \geq 1$,*

$$\mathrm{PoS}^{\mathrm{loc}}_{\mathrm{lin}}(M) \leq \frac{1}{2}\,\|G - \mathrm{Pinch}_{\mathcal{A}}\,G\|_{(M-1)}\,.$$

Proof. By the Ky Fan triangle inequality,

$$\|G\|_{(r)} \leq \|\mathrm{Pinch}_{\mathcal{A}}\,G\|_{(r)} + \|G - \mathrm{Pinch}_{\mathcal{A}}\,G\|_{(r)}\,.$$

Rearranging and multiplying by $1/2$ gives the claim. □

Remark 10.10 (What the Coupling Bound Measures)**.** The quantity $\frac{1}{2}\,\|G - \mathrm{Pinch}_{\mathcal{A}}\,G\|_{(M-1)}$ is the Ky Fan size of the cross-block coupling removed by safety pinching, and it bounds the linear surrogate from above. The surrogate may be strictly smaller: the bound is tight only when the top $(M-1)$-dimensional eigenspaces of G and $\mathrm{Pinch}_{\mathcal{A}}\,G$ are aligned, and misalignment strictly decreases $\mathrm{PoS}^{\mathrm{loc}}_{\mathrm{lin}}(M)$ relative to the coupling term. The quantity is therefore a design diagnostic — it identifies which cross-block interactions safety destroys — rather than an evaluation of the surrogate.

Proposition 10.11 (Relaxation-Gap Decomposition of the Price of Safety)**.** *Let* $\mathrm{Free}_{\mathrm{quad}}(M)$ *and* $\mathrm{Safe}_{\mathrm{quad}}(M)$ *denote the free and safe optima of the discrete right-congruence problem at budget* M*, safety being understood in the sense of Definition 10.17 and* $M \geq r$ *so that both are finite (Proposition 10.18), and let* $\mathrm{Free}_{\mathrm{lin}}(M)$ *and* $\mathrm{Safe}^{\mathrm{loc}}_{\mathrm{lin}}(M)$ *denote the corresponding linear block-local surrogates, so that*

$$\mathrm{PoS}_{\mathrm{quad}}(M) = \mathrm{Free}_{\mathrm{quad}}(M) - \mathrm{Safe}_{\mathrm{quad}}(M), \qquad \mathrm{PoS}^{\mathrm{loc}}_{\mathrm{lin}}(M) = \mathrm{Free}_{\mathrm{lin}}(M) - \mathrm{Safe}^{\mathrm{loc}}_{\mathrm{lin}}(M).$$

Define the signed discrepancies

$$\rho_{\mathrm{free}}(M) = \mathrm{Free}_{\mathrm{lin}}(M) - \mathrm{Free}_{\mathrm{quad}}(M), \qquad \rho_{\mathrm{safe}}(M) = \mathrm{Safe}^{\mathrm{loc}}_{\mathrm{lin}}(M) - \mathrm{Safe}_{\mathrm{quad}}(M).$$

Neither is a relaxation gap: the block-local linear safe value is neither an upper nor a lower bound on the discrete safe value, so ρ_{safe} *may take either sign. In the two-state singleton example of Remark 10.21 one has* $\rho_{\mathrm{safe}}(2) = -\frac{1}{4}$*. Then, identically,*

$$\mathrm{PoS}_{\mathrm{quad}}(M) = \mathrm{PoS}^{\mathrm{loc}}_{\mathrm{lin}}(M) + \rho_{\mathrm{safe}}(M) - \rho_{\mathrm{free}}(M).$$

Consequently:

(i) the surrogate is exact, $\mathrm{PoS}_{\mathrm{quad}}(M) = \mathrm{PoS}^{\mathrm{loc}}_{\mathrm{lin}}(M)$*, precisely when* $\rho_{\mathrm{free}}(M) = \rho_{\mathrm{safe}}(M)$*;*

(ii) in general $\left|\mathrm{PoS}_{\mathrm{quad}}(M) - \mathrm{PoS}^{\mathrm{loc}}_{\mathrm{lin}}(M)\right| = \left|\rho_{\mathrm{safe}}(M) - \rho_{\mathrm{free}}(M)\right|$*, so a two-sided bound on the two discrepancies transfers to a bound on the surrogate error. No one-sided bound follows from either discrepancy alone, since both may be negative.*

This is an algebraic identity together with its immediate consequences; it supplies no unconditional comparison between $\mathrm{PoS}_{\mathrm{quad}}$ *and* $\mathrm{PoS}^{\mathrm{loc}}_{\mathrm{lin}}$*.*

Proof. The identity is immediate from the definitions:

$$\begin{aligned}\mathrm{PoS}^{\mathrm{loc}}_{\mathrm{lin}} + \rho_{\mathrm{safe}} - \rho_{\mathrm{free}} &= (\mathrm{Free}_{\mathrm{lin}} - \mathrm{Safe}^{\mathrm{loc}}_{\mathrm{lin}}) + (\mathrm{Safe}^{\mathrm{loc}}_{\mathrm{lin}} - \mathrm{Safe}_{\mathrm{quad}}) \\ &\qquad - (\mathrm{Free}_{\mathrm{lin}} - \mathrm{Free}_{\mathrm{quad}}) = \mathrm{Free}_{\mathrm{quad}} - \mathrm{Safe}_{\mathrm{quad}} = \mathrm{PoS}_{\mathrm{quad}}\,.\end{aligned}$$

Claims (i) and (ii) are immediate from the identity, since $\mathrm{PoS}_{\mathrm{quad}} - \mathrm{PoS}^{\mathrm{loc}}_{\mathrm{lin}} = \rho_{\mathrm{safe}} - \rho_{\mathrm{free}}$. □

Remark 10.12 (What This Reduces the Open Problem To)**.** Proposition 10.11 does not evaluate $\mathrm{PoS}_{\mathrm{quad}}$. It reduces the comparison between the discrete Price of Safety and its linear surrogate to a single signed quantity, $\rho_{\mathrm{safe}} - \rho_{\mathrm{free}}$. Because neither discrepancy has a determined sign, this is a two-sided reduction only, and no unconditional inequality between $\mathrm{PoS}_{\mathrm{quad}}$ and $\mathrm{PoS}^{\mathrm{loc}}_{\mathrm{lin}}$ follows. What is unconditional in this section is $\mathrm{PoS}^{\mathrm{loc}}_{\mathrm{lin}}(M) \geq 0$ by pinching majorization (Theorem 10.8) and the Ky Fan coupling bound (Corollary 10.9). Whether $\rho_{\mathrm{safe}} - \rho_{\mathrm{free}}$ admits a bound in terms of spectral data of G is open.

10.4 Equality Conditions and Free-Lunch Criteria

Lemma 10.13 (Equality in Pinching Majorization)**.** *Let X be Hermitian. If*

$$\lambda(\operatorname{Pinch}_{\mathcal{A}} X) = \lambda(X)$$

as multisets, then

$$X = \operatorname{Pinch}_{\mathcal{A}} X.$$

Proof. Equality of spectra implies equality of traces of squares:

$$\operatorname{tr}(X^2) = \operatorname{tr}((\operatorname{Pinch}_{\mathcal{A}} X)^2).$$

Write $X_{bc} = P_b X P_c$. Then

$$\operatorname{tr}(X^2) = \sum_{b,c} \operatorname{tr}(X_{bc} X_{cb}) = \sum_{b} \|X_{bb}\|_F^2 + \sum_{b \neq c} \|X_{bc}\|_F^2 ,$$

while

$$\operatorname{tr}((\operatorname{Pinch}_{\mathcal{A}} X)^2) = \sum_{b} \|X_{bb}\|_F^2 .$$

Equality forces $\|X_{bc}\|_F = 0$ for all $b \neq c$, hence

$$X = \sum_{b} X_{bb} = \operatorname{Pinch}_{\mathcal{A}} X.$$

□

Theorem 10.14 (Free Lunch for All Budgets)**.** *The following are equivalent:*

(i) $\operatorname{PoS}^{\mathrm{loc}}_{\mathrm{lin}}(M) = 0$ *for every* $M \geq 1$*;*

(ii) $\|G\|_{(r)} = \|\operatorname{Pinch}_{\mathcal{A}} G\|_{(r)}$ *for every* $r \geq 0$*;*

(iii) $\lambda(G) = \lambda(\operatorname{Pinch}_{\mathcal{A}} G)$*;*

(iv) $G = \operatorname{Pinch}_{\mathcal{A}} G$*, i.e.* G *is block diagonal with respect to the safety partition.*

Proof. The equivalence of (i) and (ii) is the definition of $\operatorname{PoS}^{\mathrm{loc}}_{\mathrm{lin}}$. Equality of all Ky Fan norms is equivalent to equality of spectra, giving (ii) ⇔ (iii). Lemma 10.13 gives (iii) ⇒ (iv), and (iv) trivially implies (i). □

Theorem 10.15 (Fixed-Budget Free Lunch)**.** *Fix $M \geq 1$ and set $r = M - 1$. Then*

$$\operatorname{PoS}^{\mathrm{loc}}_{\mathrm{lin}}(M) = 0$$

if and only if

$$\|G\|_{(r)} = \|\operatorname{Pinch}_{\mathcal{A}} G\|_{(r)} .$$

Equivalently, there exists a block-local projection $P \in \mathcal{A}$ of rank at most r such that

$$\operatorname{tr}(PG) = \|G\|_{(r)} .$$

Thus a free optimal linear summary can be chosen block-local. If the top eigenspaces have multiplicities, this statement is existential: not every free optimal summary need be block-local.

Proof. If $\mathrm{PoS}^{\mathrm{loc}}_{\mathrm{lin}}(M) = 0$, then the Ky Fan norms are equal. Let P be a top spectral projection of $\mathrm{Pinch}_{\mathcal{A}}\, G$ of rank r. Since $\mathrm{Pinch}_{\mathcal{A}}\, G$ is block diagonal, P may be chosen in $\mathcal{A}$. Then

$$\mathrm{tr}(PG) = \mathrm{tr}(P\, \mathrm{Pinch}_{\mathcal{A}}\, G) = \left\|\mathrm{Pinch}_{\mathcal{A}}\, G\right\|_{(r)} = \|G\|_{(r)}\,,$$

so P is a free optimizer. The converse is immediate: if such a block-local free optimizer exists, then

$$\left\|\mathrm{Pinch}_{\mathcal{A}}\, G\right\|_{(r)} \geq \mathrm{tr}(P\, \mathrm{Pinch}_{\mathcal{A}}\, G) = \mathrm{tr}(PG) = \|G\|_{(r)}\,,$$

and the reverse inequality is Theorem 10.8. $\square$

Remark 10.16 (Block Diagonality Is Sufficient but Not Necessary at Fixed M)**.** For a fixed budget M, safety can be free even if G has cross-block coupling, provided that coupling lies outside the top $M-1$ spectral directions relevant to the free optimum. Full block diagonality is the necessary and sufficient condition for safety to be free simultaneously for all budgets.

10.5 Relation to the Discrete Price of Safety

The discrete Price of Safety is an optimization over right congruences. In the quadratic surrogate, a partition $\mathcal{C} = \{C_1, \ldots, C_K\}$ of states has cluster weights

$$\Pi_k = \sum_{s \in C_k} \pi_s$$

and centroids

$$c_k = \Pi_k^{-1} \sum_{s \in C_k} \pi_s y_s.$$

Its second-order retained information is

$$I_{\mathrm{quad}}(\mathcal{C}; Z) = \frac{1}{2} \sum_{k=1}^{K} \Pi_k \left\|c_k - \bar{y}\right\|^2.$$

Define the quadratic discrete free and safe values by

$$\mathrm{Free}_{\mathrm{quad}}(M) = \max_{\substack{\text{free right congruences}\\ \text{index} \leq M}} I_{\mathrm{quad}}(\mathcal{C}; Z),$$

and

$$\mathrm{Safe}_{\mathrm{quad}}(M) = \max_{\substack{\text{safe right congruences}\\ \text{index} \leq M}} I_{\mathrm{quad}}(\mathcal{C}; Z).$$

The quadratic Price of Safety is

$$\mathrm{PoS}_{\mathrm{quad}}(M) = \mathrm{Free}_{\mathrm{quad}}(M) - \mathrm{Safe}_{\mathrm{quad}}(M).$$

The safe feasible set can be empty, so the second maximum requires a convention. The notion of safety used here is the one described in Remark 10.6, stated for reference.

Definition 10.17 (Safe Right Congruence)**.** Let $\mathcal{B} = \{B_1, \dots, B_r\}$ be a safety partition of $\mathcal{S}^+$, with r the number of blocks of positive stationary mass. A right congruence $\sim$ on $\mathcal{S}^+$ is **safe** if every $\sim$-class is contained in some B_b; equivalently, if $\sim$ refines the safety equivalence $s \sim_{\mathcal{B}} s' \iff s, s'$ lie in a common B_b. When no safe right congruence of index at most M exists we set $\mathrm{Safe}_{\mathrm{quad}}(M) = -\infty$, so that $\mathrm{PoS}_{\mathrm{quad}}(M) = +\infty$.

Convention. Throughout this subsection the free and safe optima are read over *partitions* of $\mathcal{S}^+$: the transition-compatibility carried by the term "right congruence" in the general theory is deliberately relaxed here, exactly as Remark 10.6 reads it, so that the discrete quadratic problem and the linear surrogate of Proposition 10.11 are posed on comparable feasible classes. Under the stricter, transition-compatible reading, the existence clause of Proposition 10.18 holds with the safety partition $\mathcal{B}$ itself as witness only when $\mathcal{B}$ is a right congruence; in all cases the singleton partition witnesses existence for $M \geq |\mathcal{S}^+|$.

Proposition 10.18 (Feasibility and Range of the Discrete Price of Safety)**.** *With Definition 10.17:*

(i) a safe right congruence of index at most M exists if and only if $M \geq r$;

(ii) $\mathrm{PoS}_{\mathrm{quad}}(M) = +\infty$ *exactly when $M < r$; in particular, for $r \geq 2$ the one-class congruence is not safe and* $\mathrm{PoS}_{\mathrm{quad}}(1) = +\infty$*;*

(iii) for $M \geq r$ one has $0 \leq \mathrm{PoS}_{\mathrm{quad}}(M) \leq \mathrm{Free}_{\mathrm{quad}}(M)$*, and* $\mathrm{PoS}_{\mathrm{quad}}$ *is finite.*

Proof. (i) If $M \geq r$, the safety partition $\mathcal{B}$ itself is safe, each class being a block, has index $r \leq M$, and is admissible under the partition convention of Definition 10.17. Conversely, a safe congruence has every class inside a single B_b, so each of the r positive-mass blocks contains at least one class, whence the index is at least r.

(ii) Immediate from (i) and the convention of Definition 10.17. For $r \geq 2$ the unique congruence of index 1 has the single class $\mathcal{S}^+$, which is contained in some B_b only if $r = 1$.

(iii) For $M \geq r$ the safe feasible set is nonempty and is contained in the free feasible set, so $\mathrm{Safe}_{\mathrm{quad}}(M) \leq \mathrm{Free}_{\mathrm{quad}}(M)$ and $\mathrm{PoS}_{\mathrm{quad}}(M) \geq 0$. Both values are finite, being maxima of I_{quad} over finitely many partitions, and $I_{\mathrm{quad}} \geq 0$ gives the upper bound. □

Remark 10.19 (The Convention Is Not Inherited from the Surrogate)**.** Definition 10.17 is a choice about the discrete problem and is not forced by the linear surrogate, which imposes the strictly stronger block-local constraint $P \in \mathcal{A}$. The two differ already on four states with safety blocks $B_1 = \{s_1, s_2\}$, $B_2 = \{s_3, s_4\}$: the congruence $\{\{s_1, s_2\}, \{s_3, s_4\}\}$ is safe, every class lying inside a block, yet its centred block contrasts separate B_1 from B_2 and are therefore not block-diagonal. This is the discrepancy recorded in Remark 10.6, and it is why the surrogate and the discrete problem are related only through the signed discrepancies of Proposition 10.11 rather than by inheritance of feasible sets. The partition-faithful linear relaxation $\mathrm{Safe}_{\mathrm{lin}}^{\mathrm{part}}$ adopts the matching feasibility convention, being declared infeasible on the same range $M < r$.

Proposition 10.20 (Free Discrete Objective Is Bounded by the Free Linear Surrogate)**.** *For every M,*

$$\mathrm{Free}_{\mathrm{quad}}(M) \le \frac{1}{2} \|G\|_{(M-1)} .$$

Proof. Let $\mathcal{C} = \{C_1, \ldots, C_K\}$ be a partition with $K \le M$, with cell weights Π_k and centroids c_k, and define the between-cluster covariance

$$B_{\mathcal{C}} = \sum_{k=1}^{K} \Pi_k (c_k - \bar{y})(c_k - \bar{y})^\top,$$

so that

$$I_{\mathrm{quad}}(\mathcal{C}; Z) = \tfrac{1}{2} \operatorname{tr}(B_{\mathcal{C}}).$$

The ANOVA decomposition gives $\boldsymbol{\Sigma}_\pi = W_{\mathcal{C}} + B_{\mathcal{C}}$ with $W_{\mathcal{C}} \succeq 0$ the within-cluster covariance, whence

$$0 \preceq B_{\mathcal{C}} \preceq \boldsymbol{\Sigma}_\pi.$$

Because $\sum_k \Pi_k (c_k - \bar{y}) = 0$, the centred centroids satisfy one linear relation, so $\operatorname{rank}(B_{\mathcal{C}}) \le K - 1 \le M - 1$. Ky Fan's maximum principle [17] therefore gives

$$\operatorname{tr}(B_{\mathcal{C}}) \;\le\; \|\boldsymbol{\Sigma}_\pi\|_{(M-1)} .$$

The nonzero spectra of $\boldsymbol{\Sigma}_\pi$ and G coincide, since G is the Gram matrix of the same weighted centred vectors, so $\|\boldsymbol{\Sigma}_\pi\|_{(M-1)} = \|G\|_{(M-1)}$ and

$$I_{\mathrm{quad}}(\mathcal{C}; Z) \;\le\; \tfrac{1}{2} \|G\|_{(M-1)} .$$

Taking the maximum over free right congruences of index at most M gives the claim. □

Remark 10.21 (Safe Discrete Objective Is Not Captured by the Block-Local Surrogate)**.** The block-local safe linear value

$$\frac{1}{2} \|\mathrm{Pinch}_{\mathcal{A}}\, G\|_{(M-1)}$$

is not generally equal to, an upper bound on, or a lower bound on $\mathrm{Safe}_{\mathrm{quad}}(M)$. Safe discrete partitions may distinguish whole safety blocks, and the corresponding contrast vectors need not be block-local. Conversely, block-local linear summaries need not correspond to any hard partition or right congruence.

For example, take two states with weights $1/2, 1/2$, scalar features $y_1 = 1$, $y_2 = -1$, and singleton safety blocks. Then

$$G = \begin{pmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{pmatrix}, \qquad \mathrm{Pinch}_{\mathcal{A}}\, G = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix}.$$

For $M = 2$, the free linear value is $1/2$, while the block-local safe linear value is $1/4$, so

$$\mathrm{PoS}^{\mathrm{loc}}_{\mathrm{lin}}(2) = 1/4.$$

But the discrete quadratic free and safe partitions both separate the two states, so

$$\mathrm{Free}_{\mathrm{quad}}(2) = \mathrm{Safe}_{\mathrm{quad}}(2) = 1/2,$$

and

$$\mathrm{PoS}_{\mathrm{quad}}(2) = 0.$$

Thus the block-local surrogate is not an exact formula for the discrete Price of Safety.

Remark 10.22 (Relaxation-Gap Identity)**.** Suppose one defines some linear safe relaxation $\mathrm{Safe}_{\mathrm{lin}}(M)$ and knows relaxation gaps

$$\rho_{\mathrm{free}}(M) = \frac{1}{2}\,\|G\|_{(M-1)} - \mathrm{Free}_{\mathrm{quad}}(M),$$

and

$$\rho_{\mathrm{safe}}(M) = \mathrm{Safe}_{\mathrm{lin}}(M) - \mathrm{Safe}_{\mathrm{quad}}(M),$$

with the latter well-defined as a genuine relaxation gap. Then

$$\mathrm{PoS}_{\mathrm{quad}}(M) = \left[\frac{1}{2}\,\|G\|_{(M-1)} - \mathrm{Safe}_{\mathrm{lin}}(M)\right] + \rho_{\mathrm{safe}}(M) - \rho_{\mathrm{free}}(M).$$

Thus the discrete Price of Safety equals the linear surrogate plus the difference of safe and free relaxation gaps. Without bounds on those gaps, the surrogate does not control the discrete Price of Safety.

10.6 Scope and Conditions

- **Exact domain of the theorem.** The identities and inequalities above are exact for the linear block-local second-order surrogate. They are not exact statements about the discrete right-congruence Price of Safety.
- **Full KL / information-bottleneck objective.** For the full KL objective, the surrogate is a second-order local model.
- **Right-congruence constraint.** The linear surrogate drops the right-congruence and lumpability constraints.
- **State-indexed covariance is canonical.** The state-indexed Gram matrix G is the object for safety pinching. The feature covariance $\mathbf{\Sigma}_\pi$ has the same nonzero spectrum and may be used for the free term, but pinching $\mathbf{\Sigma}_\pi$ by feature-coordinate blocks is not invariant under orthogonal reparameterization and is not canonically induced by a safety partition of states.

11 The Aggregation Exponent Vertex Correspondence

This section states the relation between Rényi order, Schatten exponent, and aggregation exponent. The claim is a vertex correspondence, not a representation theorem identifying the manuscript's costs with raw Rényi vertices.

11.1 Layers and Scope

There are three distinct exponent-like layers:

(1) **Rényi order** α, which indexes a family of divergences;

(2) **Schatten exponent** p, which indexes operator-norm tails used in spectral converses;

(3) **Aggregation exponent or vertex label** e, which informally distinguishes counting/support, averaging/entropy, and worst-case/supremum behavior.

The sandwiched Rényi identity shows that a Rényi divergence of order α can be written as a logarithmic transform of a Schatten-α norm (a quasi-norm when $0 < \alpha < 1$) of an auxiliary α-dependent operator. That is a genuine algebraic fact. But it does not by itself prove that the Boolean commitment cost is D_0, nor that the operator-norm grounding cost is D_∞, nor that the converse exponent p is derived from the Rényi order α.

The correspondence is as follows:

- retention is exactly the $\alpha = 1$ Rényi/KL vertex;
- commitment is recovered exactly after symmetrizing the $\alpha = 0$ support test and applying a Boolean valuation;
- grounding is organized by the $\alpha = \infty$ worst-case/supremum vertex, but the exact grounding cost is the C^*- or operator-norm distance, not the max-divergence.

11.2 Rényi Divergences and the Schatten Representation

Work first on finite alphabets or finite-dimensional Hilbert spaces. Domain conditions are stated explicitly.

Definition 11.1 (Classical Rényi Divergence)**.** For probability vectors p, q on a finite alphabet and $\alpha \in (0,1) \cup (1,\infty)$,

$$D_\alpha(p\|q) = \frac{1}{\alpha - 1} \log \sum_i p_i^\alpha q_i^{1-\alpha}.$$

The vertices are defined by limits:

$$D_1(p\|q) = D_{\mathrm{KL}}(p\|q),$$

$$D_0(p\|q) = -\log q(\operatorname{supp} p),$$

and

$$D_\infty(p\|q) = \log \max_i \frac{p_i}{q_i},$$

with the usual extended-value conventions.

Definition 11.2 (Sandwiched Rényi Divergence)**.** Let $\rho, \sigma \succeq 0$ with $\operatorname{Tr}\rho = \operatorname{Tr}\sigma = 1$. For $\alpha \in (0,1) \cup (1,\infty)$, define

$$\widetilde{D}_\alpha(\rho\|\sigma) = \frac{1}{\alpha - 1} \log \operatorname{Tr}\left[\left(\sigma^{\frac{1-\alpha}{2\alpha}} \rho \sigma^{\frac{1-\alpha}{2\alpha}}\right)^\alpha\right],$$

where for $\alpha > 1$ one assumes $\operatorname{supp}\rho \subseteq \operatorname{supp}\sigma$ for finiteness, and where generalized inverses or support projections are used when σ is not full rank.

Remark 11.3 (Commuting States)**.** If ρ and σ commute they are simultaneously diagonalizable, the conjugation by $\sigma^{(1-\alpha)/2\alpha}$ acts diagonally, and $\widetilde{D}_\alpha$ reduces to the classical Rényi divergence of the two eigenvalue distributions. The noncommutative content of Definition 11.2 is therefore confined to the noncommuting case, and every classical statement in this section is the commuting specialization.

Proposition 11.4 (Schatten-Norm Representation)**.** *For $\alpha \neq 1$, let*

$$K_\alpha = \sigma^{\frac{1-\alpha}{2\alpha}} \rho \sigma^{\frac{1-\alpha}{2\alpha}} \succeq 0.$$

Then

$$\widetilde{D}_\alpha(\rho\|\sigma) = \frac{\alpha}{\alpha-1} \log \|K_\alpha\|_{\mathcal{S}_\alpha},$$

where

$$\|K_\alpha\|_{\mathcal{S}_\alpha} = (\operatorname{Tr}|K_\alpha|^\alpha)^{1/\alpha}$$

is the Schatten-α norm for $\alpha \geq 1$, and the Schatten-α quasi-norm *for $0 < \alpha < 1$, where the triangle inequality fails and only a quasi-triangle inequality holds. The identity above is an algebraic identity valid in both ranges; no norm property is used.*

Proof. Since $K_\alpha \succeq 0$,

$$\operatorname{Tr} K_\alpha^\alpha = \|K_\alpha\|_{\mathcal{S}_\alpha}^\alpha.$$

Taking logarithms and multiplying by $1/(\alpha-1)$ gives

$$\frac{1}{\alpha-1} \log \operatorname{Tr} K_\alpha^\alpha = \frac{\alpha}{\alpha-1} \log \|K_\alpha\|_{\mathcal{S}_\alpha}.$$

□

Remark 11.5 (Scope of the Schatten-Norm Representation)**.** The sandwiched Rényi divergence is not a norm. It is not homogeneous and not a distance. The operator K_α depends on α, ρ, and σ, and is not the response-operator difference used in the Schatten converse template. The proposition shows that the Rényi order and a Schatten exponent appear in the same algebraic expression. It does not by itself prove that the Rényi order explains the converse exponent.

Proposition 11.6 (Standard Vertex Limits)**.** *The following limits hold under the usual support and positivity conventions.*

(i) As $\alpha \to 1$,

$$\widetilde{D}_\alpha(\rho\|\sigma) \to \operatorname{Tr}\rho(\log\rho - \log\sigma),$$

the Umegaki relative entropy. On commuting states this is $D_{\mathrm{KL}}(p\|q)$.

(ii) As $\alpha \to 0$ on the classical face,

$$D_0(p\|q) = -\log q(\operatorname{supp} p).$$

This tests one-sided support inclusion:

$$D_0(p\|q) = 0 \iff q(\operatorname{supp} p) = 1 \iff \operatorname{supp} q \subseteq \operatorname{supp} p$$

up to null sets.

(iii) As $\alpha \to \infty$,

$$\widetilde{D}_\alpha(\rho\|\sigma) \to D_{\max}(\rho\|\sigma) = \log \left\|\sigma^{-1/2}\rho\sigma^{-1/2}\right\|_\infty,$$

the max-relative divergence. On commuting states this is

$$D_{\max}(p\|q) = \log \max_i \frac{p_i}{q_i}.$$

Proof. For $\alpha \to 1$, write $D_\alpha(p\|q) = \frac{1}{\alpha-1}\log Z(\alpha)$ with $Z(\alpha) = \sum_i p_i^\alpha q_i^{1-\alpha}$. Since $Z(1) = 1$, both numerator and denominator vanish at $\alpha = 1$, and

$$Z'(\alpha) = \sum_i p_i^\alpha q_i^{1-\alpha} \log \frac{p_i}{q_i}, \qquad Z'(1) = \sum_i p_i \log \frac{p_i}{q_i} = D_{\mathrm{KL}}(p\|q).$$

L'Hôpital's rule therefore gives $D_\alpha \to Z'(1)/Z(1) = D_{\mathrm{KL}}(p\|q)$. The corresponding statement for the sandwiched quantum family is Umegaki's relative entropy limit, for which we refer to the literature rather than reproduce the argument.

For $\alpha \to \infty$, the operator $K_\alpha = \sigma^{\frac{1-\alpha}{2\alpha}}\rho\,\sigma^{\frac{1-\alpha}{2\alpha}}$ itself depends on α, so the monotone-convergence argument for a fixed singular-value sequence does not apply to it directly; the limit holds by a two-step continuity argument. First, on the support of σ — where σ is positive definite, and $\operatorname{supp}\rho \subseteq \operatorname{supp}\sigma$ is assumed for finite divergence — the map $\alpha \mapsto K_\alpha$ is continuous, by continuity of the matrix power in the functional calculus, so $K_\alpha \to K_\infty = \sigma^{-1/2}\rho\,\sigma^{-1/2}$ and $\|K_\alpha - K_\infty\|_{\mathcal{S}_1} \to 0$; for $\alpha \geq 1$ the Schatten norms obey the triangle inequality and decrease in the exponent, so

$$\big|\|K_\alpha\|_{\mathcal{S}_\alpha} - \|K_\infty\|_{\mathcal{S}_\alpha}\big| \leq \|K_\alpha - K_\infty\|_{\mathcal{S}_\alpha} \leq \|K_\alpha - K_\infty\|_{\mathcal{S}_1} \to 0.$$

Second, for the *fixed* positive operator K_∞, monotone convergence of ℓ^α norms of its singular-value sequence gives $\|K_\infty\|_{\mathcal{S}_\alpha} \to \|K_\infty\|_\infty$. Combining the two steps gives $\|K_\alpha\|_{\mathcal{S}_\alpha} \to \left\|\sigma^{-1/2}\rho\sigma^{-1/2}\right\|_\infty$, and $\alpha/(\alpha-1) \to 1$, whence $D_{\max}$; this is the standard sandwiched-Rényi $\alpha \to \infty$ limit [20, 21].

For $\alpha \to 0$ in the classical case, $p_i^\alpha \to \mathbf{1}\{p_i > 0\}$, so $Z(\alpha) \to q(\operatorname{supp} p)$ and $\frac{1}{\alpha-1}\log Z(\alpha) \to -\log q(\operatorname{supp} p) = D_0(p\|q)$.

The operator-face limit as $\alpha \to 0$ requires separate treatment because the relevant supports need not be nested; it is not used in what follows. □

11.3 Regime Vertices

11.3.1 Retention: Exact $\alpha = 1$ Vertex

Theorem 11.7 (Retention Is the $\alpha = 1$ Vertex)**.** *The retention cost is KL divergence between predictive distributions. Hence on the classical measure face it is exactly*

$$\textit{retention cost} = D_1(p\|q) = D_{\mathrm{KL}}(p\|q).$$

On the noncommutative face, under the usual support conditions, it is the Umegaki relative entropy

$$\operatorname{Tr} \rho(\log \rho - \log \sigma).$$

Remark 11.8 (Cost versus Converse)**.** The retention cost is KL / relative entropy. The retention spectral converse in the Schatten template uses a trace-norm / Schatten-1 tail:

$$\Delta_{\mathrm{ret}}^{\mathrm{quad}}(M) \geq \frac{1}{2} \sum_{i \geq M} \lambda_i(\mathbf{\Sigma}_\pi) = \frac{1}{2} \Phi_{M-1}^{(1)}(\mathbf{\Sigma}_\pi),$$

with the effective rank $M - 1$. The trace norm appears in the converse, not as the retention cost itself.

11.3.2 Commitment: Booleanized Symmetrized $\alpha = 0$ Vertex

The raw order-0 Rényi divergence is not the commitment cost. It is one-sided and not Boolean-valued. The construction is a symmetrized support test followed by a Boolean valuation.

Definition 11.9 (Symmetrized Support Mismatch)**.** For probability vectors p, q, define

$$S_0(p, q) = D_0(p\|q) + D_0(q\|p),$$

with extended value $+\infty$ when either support overlap has zero mass. Define the Boolean valuation

$$v(t) = \begin{cases} 0, & t = 0, \\ 1, & t > 0, \end{cases}$$

or, in extended convention, $v(t) = \infty$ for $t > 0$. The Booleanized support cost is

$$B_0(p, q) = v(S_0(p, q)).$$

Proposition 11.10 (Boolean Support Equality)**.** *For probability vectors p, q,*

$$B_0(p, q) = 0 \iff \operatorname{supp} p = \operatorname{supp} q$$

up to null sets.

Proof. Since

$$D_0(p\|q) = 0 \iff q(\operatorname{supp} p) = 1 \iff \operatorname{supp} q \subseteq \operatorname{supp} p,$$

we have

$$S_0(p,q) = 0 \iff \operatorname{supp} q \subseteq \operatorname{supp} p \text{ and } \operatorname{supp} p \subseteq \operatorname{supp} q \iff \operatorname{supp} p = \operatorname{supp} q.$$

Applying v preserves the zero set. □

Theorem 11.11 (Commitment as Booleanized $\alpha = 0$ Vertex)**.** *Let the commitment residual rows be Boolean functions*

$$r_x : \Sigma^* \to \{0,1\}.$$

Define their supports by

$$\operatorname{supp} r_x = \{w \in \Sigma^* : r_x(w) = 1\}.$$

The Booleanized support cost

$$B_0(r_x, r_y) = v(\mathbf{1}_{\{\operatorname{supp} r_x \neq \operatorname{supp} r_y\}})$$

satisfies

$$B_0(r_x, r_y) = 0 \iff r_x = r_y.$$

*Equivalently, when Boolean rows are represented as normalized indicator vectors on a finite continuation set, this is the Booleanization of the symmetrized order-*0 *Rényi support test.*

Consequently, if κ_{MN} is the Myhill–Nerode index, the commitment gap satisfies

$$\Delta_{\mathrm{com}}(M) = 0 \iff M \geq \kappa_{\mathrm{MN}},$$

and $\Delta_{\mathrm{com}}(M) = 1$ otherwise under the 0/1 *convention, or ∞ under the extended Boolean convention.*

Proof. For Boolean rows, the support is exactly the set of continuations accepted by the residual language. Equality of Boolean functions is equality of their supports. Proposition 11.10 gives the support-equality test in the probabilistic encoding, and the direct Boolean definition gives the same zero set without normalization. The Myhill–Nerode theorem identifies the number of distinct residuals with the minimal deterministic automaton size. Hence a budget-M machine realizes the language exactly iff $M \geq \kappa_{\mathrm{MN}}$. □

Remark 11.12 (Scope of the Boolean Vertex)**.** The scope restrictions of Remark 6.17 apply verbatim at this vertex: raw one-sided D_0 supports inclusion rather than equality and is not Boolean-valued, so the construction uses symmetrization and Boolean valuation, and the operative object is Boolean residual equality, that is the Myhill–Nerode index, not ordinary real or complex rank.

11.3.3 Grounding: Supremum Vertex and Operator-Norm Distance

The $\alpha = \infty$ Rényi vertex is max-divergence. The grounding cost is an operator-norm / C^*-norm distance. These are not the same functional.

Definition 11.13 (Symmetrized Max-Divergence)**.** An anchor for the exponent layer, retained so that Subsection 11.3.3 can be read in isolation: the max-divergence $D_{\max}$ and its symmetrization $D_{\max}^{\mathrm{sym}}$ are exactly the objects of Definition 7.20 on finite-dimensional positive definite states, with the same conventions there, and no independent content is asserted at this vertex. All properties used below are those of Definition 7.20 and Proposition 7.21.

The local equivalence with operator-norm distance was proved in Proposition 7.21.

Theorem 11.14 (Grounding Vertex)**.** *This is Theorem 7.22, restated at the exponent vertex: the grounding cost is the C^*-norm / Schatten-∞ distance $\|a - b\|_\infty$ between residual operators or kernels, the $\alpha = \infty$ Rényi vertex is the max-divergence $D_{\max}$ or its symmetrization $D_{\max}^{\mathrm{sym}}$, and the two are distinct functionals, related locally under the bounded-overlap hypotheses of Proposition 7.21. The canonical statement, its diagonal-state comparison, and its scope restrictions are those of Theorem 7.22, which is the single maintained copy.*

Remark 11.15 (Two Possible Grounding Interpretations)**.** One may either:

(1) keep the grounding regime as the operator-norm distance regime, and regard $D_{\max}$ as the divergence-face analogue of its supremum aggregation; or

(2) define a separate max-divergence grounding regime, distinct from the C^*-norm grounding regime.

The manuscript uses interpretation (1). The exact spectral converse is the unrestricted linear finite-rank Hankel relaxation gap of Theorem 7.3, the Hankel-structured gap satisfying the lower bound recorded there, with equality only under additional structural hypotheses.

11.4 Data Processing and Monotonicity

Proposition 11.16 (Data Processing, Scoped)**.** *The following data-processing statements are classical and are quoted without proof; see [1] for the classical face and [22] for the divergence-theoretic background.*

(a) ***Classical measure face.*** *For finite alphabets and stochastic maps Φ, classical Rényi divergence satisfies*
$$D_\alpha(\Phi p \| \Phi q) \le D_\alpha(p \| q)$$
in the standard ranges, in particular for $\alpha = 1$ and $\alpha = \infty$, and for $\alpha = 0$ under the usual support conventions.

(b) ***Sandwiched operator face.*** *For completely positive trace-preserving maps Φ, the sandwiched Rényi divergence satisfies*
$$\widetilde{D}_\alpha(\Phi\rho \| \Phi\sigma) \le \widetilde{D}_\alpha(\rho \| \sigma)$$
for $\alpha \in [1/2, \infty]$, under the usual support/domain conditions.

Remark 11.17 (Monotonicity Meta-Theorem Is Not Exactly DPI)**.** The monotonicity meta-theorem says that the gap is nonincreasing in the state budget M. The primary reason is nested feasible sets:

$$\{\text{machines of index } \leq M\} \subseteq \{\text{machines of index } \leq M+1\}.$$

Therefore the optimized deficit cannot increase. In the retention regime there is an additional information-processing interpretation: if a coarser machine label is a deterministic function of a finer one, then

$$I(K_{\text{fine}}; Z) \geq I(K_{\text{coarse}}; Z).$$

Thus retention monotonicity can be viewed as DPI for mutual information / KL. But the full meta-theorem across all regimes is not literally the DPI of the Rényi family.

11.5 Relation to the Schatten Converse Exponent

The Schatten template assigns analytic converses to Schatten exponents:

$$\text{grounding: } p = \infty, \qquad \text{retention: } p = 1, \qquad \text{commitment: } p = 0,$$

with the following qualifications:

- grounding is the unrestricted linear finite-rank relaxation, for which

  $$\Delta_{\text{grd}}^{\text{unres}}(M) = \sigma_{M+1}(H_\nu),$$

  while the Hankel-structured finite-rank realization gap satisfies

  $$\Delta_{\text{grd}}^{\text{Hank,str}}(M) \geq \sigma_{M+1}(H_\nu);$$

- retention uses the effective rank $M - 1$;
- commitment is a Boolean-rank / Myhill–Nerode threshold, not an ordinary Schatten-rank limit.

The present section assigns cost vertices as follows:

$$\text{grounding: } \alpha = \infty, \qquad \text{retention: } \alpha = 1, \qquad \text{commitment: } \alpha = 0,$$

again with qualifications: commitment is Booleanized symmetrized D_0, and grounding's exact cost is operator-norm distance rather than max-divergence.

Theorem 11.18 (Formal Vertex Consistency, Not Derivation)**.** *The labels α and p agree at the three vertices:*

$$\alpha = p \in \{0, 1, \infty\}.$$

This is a formal consistency between the cost-face vertex correspondence and the Schatten converse correspondence. It is not a derivation of the Schatten converse theorem from the Rényi identity.

Proof. The Rényi cost-face analysis gives the vertex labels $\alpha \in \{0, 1, \infty\}$. The Schatten template gives the converse-face labels $p \in \{0, 1, \infty\}$. The labels match vertex by vertex. However, the sandwiched Rényi identity uses an auxiliary α-dependent operator K_α, whereas the Mirsky converse uses a regime-specific response operator difference $A_\delta - A_{\hat{\delta}}$. No implication from one to the other is established without additional domination or achievability hypotheses. □

Remark 11.19 (What the Matching Labels Do Explain)**.** Although the correspondence is not a derivation, it is not a coincidence either, and Definition 8.9 already isolates the reason. A Schatten-p response representation consists of two independent data: a *rank* condition, clause (a), asserting $\operatorname{rank}(A_{\hat{\delta}}) \leq r_{\mathbb{T}}(M)$, and a *norm* condition, clause (b), asserting that the task cost dominates $\left\|A_\delta - A_{\hat{\delta}}\right\|_{\mathcal{S}_p}$. Theorem 8.11 shows that once both are supplied the value is forced, by Eckart–Young–Mirsky, to be the Schatten-p tail $\Phi^{(p)}_{r_{\mathbb{T}}(M)}(A_\delta)$.

The three converses therefore have parallel form because they share the optimization ingredient (a) and differ only in the norm selected by (b): grounding's cost *is* an operator norm, giving $p = \infty$; retention's cost dominates a trace-norm tail of the covariance residual, giving $p = 1$; and commitment's threshold is Boolean rank, the $p \to 0$ vertex. The equality $\alpha = p$ records that the Rényi order selecting the divergence and the Schatten exponent selecting the tail agree at the three vertices. What remains unavailable, and what clause (e) of Meta-Theorem 4.18 and Theorem 8.22 record, is any transfer between regimes: the operators H_ν, $\boldsymbol{\Sigma}_\pi$, N_Λ, the rank budgets M, $M-1$, M and the moduli are unrelated, so the shared form yields no shared value.

11.6 Scope and Conditions

- **No cross-regime gap inequality.** The vertex correspondence unifies functional organization at isolated orders. It yields no inequality relating the numerical grounding, retention, and commitment gaps. The Independence Theorem remains intact.
- **Scope of the Rényi family.** The operational regimes identified here occupy the vertices $\alpha \in \{0, 1, \infty\}$. Intermediate orders $\alpha \in (0, \infty) \setminus \{1\}$ serve as interpolation inequalities and diagnostics; promoting one to a finite-state approximation regime requires a task theory and a domination bridge supplied for that order.
- **Domain regularity.** For $\alpha > 1$, sandwiched Rényi divergence requires support conditions, typically $\operatorname{supp} \rho \subseteq \operatorname{supp} \sigma$, or generalized inverses. The operator-face $\alpha \to 0$ limit is subtle and is not used to define commitment.
- **Inherited conditionality.** The alignment with the Schatten converse exponent is conditional on the Schatten template. If the Schatten correspondence is a conditional template rather than an unconditional theorem, then the present vertex consistency is likewise conditional.
- **Conditional status.** This section is a vertex correspondence. Its validity depends on the hypotheses of the preceding sections.

12 Joint, Comparative, Interaction, and Independence Results

The regime-specific gaps can be composed, compared, and constrained jointly. This section develops the joint zero-error memory theorem, the comparative state-complexity sandwich, the independence theorem, and the interaction between exact commitment and approximate retention.

12.1 Joint Zero-Error Memory

Suppose task specifications $T_1, \dots, T_m$ are posed on the same history space and induce Myhill–Nerode right congruences

$$\sim_1, \dots, \sim_m .$$

A joint zero-error approximant must be exact for every component task.

Define the meet of the congruences by intersection:

$$\bigwedge_{i=1}^{m} \sim_i := \bigcap_{i=1}^{m} \sim_i .$$

This is the finest right congruence refining every $\sim_i$.

Theorem 12.1 (Exact Joint Zero-Error Memory)**.** *Let task specifications $T_1, \dots, T_m$ induce right congruences $\sim_1, \dots, \sim_m$. Then*

$$\Delta_{T_1,\dots,T_m}(M) = 0 \iff M \geq \operatorname{index}\left(\bigwedge_{i=1}^{m} \sim_i\right).$$

Moreover,

$$\operatorname{index}\left(\bigwedge_i \sim_i\right) \geq \max_i \operatorname{index}(\sim_i),$$

and in the worst case equals the product

$$\prod_i \operatorname{index}(\sim_i).$$

Proof. A joint zero-error approximant must assign the same state only to histories that are equivalent for every component task. Therefore its congruence must refine each $\sim_i$, equivalently it must refine the meet $\bigwedge_i \sim_i$. Hence zero error is possible only if

$$M \geq \operatorname{index}\left(\bigwedge_i \sim_i\right).$$

Conversely, quotienting by the meet gives a deterministic finite-state machine that refines every component Myhill–Nerode congruence and therefore realizes every component task exactly.

The inequality

$$\operatorname{index}\left(\bigwedge_i \sim_i\right) \geq \max_i \operatorname{index}(\sim_i)$$

is immediate because the meet refines each $\sim_i$. The map

$$[u]_{\bigwedge_i \sim_i} \mapsto \big([u]_{\sim_1}, \dots, [u]_{\sim_m}\big)$$

is injective, so

$$\operatorname{index}\left(\bigwedge_i \sim_i\right) \leq \prod_i \operatorname{index}(\sim_i).$$

The upper bound is tight when the component congruences are independent, for example independent modular counters with relatively prime moduli. $\square$

12.2 Comparative State Complexity

For each regime or component task i, define the approximate state complexity

$$M_i(\varepsilon_i) = \min\left\{\operatorname{index}(\sim) : \Delta_{\mathbb{T}_i}(\sim) \leq \varepsilon_i\right\}.$$

For a vector of distortion budgets

$$\vec{\varepsilon} = (\varepsilon_1, \dots, \varepsilon_m),$$

define the joint state complexity

$$M_{\text{joint}}(\vec{\varepsilon}) = \min\left\{\operatorname{index}(\sim) : \Delta_{\mathbb{T}_i}(\sim) \leq \varepsilon_i \text{ for all } i\right\}.$$

The following product bound concerns only component tasks whose admissible approximants are deterministic right-congruence quotients of one common history system and whose representatives can be combined componentwise. It does *not* apply to the unrestricted linear finite-rank Hankel relaxation of Section 7, whose resource is operator rank rather than quotient index, and for which no representative map of the required componentwise-combinable form is available; see Remark 12.3 below. It applies directly to commitment and retention, and to symbolic grounding once the implementability condition of Proposition 7.37 is imposed so that symbolic-grounding approximants are themselves right-congruence quotients.

Theorem 12.2 (Comparative State Complexity)**.** *Assume every component task $i = 1, \dots, m$ has admissible approximants given by right-congruence quotients of a common history system, that each $M_i(\varepsilon_i)$ is attained by some congruence $\sim_i$ and representative map Ψ_i, and that the product representative remains admissible with componentwise losses unchanged, namely*

$$\Psi([u]_\sim) = \big(\Psi_1([u]_{\sim_1}), \dots, \Psi_m([u]_{\sim_m})\big)$$

is itself an admissible approximant of loss $\Delta_{\mathbb{T}_i}(\sim_i)$ in each coordinate i. Then for every $\vec{\varepsilon} = (\varepsilon_1, \dots, \varepsilon_m)$,

$$\max_i M_i(\varepsilon_i) \leq M_{\text{joint}}(\vec{\varepsilon}) \leq \prod_i M_i(\varepsilon_i).$$

Without attainment the upper bound holds with each ε_i replaced by $\varepsilon_i + \eta_i$, for arbitrary $\eta_i > 0$.

Remark 12.3 (Why the Unrestricted Linear Relaxation Is Excluded)**.** The proof of Theorem 12.2 builds the joint approximant as the quotient by the meet $\bigwedge_i \sim_i$ of the component right-congruences, and controls the index of that meet by the product of component indices, as in Theorem 12.1. The unrestricted linear finite-rank grounding gap $\Delta^{\mathrm{unres}}_{\mathrm{grd}}(M)$ of Theorem 7.3 is a statement about compact-operator rank in $\ell^2(\Sigma^*)$, not about a right congruence on input histories; "meeting" two rank-M conditions has no analogue of Theorem 12.1's exact index formula, and no product construction combining two finite-rank Hankel approximants into one of controlled rank is supplied here. Consequently Theorem 12.2 and Theorem 12.6 say nothing about joint budgets that mix an unrestricted linear grounding coordinate with a commitment or retention coordinate, absent a bridge of the kind discussed in Section 9.

Proof. The lower bound is immediate because a joint approximant is, in particular, an individual approximant for each regime.

For the upper bound, choose congruences $\sim_i$ of index $M_i(\varepsilon_i)$ attaining (or, absent attainment, η_i-approximating) the individual optima. Let

$$u \sim v \iff \bigwedge_i u \sim_i v.$$

Then

$$\mathrm{index}(\sim) \le \prod_i \mathrm{index}(\sim_i) = \prod_i M_i(\varepsilon_i).$$

The quotient by $\sim$ maps into the product of the component quotients. If Ψ_i is the optimal representative map for component i, define the product representative

$$\Psi([u]_\sim) = \big(\Psi_1([u]_{\sim_1}), \ldots, \Psi_m([u]_{\sim_m})\big).$$

This realizes all componentwise approximations simultaneously. ☐

Remark 12.4 (Tightness)**.** The lower and upper bounds can both be tight. The lower bound is tight when a single congruence simultaneously optimizes all components. The upper bound is tight, up to constants, when the component congruences are independent and the joint state must encode the tuple of component states.

Remark 12.5 (Algorithmic Consequence)**.** The product construction gives a universal joint approximant, but it may incur exponential blowup. Optimizing the joint congruence directly can be computationally hard, as shown by the interaction complexity theorem below.

12.3 Independence of Regime Obstructions

The comparative sandwich does not impose a functional relation among the individual regime gaps. The following independence theorem shows that the regime obstructions can be varied independently.

Theorem 12.6 (Independence of Regime Obstructions)**.** *There exist product sequential tasks with designated retention, commitment, and grounding coordinates such that each regime's gap or obstruction depends only on its designated coordinate. In particular:*

(i) Retention can be zero while commitment is infinite.

(ii) Commitment can be zero while retention is positive.

(iii) Retention and commitment can be zero while deterministic symbolic grounding has a positive stochasticity floor.

(iv) Commitment complexity can be made any finite integer N or infinite.

(v) Retention gap can be made zero or positive, and can be varied by varying predictive separation.

(vi) Deterministic grounding obstruction can be made zero or positive by varying intrinsic stochasticity.

Consequently, no nontrivial universal inequality or functional dependence holds among the grounding, retention, and commitment obstructions across all tasks.

Proof. We construct product tasks whose output stream is a tuple of independent coordinates. Each regime is measured on its designated coordinate.

Retention zero, commitment infinite. Let the retention coordinate be a memoryless fair coin:

$$Y_t^{\text{ret}} \sim \text{Bernoulli}(1/2),$$

independent of history. Its predictive distribution is the same after every history, so it has one causal state and

$$\Delta_{\text{ret}}^{\text{KL}}(1) = 0.$$

Let the commitment coordinate be the square-counting language after a delimiter #:

$$L_{\text{sq}} = \{\#1^n : n \text{ is a perfect square}\}.$$

For histories $\#1^n$ and $\#1^m$ with $n < m$, choose

$$t = m^2 - n.$$

Then $n + t = m^2$ is square, while

$$m + t = m^2 + (m - n)$$

lies strictly between m^2 and $(m+1)^2$, because

$$0 < m - n < m < 2m + 1.$$

Thus $m + t$ is not square. Hence the residuals after $\#1^n$ and $\#1^m$ are distinct for every $n \neq m$, so

$$\kappa_{\text{MN}}(L_{\text{sq}}) = \infty.$$

The product task has zero one-state retention gap but infinite exact commitment complexity.

Commitment zero, retention positive. Let the commitment coordinate be the constant safe output, with

$$\kappa_{\mathrm{MN}} = 1.$$

Let the retention coordinate be the running-parity process with fair i.i.d. inputs and two causal states, as in Example 5.3. Then

$$\Delta_{\mathrm{ret}}^{\mathrm{KL}}(2) = 0, \qquad \Delta_{\mathrm{ret}}^{\mathrm{KL}}(1) = \log 2.$$

The product task has exact commitment with one state but positive one-state retention gap.

Retention and commitment zero, deterministic grounding obstruction positive. Let the output be a memoryless fair coin independent of input, and impose no nontrivial commitment constraint. The retention coordinate has one causal state, so retention is zero. The commitment coordinate is trivial, so commitment is zero. However, a deterministic Mealy machine emits a deterministic output bit at each time. For a memoryless fair coin, the local stochastic spread is

$$s(h_t) = 1 - \max\{1/2, 1/2\} = 1/2$$

at every time. Therefore the stochasticity floor of Theorem 7.28 is

$$\sigma_\gamma = \sum_{t=1}^{\infty} \gamma^{t-1} \frac{1}{2} = \frac{1}{2(1-\gamma)} > 0.$$

Thus deterministic symbolic grounding has a positive obstruction even though retention and commitment vanish. The associated linear Hankel object has rank one, illustrating the separation between linear finite-rank grounding and deterministic symbolic grounding.

Finite and tunable complexity. Replacing L_{sq} by the delimiter-free modular language

$$L_N = \{1^n : n \equiv 0 \pmod{N}\}$$

over the unary alphabet gives commitment complexity exactly N: the Myhill–Nerode classes of L_N are the N residue classes, since the residual after 1^n is $\{1^k : n + k \equiv 0 \pmod{N}\}$, which distinguishes the N residues of n. Varying the predictive separation in a two-state retention process varies the retention gap continuously from 0 to its maximum. Varying the Bernoulli bias p in a memoryless grounding coordinate varies the stochasticity floor as

$$\sigma_\gamma(p) = \frac{\min\{p, 1-p\}}{1-\gamma}.$$

Thus the coordinates can be tuned independently. □

Remark 12.7 (Consequence for Cross-Regime Comparisons)**.** The independence theorem is the structural reason why the unified Schatten template and the exponent vertex correspondence cannot produce unconditional cross-regime inequalities. They unify functional forms under shared response representations; they do not unify the numerical gaps. The product construction measures each regime on its designated coordinate; it does not identify the machine types.

12.4 Interaction of Exact Commitment and Approximate Retention

Let commitment be exact and retention be approximate. Let $\sim_{\text{com}}$ be the commitment congruence, with index

$$\kappa = \text{index}(\sim_{\text{com}}).$$

Exact commitment requires the machine congruence to refine $\sim_{\text{com}}$:

$$\sim \subseteq \sim_{\text{com}} .$$

Theorem 12.8 (Exact Commitment with Approximate Retention)**.** *For every $\varepsilon \geq 0$,*

$$M_{\text{joint}}(0, \varepsilon) = \min \left\{ \text{index}(\sim) : \sim \subseteq \sim_{\text{com}},\ \Delta_{\text{ret}}(\sim) \leq \varepsilon \right\}.$$

Proof. Exact commitment requires $\sim \subseteq \sim_{\text{com}}$. The same congruence $\sim$ governs retention. The result follows by minimizing the retention cost over safe congruences. □

Lemma 12.9 (Constrained Information-Bottleneck Form)**.** *In the synchronized finite-history subclass, suppose*

$$\sim_\delta \subseteq \sim \subseteq \sim_{\text{com}},$$

so that $\sim$ is coarser than causal equivalence and refines the commitment congruence. Then

$$\Delta_{\text{ret}}^{\text{KL}}(\sim) = I(S; Z \mid K_\sim) = I(S; Z) - I(K_\sim; Z).$$

Consequently,

$$\Delta_{\text{ret}}^{\text{safe}}(M) = I(S; Z) - \max_{\substack{\sim \subseteq \sim_{\text{com}} \\ \text{index}(\sim) \leq M}} I(K_\sim; Z).$$

In the stationary formulation of Definition 3.14, the same identity holds with the feasible set written as safe lumpable quotients

$$\phi : \mathcal{S}^+ \to \mathcal{K}$$

whose induced quotient refines the commitment congruence:

$$\Delta_{\text{ret}}^{\text{safe}}(M) = I(S; Z) - \max_{\substack{\phi \ \textit{safe lumpable} \\ |\mathcal{K}| \leq M}} I(K_\phi; Z).$$

Proof. This is Meta-Theorem 4.14, with the feasible set restricted to safe congruences $\sim \subseteq \sim_{\text{com}}$ in the synchronized subclass, or equivalently to safe lumpable quotients of the stationary support $\mathcal{S}^+$ in the stationary formulation. □

Remark 12.10 (Why the Lower Constraint Is Only Sufficient)**.** Lemma 12.9 assumes

$$\sim_\delta \subseteq \sim \subseteq \sim_{\text{com}},$$

which guarantees that the machine state is a function of the causal state, so that $K_\sim$ is a quotient of S. This is a *sufficient* condition for the information-bottleneck form, not a constraint that may be imposed in general.

Requiring $\sim_\delta \subseteq \sim$ in the definition of the joint problem would be incorrect. A joint machine may need to *split* a causal state in order to carry commitment information, since exact commitment is a constraint on histories rather than on predictive laws. In that case the machine state is not a function of S, and the feasible set of Theorem 12.8 is strictly larger than the set of quotients of S.

The correct general formulation therefore optimizes over finite-state history factors K that satisfy exact commitment. If K is such a factor and S is predictively sufficient with $Z \perp K \mid S$, then

$$\mathbb{E}\, D_{\mathrm{KL}}(P_{Z|S} \| P_{Z|K}) = I(S; Z \mid K) = I(S; Z) - I(K; Z),$$

so the safe retention cost is

$$\Delta_{\mathrm{ret}}^{\mathrm{safe}}(M) = I(S; Z) - \sup_K I(K; Z),$$

the supremum being over at-most-M-state history factors whose state congruence refines the commitment congruence. When commitment is constant on each causal state, this reduces to the quotient formulation above.

Corollary 12.11 (Equality Condition)**.** *Let*

$$M_{\mathrm{ret}}(\varepsilon) = \min \left\{ \mathrm{index}(\sim) : \Delta_{\mathrm{ret}}(\sim) \leq \varepsilon \right\}.$$

Then

$$M_{\mathrm{joint}}(0, \varepsilon) \geq \max(\kappa, M_{\mathrm{ret}}(\varepsilon)).$$

Equality holds iff there exists a safe congruence $\sim \subseteq \sim_{\mathrm{com}}$ *with*

$$\mathrm{index}(\sim) = \max(\kappa, M_{\mathrm{ret}}(\varepsilon))$$

and

$$\Delta_{\mathrm{ret}}(\sim) \leq \varepsilon.$$

Equivalently:

(1) If $M_{\mathrm{ret}}(\varepsilon) \leq \kappa$*, equality holds iff*

$$\Delta_{\mathrm{ret}}(\sim_{\mathrm{com}}) \leq \varepsilon.$$

(2) If $M_{\mathrm{ret}}(\varepsilon) > \kappa$*, equality holds iff there exists an optimal retention congruence* $\sim^*$ *with*

$$\mathrm{index}(\sim^*) = M_{\mathrm{ret}}(\varepsilon),$$

$$\Delta_{\mathrm{ret}}(\sim^*) \leq \varepsilon,$$

and

$$\sim^* \subseteq \sim_{\mathrm{com}}.$$

Proof. The lower bound follows because any safe congruence has index at least κ and any feasible retention congruence has index at least $M_{\mathrm{ret}}(\varepsilon)$.

If $M_{\mathrm{ret}}(\varepsilon) \le \kappa$, then a safe congruence of index κ must be exactly $\sim_{\mathrm{com}}$, because any refinement of $\sim_{\mathrm{com}}$ has index at least κ, and equality of finite index forces equality of partitions. Hence equality holds exactly when the commitment congruence itself is retention-feasible.

If $M_{\mathrm{ret}}(\varepsilon) > \kappa$, then equality requires a retention optimal congruence of index $M_{\mathrm{ret}}(\varepsilon)$ that is also safe. □

Corollary 12.12 (Price of Safety)**.** *Let K range over at-most-M-state deterministic history factors whose state congruence refines the exact commitment congruence, and assume S is predictively sufficient, so that $Z \perp K \mid S$. Then*

$$\Delta_{\mathrm{ret}}^{\mathrm{safe}}(M) = I(S;Z) - \sup_{K} I(K;Z).$$

If commitment is constant on every predictive state, the feasible factors may be represented as safe lumpable quotients $K_\sim$ of S, and the supremum reduces to the quotient form

$$\Delta_{\mathrm{ret}}^{\mathrm{safe}}(M) = I(S;Z) - \max_{\substack{\sim \subseteq \sim_{\mathrm{com}} \\ \mathrm{index}(\sim) \le M}} I(K_\sim;Z).$$

See Remark 12.10. The Price of Safety is

$$\mathrm{PoS}(M) = \Delta_{\mathrm{ret}}^{\mathrm{safe}}(M) - \Delta_{\mathrm{ret}}^{\mathrm{free}}(M) \ge 0.$$

Proof. This follows from Lemma 12.9. The safe feasible set is a subset of the free feasible set, so the constrained maximum of $I(K_\sim;Z)$ is no larger than the free maximum. Therefore the safe retention gap is no smaller than the free retention gap. □

Remark 12.13 (No Universal Strict Overhead)**.** Overhead beyond

$$\max(\kappa, M_{\mathrm{ret}}(\varepsilon))$$

is not structurally forced. It occurs exactly when no safe congruence of index

$$\max(\kappa, M_{\mathrm{ret}}(\varepsilon))$$

satisfies the retention distortion constraint. In particular, if

$$M_{\mathrm{ret}}(\varepsilon) \le \kappa,$$

equality requires

$$\Delta_{\mathrm{ret}}(\sim_{\mathrm{com}}) \le \varepsilon.$$

If that fails, then

$$M_{\mathrm{joint}}(0,\varepsilon) > \kappa$$

even though

$$M_{\mathrm{ret}}(\varepsilon) < \kappa.$$

Theorem 12.14 (Complexity of Interaction)**.** *For rational data — rational predictive means and a rational tolerance ε — the decision problem*

$$M_{\mathrm{joint}}(0, \varepsilon) \leq M$$

is NP-hard, even for trivial safety constraints $\kappa = 1$, in the Gaussian quadratic restricted retention regime. The qualifier is inherited from Lemma 5.54 and Theorems 5.56–5.57, on which the reduction below depends, and without it the threshold ε constructed in the proof need not be representable in polynomial bit complexity.

Proof. For $\kappa = 1$, the commitment constraint is vacuous: every congruence refines the one-class commitment congruence, so $M_{\mathrm{joint}}(0, \varepsilon) \leq k$ holds exactly when some congruence of index at most k has retention gap at most ε.

The joint problem is stated with a non-strict threshold and the retention problem with a strict one, so the two are not interchangeable at a common θ; a threshold strictly separating the two cases must be chosen. The gap-amplification lemma supplies one. By Lemma 5.54 the transformed instances satisfy the promise

$$\mathrm{OPT}_a \leq T \qquad \text{or} \qquad \mathrm{OPT}_a \geq T + 2n,$$

and the reductions of Theorems 5.56 and 5.57 carry this separation to the retention objective, giving rational values $\theta_{\mathrm{yes}} < \theta_{\mathrm{no}}$ with $\Delta^{\mathrm{quad}}_{\mathrm{ret}}(k) \leq \theta_{\mathrm{yes}}$ in the YES case and $\Delta^{\mathrm{quad}}_{\mathrm{ret}}(k) \geq \theta_{\mathrm{no}}$ in the NO case. Set $M = k$ and

$$\varepsilon = \tfrac{1}{2}\big(\theta_{\mathrm{yes}} + \theta_{\mathrm{no}}\big),$$

a rational of polynomial bit complexity. Then $M_{\mathrm{joint}}(0, \varepsilon) \leq k$ holds in the YES case and fails in the NO case, so the reduction is correct, and it is computable in polynomial time. NP-hardness of the retention problem in the Gaussian quadratic restricted regime is Theorems 5.56 and 5.57. □

Remark 12.15 (Scope of the Interaction Complexity)**.** The theorem is a theorem of the Gaussian quadratic restricted retention regime. It does not establish NP-hardness for unrestricted full-KL retention. It also does not assert that the exact Price of Safety is given by the linear pinching surrogate of Section 10.

13 The Temporal Axis

The offline approximation gaps and the joint/interaction results are static. This section adds the temporal axis: online learning of finite-state predictors. The theory distinguishes passive/adversarial-input models from active/query models. It also separates realizable mistake bounds from agnostic regret bounds.

The temporal results supply the estimation rates used in the oracle bridge of Section 9.

13.1 Passive and Active Models

Let $\mathcal{H}_M$ denote the class of input-output functions realized by deterministic Mealy machines with at most M states over fixed finite alphabets $\mathcal{I}$ and $\mathcal{O}$, with $|\mathcal{I}| \geq 2$ and $|\mathcal{O}| \geq 2$.

The temporal theory distinguishes two information structures.

- **Passive / adversarial-input model.** The learner does not choose inputs. At each round t, an adversary chooses $x_t \in \mathcal{I}$, the learner predicts an output $\widehat{y}_t$, and then the true output y_t is revealed. The input sequence may be adversarial.
- **Active / query model.** The learner chooses inputs x_t and may apply synchronizing or distinguishing experiments. The learner still predicts the output before observing it, and mistakes are counted in the same way.

The two models have different synchronization properties. In the passive model, the learner cannot force the system to reveal its state. In the active model, the learner may be able to synchronize or identify the initial state by choosing suitable input words.

13.2 Littlestone Dimension of Finite-State Machines

Lemma 13.1 (Littlestone Dimension)**.** *For fixed alphabets with $|\mathcal{I}| \geq 2$ and $|\mathcal{O}| \geq 2$,*

$$\mathrm{Ldim}(\mathcal{H}_M) = \Theta(M \log M),$$

with constants depending only on $|\mathcal{I}|$ and $|\mathcal{O}|$, not on M or the horizon T.

Here $\mathcal{H}_M$ carries the reset-word *protocol: each instance is a fresh word evaluated from the designated initial state. Write $\mathcal{H}_M^{\text{reset}}$ when the distinction matters, and $\mathcal{H}_M^{\text{stream}}$ for the same machine family evaluated along a single persistent stream, as in Corollary 13.57. These are different online classes and their sequential dimensions are established by different arguments.*

Remark 13.2 (From Word Classifiers to Transducers)**.** The VC estimate of [8] concerns deterministic *acceptors*: n-state automata classifying independently presented words. Its transfer to Mealy transductions uses the following reduction. Given an acceptor $(\mathcal{S}, s_0, \delta, F)$ with $|\mathcal{S}| = n$, adjoin a delimiter $\# \notin \mathcal{I}$ and define a Mealy machine on the same state set with

$$\lambda(s, x) = 0 \;\; (x \in \mathcal{I}), \qquad \lambda(s, \#) = \mathbf{1}_{\{s \in F\}}, \qquad \tau(s, \#) = s_0,$$

and $\tau = \delta$ on $\mathcal{I}$. Reading $u\#$ from s_0 emits the classification of u on the final symbol and returns to s_0, so a shattered set of words for the acceptor yields a shattered set of instances $u\#$ for the transducer with the same cardinality and n states.

If the delimiter is not admitted into the alphabet, encode $\mathcal{I} \cup \{\#\}$ into fixed-length binary blocks; this multiplies the state count by a constant depending only on $|\mathcal{I}|$, which is absorbed into the constants of the lemma.

Proof of Lemma 13.1. Upper bound. A member of $\mathcal{H}_M$ is specified by a designated initial state, a transition table $\tau : Q \times \mathcal{I} \to Q$, and an output table $\lambda : Q \times \mathcal{I} \to \mathcal{O}$, so

$$|\mathcal{H}_M| \;\leq\; M \cdot M^{M|\mathcal{I}|} \cdot |\mathcal{O}|^{M|\mathcal{I}|}, \qquad \log_2 |\mathcal{H}_M| \;\leq\; \big(M|\mathcal{I}| + 1\big) \log_2 M + M|\mathcal{I}| \log_2 |\mathcal{O}|,$$

which is $O(M \log M)$ for fixed alphabets. The inequality is not an equality, since distinct labelled tables may realize the same transduction; only the upper bound is needed. The halving algorithm predicts by majority over the version space and at least halves it on every mistake, so no shattered mistake tree has depth exceeding $\log_2 |\mathcal{H}_M|$. Hence $\mathrm{Ldim}(\mathcal{H}_M) = O(M \log M)$.

Lower bound. Every shattered set yields a shattered tree, presenting its points in a fixed order, so Ldim $\geq$ VCdim. The automata VC estimate of [8] gives VCdim $= \Omega(n \log n)$ for n-state acceptors over fixed alphabets, and Remark 13.2 converts an n-state acceptor into an $O(n)$-state Mealy machine preserving shattering, so $\mathrm{VCdim}(\mathcal{H}_M^{\mathrm{reset}}) = \Omega(M \log M)$.

That estimate concerns classification of *separately presented* words and does not by itself bound the sequential complexity of a single persistent transducer stream, which is a different protocol. The lower bound appropriate to the persistent-stream protocol is the explicit forcing construction of Theorem 13.5, which is proved without reference to this lemma. The present lemma is therefore stated for the ordinary Littlestone dimension in the reset-word protocol; persistent-stream statements rest on the explicit construction rather than on this lemma.

For $|\mathcal{O}| > 2$ the halving step is the multiclass version [11, 27], which changes only the alphabet-dependent constants. □

Remark 13.3 (Protocol Scope of Lemma 13.1)**.** The upper bound is a counting bound and is protocol independent. The lower bound is inherited from the reset-word VC construction [8], in which each instance is a fresh word evaluated from the designated initial state. Consequently $\mathrm{Ldim}(\mathcal{H}_M) = \Theta(M \log M)$ should be read in the reset-word protocol. Transfer to a single persistent stream requires a mistake tree realizable *inside one input stream*; Theorem 13.5 constructs such a tree, so the transfer is valid and the persistent-stream bound is also $\Theta(M \log M)$. See Remark 13.6.

The constants depend on the alphabets but not on M or the horizon T.

13.3 Passive Realizable Setting

In the passive realizable setting, the target input-output function $f^\star$ belongs to $\mathcal{H}_M$, and the learner makes predictions against an adversarial input sequence.

Theorem 13.4 (Passive Realizable Mistakes)**.** *In the passive realizable setting, with known or unknown initial state, the minimax mistake count on a single persistent stream satisfies the unconditional upper bound*

$$\inf_{\mathcal{A}} \sup_{f^\star \in \mathcal{H}_M} \sup_{x \in \mathcal{I}^{\mathbb{N}}} \sum_{t=1}^{T} \mathbf{1}\{y_t \neq f^\star(x)_t\} = O(M \log M).$$

Moreover the matching lower bound holds in the same *persistent-stream protocol: by Theorem 13.5 there are fixed alphabets with $|\mathcal{I}| = 4$, $|\mathcal{O}| = 2$ and, for infinitely many M, an* *adversary strategy on one never-reset stream forcing $\Omega(M \log M)$ mistakes. Hence*

$$\inf_{\mathcal{A}} \sup_{f^\star \in \mathcal{H}_M} \sup_{x \in \mathcal{I}^{\mathbb{N}}} \sum_{t=1}^{T} \mathbf{1}\{y_t \neq f^\star(x)_t\} = \Theta(M \log M)$$

for T at least the length of the forcing stream and all sufficiently large M, the passage from the subsequences $M = 2^L$ and $M' = 3 \cdot 2^L$ on which the forcing constructions act to all large budgets being Lemma 13.9, as recorded in Corollary 13.10. The same conclusion holds a fortiori in the reset-word protocol, where a reset is a special case of the input letter `r` *below.*

Proof. For known initial state, the optimal mistake bound is the Littlestone dimension of $\mathcal{H}_M$ by Littlestone's theorem [9]. By Lemma 13.1, this is $O(M \log M)$. The matching lower bound is Theorem 13.5, proved below; its proof is independent of the present theorem, so there is no circularity.

For unknown initial state, the effective hypothesis class is enlarged by at most a factor M, because each M-state machine has at most M possible initial states. The logarithm of the enlarged class size increases by at most $\log M$, which is lower order relative to $M \log M$. More formally, the Littlestone dimension of a finite union of M classes of Littlestone dimension $O(M \log M)$ remains $O(M \log M)$, while the lower bound remains $\Omega(M \log M)$ because the known-initial-state subclass is contained in the unknown-initial-state class. Hence the upper bound $O(M \log M)$ holds in general, and it is tight in the reset-word protocol. □

Theorem 13.5 (Persistent-Stream Mistake Lower Bound)**.** *Fix $\mathcal{I} = \{\mathtt{r}, \mathtt{e}, \mathtt{d}, \mathtt{c}\}$, so $|\mathcal{I}| = 4$, and $\mathcal{O} = \{0, 1\}$. For every $L \geq 1$ and $M = 2^L$ there is an adversary strategy on a* single persistent input stream, *never reset by any external mechanism, and a target $f^\star \in \mathcal{H}_M$ consistent with all revealed outputs, such that:*

(i) every deterministic *learner makes at least*

$$M \log_2 M = \Omega(M \log M)$$

mistakes;

(ii) for every randomized *learner there is a fixed target in the family with*

$$\mathbb{E}[\textit{mistakes}] \;\geq\; \tfrac{1}{2} M \log_2 M = \Omega(M \log M),$$

the expectation being over the learner's internal randomness alone. This follows from the bound $\mathbb{E}_g \mathbb{E}[\textit{mistakes}] \geq \frac{1}{2} M \log_2 M$ under the uniform target distribution on Q^Q, since an average is attained by some target. Under that same uniform distribution, for every $\eta \in (0, 1)$

$$\Pr\left[\textit{mistakes} \;\geq\; \left(\tfrac{1}{2} - \eta\right) M \log_2 M\right] \;\geq\; 1 - \exp\left(-\tfrac{1}{2}\eta^2 M \log_2 M\right).$$

The forcing stream has length $O(M \log M)$. The construction is stated for $|\mathcal{I}| = 4$; Theorem 13.8 extends it to binary input by block encoding, so the bound holds for all $|\mathcal{I}| \geq 2$, $|\mathcal{O}| \geq 2$.

Proof. **The machine class.** Let $Q = \{0,1\}^L$, so $|Q| = M = 2^L$. We exhibit a subfamily of $\mathcal{H}_M$ indexed by an arbitrary function $g : Q \to Q$. Write a state as a bit string $v = v_1 \cdots v_L$. Transitions and Mealy outputs are

$$\begin{array}{llll}
\mathtt{r}: & \tau(v,\mathtt{r}) = 0^L, & \lambda(v,\mathtt{r}) = 0 & \text{(return to a fixed state)},\\
\mathtt{e}: & \tau(v,\mathtt{e}) = v \oplus 0^{L-1}1, & \lambda(v,\mathtt{e}) = 0 & \text{(flip the last bit)},\\
\mathtt{d}: & \tau(v,\mathtt{d}) = v_2 \cdots v_L v_1, & \lambda(v,\mathtt{d}) = v_1 & \text{(cyclic shift, emit the bit shifted out)},\\
\mathtt{c}: & \tau(v,\mathtt{c}) = g(v), & \lambda(v,\mathtt{c}) = 0 & \text{(the unknown map)}.
\end{array}$$

Only $\mathtt{c}$ depends on g; the letters $\mathtt{r}, \mathtt{e}, \mathtt{d}$ are the same in every member of the family. Each such machine has exactly M states, so the family is a subset of $\mathcal{H}_M$, and it has M^M members.

Two elementary facts. First, the word $\mathtt{d}^L$ acts as the identity on states, since L cyclic shifts restore v, and its output sequence is exactly $v_1 v_2 \cdots v_L$. Thus $\mathtt{d}^L$ *reads out* the current state without changing it. Second, from any state the word

$$w_v = \mathtt{r} \prod_{i=1}^{L} \big(\mathtt{d}\,\mathtt{e}^{v_i}\big)$$

drives the machine to v: starting from 0^L, each round shifts left and optionally sets the new last bit, so after L rounds the state is $v_1 \cdots v_L$. All outputs along w_v are determined by $\mathtt{r}, \mathtt{e}, \mathtt{d}$ alone, hence known to the learner; they are not counted as forced mistakes. Note $|w_v| \le 2L + 1$.

The adversary. Enumerate $Q = \{v^{(1)}, \ldots, v^{(M)}\}$. The adversary maintains a partial function g, initially empty, and processes the states in order. For each $v = v^{(k)}$ it plays the block

$$w_v\ \mathtt{c}\ \mathtt{d}^L.$$

The prefix w_v drives the machine to v; the letter $\mathtt{c}$ moves it to the still-unspecified state $g(v)$; the suffix $\mathtt{d}^L$ reads out the L bits of $g(v)$ one per round, returning to $g(v)$.

During those L readout rounds the learner must predict a bit before seeing it. The adversary answers with the *opposite* bit each time, and records those L answers as the bits of $g(v)$. Every answer is legal: at the moment of answering, the corresponding bit of $g(v)$ has not been constrained by any earlier round, because $g(v)$ is queried only in this block and g is an arbitrary function. Hence the learner is forced into L mistakes per block.

After the block the machine sits at $g(v)$; the next block begins with $\mathtt{r}$, which returns it to 0^L.

The precise content of the "no reset" claim should be stated carefully, since the letter $\mathtt{r}$ is a synchronizing word for the family. What is excluded is an *external* restart: there is no mechanism outside the interaction that reinitializes the machine, no episode boundary, and no point at which the learner's state or the mistake count is cleared. The stream is one uninterrupted sequence of rounds, every round is scored, and the learner's knowledge carries across block boundaries. What is *not* claimed is that the input alphabet lacks a synchronizing letter: $\mathtt{r}$ is an ordinary input symbol that happens to send every state to 0^L, and the learner may play it or observe it like any other. The lower bound is therefore a statement about the persistent-stream protocol with a synchronizing input available, not about families in which no synchronizing word exists. This is the stronger reading for the purpose at hand, since the adversary's power is not derived from resets: the $M \log_2 M$ forced mistakes occur in the readout rounds $\mathtt{d}^L$, and the $\mathtt{r}$ letters cost nothing and reveal nothing.

Consistency. After all M blocks the adversary has defined g on all of Q, and every $g(v) \in Q$. The resulting machine $f^\star$ lies in the family, hence in $\mathcal{H}_M$, and by construction reproduces every output revealed during the stream: the $\mathtt{r}, \mathtt{e}, \mathtt{d}$ outputs are forced by the fixed part of the transition table, and the $\mathtt{d}^L$ readouts are exactly the bits of $g(v)$ as recorded. The interaction is therefore realizable, and the protocol is realizable rather than agnostic.

Count. Each of the M blocks forces L mistakes, giving

$$M \cdot L = M \log_2 M$$

mistakes in total. The stream length is $M(|w_v| + 1 + L) \leq M(3L + 2) = O(M \log M)$.

For randomized learners we use Yao's minimax principle rather than an adaptive adversary, which avoids any independence assumption about the learner's predictions. Draw the target map $g : Q \to Q$ uniformly at random from the M^M possibilities and fix in advance the deterministic input stream consisting of the blocks $w_v\, \mathtt{c}\, \mathtt{d}^L$ in a fixed enumeration of Q. This stream does not depend on the learner.

Enumerate the readout rounds $i = 1, \ldots, N$ with $N = M \log_2 M$, and let $\mathcal{F}_{i-1}$ be the σ-algebra generated by the transcript before round i together with the learner's internal randomness. The bit revealed at round i is a specific bit of $g(v)$ for the block currently being processed, and each such bit is queried exactly once along the stream. Under the uniform law on Q^Q the bits of g are i.i.d. fair bits, so the bit revealed at round i is a fair bit *independent of* $\mathcal{F}_{i-1}$. The learner's prediction at round i is $\mathcal{F}_{i-1}$-measurable, hence independent of that bit, so writing X_i for the mistake indicator,

$$\mathbb{E}\big[X_i \mid \mathcal{F}_{i-1}\big] = \tfrac{1}{2} \qquad \text{almost surely.}$$

Note that this conditional statement is all that is required; the X_i themselves need not be independent. Summing,

$$\mathbb{E}\Big[\sum_{i=1}^{N} X_i\Big] = \frac{N}{2},$$

and since the average over g is at most the worst case over g, some fixed $g \in Q^Q$ forces expected mistakes at least $N/2$ for the given learner.

For concentration, set $Y_i = X_i - \mathbb{E}[X_i \mid \mathcal{F}_{i-1}]$. Then (Y_i) is a martingale difference sequence with $|Y_i| \leq 1$, so Azuma–Hoeffding [26, Ch. 2] gives, for $\eta \in (0, 1)$,

$$\Pr\Big[\sum_{i=1}^{N} X_i \leq \big(\tfrac{1}{2} - \eta\big) N\Big] = \Pr\Big[\sum_{i=1}^{N} Y_i \leq -\eta N\Big] \leq \exp\!\left(-\frac{\eta^2 N}{2}\right).$$

This is the stated high-probability bound, with no independence assumption on the rounds.

No deterministic per-run bound is available for randomized learners: a learner that happens to guess every bit correctly makes no mistakes on that run, an event of probability $2^{-N} > 0$. Claim (i) is therefore restricted to deterministic learners, and (ii) is the correct randomized statement.

Finally, for M not a power of two, apply the construction with $L = \lfloor \log_2 M \rfloor$ and leave the remaining fewer than $M/2$ states unreachable; the bound changes only by a constant factor. □

Remark 13.6 (What the Construction Supplies)**.** The obstruction identified in Remark 13.3 is that a reset-word shattering argument does not embed into one persistent stream without a mechanism for changing state, since the learner cannot be repositioned between instances. The construction above removes that obstruction by making navigation part of the input alphabet: `r` and `e`, `d` furnish an explicit *transport system* that reaches any state from any state using only outputs the learner can already predict, so the transport rounds are free. The $\Omega(M \log M)$ information is then extracted by the readout word $\mathtt{d}^L$, which is output-revealing and state-preserving. In effect the persistent stream simulates the reset protocol at no mistake cost, which is exactly what the transfer required.

Remark 13.7 (Alphabet Size)**.** The construction uses $|\mathcal{I}| = 4$ and $|\mathcal{O}| = 2$, and is unchanged for any larger alphabets, so the bound holds whenever $|\mathcal{I}| \geq 4$ and $|\mathcal{O}| \geq 2$. The binary-input case is handled separately in Theorem 13.8 by block encoding, at the cost of a constant factor.

Theorem 13.8 (Binary-Input Persistent-Stream Lower Bound)**.** *Let $\mathcal{I} = \{0,1\}$ and $\mathcal{O} = \{0,1\}$. There is a constant $c > 0$ such that for infinitely many M' there is an adversary on a single persistent, never-reset binary input stream forcing every deterministic learner to make at least*

$$c\, M' \log_2 M' = \Omega(M' \log M')$$

mistakes, and every randomized learner to make that many in expectation up to a factor $\frac{1}{2}$. One may take any $c < \frac{1}{3}$ for M' large.

Proof. Encode the four letters of Theorem 13.5 as two-bit blocks

$$\mathtt{r} \mapsto 00, \qquad \mathtt{e} \mapsto 01, \qquad \mathtt{d} \mapsto 10, \qquad \mathtt{c} \mapsto 11.$$

Let A be an M-state machine of the family $\mathcal{G}_M$, $M = 2^L$. Define a binary-input Mealy machine A' with state set

$$Q' = Q \times \{-, 0, 1\},$$

where the second coordinate buffers the first bit of the current block. On reading the first bit b of a block, A' moves from $(q, -)$ to (q, b) and emits 0; on reading the second bit b', it applies the letter encoded by (b, b'), moves to $(\tau(q, a), -)$, and emits $\lambda(q, a)$. Then $|Q'| = 3M$, and A' reproduces exactly the output of A on the second bit of each block, emitting the uninformative symbol 0 on every first bit.

The adversary runs the strategy of Theorem 13.5 through this encoding. First bits are perfectly predictable, so they force no mistakes and cost nothing; each readout of A still occurs on exactly one round of the binary stream, namely the second bit of a `d` block, and is still a forced mistake by the same lazy-adversary argument. Hence the number of forced mistakes is unchanged at $M \log_2 M$.

Writing $M' = 3M$ for the state count of the binary machine,

$$\frac{M \log_2 M}{M' \log_2 M'} = \frac{M \log_2 M}{3M(\log_2 M + \log_2 3)} = \frac{1}{3} \cdot \frac{L}{L + \log_2 3} \xrightarrow[L\to\infty]{} \frac{1}{3},$$

and the ratio is increasing in L, exceeding 0.23 already at $L = 4$. Thus the forced mistake count is at least $c\, M' \log_2 M'$ for any $c < \frac{1}{3}$ once M' is large, which is $\Omega(M' \log M')$. The randomized statement follows exactly as in Theorem 13.5(ii). $\square$

Lemma 13.9 (Subsequence-to-All-M Extension)**.** *Let $C : \mathbb{N} \to [0, \infty)$ be nondecreasing. Suppose there are $M_1 < M_2 < \cdots$ with $M_{j+1} \le \alpha M_j$ for a constant $\alpha > 1$, and $C(M_j) \ge \beta M_j \log M_j$ for all j. Then there is $\beta' > 0$ with*

$$C(M) \;\ge\; \beta' M \log M$$

for all sufficiently large M.

Proof. Given large M, pick j with $M_j \le M < M_{j+1}$; then $M_j \ge M/\alpha$. Monotonicity gives

$$C(M) \;\ge\; C(M_j) \;\ge\; \beta M_j \log M_j \;\ge\; \frac{\beta}{\alpha} M\bigl(\log M - \log \alpha\bigr).$$

For $M \ge \alpha^2$ we have $\log M - \log \alpha \ge \frac{1}{2} \log M$, so $C(M) \ge \frac{\beta}{2\alpha} M \log M$. $\square$

Corollary 13.10 (Persistent-Stream Bound for All M)**.** *The minimax realizable mistake complexity on a persistent stream is nondecreasing in M, since $\mathcal{H}_M \subseteq \mathcal{H}_{M+1}$. The constructions of Theorems 13.5 and 13.8 give lower bounds along the subsequences $M = 2^L$ and $M' = 3 \cdot 2^L$ respectively, each with ratio $\alpha = 2$. By Lemma 13.9, therefore,*

$$\mathrm{Mistakes}^{\mathrm{stream}}_{\mathrm{realizable}}(M) = \Omega(M \log M)$$

for all *sufficiently large M, and hence, with the counting upper bound, $\Theta(M \log M)$ for all sufficiently large M and every fixed alphabet pair with $|\mathcal{I}| \ge 2$, $|\mathcal{O}| \ge 2$.*

Remark 13.11 (Binary Input Closes the Alphabet Gap)**.** Theorems 13.5 and 13.8 together give $\Omega(M \log M)$ on a persistent stream for *every* fixed alphabet pair with $|\mathcal{I}| \ge 2$ and $|\mathcal{O}| \ge 2$, matching the finite-class upper bound. The block encoding costs only the constant factor 3 in state count and nothing in forced mistakes, because the padding rounds are perfectly predictable and therefore free.

Remark 13.12 (Scope of the Streaming Lower Bound)**.** The upper bound above is a finite-class bound and is unconditional. The lower bound is inherited from $\mathrm{Ldim}(\mathcal{H}_M)$, and standard VC-dimension lower bounds for automata are stated for classification of independently presented words with resets. A single persistent transducer stream is a different protocol: the learner cannot reset the machine, and successive rounds are dependent. Transferring the lower bound therefore requires a shattering or mistake-tree construction realized *within one input stream*. Absent such a construction, Theorem 13.4 yields an unconditional $O(M \log M)$ upper bound together with an $\Omega(M \log M)$ lower bound for the reset-word protocol only.

Remark 13.13 (No Passive Synchronization Assumption)**.** The unknown-initial-state proof does not assume that the learner can force synchronization. In the passive model, inputs are adversarial, and the learner cannot in general apply a synchronizing word. The bound follows from the size and Littlestone dimension of the enlarged hypothesis class, not from active state identification.

Remark 13.14 (Identification versus State Tracking)**.** What is identified or predicted is the correct input-output function, not necessarily the future internal state trajectory. In deterministic realizable prediction, correct output prediction over all continuations may require identifying the residual equivalence class of the current history, but the learner is not required to track the machine's internal state label.

13.4 Active Realizable Setting

In the active realizable setting, the learner chooses inputs. This can reduce or eliminate the unknown-initial-state obstruction if the machine admits an output-aware synchronizing or distinguishing experiment.

Definition 13.15 (Active Identification Objectives)**.** Three objectives must be separated, because they carry different costs and are achieved against different prior knowledge. In each case the learner is said to *attain* the objective at the first time τ at which the stated quantity is determined by the transcript up to τ, uniformly over all targets in the class consistent with that transcript.

(SI) State identification in a known skeleton. The transition and output tables are given; only the initial state is unknown. The learner attains (SI) when it can name the current state *up to observational equivalence*, that is, when all states consistent with the transcript have the same continuation function. The qualification is needed because a non-minimal skeleton may contain distinct states with identical behaviour, which no input-output experiment can separate; for a minimal skeleton it is the same as naming the state.

(RI) Residual identification. The machine is unknown. The learner attains (RI) when it can identify the current Myhill–Nerode residual class, equivalently the full continuation function from the current history.

(MI) Model identification from a known initial state. The initial state is given; the tables are unknown. The learner attains (MI) when it can name the residual class of every reachable history, not merely of the current one.

Each objective is *prediction-closing*: once attained, the continuation function from the current history is determined, so under the realizable assumption every subsequent output is predictable with no further mistakes. For (SI) this holds because the tables are given and the state is now known; for (RI) and (MI) it is immediate from the definition. Consequently no objective admits a nontrivial post-attainment continuation term, and the total cost of a learner meeting the objective is exactly the cost of attaining it.

Throughout,

$$\mathrm{Mistakes}_{\mathrm{active,RI}}(M) = \inf_{\mathcal{A}\in\mathcal{R}} \sup_{A\in\mathcal{H}_M,\ q\in Q_A} \mathrm{tot}(\mathcal{A}, A, q),$$

where tot counts *all* prediction mistakes over the whole interaction and $\mathcal{R}$ is the set of active learners that attain objective (RI) at some finite time on every instance. Both $\mathrm{Mistakes}_{\mathrm{active,RI}}$ and E_{sync} are minimax over the same learner set; they differ in what they count. $E_{\mathrm{sync}}(M)$ of Definition 13.18 counts only the mistakes made *before* attainment, so

$$\mathrm{Mistakes}_{\mathrm{active,RI}}(M) \ \geq\ E_{\mathrm{sync}}(M)$$

holds by definition, a learner being free a priori to err after attaining (RI). The reverse inequality is not definitional: it is Theorem 13.27(I), and its proof turns on the prediction-closing property of Definition 13.15, which forbids post-attainment mistakes. Were (RI) replaced by an objective that is not prediction-closing, the two quantities would separate.

The requirement is not vacuous: as Remark 13.50 observes, a learner that repeatedly plays an input with known constant output makes no mistakes and learns nothing, so without an identification requirement the unconstrained quantity is 0. All active lower bounds in this subsection are therefore identification complexities.

The objectives differ in what must be paid *before* attainment, and are not ordered by inclusion: (SI) presupposes the tables and asks only for the state, whereas (MI) presupposes the state and asks for the tables. (RI) presupposes neither, and is the objective against which Definition 13.18 and all lower bounds of this subsection are stated.

Remark 13.16 (Which Objective Each Bound Uses)**.** Because all three objectives of Definition 13.15 are prediction-closing, a mistake bound in this subsection is always a bound on the cost of *attainment*, never a sum of an attainment cost and a subsequent learning cost. In particular no $O(M \log M)$ continuation term is appended after attainment under any of the three objectives; where such a term appears it is part of the attainment cost itself.

The complexity $E_{\mathrm{sync}}(M)$ of Definition 13.18 is defined against (RI), and the upper bound it yields is $O(E_{\mathrm{sync}}(M))$ (Theorem 13.27(I)). The length-form bound of Proposition 13.28 is also stated against (RI); the $M \log M$ term there arises because the universal experiment realising $L^{\mathrm{univ}}_{\mathrm{sync}}(M)$ certifies the residual class only after a further model-search phase, and is therefore part of the attainment cost, not a continuation term. The lower-bound constructions of Theorems 13.45 and 13.48 are likewise stated against (RI); their additive content lies in the two *independent unknown maps*, not in a synchronization-plus-Littlestone split.

Definition 13.17 (Adaptive Output-Aware Synchronization Length)**.** For a deterministic Mealy machine A with unknown initial state, an adaptive output-aware synchronization experiment is a finite decision tree. Each internal node is labeled by an input $x \in \mathcal{I}$; each outgoing edge is labeled by an observed output $y \in \mathcal{O}$. A leaf is reached after observing the corresponding input-output transcript.

The experiment identifies the residual class if every leaf corresponds to a single Myhill–Nerode residual class of A. Let

$$L_{\mathrm{sync}}^{\mathrm{adapt}}(A)$$

be the minimum worst-case depth of such an experiment *for the machine* A. Define

$$L_{\mathrm{sync}}^{\mathrm{adapt}}(M) = \sup_{A \in \mathcal{H}_M} L_{\mathrm{sync}}^{\mathrm{adapt}}(A),$$

with value ∞ if some M-state machine admits no finite adaptive synchronization experiment.

This quantity is *machine-specific*: the experiment realizing $L_{\mathrm{sync}}^{\mathrm{adapt}}(A)$ may depend on A. A learner that does not know A cannot in general run it. For such a learner define instead the **universal adaptive synchronization depth**

$$L_{\mathrm{sync}}^{\mathrm{univ}}(M) = \min_{\mathcal{T}} \sup_{A \in \mathcal{H}_M} \mathrm{depth}_A(\mathcal{T}),$$

the minimum over *single* decision trees $\mathcal{T}$, implementable without knowledge of A, of the worst-case depth at which the transcript produced by $\mathcal{T}$ determines the *current state* of A within each machine still consistent with it.

The two quantities certify different things, and the distinction is what makes the length-form bound of Proposition 13.28 come out as it does. $L_{\mathrm{sync}}^{\mathrm{adapt}}(A)$ is defined for a known machine A and certifies its residual class, which for known tables is the same as certifying the state. A universal tree faces an unknown machine, so the strongest thing it can certify from an output transcript is the state within each consistent candidate; the tables, and hence the residual class, remain to be resolved. This is objective (SI) of Definition 13.15 relativized to the consistent set, not objective (RI), and it is why a subsequent model-search phase appears in the proof.

Always

$$L_{\mathrm{sync}}^{\mathrm{adapt}}(M) \;\leq\; L_{\mathrm{sync}}^{\mathrm{univ}}(M),$$

and the inequality may be strict, since the left side optimizes the experiment per machine while the right side must fix one experiment in advance.

Identification is up to observational equivalence. The phrase "determines the current state" must be read modulo Myhill–Nerode equivalence. The class $\mathcal{H}_M$ contains non-minimal machines carrying distinct states $s \neq s'$ with $\lambda(s, w) = \lambda(s', w)$ for every input word w. No experiment whatsoever can separate such a pair, since separation would require an observable difference and there is none. Under a literal state-identification reading one would therefore have $L_{\mathrm{sync}}^{\mathrm{univ}}(M) = \infty$ for trivial reasons, and the quantity would carry no content.

Accordingly, a transcript is said to determine the current state when it determines its class under the observational equivalence

$$s \sim s' \iff \lambda(s, w) = \lambda(s', w) \text{ for all } w \in \mathcal{I}^*,$$

equivalently when it determines the state of the minimal machine equivalent to A. On minimal machines the two readings agree, since there $\sim$ is the identity; the same convention applies to $L_{\mathrm{sync}}^{\mathrm{adapt}}$, to objective (SI) of Definition 13.15, and to $E_{\mathrm{sync}}^{\mathrm{SI}}$. With this reading $L_{\mathrm{sync}}^{\mathrm{univ}}(M)$ is finite whenever some universal experiment exists, and Lemma 13.40 supplies the separating words that make it so.

Definition 13.18 (Synchronization Mistake Complexity)**.** For an active learner $\mathcal{A}$, a machine $A \in \mathcal{H}_M$ and an initial state q, let $\mathrm{mis}(\mathcal{A}, A, q)$ denote the number of prediction mistakes $\mathcal{A}$ incurs before it attains objective (RI) of Definition 13.15, with the value $+\infty$ if it never does. Define the *synchronization mistake complexity*

$$E_{\mathrm{sync}}(M) = \inf_{\mathcal{A}} \sup_{A \in \mathcal{H}_M,\ q \in Q_A} \mathrm{mis}(\mathcal{A}, A, q),$$

the infimum being over all active learners, deterministic or randomized, with mis replaced by its expectation in the randomized case.

The order of the operators is the substance of the definition. The infimum comes first, so $E_{\mathrm{sync}}(M)$ is the cost incurred by the *best* learner against the worst instance: it is simultaneously an upper bound achievable by some learner and a lower bound valid for every learner. This two-sided reading is what licenses its use on both sides of the bounds below. Reversing the operators would define a different and much larger quantity.

Despite the subscript, $E_{\mathrm{sync}}(M)$ is the cost of attaining objective (RI), which on an unknown machine includes identifying the continuation function and not merely synchronizing the state. It reduces to a state-synchronization cost exactly when the tables are known, as in objective (SI).

For a subclass $\mathcal{C}_M \subseteq \mathcal{H}_M$ write

$$E_{\mathrm{sync}}(\mathcal{C}_M) = \inf_{\mathcal{A}} \sup_{A \in \mathcal{C}_M,\ q \in Q_A} \mathrm{mis}(\mathcal{A}, A, q),$$

so that $E_{\mathrm{sync}}(M) = E_{\mathrm{sync}}(\mathcal{H}_M)$.

Two distinct complexities are in play and must not be conflated. Alongside the residual-identification quantity above, define the *state-identification complexity*

$$E_{\mathrm{sync}}^{\mathrm{SI}}(\mathcal{C}_M) = \inf_{\mathcal{A}} \sup_{A \in \mathcal{C}_M,\ q \in Q_A} \mathrm{mis}^{\mathrm{SI}}(\mathcal{A}, A, q),$$

where $\mathrm{mis}^{\mathrm{SI}}$ counts mistakes before objective (SI) is attained, the tables being given. Since attaining (RI) on an unknown machine requires resolving the tables as well as the state,

$$E_{\mathrm{sync}}^{\mathrm{SI}}(\mathcal{C}_M) \ \leq\ E_{\mathrm{sync}}(\mathcal{C}_M),$$

and the two may differ by the cost of model search. The additive statements below decompose E_{sync} into a state-identification part and a continuation part, so it is $E_{\mathrm{sync}}^{\mathrm{SI}}$ that pairs additively with the continuation cost. The distinction matters because on the full class the complexity is pinned down outright (Corollary 13.47), so the additive form $M \log M + E_{\mathrm{sync}}(M)$ collapses to $\Theta(E_{\mathrm{sync}}(M))$ there; it is on subclasses, where the counting bound $\log_2 |\mathcal{C}_M|$ need not be $O(M \log M)$, that the two terms can genuinely separate. Statements below carrying a direct-sum or operational-equivalence hypothesis are to be read with E_{sync} evaluated on the relevant subclass.

Proposition 13.19 (State Identification Costs Only Logarithmically Many Mistakes)**.** *Let the transition and output tables of a* minimal *M-state machine be given and only the initial state unknown, as in objective (SI). Then*

$$E^{\mathrm{SI}}_{\mathrm{sync}}(M) \;\le\; \lfloor \log_2 M \rfloor,$$

for every *output alphabet with $|\mathcal{O}| \ge 2$. In particular the bound carries no dependence on $|\mathcal{O}|$.*

Proof. Maintain the version space $V \subseteq Q$ of states consistent with the transcript, initially $V = Q$, so $|V| \le M$. While $|V| > 1$, minimality guarantees that some input word separates two survivors; let x be the first letter of such a word on which the survivors do not all agree in output, which exists after advancing V along finitely many letters with agreed output. Predict the output symbol receiving a plurality of votes among $\{\lambda(s,x) : s \in V\}$ and play x.

Partition V by the value of $\lambda(\cdot,x)$ into output classes of sizes $c_1 \ge c_2 \ge \cdots \ge c_r$, where $r \le |\mathcal{O}|$ and $\sum_i c_i = |V|$, and let the learner predict the plurality symbol, the one realizing c_1.

If the prediction is wrong, the observed symbol realizes some class c_j with $j \ge 2$, and the survivors are exactly that one class, of size $c_j \le c_2$. The key point is that the survivors form a *single* class rather than the union of all non-predicted classes. Since $c_1 \ge c_2$ and $c_1 + c_2 \le |V|$,

$$2c_2 \;\le\; c_1 + c_2 \;\le\; |V|, \qquad \text{so} \qquad c_2 \;\le\; \tfrac{1}{2}|V|.$$

Every mistake therefore at least halves the version space, whatever the size of the output alphabet. If the prediction is right, V can only shrink. Advancing every survivor along x preserves $|V|$ or decreases it, since $\tau(\cdot,x)$ is a function.

Starting from $|V| \le M$ and halving at each mistake, with $|V| \ge 1$ throughout, the number of mistakes is at most $\lfloor \log_2 M \rfloor$. When $|V| = 1$ the current state is determined, which is objective (SI); since $E^{\mathrm{SI}}_{\mathrm{sync}}$ is an infimum over learners, the bound follows. □

Remark 13.20 (Why the Alphabet Drops Out)**.** A coarser count replaces "the survivors are one class of size c_2" by "the survivors are everything not predicted", giving the weaker factor $1 - 1/|\mathcal{O}|$ and the bound $\log_{|\mathcal{O}|/(|\mathcal{O}|-1)} M$, which degrades as the alphabet grows: at $M = 1024$ it reads 10.0, 17.1, 24.1 and 51.9 for $|\mathcal{O}| = 2, 3, 4, 8$. That is an artefact of the counting, not of the problem. Because a deterministic machine sends each surviving state to exactly one output symbol, an erring learner is left with a single class, and the plurality rule forces that class to be no larger than half the version space.

The same counting applies wherever the version space consists of candidates each of which predicts a single deterministic symbol. In particular it applies to the machine–state pairs of Theorem 13.41, since a deterministic machine paired with a state emits one symbol under a given input; the alphabet-dependent factor is unnecessary there as well. Enumerating all integer class profiles with $|V| \le 25$ and $r \le 5$ confirms $c_2 \le |V|/2$ without exception, with equality exactly when $c_1 = c_2 = |V|/2$ and $r = 2$; larger alphabets only fragment the complement further and cannot help the adversary.

Theorem 13.21 (State Identification Costs Logarithmically Many Mistakes, Exactly)**.** *For every output alphabet with $|\mathcal{O}| \geq 2$ and every input alphabet with $|\mathcal{I}| \geq 1$,*

$$E^{\mathrm{SI}}_{\mathrm{sync}}(M) \;=\; \lfloor \log_2 M \rfloor,$$

with no *dependence on $|\mathcal{O}|$ or $|\mathcal{I}|$ beyond the requirement $|\mathcal{O}| \geq 2$. In particular $E^{\mathrm{SI}}_{\mathrm{sync}}(2^L) = L$.*

The lower bound is witnessed by a minimal M-state machine on which every deterministic learner is forced to make $\lfloor \log_2 M \rfloor$ mistakes before identifying the current state; every randomized learner makes at least $\frac{1}{2}\lfloor \log_2 M \rfloor$ mistakes in expectation against a fixed initial state.

Proof. The upper bound is Proposition 13.19, which gives $\lfloor \log_2 M \rfloor$ for every $|\mathcal{O}| \geq 2$. For the lower bound fix $L \geq 1$, put $M = 2^L$, and let

$$Q = \{0,1\}^L, \qquad \mathcal{O} = \{0,1\}.$$

Use a single input letter $\mathtt{d}$ acting as the left cyclic shift,

$$\tau(v_1 v_2 \cdots v_L, \mathtt{d}) = v_2 \cdots v_L v_1, \qquad \lambda(v_1 v_2 \cdots v_L, \mathtt{d}) = v_1,$$

adjoining a second letter acting as the identity with constant output if $|\mathcal{I}| \geq 2$ is required; that letter reveals nothing and can be ignored. The tables are known to the learner and only the initial state is unknown, as objective (SI) requires. The construction uses only two of the available output symbols, so it is available verbatim for every $|\mathcal{O}| \geq 2$; a larger alphabet neither helps the learner nor the adversary here.

Minimality. Two distinct states $v \neq w$ differ in some coordinate j, and the word $\mathtt{d}^{j-1}$ drives them to states whose first coordinates are $v_j \neq w_j$, so the next output separates them. Hence all M states are pairwise distinguishable and the machine is minimal.

The transcript is exactly the initial state. Applying $\mathtt{d}$ repeatedly from v emits $v_1, v_2, \ldots, v_L$ in order and returns to v. Thus after L rounds the state is determined, and before round $t \leq L$ the transcript constrains only $v_1, \ldots, v_{t-1}$: the remaining 2^{L-t+1} states consistent with the transcript are exactly those with the observed prefix, and every completion is a legal initial state.

Deterministic lower bound. Run the adversary argument on the version space. At round $t \leq L$ the learner must commit to a prediction $\widehat{y}_t \in \{0,1\}$ for the bit v_t. Both completions $v_t = 0$ and $v_t = 1$ remain consistent, so the adversary sets $v_t = 1 - \widehat{y}_t$, forcing a mistake, and records the bit. After L rounds the adversary has specified an initial state $v^\star \in Q$ consistent with every revealed output, and the learner has made $L = \log_2 M$ mistakes. Until round L the version space has size at least 2, so the learner has not attained (SI); the mistakes are therefore all incurred before identification.

Randomized lower bound. Let the initial state be uniform on Q. Conditioned on the transcript through round $t-1$, the bit v_t is a fair coin independent of the learner's internal randomness, so any prediction errs with probability exactly $\frac{1}{2}$ and the expected number of mistakes over the first L rounds is $L/2$. Averaging over the prior, some fixed $v^\star \in Q$ forces at least $\frac{1}{2}\log_2 M$ expected mistakes against the learner's own randomness.

For general M no padding is required, because $\mathcal{H}_M$ consists of the machines with *at most* M states. Put $L = \lfloor \log_2 M \rfloor$, so $2^L \le M$. The cyclic-shift machine above has exactly 2^L states and is minimal, hence belongs to $\mathcal{H}_M$, and against it every deterministic learner is forced into L mistakes. Since $E^{\mathrm{SI}}_{\mathrm{sync}}(M)$ is a supremum over $\mathcal{H}_M$, this gives $E^{\mathrm{SI}}_{\mathrm{sync}}(M) \ge L = \lfloor \log_2 M \rfloor$, matching the upper bound. $\square$

Remark 13.22 (Why No Padding Is Needed)**.** Equality at intermediate budgets is a consequence of the "at most M states" convention for $\mathcal{H}_M$, not of an interpolating construction. The witness is built at 2^L states, and for any M with $2^L \le M$ that same machine is already a member of $\mathcal{H}_M$; no padding to exactly M states is required, and none is available, since adjoining duplicate states destroys minimality while adjoining fresh reachable ones destroys the fair-bit property.

Were $\mathcal{H}_M$ instead read as machines with *exactly* M states, the argument would give only $E^{\mathrm{SI}}_{\mathrm{sync}}(M) = \Theta(\log M)$, with the constant lost to monotonicity. The two readings agree on the powers of two.

Remark 13.23 (Both Bounds Are Attained by the Same Mechanism)**.** The matching bounds have a common explanation. The halving argument of Proposition 13.19 shows that each mistake removes at least half of the version space, so $\lfloor \log_2 M \rfloor$ mistakes suffice; the cyclic-shift family of Theorem 13.21 is the case in which each mistake removes exactly half and no more, because each round exposes one fresh, unconstrained bit of the initial state. The family is thus extremal for the halving bound, and the two sides meet exactly rather than up to a constant.

Solving the associated games exactly confirms both sides. For $L = 1, \ldots, 10$ the deterministic minimax mistake count on the cyclic-shift family is exactly $L = \log_2 M$ and the Bayes-optimal expected count under the uniform prior is exactly $L/2$, with all M states pairwise separated so that the family is minimal. In the opposite direction, exhausting all minimal Mealy machines across nine signatures with $|\mathcal{S}^+| \le 4$, $|\mathcal{I}| \le 2$ and $|\mathcal{O}| \le 4$ — 46,656 table pairs in the largest, of which 35,640 are minimal, counted under the conventions of Remark 5.29 — the worst-case state-identification cost never exceeds $\lfloor \log_2 M \rfloor$, and attains it at $M = 2$ and $M = 4$. Holding $|\mathcal{S}^+|$ fixed and enlarging $|\mathcal{O}|$ from 2 to 4 leaves the worst case unchanged, as Remark 13.20 predicts.

Two consequences follow for the structure of the theory. First, the collapse recorded in Remark 13.24 is now unconditional: it rests on a two-sided estimate rather than on an upper bound alone, so no choice of subclass can rescue a mistake-currency additive decomposition. Second, there is no dependence on $|\mathcal{O}|$ at all. Both bounds equal $\lfloor \log_2 M \rfloor$: the upper bound because an erring learner is left with a single output class of size at most half the version space (Remark 13.20), the lower bound because the cyclic-shift witness needs only two output symbols. Enlarging the alphabet neither speeds up identification nor slows it down, and the base of the logarithm is 2 irrespective of $|\mathcal{O}|$.

Remark 13.24 (The Additive Form Collapses)**.** Proposition 13.19, sharpened to an equality by Theorem 13.21, has a structural consequence for the additive decomposition. Since $E^{\mathrm{SI}}_{\mathrm{sync}}(\mathcal{C}_M) = O(\log M)$ for every subclass of minimal machines,

$$M \log M + E^{\mathrm{SI}}_{\mathrm{sync}}(\mathcal{C}_M) = \Theta(M \log M),$$

so the two terms never separate and the additive form carries no information beyond $\Theta(M \log M)$. Accordingly no additive theorem is asserted with a state-identification second term.

A genuine two-term decomposition would require measuring the synchronization component in a currency other than mistakes — experiment *length*, as in $L^{\mathrm{univ}}_{\mathrm{sync}}$ of Definition 13.17, is the natural candidate, and Proposition 13.28 is stated in that currency. The mistake-counting decomposition is degenerate.

Lemma 13.25 (Operational Equivalence)**.** *Let the class be a set of deterministic machines and let the protocol be realizable. Then a learner guarantees zero future mistakes on all continuations from a given history if and only if it has attained objective (RI) of Definition 13.15 at that history.*

For deterministic realizable classes this is therefore a theorem, not a hypothesis; it is stated separately because the results below that invoke it are also meaningful for stochastic classes, where it must be assumed.

Proof. (RI) implies zero future mistakes. This is the prediction-closing property recorded in Definition 13.15: the continuation function from the current history is determined by the transcript, uniformly over consistent targets, and realizability places the true target among them, so predicting by that common function is correct on every subsequent round.

Zero future mistakes implies (RI). Suppose (RI) has not been attained, so two machines A, A' in the class are consistent with the transcript and their continuation functions from the current histories differ. Let w be a shortest word on which the two continuation functions disagree, which exists by Lemma 13.40, and let j be the first position at which the emitted symbols differ. The learner's prediction at that round is a function of the transcript alone, hence identical in the run against A and in the run against A', while the two true symbols differ. Both runs are realizable, since A and A' both lie in the class and both are consistent with everything observed. The learner therefore errs in at least one of them, contradicting the guarantee of zero future mistakes on all continuations. □

Definition 13.26 (Direct-Sum Active Instance)**.** A **direct-sum active instance** of budget M consists of an $O(M)$-state machine family together with a decomposition of the learning task into two components satisfying:

(i) a synchronization component whose minimax mistake complexity is S_M, measuring *state identification only* and so comparable to $E^{\mathrm{SI}}_{\mathrm{sync}}$ rather than to E_{sync};

(ii) a continuation component whose active realizable mistake complexity is C_M, measuring the residual model search that remains once the state is known;

(iii) **disjoint accounting**: the rounds on which the two components can force mistakes are disjoint;

(iv) **unavoidability**: the objective cannot be met without meeting it for *both* components, and this is a property of the machine family rather than an external restriction on the protocol;

(v) **product structure**: the family is parameterized by a pair (θ_1, θ_2) of unknowns, one per component, and *every* combination occurs, so the family is the full product of its two coordinate ranges;

(vi) **coordinate locality**: the outputs on the rounds of R_j are determined by θ_j alone, independently of θ_{3-j}.

Conditions (iii)–(vi) together constitute the *component-restriction property*. It is exactly what the reduction in Theorem 13.52 consumes: (vi) makes a component's observations well defined once its own parameter is fixed, (v) guarantees that an arbitrary dummy value of the other parameter yields a legitimate composite target, (iii) keeps the two mistake counts from overlapping, and (iv) forces both components to be paid for. Condition (iv) is essential in the active protocol, where the learner chooses its own inputs. A family in which one component can be bypassed – for example by an irreversible transition into the other component – does not satisfy (iv), and for such a family the additive bound fails. Theorem 13.48 constructs a family satisfying (iv) by making the two components mutually reachable.

Theorem 13.27 (Active Realizable Certified Bounds)**.** *For active realizable learning over M-state deterministic machines, the following three statements hold under* separate *hypotheses, which are not to be combined implicitly.*

(I) Upper bound (unconditional). *Under objective (RI) of Definition 13.15,*

$$\mathrm{Mistakes}_{\mathrm{active,RI}}(M) \;\le\; O\big(E_{\mathrm{sync}}(M)\big),$$

and this is sharp: by Definition 13.18 some learner attains (RI) within $E_{\mathrm{sync}}(M)$ mistakes, and (RI) is prediction-closing, so no mistake occurs afterwards. No continuation term is appended.

(II) Synchronization lower bound (needs Lemma 13.25 alone). *If any learner guaranteeing zero future mistakes must first acquire guaranteed residual-class knowledge, then*

$$\mathrm{Mistakes}_{\mathrm{active,RI}}(M) \ge \Omega\big(E_{\mathrm{sync}}(M)\big).$$

This gives only *the E_{sync} term; it does not by itself supply the $M \log M$ term.*

(III) Disjoint-component lower bound (needs direct-sum saturation, Definition 13.26). *On a subclass $\mathcal{C}_M$ carrying a direct-sum instance with component complexities S_M and C_M,*

$$\mathrm{Mistakes}_{\mathrm{active,RI}}(M) \;\ge\; S_M + C_M.$$

This is a lower bound only; no matching upper bound is asserted, and no identification of S_M with a state-identification complexity is made. By Proposition 13.19 the state-identification *complexity is $O(\log M)$, so a decomposition with $E^{\mathrm{SI}}_{\mathrm{sync}}$ as its second term would be degenerate (Remark 13.24); the components realized by the explicit construction of Theorem 13.48 are both model-identification components.*

Parts (II) and (III) are stated in terms of the abstract complexity E_{sync} and are the form that remains informative on subclasses $\mathcal{C}_M \subseteq \mathcal{H}_M$*, where the two terms of the additive form can separate. On the full class $\mathcal{H}_M$ no hypothesis is needed at all: Corollary 13.47 gives* $\mathrm{Mistakes}_{\mathrm{active,RI}}(M) = \Theta(M \log M)$ *outright, and $M \log M + E_{\mathrm{sync}}(M) = \Theta(E_{\mathrm{sync}}(M))$ there. An explicit family satisfying the hypothesis of (III) is constructed in Theorem 13.48, so (III) is nonvacuous.*

Proof. Part (I). Fix $\varepsilon > 0$. By Definition 13.18 the infimum over learners is attained to within ε, so there is an active learner $\mathcal{A}_\varepsilon$ which, on every $A \in \mathcal{H}_M$ and every initial state, attains objective (RI) after at most $E_{\mathrm{sync}}(M) + \varepsilon$ mistakes. Run $\mathcal{A}_\varepsilon$. At the moment of attainment the continuation function from the current history is determined by the transcript, uniformly over all targets still consistent with it; since the protocol is realizable, the true target is among them, and predicting according to that common continuation function is correct on every subsequent round. Hence the total number of mistakes is at most $E_{\mathrm{sync}}(M) + \varepsilon$. As $\varepsilon > 0$ was arbitrary and the mistake count is integer valued, the bound $O(E_{\mathrm{sync}}(M))$ follows.

No continuation term appears, since objective (RI) already fixes the continuation function from the current history.

Part (II). By Lemma 13.25, any learner that eventually guarantees correct prediction on all continuations must first attain objective (RI). By Definition 13.18 the infimum over learners of the worst-case number of mistakes before that point is $E_{\mathrm{sync}}(M)$, so every learner incurs at least $E_{\mathrm{sync}}(M)$ mistakes in the worst case:

$$\mathrm{Mistakes}_{\mathrm{active,RI}}(M) \;\geq\; \Omega(E_{\mathrm{sync}}(M)).$$

This uses no direct-sum hypothesis and yields *only* the E_{sync} term.

Part (III). Assume the direct-sum saturation condition of Definition 13.26. The Littlestone lower bound is proved against adversarial inputs and does not transfer automatically to the active protocol (Remark 13.43), so each component's cost must be supplied by that hypothesis rather than by counting.

The bound is $S_M + C_M$, not $\max\{S_M, C_M\}$, and the additivity comes from disjointness rather than from any inequality between maxima and sums. By condition (iv) the objective forces both components; by condition (iii) the rounds on which they force mistakes are disjoint; and by Theorem 13.52, whose restriction argument uses conditions (v) and (vi), each component contributes at least its own minimax complexity on those rounds. Summing over the two disjoint round sets gives

$$\mathrm{Mistakes}_{\mathrm{active,RI}}(M) \;\geq\; S_M + C_M.$$

No matching upper bound is claimed: Part (I) bounds $\mathrm{Mistakes}_{\mathrm{active,RI}}(M)$ by $O(E_{\mathrm{sync}}(\mathcal{C}_M))$, and identifying $E_{\mathrm{sync}}(\mathcal{C}_M)$ with $S_M + C_M$ would require an explicit composite learner, which is not constructed here. $\square$

Proposition 13.28 (Length-Based Active Upper Bound)**.** *For fixed finite alphabets — the hypothesis $L^{\mathrm{univ}}_{\mathrm{sync}}(M) < \infty$ being automatic by Proposition 13.30 —*

$$\mathrm{Mistakes}_{\mathrm{active,RI}}(M) \le O\big(M \log M + L^{\mathrm{univ}}_{\mathrm{sync}}(M)\big),$$

and in particular

$$E_{\mathrm{sync}}(M) \;\le\; O\big(M \log M + L^{\mathrm{univ}}_{\mathrm{sync}}(M)\big).$$

The universal depth, not the machine-specific $L^{\mathrm{adapt}}_{\mathrm{sync}}(M)$, is the correct quantity here: a learner facing an unknown machine must fix its experiment in advance, so it cannot in general realize $L^{\mathrm{adapt}}_{\mathrm{sync}}(A)$ for the particular A it faces. The two terms are not redundant: $L^{\mathrm{univ}}_{\mathrm{sync}}(M)$ pays for state identification and the $M \log M$ term for the subsequent model search, these being the two things a universal experiment cannot deliver at once (Definition 13.17). If the machine is known except for its initial state, the model-search phase is vacuous and the bound reads $O(L^{\mathrm{adapt}}_{\mathrm{sync}}(M))$.

In order of magnitude this bound is subsumed by Theorem 13.41, which attains the same objective in $O(M \log M)$ mistakes without any synchronization experiment; even for $L^{\mathrm{univ}}_{\mathrm{sync}}(M) = o(M \log M)$ the total here remains $O(M \log M)$. Its interest is structural. It separates the contribution of the synchronization experiment from that of the model search, and its first phase is governed by an experiment fixable in advance, which is the operative quantity when the model-search phase is externally curtailed. No lower bound is asserted here; see Remark 13.43.

Proof. For the upper bound, the learner fixes in advance a universal adaptive synchronization tree of worst-case depth $L^{\mathrm{univ}}_{\mathrm{sync}}(M)$, as in Definition 13.17; this requires no knowledge of the target machine. During this phase it makes at most $L^{\mathrm{univ}}_{\mathrm{sync}}(M)$ prediction mistakes, one per step in the worst case. By Definition 13.17 the observed transcript then determines the current state of the target within each machine still consistent with it; the tables, and hence the residual class, are not yet determined, so objective (RI) has not been attained and a second phase is required.

The learner then runs the standard halving algorithm over the set of machines in $\mathcal{H}_M$ consistent with the transcript so far, of cardinality at most $|\mathcal{H}_M|$. Each mistake halves that set, so at most $\log_2 |\mathcal{H}_M| = O(M \log M)$ further mistakes occur before a single residual class remains, at which point objective (RI) is attained and, being prediction-closing, no mistake occurs afterwards. The two phases together give

$$O(M \log M + L^{\mathrm{univ}}_{\mathrm{sync}}(M)).$$

Here the $M \log M$ term is part of the cost of *attaining* (RI), not a continuation term appended after attainment; this is the distinction drawn in Remark 13.16.

For the second display, the composite strategy just described is one admissible learner attaining (RI), and it is implementable without knowing the target. Since $E_{\mathrm{sync}}(M)$ is an infimum over such learners (Definition 13.18), the bound $E_{\mathrm{sync}}(M) \le O(M \log M + L^{\mathrm{univ}}_{\mathrm{sync}}(M))$ follows; when $L^{\mathrm{univ}}_{\mathrm{sync}}(M) = \Omega(M \log M)$ this reads $E_{\mathrm{sync}}(M) = O(L^{\mathrm{univ}}_{\mathrm{sync}}(M))$.

□

Proposition 13.29 (Machine-Specific Synchronization Depth Is Quadratic)**.** *For fixed finite alphabets and every $M \geq 2$,*

$$L_{\text{sync}}^{\text{adapt}}(M) \;\leq\; (M-1)^2 \;=\; O(M^2).$$

The bound is on the machine-specific *depth $L_{\text{sync}}^{\text{adapt}}$, in which the experiment may depend on the target; see Remark 13.33 for why it does not transfer to the universal depth $L_{\text{sync}}^{\text{univ}}$.*

Proof. Fix $A \in \mathcal{H}_M$, minimal without loss of generality by the convention of Definition 13.17, and let $U \subseteq Q_A$ be the set of states consistent with the transcript so far, initially $U = Q_A$ with $|U| \leq M$. We exhibit a strategy, fixable in advance, that drives U to a single observational class.

Step: one separation episode. Suppose U contains two states $s \neq t$ that are not observationally equivalent, so their continuation functions in A differ. Lemma 13.34 with $U = \{s,t\}$ bounds the length of the shortest word separating them by

$$d(\{s,t\}) \;\leq\; M - |\{s,t\}| + 1 \;=\; M-1,$$

via Moore partition refinement; Lemma 13.40 is the cross-machine form of the same bound. Let w be such a separating word, $|w| \leq M-1$.

In pair-automaton terms: from $\{u,v\}$ and input x there is an edge to $\{\tau(u,x), \tau(v,x)\}$, and $\{u,v\}$ is *immediately separated* by x when $\lambda(u,x) \neq \lambda(v,x)$. Breadth-first search from $\{s,t\}$ reaches an immediately separated pair, since $s \not\sim t$; but the distance bound is the Moore bound $M-1$ above, not a distinctness-of-pair-nodes argument — distinct pair nodes number at most $\binom{M}{2}$, which would bound the path by $\binom{M}{2}$, not by $M-1$.

Feed w. Transitions are deterministic, so applying an input maps U forward and $|U|$ never increases. At the final letter of w the two survivors descended from s and t emit different outputs, so whichever output is observed at least one of the two branches is eliminated and $|U|$ strictly decreases.

Aggregation. Each episode costs at most $M-1$ inputs and strictly decreases $|U|$. Since $|U|$ starts at most M and terminates when all surviving states are observationally equivalent, at most $M-1$ episodes occur, giving total depth at most $(M-1)^2$.

The strategy runs breadth-first search on the pair automaton *of the target* A, so it presupposes knowledge of A and is admissible for $L_{\text{sync}}^{\text{adapt}}(A)$. Taking the supremum over $A \in \mathcal{H}_M$ gives the stated bound on $L_{\text{sync}}^{\text{adapt}}(M)$. $\square$

Proposition 13.30 (Universal Synchronization Depth, Version-Space Form)**.** *Let*

$$N_M = \big|\{(A,q) : A \in \mathcal{H}_M,\; q \in Q_A\}\big| \;\leq\; M\,|\mathcal{H}_M|.$$

Then

$$L_{\text{sync}}^{\text{adapt}}(M) \;\leq\; L_{\text{sync}}^{\text{univ}}(M) \;\leq\; (M-1)(N_M-1) \;<\; \infty.$$

In particular $L_{\text{sync}}^{\text{univ}}(M)$ is finite for fixed finite alphabets and fixed M, so the hypothesis $L_{\text{sync}}^{\text{univ}}(M) < \infty$ of Proposition 13.28 is automatic.

Proof. Maintain the version space $V \subseteq \{(A,q) : A \in \mathcal{H}_M,\ q \in Q_A\}$ of machine–state pairs consistent with the transcript so far, initially all of them. The whole strategy is a function of the transcript, since V is, and it is therefore a single decision tree in the sense of Definition 13.17; this is the point on which the machine-specific bounds of Proposition 13.29 fail and this one does not.

While some machine A occurring in V carries two candidates (A,q) and (A,q') in V whose states are not observationally equivalent in A, select one such triple. The learner may compute inside A because A is a *candidate*, named by the version space, not the unknown target. Both q and q' are states of the same M-state machine, so Lemma 13.40 applied within A — rather than to a disjoint union of two machines — supplies a word w with $\lambda_A(q,w) \neq \lambda_A(q',w)$ and $|w| \leq M-1$.

Feed w and delete from V every candidate whose predicted output sequence disagrees with the observation, advancing the survivors along w. At least one of (A,q), (A,q') is deleted: their predicted sequences differ, so at most one can agree with whatever is observed. Hence $|V|$ strictly decreases in each episode. Starting from $|V| = N_M$ and terminating when no consistent machine carries two observationally distinct candidate states, at most $N_M - 1$ episodes occur, each of length at most $M-1$.

At termination the transcript determines the current state of the target within every machine still consistent with it, which is precisely the objective of Definition 13.17. □

Remark 13.31 (Finiteness Is Now Unconditional; the Rate Is Not)**.** Proposition 13.30 settles finiteness, which Remark 13.33 had left open once $|\mathcal{I}| \geq 2$, but it does not settle the rate. The bound is polynomial in the number of candidates $N_M \leq M|\mathcal{H}_M|$ and hence exponential in M, since $|\mathcal{H}_M|$ is itself exponential; it is therefore far weaker in M than the machine-specific $\binom{M}{2}$. Whether $L_{\mathrm{sync}}^{\mathrm{univ}}(M)$ is polynomial in M is Conjecture 13.32, and whether it is $O(M \log M)$ is the surviving branch of Open Problem 13.49.

Conjecture 13.32 (Polynomial Universal Depth)**.** $L_{\mathrm{sync}}^{\mathrm{univ}}(M) = O(M^2)$ for fixed finite alphabets.

Remark 13.33 (The Machine-Specific Bounds Are on $L_{\mathrm{sync}}^{\mathrm{adapt}}$, Not on $L_{\mathrm{sync}}^{\mathrm{univ}}$)**.** Propositions 13.29 and 13.35 bound the machine-specific depth, in which the separating word is chosen by searching the pair automaton of the target. A learner realizing $L_{\mathrm{sync}}^{\mathrm{univ}}$ must fix *one* decision tree in advance and run it against every member of $\mathcal{H}_M$, so it cannot perform that search, and the bounds do not transfer.

The two quantities genuinely differ, already at $M=2$. Over $\mathcal{I} = \{0,1\}$ and $\mathcal{O} = \{0,1\}$ take

$$A_1 : \ \lambda(s,0) = s,\ \lambda(s,1) = 0, \qquad A_2 : \ \lambda(s,0) = 0,\ \lambda(s,1) = s,$$

both with $\tau(s,x) = s$. Each is minimal and each is synchronized by a single input — input 0 for A_1, input 1 for A_2 — so $L_{\mathrm{sync}}^{\mathrm{adapt}} = 1$ on the pair. A universal tree must commit to its first input before learning which machine it faces; whichever it plays, the other machine emits the constant output 0 and nothing is learned, so a second input is required and $L_{\mathrm{sync}}^{\mathrm{univ}} = 2$ on the same pair.

Consequently the finiteness of $L_{\text{sync}}^{\text{univ}}(M)$, and its rate, are *not* settled by these propositions. Finiteness is settled separately, by the version-space strategy of Proposition 13.30, whose input choices depend on the transcript alone; the rate remains open.

Lemma 13.34 (Separation–Size Tension)**.** *Let A be a minimal machine with M states and let $U \subseteq Q_A$ with $|U| \geq 2$. Write*

$$d(U) = \min_{s \neq t \in U} \min\{|w| : \lambda(s,w) \neq \lambda(t,w)\}$$

for the length of the shortest word separating some pair inside U. Then

$$d(U) \;\leq\; M - |U| + 1.$$

Proof. Write $\sim_k$ for the Moore partition after k rounds, so $\sim_0$ is the single block Q_A and $\sim_{k+1}$ refines $\sim_k$ by the value of $\lambda(\cdot, x)$ together with the $\sim_k$-class of $\tau(\cdot, x)$, over all $x \in \mathcal{I}$. By construction $s \sim_k t$ exactly when no word of length at most k separates s from t.

Put $k = d(U)$. No word of length $k-1$ separates any pair inside U, so all of U lies in a single block of $\sim_{k-1}$.

The refinement is strictly increasing until it stabilizes: if a round fails to split then the partition is fixed forever, and minimality forces the stable partition to be discrete. Since $k-1$ rounds have been taken and the process has not yet stabilized — some pair inside U is still unseparated, while minimality guarantees it will be — the block count satisfies

$$\big|Q_A/{\sim_{k-1}}\big| \;\geq\; k.$$

On the other hand U occupies one block and the remaining $M - |U|$ states occupy at most $M - |U|$ further blocks, so

$$\big|Q_A/{\sim_{k-1}}\big| \;\leq\; 1 + \big(M - |U|\big).$$

Combining the two displays gives $k \leq M - |U| + 1$. □

Proposition 13.35 (Machine-Specific Depth Is at Most $\binom{M}{2}$)**.** *For fixed finite alphabets and every $M \geq 2$,*

$$L_{\text{sync}}^{\text{adapt}}(M) \;\leq\; \frac{M(M-1)}{2},$$

The bound is attained at $M = 3$ and $M = 4$ — there is a minimal 4-state machine over binary alphabets with $L_{\text{sync}}^{\text{adapt}}(A) = 6 = \binom{4}{2}$ — but not beyond: see Remark 13.36.

Proof. Run the strategy of Proposition 13.29, but account for each episode with Lemma 13.34 rather than with the crude bound $M - 1$.

Let U be the current uncertainty set. While U contains two states that are not observationally equivalent, feed a shortest word realizing $d(U)$. As in Proposition 13.29, deterministic transitions never increase $|U|$, and at the final letter of that word the observed output eliminates at least one branch, so $|U|$ strictly decreases. The episode costs $d(U) \leq M - |U| + 1$ inputs by Lemma 13.34.

The successive sizes are therefore at worst $M, M-1, \ldots, 2$, and the total is at most

$$\sum_{m=2}^{M} \bigl(M - m + 1\bigr) = \sum_{j=1}^{M-1} j = \frac{M(M-1)}{2}.$$

As in Proposition 13.29 the separating word is found by searching the pair automaton of the target, so the bound is on $L_{\mathrm{sync}}^{\mathrm{adapt}}$ (Remark 13.33).

Attainment at $M = 3, 4$ is a computational observation rather than a proved lower bound: exhaustive search over all minimal machines with $|\mathcal{I}| = |\mathcal{O}| = 2$, under the conventions of Remark 5.29, returns maximum adaptive depth exactly 3 and 6 respectively, the latter realized by 3072 machines. An explicit witness at $M = 4$ is

$$\tau(\cdot, 0) = (0, 0, 3, 2), \quad \tau(\cdot, 1) = (0, 2, 3, 1), \quad \lambda(\cdot, 0) = (0, 1, 0, 0), \quad \lambda(\cdot, 1) \equiv 0,$$

on state set $\{0, 1, 2, 3\}$; the accompanying search program reports its optimal adaptive depth as 6. No hand proof of that lower bound is given here. □

Remark 13.36 (Attainment Is Sporadic, and the Extremal Structure Is Linear)**.** The bound $\binom{M}{2}$ is attained at $M = 3, 4$ and, on the evidence, at no larger M. The distinction matters for Open Problem 13.49, because attainment for all M would give $L_{\mathrm{sync}}^{\mathrm{univ}}(M) = \Theta(M^2) = \omega(M \log M)$ and settle it affirmatively.

Every machine realizing the bound at $M = 3, 4$ has the same shape. One input is a permutation fixing a sink state and cycling the others; the second input is collapsing and carries the *only* informative output, emitted from a single *probe* state. The learner can therefore ask just one question — "is the true state at the probe now?" — and must rotate candidates past the probe between questions. Tracing the optimal play at $M = 4$ makes the cost explicit: the uncertainty set moves $\{0, 1, 2, 3\} \to \{0, 2, 3\} \to \{0, 1, 3\} \to \{0, 2\} \to \{0, 3\} \to \{0, 1\} \to \{0\}$, alternating probe and rotate.

That mechanism does not scale quadratically. Generalizing it to arbitrary M — sink, an $(M-1)$-cycle, and a single probe — gives adaptive depth exactly $2M - 2$, since each elimination costs one rotation step plus one probe step, and the ratio to $M \log_2 M$ *decreases* in M. Exhaustive search at $M = 5$ over the 2,839,200 minimal machines whose first input is a permutation and whose output function has a single probe, in the sense of Remark 5.29, returns maximum depth 9, short of $\binom{5}{2} = 10$; hill-climbing over unrestricted minimal machines returns 9, 14 and 14 at $M = 6, 7, 8$ against $\binom{M}{2} = 15$, 21, 28, a gap that widens.

The evidence therefore points away from a quadratic family and towards $L_{\mathrm{sync}}^{\mathrm{univ}}(M) = O(M)$ for the worst case, which would collapse the length currency exactly as the mistake currency collapsed in Remark 13.24. This is evidence, not proof: the search at $M \geq 6$ is heuristic, and the exhaustive $M = 5$ search covers a structured subclass rather than all minimal machines.

Proposition 13.37 (Single-Input Machines Are Linear)**.** *Let $|\mathcal{I}| = 1$. Then for every minimal $A \in \mathcal{H}_M$ the adaptive synchronization depth satisfies*

$$L_{\mathrm{sync}}^{\mathrm{adapt}}(A) \;\leq\; M - 1,$$

and consequently $L^{\mathrm{adapt}}_{\mathrm{sync}}(M) = L^{\mathrm{univ}}_{\mathrm{sync}}(M) \le M-1$ *for unary input alphabets. In particular* $L^{\mathrm{univ}}_{\mathrm{sync}}(M) = O(M) = o(M \log M)$ *there.*

Proof. With a single input letter the learner has no choice of experiment: the only strategy is to feed that letter repeatedly, so the adaptive tree degenerates to a path and $L^{\mathrm{adapt}}_{\mathrm{sync}}$ and $L^{\mathrm{univ}}_{\mathrm{sync}}$ coincide. This is the one case in which the distinction of Remark 13.33 is vacuous, because there is only one universal tree.

Write a for the letter and define the equivalence relations

$$s \sim_t u \iff \lambda(s, a^k) = \lambda(u, a^k) \ \text{ for all } k \le t,$$

so that $\sim_1$ is the partition of Q_A by the value of $\lambda(\cdot, a)$ and $\sim_{t+1}$ refines $\sim_t$. This is exactly Moore's partition refinement for the unary automaton.

Two observations finish the argument. First, the refinement is monotone and *stabilizes at the first round that fails to split*: if $\sim_{t+1}=\sim_t$ then $\sim_{t'}=\sim_t$ for all $t' \ge t$, since the refinement step is a function of the current partition alone. Second, minimality of A means the stable partition is discrete, with M blocks.

Let R be the number of rounds until stabilization. The block count is at least 1 initially, strictly increases at each of the first $R-1$ rounds by the stabilization property, and equals M at the end; hence $R \le M-1$. After R applications of a the observed output prefix determines the block of $\sim_R$ containing the initial state, and that block is a singleton, so the current state is determined. Therefore $L^{\mathrm{adapt}}_{\mathrm{sync}}(A) \le R \le M-1$. □

Remark 13.38 (The Search for a Witness Needs a Nontrivial Input Alphabet)**.** Proposition 13.37 removes an entire alphabet class from consideration: any family witnessing $L^{\mathrm{univ}}_{\mathrm{sync}}(\mathcal{C}_M) = \omega(M \log M)$ must have $|\mathcal{I}| \ge 2$. The reason is structural rather than quantitative. With one letter there is no adaptivity to exploit — the transcript is a fixed function of the initial state — so the process is pure partition refinement, and refinement can happen at most $M-1$ times because each effective round splits at least one block of a partition of an M-element set. Adaptive depth beyond $M-1$ therefore requires the learner's *choice* of next input to matter, which needs at least two letters.

Exhaustive computation supports this and locates the extremal behaviour precisely. Over all minimal single-input machines with $M \le 6$ and binary output — 8, 72, 960, 16,800 and 362,880 machines for $M = 2, \dots, 6$ — the depth equals the number of refinement rounds in every instance, and the maximum is exactly $M-1$; the block count increases strictly at each round, verified over 9,313,920 minimal machines at $M = 7$ (Remark 5.29).

Remark 13.39 (Consequences, and What Remains Open)**.** Two consequences follow, and one question is sharpened rather than settled.

First, the finiteness hypothesis in Proposition 13.28 is not discharged by *these* bounds, since they concern $L^{\mathrm{adapt}}_{\mathrm{sync}}$ rather than $L^{\mathrm{univ}}_{\mathrm{sync}}$ (Remark 13.33); it is discharged in general by Proposition 13.30, and with a bound polynomial in M rather than in N_M for unary input alphabets by Proposition 13.37.

Second, the quadratic bound does *not* by itself decide whether the length-form bound is dominated by the halving bound of Theorem 13.41. The two rates cross: even the sharpened $\binom{M}{2}$ of Proposition 13.35 exceeds $M \log_2 M$ for $M \geq 7$, so an upper bound of order M^2 leaves room for a family with $L_{\mathrm{sync}}^{\mathrm{univ}}(\mathcal{C}_M) = \omega(M \log M)$. Proposition 13.35 fixes the constant at $\frac{1}{2}$ for $L_{\mathrm{sync}}^{\mathrm{adapt}}$, and the bound is achieved at $M = 3, 4$. It is not achieved beyond (Remark 13.36), and the extremal mechanism at those two values generalizes to a family of depth $2M - 2$ rather than to a quadratic one, so the evidence favours $L_{\mathrm{sync}}^{\mathrm{adapt}}(M) = \Theta(M)$. None of this bounds $L_{\mathrm{sync}}^{\mathrm{univ}}$, by Remark 13.33.

What the evidence suggests is that no such family exists. Exhaustive computation over all minimal machines with $M \leq 5$ states and $|\mathcal{I}|, |\mathcal{O}| \leq 3$, under the conventions of Remark 5.29, gives maximum adaptive depth $M-1$ for single-input machines and never more than $\binom{M}{2}$; the natural candidates — cyclic counters with a single marked state, and the cyclic-shift family of Theorem 13.21 — realize $M - 1$ and $\log_2 M$ respectively, both $o(M \log M)$. Hill-climbing on the ratio $L_{\mathrm{sync}}^{\mathrm{adapt}}(A)/(M \log_2 M)$ over minimal machines up to $M = 20$ never exceeds 0.78 and trends downward. Whether $L_{\mathrm{sync}}^{\mathrm{univ}}(M) = O(M \log M)$ holds in general remains open (Open Problem 13.49); what is now settled is that the answer lies between $M - 1$ and $(M - 1)^2$, so the question is a question about the exponent and not about finiteness.

Lemma 13.40 (Separation Across Two Mealy Machines)**.** *Let A_1, A_2 be deterministic Mealy machines over the same alphabets, with N states in total, and let q_1, q_2 be states of A_1, A_2 respectively whose continuation functions differ. Then some word of length at most $N - 1$ produces different output sequences from q_1 and q_2.*

Proof. Work in the disjoint union $A_1 \sqcup A_2$, a single Mealy machine with N states. For $k \geq 0$ let $\equiv_k$ identify two states when their output sequences agree on every input word of length at most k. Each $\equiv_k$ is an equivalence relation, $\equiv_{k+1}$ refines $\equiv_k$, and the refinement is the standard Moore construction [2, Ch. II].

If $\equiv_{k+1} = \equiv_k$ for some k, then the sequence is stationary from k onwards, since $\equiv_{k+2}$ is determined by $\equiv_{k+1}$ together with the one-step transition and output maps. Hence the relations strictly refine until they stabilize, and $\equiv_0$ has at least one block, so stabilization occurs at some $k^* \leq N - 1$: each strict refinement increases the block count by at least one, and the count cannot exceed N.

By hypothesis $q_1 \not\equiv_\infty q_2$, so $q_1 \not\equiv_{k^*} q_2$, and a word of length at most $k^* \leq N - 1$ witnesses the difference. □

Theorem 13.41 (Unconditional Active Attainment Bound)**.** *Fix finite alphabets $\mathcal{I}, \mathcal{O}$ with $|\mathcal{I}|, |\mathcal{O}| \geq 2$. There is an active learner that, on every $A \in \mathcal{H}_M$ and every initial state, attains objective (RI) of Definition 13.15 after at most*

$$\log_2 \left| \mathcal{H}_M \times Q \right| = O(M \log M)$$

prediction mistakes, the implied constant depending only on the alphabets. Consequently

$$E_{\mathrm{sync}}(M) \;=\; O(M \log M),$$

with no synchronizability, operational-equivalence, or direct-sum hypothesis.

Proof. Let V be the version space: the set of pairs (A, q) with $A \in \mathcal{H}_M$ and q a state of A, consistent with the transcript so far, where the second coordinate is advanced along the observed inputs. Initially $|V| = |\mathcal{H}_M \times Q|$, which is finite.

Call two elements of V *residually distinct* if their continuation functions differ, i.e. if some input word separates their output sequences. Two residually distinct elements (A, q) and (B, p) of V are separated by a word of length at most $2M - 1$, by Lemma 13.40 applied to the disjoint union of A and B, which has at most $2M$ states. The familiar $M - 1$ figure is not available here, since it separates two states of a *single* M-state machine whereas the two elements of V may come from different machines; the disjoint-union bound $2M - 1$ is the correct one. The exact length is in any case immaterial to the mistake count, which is driven by halving and not by word length; only finiteness is used.

The learner repeats the following until all elements of V share one continuation function. Choose two residually distinct elements of V and a shortest word w separating them. Play w one letter at a time; before each letter, predict the output symbol receiving a plurality of votes among the elements of V; after observing the true symbol, delete from V every element predicting a different symbol, and advance the states of the survivors.

Two facts complete the argument. First, each mistake at least halves $|V|$. Fix the input x played at the current step. Every element (A, q) of the version space is a *deterministic* machine together with a state, so it emits the single symbol $\lambda_A(q, x)$; the version space therefore partitions into output classes of sizes $c_1 \geq c_2 \geq \dots$, and the learner predicts the plurality symbol. If the prediction is wrong, the observed symbol identifies one class other than the plurality class, and the survivors are *that single class*, of size at most c_2 — not the union of all non-predicted classes. Since $c_1 \geq c_2$ and $c_1 + c_2 \leq |V|$,

$$2c_2 \;\leq\; c_1 + c_2 \;\leq\; |V|, \qquad \text{so} \qquad c_2 \;\leq\; \tfrac{1}{2}|V|.$$

This is the same counting as in Proposition 13.19, and it applies verbatim here: what it needs is that each surviving candidate predicts one deterministic symbol, which holds for machine–state pairs exactly as it does for states of a known machine. Consequently the number of mistakes is at most $\lfloor \log_2 |\mathcal{H}_M \times Q| \rfloor$, with no dependence on $|\mathcal{O}|$. Second, the procedure terminates: playing a separating word w deletes at least one of the two chosen elements, so $|V|$ strictly decreases on each iteration, and V is finite and never empty since the realizable target is always retained.

At termination all surviving elements have the same continuation function, and the target is among them, so the continuation function from the current history is determined by the transcript: objective (RI) is attained. Being prediction-closing, it incurs no further mistakes. Finally $\log_2 |\mathcal{H}_M \times Q| = O\big(M|\mathcal{I}| \log(M|\mathcal{O}|)\big) = O(M \log M)$ for fixed alphabets, and $E_{\text{sync}}(M)$ is an infimum over learners by Definition 13.18, so $E_{\text{sync}}(M) \leq \log_2 |\mathcal{H}_M \times Q| = O(M \log M)$. □

Remark 13.42 (What Becomes Unconditional, and What Does Not)**.** Corollary 13.47 settles the *order* of the active realizable mistake complexity without any of the auxiliary hypotheses used earlier in this subsection. In particular Lemma 13.25 and the direct-sum saturation condition are not needed for the $\Theta(M \log M)$ statement, and the finiteness of $L_{\mathrm{sync}}^{\mathrm{univ}}(M)$ is not needed either: the halving learner of Theorem 13.41 never performs a synchronization experiment and is well defined even when no machine in $\mathcal{H}_M$ admits one.

What those hypotheses still buy is finer information. Parts (II) and (III) of Theorem 13.27 are statements about $E_{\mathrm{sync}}(M)$ as an abstract complexity, and remain the right form when E_{sync} is replaced by the corresponding quantity for a *subclass* of $\mathcal{H}_M$, where the counting bound $\log_2 |\mathcal{H}_M|$ need not be $O(M \log M)$ and the two terms of the additive form can separate. The additive form is therefore not vacuous, but on the full class $\mathcal{H}_M$ it collapses: $M \log M + E_{\mathrm{sync}}(M) = \Theta(E_{\mathrm{sync}}(M))$.

Remark 13.43 (No Automatic Active Lower Bound)**.** An active lower bound does not follow from the passive theory, and must be established by an explicit adversarial construction. Two standard arguments fail here. First, ordinary Littlestone lower bounds are proved against an *adversarial* input sequence; when the learner chooses the inputs, the adversary no longer controls the instance selection game, so those constructions do not transfer. Second, a counting argument of the form "each mistake conveys at most a constant amount of information" is invalid, because a *correctly predicted* output also reveals information about the target and costs the learner nothing.

The obstruction is to those two methods, not to the conclusion. The gated family of Definition 13.44 supplies an explicit construction and yields the unconditional bound $\Omega(M \log M)$ of Corollary 13.47. What requires a further hypothesis is the *separation* of the two terms in the additive form, which uses the direct-sum saturation condition of Definition 13.26.

Definition 13.44 (Gated Active Family $\mathcal{G}_M^{\mathrm{act}}$)**.** Let $Q = \{0,1\}^L$, $M = 2^L$, and let the state set be $Q \times \{\mathsf{f}, \mathsf{r}\}$, of size $2M$. Write $\mathrm{rot}(v) = v_2 \cdots v_L v_1$ and let $\mathrm{flip}(v)$ flip the last bit of v. For an arbitrary map $g : Q \to Q$ the machine $A_g \in \mathcal{G}_M^{\mathrm{act}}$ has transitions and Mealy outputs

$$\begin{aligned}
(\mathsf{f}, v) :\quad & \mathtt{r} \mapsto (\mathsf{f}, 0^L)/0, \quad \mathtt{e} \mapsto (\mathsf{f}, \mathrm{flip}(v))/0, \quad \mathtt{d} \mapsto (\mathsf{f}, \mathrm{rot}(v))/v_1, \quad \mathtt{c} \mapsto (\mathsf{r}, g(v))/0,\\
(\mathsf{r}, u) :\quad & \mathtt{r} \mapsto (\mathsf{f}, 0^L)/0, \quad \mathtt{e} \mapsto (\mathsf{r}, u)/0, \qquad\quad\;\; \mathtt{d} \mapsto (\mathsf{r}, \mathrm{rot}(u))/u_1, \quad \mathtt{c} \mapsto (\mathsf{r}, u)/0.
\end{aligned}$$

The single structural change from the family of Theorem 13.5 is the last entry: in read mode $\mathtt{c}$ is a *no-op*. The mode is visible in the output alphabet if desired; this only helps the learner. Only $\mathtt{c}$ from free mode depends on g, so $|\mathcal{G}_M^{\mathrm{act}}| = M^M$ and $\mathcal{G}_M^{\mathrm{act}} \subseteq \mathcal{H}_{2M}$.

The *gating property* is the following. Since read mode is entered only by playing $\mathtt{c}$ from free mode, and $\mathtt{c}$ does nothing thereafter, the register in read mode is always a cyclic rotation of $g(v)$ for the single argument v that was the free-mode state when $\mathtt{c}$ was played. That argument was reached by the learner's own inputs from the fixed state 0^L using only the g-independent letters $\mathtt{r}, \mathtt{e}, \mathtt{d}$, and is therefore known to the learner. Consequently every g-dependent output round reports a bit of g at a learner-known argument, and the set of maps consistent with any transcript is exactly a product

$$\prod_{v \in Q} S_v, \qquad S_v \subseteq Q,$$

of independent per-argument constraints.

Theorem 13.45 (Gated Family and Unconditional Active Lower Bound)**.** *Let $\mathcal{I} = \{\mathtt{r}, \mathtt{e}, \mathtt{d}, \mathtt{c}\}$ and $\mathcal{O} = \{0, 1\}$, and let $\mathcal{G}_M^{\mathrm{act}}$ be the* gated *family of Definition 13.44 on $2M$ states, $M = 2^L$. Consider an active learner that chooses all inputs and is required to attain objective (RI) of Definition 13.15. Then every* deterministic *such learner incurs at least*

$$M \log_2 M = \Omega(M \log M)$$

mistakes in the worst case over $\mathcal{G}_M^{\mathrm{act}}$. For every randomized *such learner there is a fixed target in $\mathcal{G}_M^{\mathrm{act}}$ attaining the bound below, by averaging over the uniform target distribution; explicitly, every randomized learner satisfies*

$$\mathbb{E}[\textit{mistakes}] \ \geq \ \tfrac{1}{2} M \log_2 M = \Omega(M \log M),$$

and, under the uniform target distribution on Q^Q, for every $\eta \in (0, 1)$,

$$\Pr\left[\textit{mistakes} \ \geq \ \left(\tfrac{1}{2} - \eta\right) M \log_2 M\right] \ \geq \ 1 - \exp\left(-\tfrac{1}{2}\eta^2 M \log_2 M\right).$$

This family supplies one *component; the two-component direct-sum instance required by Definition 13.26 is assembled from two independent copies in Theorem 13.48. The unknown datum here is the map $g : Q \to Q$, so what is forced is* model *identification, not state identification within a known skeleton; accordingly $\mathcal{G}_M^{\mathrm{act}}$ realizes a component of complexity*

$$S_M = \Omega(M \log M)$$

against objective (RI), and not the state-identification quantity $E_{\mathrm{sync}}^{\mathrm{SI}}$ of Definition 13.18, and $E_{\mathrm{sync}}(M) = \Omega(M \log M)$ on this family. A two-phase instance, exhibiting a separate continuation cost C_M, is constructed in Theorem 13.48.

Gating doubles the state count, so $\mathcal{G}_M^{\mathrm{act}}$ lies in $\mathcal{H}_N$ with $N = 2M$. The bound is unaffected in order: writing it against the true budget N gives

$$\tfrac{N}{2} \log_2 \tfrac{N}{2} = \Omega(N \log N),$$

since $\log_2(N/2) = \log_2 N - 1 \geq \frac{1}{2} \log_2 N$ for $N \geq 4$. All statements below are therefore valid verbatim with M read as the state budget, at the cost of absolute constants only.

Proof. Throughout, the learner chooses every input; the adversary answers.

Step 1: attaining (RI) requires knowing g everywhere. Suppose $g(v)$ is undetermined by the transcript for some $v \in Q$. From the current state the learner may play $\mathtt{r}\, w_v\, \mathtt{c}\, \mathtt{d}^L$, where w_v is the transport word of Theorem 13.5; this reaches (f, v) using only g-independent letters, enters read mode at $(\mathsf{r}, g(v))$, and reads out all L bits of $g(v)$. Two members of $\mathcal{G}_M^{\mathrm{act}}$ consistent with the transcript disagree on that continuation, so the continuation function is not determined and (RI) has not been attained. Since $|\mathcal{G}_M^{\mathrm{act}}| = M^M$, attaining (RI) requires fixing all $M \log_2 M$ bits of g.

Step 2: every g-dependent round is a readout at a known argument. The letters $\mathtt{r}, \mathtt{e}, \mathtt{d}$ have g-independent transitions and outputs in both modes, and $\mathtt{c}$ emits the constant 0 in both modes, so no round using them at a g-independent state reveals anything about

g. The only g-dependent outputs are the `d` rounds in read mode, each emitting one bit of the register. By the gating property of Definition 13.44 the register is a rotation of $g(v)$ for a single argument v determined by the learner's own inputs, hence known to it. This is the only step at which gating is used, and Remark 13.46 shows it fails without it.

Step 3: the lazy adversary is well defined. Maintain a partial assignment recording, for each $v \in Q$ and each coordinate $j \leq L$, either a fixed bit or the symbol $*$. When the learner plays `d` in read mode with register $\mathrm{rot}^o(g(v))$ and coordinate $j = o \bmod L$ still $*$, the adversary answers with the complement of the learner's prediction and records that bit. The answer is legal precisely because, by Step 2, the consistent set is at every moment the product $\prod_v S_v$ of per-argument constraints, so fixing one coordinate of $g(v)$ leaves every completion at every other argument available; no earlier round constrained it. Each of the $M \log_2 M$ bits is therefore revealed on a round that is a forced mistake, and no bit is ever revealed on a correctly predicted round. Rounds where the coordinate is already recorded are predicted correctly and cost nothing, so no bit is paid for twice.

By Step 1 the learner must reach the all-recorded assignment, so it incurs at least $M \log_2 M$ mistakes. The learner's control of the input sequence determines only *when* each bit is paid for, not *whether*.

The argument does not proceed by counting information per mistake, which would be invalid in the active protocol because correctly predicted rounds may also be informative (Remark 13.43). It rests instead on the structural property of Step 2: in this family every information-bearing round is a forced mistake.

Step 4: randomized learners. Here the fixed-stream form of Yao's principle used in Theorem 13.5(ii) is *not* available, since an active learner chooses its own inputs and a fixed stream lower-bounds only those learners that happen to follow it. We argue directly on the adaptive transcript.

We first fix the accounting convention for learners that fail. By Definition 13.18 a learner that never attains (RI) is charged $\mathrm{mis} = +\infty$, so such learners cannot lower the infimum and may be excluded; equivalently, the bound below is asserted for learners attaining the objective almost surely, and is vacuously true otherwise. Without such a convention no unconditional expectation bound could hold, since a learner could attain (RI) only on an event of small probability and idle otherwise.

Draw g uniformly from Q^Q; equivalently the M values $g(v)$ are independent and uniform on Q, so the $M \log_2 M$ bits of g are i.i.d. fair. Fix any randomized active learner attaining (RI) almost surely, and let $\mathcal{F}_{t-1}$ be the σ-field generated by the transcript through round $t-1$ together with the learner's internal randomness. Call round t a *first emission* if it is a read-mode `d` round whose coordinate (v, j) has not been emitted at any earlier round, and let N be the set of first-emission rounds.

By Step 2 the coordinate (v, j) read at round t is $\mathcal{F}_{t-1}$-measurable, since the argument v is determined by the learner's own inputs and j by the number of read-mode shifts since `c`. By the product structure of Step 2, the transcript through round $t-1$ constrains only coordinates already emitted, and (v, j) is not among them; hence $g(v)_j$ is independent of

$\mathcal{F}_{t-1}$ and uniform on $\{0,1\}$. The prediction $\hat{y}_t$ is $\mathcal{F}_{t-1}$-measurable, so on the event $\{t \in N\}$,

$$\Pr\left[\hat{y}_t \neq g(v)_j \,\middle|\, \mathcal{F}_{t-1}\right] = \tfrac{1}{2}.$$

By Step 1, attaining (RI) forces every one of the $m = M \log_2 M$ coordinates to be emitted, so N has exactly m elements. The rounds of N occur at random times, so we index them in order of occurrence rather than by t: let $\sigma_1 < \sigma_2 < \cdots < \sigma_m$ enumerate N, each σ_k being a stopping time with respect to $(\mathcal{F}_t)$, and set $\mathcal{G}_k = \mathcal{F}_{\sigma_k - 1}$. The σ-fields $\mathcal{G}_1 \subseteq \cdots \subseteq \mathcal{G}_m$ form a filtration by the optional stopping theorem for discrete filtrations. Put

$$Y_k = \mathbf{1}\{\hat{y}_{\sigma_k} \neq g(v)_j \text{ at round } \sigma_k\} - \tfrac{1}{2}, \qquad k = 1, \ldots, m.$$

By the display above, $\mathbb{E}[Y_k \mid \mathcal{G}_k] = 0$ and $|Y_k| \leq \frac{1}{2}$, so $(Y_k)_{k \leq m}$ is a bounded martingale difference sequence indexed by a *deterministic* range $1, \ldots, m$. The total number of mistakes is at least $\sum_{k=1}^{m} Y_k + \frac{m}{2}$, so taking expectations gives $\mathbb{E}[\text{mistakes}] \geq \frac{m}{2} = \frac{1}{2} M \log_2 M$. Azuma–Hoeffding applied to $\sum_{k=1}^{m} Y_k$ [26, Ch. 6] gives, for every $\eta \in (0,1)$,

$$\Pr\left[\text{mistakes} < \left(\tfrac{1}{2} - \eta\right) M \log_2 M\right] \leq \exp\left(-\tfrac{1}{2}\eta^2 M \log_2 M\right),$$

which is the stated high-probability form. No independence among rounds is assumed, and no input stream is fixed in advance.

Step 5: disjoint accounting. The readout rounds carrying g-information occur in read mode, and the transport rounds w_v occur in free mode, so the two sets of rounds are disjoint. This is condition (iii) of Definition 13.26. □

Remark 13.46 (Why the Passive Family Must Be Gated)**.** Gating is what separates the active setting from the passive one. In the passive protocol the adversary fixes the input stream, and in the stream of Theorem 13.5 the letter c is always issued from a state v that the transport word w_v has just established, so every readout reports g at a known argument. An active learner is under no such discipline: playing c twice in succession moves the machine to $g(g(v))$, and a readout there constrains g at an argument that is itself undetermined.

Under such chaining the set of maps consistent with a transcript is not a product of independent per-argument constraints. For $L = 2$ a single chained readout $g(g(0^L)) = u$ leaves 48 of the 256 maps consistent, whereas the product of the per-argument projections has $3 \cdot 4 \cdot 4 \cdot 4 = 192$ elements. Both the per-bit adversary, which fixes one bit of one $g(v)$ per readout round and requires every completion to remain available, and the conditional-uniformity step of the Yao argument, which requires each queried bit to be fair given the transcript, depend on that product structure.

Making c a no-op in read mode, as in Definition 13.44, confines every readout argument to the free-mode state the learner transported to, at the cost of doubling the state count. The product structure holds throughout, and with it both arguments.

Corollary 13.47 (Active Realizable Mistake Complexity, Unconditionally)**.** *For fixed alphabets with $|\mathcal{I}| \geq 4$ and $|\mathcal{O}| \geq 2$, and for* all *sufficiently large M,*

$$E_{\text{sync}}(M) = \Theta(M \log M) \qquad \textit{and} \qquad \text{Mistakes}_{\text{active,RI}}(M) = \Theta(M \log M),$$

with no auxiliary hypothesis. The same holds for every $|\mathcal{I}| \geq 2$ by the block encoding of Theorem 13.8.

Proof. The upper bound $E_{\text{sync}}(M) = O(M \log M)$ is Theorem 13.41, and it holds for every M.

For the lower bound, first note that the gated family is defined only for $M' = 2^L$ and occupies $2M'$ states, so it yields a bound along the subsequence

$$N_L = 2^{L+1}, \qquad L = 1, 2, \ldots,$$

and not immediately at every budget. Along that subsequence, $\mathcal{G}^{\text{act}}_{2^L} \subseteq \mathcal{H}_{N_L}$ forces every active learner attaining (RI) to make at least $2^L \log_2 2^L = \frac{1}{2} N_L(\log_2 N_L - 1)$ mistakes by Theorem 13.45. Since E_{sync} is an infimum over learners of a supremum over the whole class, and $\mathcal{G}^{\text{act}}_{2^L}$ is one subfamily, $E_{\text{sync}}(N_L) \geq \frac{1}{4} N_L \log_2 N_L$ for $N_L \geq 4$.

To pass from the subsequence to all budgets, observe that $\mathcal{H}_M \subseteq \mathcal{H}_{M+1}$, so E_{sync} is nondecreasing in M, and that $N_{L+1}/N_L = 2$, a bounded ratio. Lemma 13.9, applied with $\alpha = 2$, therefore gives a constant $\beta' > 0$ with

$$E_{\text{sync}}(M) \;\geq\; \beta' M \log M$$

for all sufficiently large M. Combined with the upper bound, $E_{\text{sync}}(M) = \Theta(M \log M)$ for all sufficiently large M.

For the mistake complexity, Theorem 13.27(I) gives $\text{Mistakes}_{\text{active,RI}}(M) \leq O(E_{\text{sync}}(M)) = O(M \log M)$, and the same family, extended by the identical monotonicity argument, gives the matching lower bound, since attaining (RI) is the objective. $\square$

Theorem 13.48 (Explicit Two-Phase Direct-Sum Instance)**.** *There is a family of $O(M)$-state machines over fixed finite alphabets, and a pair of switch letters $\mathtt{s}_1, \mathtt{s}_2$, realizing a direct-sum active instance in the sense of Definition 13.26: both*

$$S_M = \Omega(M \log M) \qquad \textit{and} \qquad C_M = \Omega(M \log M)$$

are forced, on disjoint sets of rounds. Consequently

$$\text{Mistakes}_{\text{active,RI}}(M) \;\geq\; S_M + C_M = \Omega(M \log M),$$

Both components are model-identification components, each of complexity $\Theta(M \log M)$, so their sum is again $\Theta(M \log M)$ and this family does not *exhibit a separation between the two terms of the additive form. What it establishes is that two independent unknown maps must each be paid for on disjoint rounds.*

Proof. Split the budget as $M = M_1 + M_2 + O(1)$ with $M_1 \asymp M_2 \asymp M$.

Component one. On the state block of size M_1, install the gated transport-readout family $\mathcal{G}^{\text{act}}_{M_1}$ of Definition 13.44, whose unknown datum is a map $g : Q_1 \to Q_1$. Determining g in full forces $S_M = \Omega(M_1 \log M_1) = \Omega(M \log M)$ mistakes, by Theorem 13.45.

Switch. Adjoin *two* letters $\mathtt{s}_1, \mathtt{s}_2$, where $\mathtt{s}_j$ forces entry into block j from anywhere:

$$\tau(q, \mathtt{s}_j) = \iota_j \quad \text{for every state } q, \qquad \lambda(q, \mathtt{s}_j) = 0,$$

ι_j being the distinguished initial state of block j. Tag the output alphabet so the learner can see which block is active; this only helps the learner and does not weaken the bound.

Both letters are idempotent and total, so for each j and each word u the composite $\mathtt{s}_j\, u$ reaches block j and executes u from ι_j irrespective of the current state. The forcing argument below uses exactly this: a single word, valid from every configuration, that drives the machine into a prescribed block. Totality is also what removes any evasion, since in the active protocol the learner chooses its own inputs and block one remains reachable at every time.

Component two. On the state block of size M_2, install a second independent copy of the gated transport-readout family, with its own unknown map $g' : Q_2 \to Q_2$. Restricted to inputs that keep the machine in Q_2, the interaction is exactly a fresh instance of Theorem 13.45 on M_2 states, forcing $C_M = \Omega(M_2 \log M_2) = \Omega(M \log M)$ mistakes.

Both components are forced. Suppose the learner has attained objective (RI) at some time t, so that it can predict every continuation with no further mistakes. Let b be a bit of g, say the j-th bit of $g(v)$ for some $v \in Q_1$, that is not determined by the transcript. From the current state – *whatever* it is, and in particular whichever block it lies in – the learner may play

$$\mathtt{s}_1\, w_v\, \mathtt{c}\, \mathtt{d}^{L_1},$$

which reaches ι_1 by the first letter, transports to (f, v) within block one by w_v, enters read mode, and reads out all L_1 bits of $g(v)$, among them b. Every letter of this word is available at every state, so no case distinction on the current block is needed. Since b is unconstrained by the transcript, two members of the family agree on everything observed so far and disagree on that continuation, so the continuation function is not determined and (RI) has not been attained – a contradiction. Hence every one of the $M_1 \log_2 M_1$ bits of g is determined by the transcript, and, by the same argument with $\mathtt{s}_2$ and g', every one of the $M_2 \log_2 M_2$ bits of g'. By Theorem 13.45, Step 3, each such bit was determined on a round that was a forced mistake.

Disjointness and additivity. A readout of g occurs only while the machine is in Q_1, and a readout of g' only while it is in Q_2; the two blocks are disjoint, so the two sets of forced-mistake rounds are disjoint. Moreover no observation in one block constrains the unknown map of the other, the two maps being independent coordinates of the target. Therefore the total is at least $S_M + C_M$.

Conditions (iv) and (v). The construction satisfies the unavoidability requirement of Definition 13.26: because $\mathtt{s}_1$ and $\mathtt{s}_2$ are total, both blocks are reachable from every state at every time, so neither component can be bypassed, and the argument of the previous paragraph shows the objective forces both. Unavoidability is here a property of the transition structure rather than a restriction imposed on the protocol, as condition (iv) requires.

Product structure, condition (v), also holds: the family is indexed by the pair $(g, g') \in Q_1^{Q_1} \times Q_2^{Q_2}$, the two maps are chosen independently, and every combination yields a machine of the family. The parameter set is therefore the full Cartesian product of its two coordinate ranges, as condition (v) of Definition 13.26 requires.

The state count is $M_1 + M_2 + O(1) = O(M)$, so this is a direct-sum active instance.

The additive conclusion here does *not* route through Theorem 13.52 and so does not need its rectangular hardness hypothesis. The adversary above is lazy on each component independently: every coordinate of g and of g' must be emitted before the objective is attained, and each coordinate's first emission is a forced mistake whatever the learner has seen on the other component, because the two maps are independent and the answering rule commits nothing about one while responding about the other. Summing the forced mistakes over the two disjoint coordinate sets gives $S_M + C_M$ directly.

No matching upper bound of the form $O(S_M + C_M)$ is claimed, and none is proved here. The available upper bound is $\mathrm{Mistakes}_{\mathrm{active,RI}}(M) \le O(E_{\mathrm{sync}}(M))$ from Theorem 13.27(I), stated in terms of the aggregate synchronization complexity; it is not an upper bound in the two components separately, and no composite learner is exhibited whose mistake count is $O(S_M + C_M)$. For the family constructed here both quantities happen to be $\Theta(M \log M)$, so the lower and upper bounds do coincide in order for this family, but that coincidence is a property of the instance and not a general additive equality. □

Open Problem 13.49 (A Non-Degenerate Two-Term Active Decomposition)**.** Theorem 13.21 shows that state identification in a known minimal skeleton costs $\Theta(\log M)$ mistakes, so no decomposition of the active complexity whose second term is a state-identification *mistake* complexity can be non-degenerate. This settles the mistake currency negatively and leaves the question of currency itself.

Is there a currency in which a genuine two-term decomposition holds? The natural candidate is experiment *length*: exhibit a family for which $L^{\mathrm{univ}}_{\mathrm{sync}}(\mathcal{C}_M) = \omega(M \log M)$, so that Proposition 13.28 is not dominated by the halving bound of Theorem 13.41, or prove that $L^{\mathrm{univ}}_{\mathrm{sync}}(M) = O(M \log M)$ always.

Propositions 13.29 and 13.35 localize the *machine-specific* quantity,

$$M - 1 \;\le\; L^{\mathrm{adapt}}_{\mathrm{sync}}(M) \;\le\; \binom{M}{2},$$

the upper bound attained at $M = 3, 4$ but apparently not beyond and the lower bound from the counter family, both in Remark 13.36, whose evidence in fact favours $L^{\mathrm{adapt}}_{\mathrm{sync}}(M) = \Theta(M)$.

For the universal quantity $L^{\mathrm{univ}}_{\mathrm{sync}}$, which is the one appearing in Proposition 13.28, none of this transfers: by Remark 13.33 a universal tree cannot search the target's pair automaton, and the two quantities already differ at $M = 2$. Finiteness holds in general by Proposition 13.30, but with a bound exponential in M; for unary alphabets Proposition 13.37 gives $M - 1$. What is open is the rate: whether $L^{\mathrm{univ}}_{\mathrm{sync}}(M)$ is polynomial in M (Conjecture 13.32), and if so whether it exceeds $M \log M$.

Proposition 13.37 narrows the search: any witness must use $|\mathcal{I}| \ge 2$, since unary input alphabets give $L^{\mathrm{univ}}_{\mathrm{sync}}(M) \le M - 1$ — the one regime where the universal and machine-specific depths coincide.

The tension between the two effects a witness must combine — many separation episodes, and a long word within an episode — is now precise rather than heuristic. Lemma 13.34 states it as $d(U) \le M - |U| + 1$: a set that resists splitting for many rounds must be *small*, because each round that fails to split it still splits something else, and there are only M states to divide. Summing the trade-off over episodes yields $\binom{M}{2}$. The remaining question is whether the worst case over a *family* can be pushed past $M \log M$, which is a statement about lower bounds and not about the counting above. The natural extremal mechanism — a single probe state with the other candidates rotated past it — realizes only $2M - 2$, and searches at $M \le 8$ find nothing quadratic (Remark 13.36), so the likely answer is that no such family exists and $L^{\mathrm{univ}}_{\mathrm{sync}}(M) = O(M)$.

Remark 13.50 (Necessity of the Identification Objective)**.** Some objective of this kind is unavoidable. In the bare active protocol, where the learner chooses every input and is scored only on the rounds it chooses, the learner may play `r` forever and make no mistakes at all while learning nothing. A nontrivial active lower bound therefore requires an identification, coverage, or guaranteed-future-prediction requirement, and Theorem 13.45 supplies the bound relative to the guaranteed-prediction objective already fixed in Definition 13.15.

Remark 13.51 (Direct-Sum Saturation Alternative)**.** Instead of Lemma 13.25, which requires determinism, one may assume an explicit direct-sum saturation condition: there exists a family of synchronization-hard machines and Littlestone-hard continuation machines whose sequential composition uses $O(M)$ states and forces any active learner to pay both costs. Under that hypothesis, the same additive lower bound follows. This alternative is made precise in Definition 13.26 and Theorem 13.52, which do not rely on Lemma 13.25.

Theorem 13.52 (Direct-Sum Additive Lower Bound)**.** *Let a direct-sum active instance of budget M have component complexities S_M and C_M, and assume in addition* rectangular hardness*: there are distributions Π_1, Π_2 on the two coordinate families such that, for every composite learner run under the product law $\Pi_1 \otimes \Pi_2$, the expected number of mistakes on component j, conditioned on the entire transcript of the other component, is at least the component bound. Then*

$$\mathrm{Mistakes}_{\mathrm{active,RI}}(M) \ge S_M + C_M.$$

The bound is a sum, not a maximum, and the additivity comes from the disjointness of the two round sets together with rectangular hardness.

No two-sided statement follows. Converting the lower bound into an equality would require an explicit composite learner attaining $S_M + C_M$, which is not constructed here. Moreover, by Proposition 13.19 the state-identification mistake complexity is $O(\log M)$, so a decomposition in which one component is a state-identification cost is necessarily degenerate (Remark 13.24); the components of Theorem 13.48 are both model-identification components, each of complexity $\Theta(M \log M)$.

Proof. No temporal ordering of the two components is assumed; in the active protocol the learner may interleave them freely.

Rectangular hardness is what converts two separate lower bounds into a sum, and it cannot be dispensed with. Conditions (iii)–(v) of Definition 13.26 give, for each component separately, a target forcing that component's bound; but those are targets of *different* runs, and a supremum over each coordinate separately yields $\max\{S_M, C_M\}$, not $S_M + C_M$. Under the product law the two conditional expectations add, by linearity of expectation applied to the conditional counts, which is the step performed below.

Let $\mathcal{A}$ be any learner attaining the objective on the composite instance. By condition (iv), unavoidability, $\mathcal{A}$ must determine the unknown datum of each component, and this is a property of the machine family rather than a restriction on the protocol, so it holds however $\mathcal{A}$ chooses its inputs.

Restriction. For $j = 1, 2$ let R_j be the set of rounds on which component j forces a mistake. We build a learner $\mathcal{A}_1$ for component 1 in isolation; the construction for $\mathcal{A}_2$ is symmetric.

Since $\mathcal{A}$ chooses its inputs adaptively using outputs from *both* components, while the outputs of component 2 are not determined by anything $\mathcal{A}_1$ observes, the simulation must supply those outputs from data available to $\mathcal{A}_1$. Condition (v), product structure, is what makes this possible. Fix once and for all an arbitrary value θ_2^0 of the second unknown. Define $\mathcal{A}_1$ on a component-1 instance θ_1 as follows: run $\mathcal{A}$ internally; whenever $\mathcal{A}$ issues an input whose round belongs to component 2, answer it from θ_2^0, which $\mathcal{A}_1$ knows; whenever the round belongs to component 1, forward it to the real interaction and return the observed output.

By condition (v) the composite target (θ_1, θ_2^0) lies in the family, so the internal run of $\mathcal{A}$ is a genuine run against a genuine target; by condition (vi) the outputs $\mathcal{A}_1$ must supply on component-2 rounds are determined by θ_2^0 alone, so it can supply them, and the outputs on component-1 rounds are determined by θ_1 alone, so they agree with its own interaction. The component-1 rounds of the internal run therefore coincide exactly with the interaction $\mathcal{A}_1$ has with its own instance. Hence $\mathcal{A}_1$ is a well-defined component-1 learner making precisely $|R_1|$ mistakes on θ_1. Since $\mathcal{A}$ attains the objective, by condition (iv) it determines θ_1, and therefore so does $\mathcal{A}_1$. By the definition of S_M and C_M as the minimax mistake complexities of the components,

$$|R_1| \;\geq\; S_M, \qquad |R_2| \;\geq\; C_M$$

in the worst case over the composite family.

Summation. By condition (iii), disjoint accounting, $R_1 \cap R_2 = \emptyset$, so the total number of forced-mistake rounds is $|R_1| + |R_2|$ regardless of the order in which they occur. Hence

$$\mathrm{Mistakes}_{\mathrm{active,RI}}(M) \geq S_M + C_M.$$

This is the stated bound. No upper bound is derived: the unconditional upper bound bounds $\mathrm{Mistakes}_{\mathrm{active,RI}}$ by $O(E_{\mathrm{sync}}(\mathcal{C}_M))$, and identifying $E_{\mathrm{sync}}(\mathcal{C}_M)$ with $S_M + C_M$ would require exhibiting a learner that attains the sum, which the restriction argument does not provide. □

Remark 13.53 (When Length Gives a Lower Bound)**.** If every step of every optimal synchronization experiment has constant output ambiguity, then a synchronization-length lower bound may be converted into a mistake lower bound. In that case, if

$$E_{\mathrm{sync}}(M) = \Omega(L_{\mathrm{sync}}^{\mathrm{adapt}}(M)),$$

then the additive theorem may be written in terms of $L^{\mathrm{adapt}}_{\mathrm{sync}}(M)$:

$$\mathrm{Mistakes}_{\mathrm{active,RI}}(M) = \Theta(M \log M + L^{\mathrm{adapt}}_{\mathrm{sync}}(M)).$$

Without such an ambiguity hypothesis, synchronization length alone does not lower-bound prediction mistakes.

Remark 13.54 (Additive Synchronization Cost)**.** On a subclass for which a direct-sum instance exists, the attainment cost is additive: it does not replace the $\Theta(M \log M)$ term, and active synchronization removes the unknown-initial-state penalty only by paying the cost of the experiment. On the full class $\mathcal{H}_M$ the two terms are of the same order by Corollary 13.47, so additivity carries no extra information there.

Remark 13.55 (Oracle Integration)**.** In the oracle bridge of Section 9, the passive realizable estimation rate is

$$\mathsf{E}^{\mathrm{passive}}_M(T) = \Theta(M \log M),$$

while on the full class the active realizable estimation rate is, by Corollary 13.47,

$$\mathsf{E}^{\mathrm{active}}_M(T) = \Theta(M \log M)$$

unconditionally, so the two protocols enter the oracle bridge with rates of the same order. On a subclass the rate is $\Theta(E_{\mathrm{sync}}(\mathcal{C}_M))$ by Theorem 13.27(I). In either case the oracle inequality itself is unchanged; only the estimation rate input is modified.

13.5 Agnostic Setting

In the agnostic setting, the target sequence need not be realized by any M-state Mealy machine. The learner competes against the best function in $\mathcal{H}_M$.

Theorem 13.56 (Agnostic Regret)**.** *For the agnostic setting,*

$$\inf_{\mathcal{A}} \sup_{(x_t, Y_t)} R^{\mathrm{agn}}_T(\mathcal{A}) = O\big(\sqrt{T\, M \log M}\big),$$

up to constants depending on the alphabets and possible logarithmic refinements. The matching lower bound also holds, in the persistent-stream protocol as well as the reset-word protocol, so

$$\inf_{\mathcal{A}} \sup_{(x_t, Y_t)} R^{\mathrm{agn}}_T(\mathcal{A}) = \Theta\big(\sqrt{T\, M \log M}\big).$$

The two-sided statement is asserted in the regime

$$T \gtrsim M \log M,$$

that is, for horizons at least of the order of the Littlestone dimension. For shorter horizons the lower bound saturates: the adversary cannot force more than one mistake per round, so the minimax regret is $\Theta(T)$ when $T = O(M \log M)$, and the correct uniform statement is $\Theta\big(\min\{T, \sqrt{T\, M \log M}\}\big)$.

Proof. By Lemma 13.1,

$$\mathrm{Ldim}(\mathcal{H}_M) = \Theta(M \log M).$$

Since $\log|\mathcal{H}_M| = O(M \log M)$, finite-expert aggregation over the labeled machines gives the unconditional upper bound

$$R_T^{\mathrm{agn}} \le \sqrt{2T \log|\mathcal{H}_M|} = O\big(\sqrt{T\, M \log M}\big).$$

For the lower bound, the minimax theorem for online learning [10, 11] gives

$$R_T^{\mathrm{agn}} = \Omega\big(\sqrt{T\ \mathrm{Ldim}(\mathcal{H}_M)}\big)$$

whenever the Littlestone dimension is witnessed by a mistake tree realizable in the protocol at hand. Theorem 13.5 supplies exactly such a tree: at each of its $M \log_2 M$ readout rounds *both* output values remain consistent with some member of $\mathcal{G}_M \subseteq \mathcal{H}_M$, because the corresponding bit of g is unconstrained at that moment. The adversary's decision tree is therefore a complete shattered mistake tree of depth $M \log_2 M$ embedded in a single never-reset stream, whence

$$\mathrm{Ldim}^{\mathrm{stream}}(\mathcal{H}_M) \ \ge\ M \log_2 M,$$

and the reverse inequality $\mathrm{Ldim} \le \log_2|\mathcal{H}_M| = O(M \log M)$ is the counting bound. Substituting $\mathrm{Ldim}^{\mathrm{stream}}(\mathcal{H}_M) = \Theta(M \log M)$ gives the claim. □

Corollary 13.57 (Stream Littlestone Dimension)**.** *For fixed alphabets with $|\mathcal{I}| \ge 4$ and $|\mathcal{O}| \ge 2$, the Littlestone dimension of $\mathcal{H}_M$ witnessed by mistake trees realizable inside a single persistent stream satisfies*

$$\mathrm{Ldim}^{\mathrm{stream}}(\mathcal{H}_M) = \Theta(M \log M),$$

matching the reset-word value. Thus the two protocols have the same sequential complexity up to constants.

Proof. The upper bound is the counting bound of Lemma 13.1, since $\mathrm{Ldim} \le \log_2|\mathcal{H}_M| = O(M \log M)$.

For the lower bound we exhibit the mistake tree explicitly rather than appeal to the forcing argument as a black box. Recall from Theorem 13.5 that the adversary processes the states $v^{(1)}, \ldots, v^{(M)}$ in order, playing the block $w_{v^{(k)}}\,\mathtt{c}\,\mathtt{d}^L$ for each, and that the L readout rounds of block k reveal the L bits of $g(v^{(k)})$.

Build a binary tree of depth $M \log_2 M$ whose levels are indexed by the pairs (k, j) with $1 \le k \le M$ and $1 \le j \le L$, ordered lexicographically. The node at level (k, j) reached by a path $\sigma \in \{0,1\}^{<M \log_2 M}$ is labelled by the input letter that the adversary plays at that round, namely the j-th $\mathtt{d}$ of block k; the transport letters of $w_{v^{(k)}}$ and the letter $\mathtt{c}$ are inserted as forced unary segments between levels, since their outputs are determined by the fixed part of the table and branch nowhere. The two outgoing edges of the node correspond to the two possible answers for the bit $g(v^{(k)})_j$.

Each root-to-leaf path σ determines a full assignment of all $M \log_2 M$ bits, hence a unique $g_\sigma : Q \to Q$ and a unique member f_σ of the family, which by the consistency step of Theorem 13.5 realizes every label along σ. The tree is therefore shattered by $\mathcal{H}_M$, and the whole labelled sequence occurs inside one persistent stream with no external reset. Its depth is $M \log_2 M = \Omega(M \log M)$, giving $\mathrm{Ldim}^{\mathrm{stream}}(\mathcal{H}_M) = \Omega(M \log M)$. $\square$

Remark 13.58 (Logarithmic Refinements)**.** If a refined theorem introduces additional logarithmic factors in T, those factors should be stated explicitly. The M-scaling

$$\sqrt{M \log M}$$

is robust. In the notation of Section 9, the agnostic estimation rate is

$$\mathsf{E}_M^{\mathrm{agn}}(T) = \widetilde{\Theta}\Big(\sqrt{T\,M}\Big),$$

with the $\widetilde{\Theta}$ hiding logarithmic factors such as $\log M$ and possible $\log T$ refinements.

Remark 13.59 (Log-Loss Bridge for Retention)**.** For retention the compatible online bridge is log-loss. The approximation term $A_T^{\mathrm{ret}}(M)$ is then a cumulative excess KL relative to the best stochastic finite-state predictor of budget M, and the estimation term is a mixability-based rate. It is not a statement about deterministic point-mass Mealy machines, whose compatible bridge is the realizable mistake bound.

Remark 13.60 (Type-Correct Temporal Rates)**.** The bounds in this section are stated for deterministic Mealy hypothesis classes, which are the type-correct online classes for commitment and for realizable deterministic prediction. Retention with KL or log-loss uses stochastic finite-state predictors and mixability-based estimation rates. Grounding spectral relaxations use linear finite-rank predictors. The oracle bridge imports the appropriate type-correct estimation rate for each regime.

14 Spectral Tail Heuristic

The exact theorems of this manuscript are regime-specific. Nevertheless, a common spectral-tail picture organizes many of the analytic converses. This section states that picture as a heuristic, not as a theorem dictated solely by aggregation.

14.1 The Heuristic

Heuristic 14.1 (Spectral Tail Heuristic)**.** In several approximation regimes, the M-state or M-rank error is controlled by the tail of a regime-specific positive operator. The precise tail form depends on:

- operator approximation versus clustering;
- linear versus affine subspaces;
- rank count versus state count;

- compactness, trace-class, domination, and quadratic-regime assumptions;
- the aggregation norm and the response representation.

Aggregation alone does not determine the tail.

The heuristic serves as a design principle and as a diagnostic. It becomes a theorem only after a regime supplies:

(1) a response operator A_δ;

(2) an effective rank budget $r_{\mathbb{T}}(M)$;

(3) a Schatten or Ky Fan domination inequality;

(4) and, for online rates, an achievability bound.

14.2 Why Different Tails Appear

The different tail forms reflect different approximation mechanisms.

Operator-norm case: Eckart–Young–Mirsky. For the unrestricted linear finite-rank grounding relaxation, the exact gap is an operator-norm approximation problem:

$$\Delta_{\mathrm{grd}}^{\mathrm{unres}}(M) = \inf_{\operatorname{rank} B \leq M} \|H_\nu - B\|_{\mathrm{op}}.$$

Eckart–Young–Mirsky gives the single singular value

$$\Delta_{\mathrm{grd}}^{\mathrm{unres}}(M) = \sigma_{M+1}(H_\nu).$$

The rank budget is M, and the error is the first omitted singular value.

The Hankel-structured finite-rank realization gap satisfies the lower bound

$$\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) \geq \sigma_{M+1}(H_\nu),$$

with equality only under additional structural hypotheses.

Nuclear / squared-error case: covariance residual. For quadratic retention, the residual object is a positive semidefinite covariance residual. An M-cluster approximation induces at most $M-1$ centered centroid directions. The residual covariance has trace equal to its nuclear norm, so the lower bound is a trace tail:

$$\Delta_{\mathrm{ret}}^{\mathrm{quad}}(M) \geq \frac{1}{2}\sum_{i \geq M} \lambda_i(\boldsymbol{\Sigma}_\pi) = \frac{1}{2}\Phi_{M-1}^{(1)}(\boldsymbol{\Sigma}_\pi).$$

The exponent 1 appears because the residual covariance is positive semidefinite, not because the original cost is an ℓ^1 aggregation.

Commitment case: combinatorial threshold. Commitment does not possess a genuine singular-value spectrum. Its exact gap is a Boolean rank / Myhill–Nerode threshold:

$$\Delta_{\mathrm{com}}(M) = 0 \iff M \geq \kappa_{\mathrm{MN}}.$$

One may view this as a combinatorial step spectrum: the zero-error gap drops from positive to zero when M crosses the Boolean realization rank. This is not an ordinary Schatten $p \to 0$ limit.

Index offset. The grounding tail begins at σ_{M+1} because a rank-M linear approximation uses an M-dimensional linear subspace. The retention tail begins at $i \geq M$ because M centroids span at most $M-1$ centered directions after the global mean is removed. Thus the same state budget M produces different effective ranks in operator approximation and clustering.

Frobenius / Schatten-2 case. If a regime satisfies a Schatten-2 domination hypothesis, the conditional template gives a Frobenius tail:

$$\Delta_{\mathbb{T}}(M) \geq \frac{1}{c_2}\left(\sum_{i>r_{\mathbb{T}}(M)} \sigma_i(A_\delta)^2\right)^{1/2}.$$

This is a conditional consequence of the Schatten template, not a verified manuscript regime unless such a domination bridge is explicitly supplied.

14.3 Conditional Cross-Term Conjectures

Some cross-term tail forms require additional hypotheses. They are not consequences of aggregation alone.

Conjecture 14.2 (Conditional Cross Terms)**.** The following are plausible under additional hypotheses:

(1) An $\mathcal{L}^1$-type grounding bound of nuclear-tail form

$$\Delta^{L^1}_{\mathrm{grd}}(M) \gtrsim \sum_{i>M} \sigma_i(H_\nu)$$

requires H_ν to be trace class and requires a nuclear-norm domination hypothesis relating the task cost to the trace norm of $H_\delta - H_{\widehat{\delta}}$.

(2) An $\mathcal{L}^\infty$-type retention bound of single-eigenvalue form

$$\Delta^{L^\infty}_{\mathrm{ret}}(M) \gtrsim \frac{1}{2}\lambda_M(\mathbf{\Sigma}_\pi)$$

requires a worst-case quadratic surrogate or additional geometric assumptions aligning the worst-case state with the top Fisher eigendirection.

(3) A Frobenius / Schatten-2 converse

$$\Delta_{\mathbb{T}}(M) \gtrsim \left(\sum_{i>r(M)} \sigma_i(A_\delta)^2 \right)^{1/2}$$

requires an explicit Schatten-2 domination bridge.

(4) Intermediate Rényi orders $\alpha \notin \{0, 1, \infty\}$ may organize interpolation inequalities, but they do not define operational finite-state regimes unless a matching task theory, domination bridge, and rank-control structure are supplied.

These conjectures are compatible with the Schatten template and the exponent correspondence. They are not asserted as theorems.

14.4 Scope and Conditions

- **Aggregation alone is insufficient.** The tail shape is influenced by aggregation, but also by affine versus linear approximation, clustering versus operator approximation, state count versus rank count, and regime-specific domination or quadratic-regime assumptions.
- **No Boolean singular values.** Commitment does not have a genuine Schatten singular-value spectrum. Its threshold is a Boolean rank / Myhill–Nerode realizability threshold.
- **No unconditional cross-regime ordering.** The spectral-tail heuristic does not imply inequalities among grounding, retention, and commitment gaps. The Independence Theorem remains intact.
- **Converses are not achievability rates.** Spectral tails are lower bounds on approximation gaps. They become online regret upper rates only under explicit spectral-rate assumptions, as in Section 9.
- **Numerical checks are illustrations.** Numerical evaluations can illustrate singular-value tails, covariance tails, and threshold behavior on finite instances. They do not prove domination hypotheses, Boolean-rank identities, relaxation gaps, or independence.

The spectral-tail picture is an organizing heuristic once a response operator, an effective rank budget, and a domination bridge are supplied. It is not a single theorem derived from aggregation alone, and it does not unify the numerical gaps across regimes.

15 Conditional Representation Theorem

The schema does not by itself force KL divergence, Boolean equality, or operator norm. Those cost functionals arise under additional analytical axioms. The following theorem records the conditional representation results that justify the canonical regime costs.

15.1 Theorem Statement

Theorem 15.1 (Conditional Representation)**.** *Let $\mathbb{T}$ be a task theory on a finite input alphabet $\mathcal{I}$.*

Clause I, Retention. *Suppose $\mathbf{R} = \mathbf{FinStoch}$, so that residual objects are finite probability distributions. Suppose $\mathcal{E}$ satisfies:*

(a) ***Faithfulness.***

$$\mathcal{E}(P\|Q) \geq 0,$$

and

$$\mathcal{E}(P\|Q) = 0 \iff P = Q.$$

(b) ***Regularity.*** *$\mathcal{E}$ is continuous on full-support pairs and lower semicontinuous on the boundary.*

(c) ***Data processing.*** *For every stochastic kernel K,*

$$\mathcal{E}(P\|Q) \geq \mathcal{E}(PK\|QK).$$

(d) ***Exact chain rule.*** *For all finite joint distributions,*

$$\mathcal{E}(P_{XY}\|Q_{XY}) = \mathcal{E}(P_X\|Q_X) + \mathbb{E}_{P_X}\big[\mathcal{E}(P_{Y|X}\|Q_{Y|X})\big].$$

(e) ***f-divergence form (independent structural axiom).*** *Assume outright that $\mathcal{E}$ lies in the class of f-divergences, i.e.*

$$\mathcal{E}(P\|Q) = \sum_i q_i f\left(\frac{p_i}{q_i}\right)$$

for a convex generator f with $f(1) = 0$, extended lower semicontinuously at the boundary. This is an assumed *hypothesis, not a consequence of clauses (a)–(d): faithfulness, regularity, and data processing are all satisfied by the Rényi divergences, which are not f-divergences (Remark 15.5). Deriving the f-divergence form from information monotonicity requires further structural axioms such as expansibility, symmetry, and sum-form decomposability [18, 19], which are not assumed here. Given clause (e), the chain rule pins down the generator by Lemma 15.2.*

Given this clause, data processing is not used in the derivation: every f-divergence with convex f satisfies it automatically. It appears in the hypothesis list because the clauses are stated as independent axioms on $\mathcal{E}$, but the proof below invokes only faithfulness, the f-divergence form, and the chain rule.

Then there exists a constant $c > 0$ such that

$$\mathcal{E}(P\|Q) = c\, D_{\mathrm{KL}}(P\|Q).$$

Clause II, Commitment. *Suppose* $\mathbf{R} = \mathbf{Set}$ *and* $\mathcal{E}$ *is faithful and* $\{0, \infty\}$*-valued:*

$$\mathcal{E}(x, y) = 0 \iff x = y.$$

Then

$$\mathcal{E}(x, y) = \begin{cases} 0, & x = y, \\ \infty, & x \neq y. \end{cases}$$

Clause III, Grounding. *Suppose* $\mathrm{ob}(\mathbf{R})$ *is a concrete* C^**-algebra of bounded operators and*

$$\mathcal{E}(T, S) = \|T - S\|,$$

where $\|\cdot\|$ *satisfies the* C^**-identity*

$$\|T^*T\| = \|T\|^2.$$

Then

$$\|T\| = \|T\|_{\mathrm{op}}.$$

Consequence. *Under these representations, the canonical gaps are respectively:*

- *the full-KL information-bottleneck objective;*
- *the Myhill–Nerode index;*
- *the Eckart–Young–Mirsky spectral lower bound for the unrestricted linear Hankel relaxation.*

15.2 Csiszár Representation Lemma

Lemma 15.2 (Csiszár Representation)**.** *The* f*-divergence class and its characterizations are due to Csiszár [18]; the failure of information monotonicity alone to pin down the class, and the role of the additional sum-form axioms, are discussed by Amari [19]. The chain-rule selection of the logarithmic generator below is stated in the form used here.*

Let $\mathcal{E}(P\|Q)$ *be a functional on pairs of probability distributions over a finite alphabet satisfying:*

(a) ***Faithfulness.*** $\mathcal{E}(P\|Q) \geq 0$*, with equality iff* $P = Q$*.*

(b) ***Regularity.*** $\mathcal{E}$ *is continuous on full-support pairs and lower semicontinuous on the boundary.*

(c) ***Data processing.*** *For every stochastic kernel* K*,*

$$\mathcal{E}(P\|Q) \geq \mathcal{E}(PK\|QK).$$

(d) ***f*-divergence form.** *$\mathcal{E}$ is an f-divergence: there is a convex $f : [0, \infty) \to \mathbb{R} \cup \{+\infty\}$ with $f(1) = 0$, extended lower semicontinuously at the boundary, such that for all finite alphabets*

$$\mathcal{E}(P\|Q) = \sum_i q_i \, f\left(\frac{p_i}{q_i}\right).$$

Recall that the generator is determined only modulo an affine term: replacing $f(t)$ by $f(t) + a(t-1)$ leaves $\mathcal{E}$ unchanged.

If, in addition, $\mathcal{E}$ satisfies the exact P-weighted chain rule for all finite joint distributions, then, modulo such an affine term, $f(t) = c\,t \log t$ for some $c > 0$; equivalently the normalized convex representative is $f(t) = c(t \log t - t + 1)$, and hence

$$\mathcal{E}(P\|Q) = c\, D_{\mathrm{KL}}(P\|Q).$$

Proof. Hypothesis (d) places $\mathcal{E}$ in the f-divergence class; it is assumed, not derived, and cannot be replaced by faithfulness, regularity, and data processing alone (Remark 15.5). Faithfulness excludes the identically zero generator and forces f to be strictly convex at $t = 1$.

For the second assertion, apply the chain rule to a joint distribution P_{XY} and its marginal-conditional factorization. The chain rule requires

$$\sum_{x,y} q_x q_{y|x} \, f\left(\frac{p_x p_{y|x}}{q_x q_{y|x}}\right) = \sum_x q_x \, f\left(\frac{p_x}{q_x}\right) + \sum_x p_x \sum_y q_{y|x} \, f\left(\frac{p_{y|x}}{q_{y|x}}\right).$$

Since the generator matters only modulo an affine term $a(t-1)$, write $f(t) = g(t) + a(t-1)$ and normalize by $g(1) = 0$.

Product additivity is not enough. Specializing to product distributions, $p_{y|x} = p_y$ and $q_{y|x} = q_y$ independent of x, yields the additivity $\mathcal{E}(p \otimes p' \| q \otimes q') = \mathcal{E}(p\|q) + \mathcal{E}(p'\|q')$, which is too weak to select the generator. The reverse Kullback–Leibler divergence, with generator $g(t) = -\log t$, is an f-divergence and is additive over products, yet it is not a multiple of D_{KL}. Any argument resting on product additivity alone therefore cannot reach the conclusion, and we must use the chain rule in its genuinely conditional form, where $q_{y|x}$ varies with x.

An exact two-variable identity from a non-product conditional. Let X be binary with $q_X = (\alpha, 1-\alpha)$ and $p_X = (u\alpha, 1 - u\alpha)$ for $u > 0$ and α small enough that both are probability vectors. Fix a conditional reference $q_{Y|X}$ that is the *same* law q' on both branches, and a conditional $p_{Y|X}$ that equals p' on the branch $x = 1$ and equals q' on the branch $x = 2$. Write $t_j = p'_j / q'_j$.

On the branch $x = 2$ the conditional divergence vanishes, since $p_{Y|x=2} = q_{Y|x=2}$ and $g(1) = 0$. The joint ratios are $u\, t_j$ on the branch $x = 1$, with weights $\alpha q'_j$, and $\rho := (1 - u\alpha)/(1-\alpha)$ on the branch $x = 2$, with weights $(1-\alpha) q'_j$. The P-weighted chain rule

$$\mathcal{E}(P_{XY}\|Q_{XY}) = \mathcal{E}(P_X\|Q_X) + \sum_x p_x \, \mathcal{E}(P_{Y|x}\|Q_{Y|x})$$

therefore reads

$$\alpha \sum_j q'_j\, g(u\,t_j) + (1-\alpha)\, g(\rho) = \Big[\alpha\, g(u) + (1-\alpha) g(\rho)\Big] + u\alpha \sum_j q'_j\, g(t_j),$$

where the bracket is $\mathcal{E}(P_X \| Q_X)$. The terms $(1-\alpha)g(\rho)$ cancel *exactly* — no expansion in α is needed — and dividing by $\alpha > 0$ leaves the identity

$$\sum_j q'_j\, g(u\,t_j) = g(u) + u \sum_j q'_j\, g(t_j), \tag{$\dagger$}$$

valid for every $u > 0$, every reference q', and every p'.

One-sided specialization; no differentiability is assumed. Take $q' = (\beta, 1-\beta)$ and $p' = (v\beta, 1 - v\beta)$ with $v > 0$ and $\beta \in (0,1)$ small enough that p' is a probability vector, so that $t_1 = v$ and $\sigma(\beta) \coloneqq t_2 = (1 - v\beta)/(1-\beta)$. Then ($\dagger$) reads

$$\beta\, g(uv) + (1-\beta)\, g(u\sigma) = g(u) + u\beta\, g(v) + u(1-\beta)\, g(\sigma). \tag{$\dagger\dagger$}$$

Both sides tend to $g(u)$ as $\beta \downarrow 0$. Subtract $g(u)$, divide by β, and let $\beta \downarrow 0$. This uses no differentiability hypothesis: a convex function on an open interval has finite one-sided derivatives at every point, and β approaches 0 from above only, so every limit occurring is one-sided. Since $\sigma = 1 + \beta(1-v) + O(\beta^2)$, the argument $u\sigma$ approaches u from above when $v < 1$ and from below when $v > 1$, and σ approaches 1 from the same side. Writing g'_+ and g'_- for the right and left derivatives,

$$g(uv) - g(u) + u(1-v)\, g'_\pm(u) = u\, g(v) + u(1-v)\, g'_\pm(1), \qquad \begin{cases} +, & v < 1, \\ -, & v > 1, \end{cases} \tag{$\ddagger$}$$

for every $u > 0$; the case $v = 1$ is trivial.

Symmetry on each side of 1. The left side of ($\ddagger$) is symmetric in u and v. For $u, v \in (0,1)$ both ($\ddagger$) with the $+$ sign and its transpose hold, and equating the two expressions for $g(uv)$ gives

$$\frac{u\, g'_+(u) - g(u) - g'_+(1)}{u - 1} = \frac{v\, g'_+(v) - g(v) - g'_+(1)}{v - 1}.$$

Each side depends on one variable only, so there is a constant c_1 with

$$u\, g'_+(u) = g(u) + c_1(u-1) + g'_+(1), \qquad u \in (0,1). \tag{$\ddagger_+$}$$

The same argument with the $-$ sign for $u, v > 1$ gives a constant c_2 with $u\, g'_-(u) = g(u) + c_2(u-1) + g'_-(1)$ on $(1, \infty)$.

Regularity bootstrap. Equation ($\ddagger_+$) exhibits $g'_+(u)$ as $[g(u) + c_1(u-1) + g'_+(1)]/u$, which is continuous on $(0,1)$ because convex functions are continuous on open intervals. A convex function whose right derivative is continuous is C^1, since then $g'_-(u) = \lim_{t \uparrow u} g'_+(t) = g'_+(u)$ at every point. Hence g is C^1 on $(0,1)$ and satisfies $u\, g'(u) - g(u) = c_1(u-1) + \alpha$ with $\alpha \coloneqq g'_+(1)$. Dividing by u^2 makes the left side $(g(u)/u)'$, and integrating with $g(1) = 0$ gives

$$g(u) = c_1\big(u \log u - u + 1\big) + \alpha(u - 1), \qquad u \in (0,1].$$

Then $g'_-(1) = \lim_{u\uparrow 1} g'(u) = \alpha$, and the same bootstrap applied on $(1,\infty)$ yields $g(u) = c_2(u\log u - u + 1) + \alpha(u-1)$ for $u \geq 1$.

Gluing across 1. Apply (‡) with the $-$ sign at $u \in (0,1)$ and $v = 1/u > 1$, so $uv = 1$ and $g(uv) = 0$. Substituting the two branch formulas and $g'_-(u) = c_1 \log u + \alpha$, the affine terms cancel and

$$0 = (c_1 - c_2)\big(\log u - u + 1\big) \qquad \text{for all } u \in (0,1).$$

Since $\log u < u - 1$ for $u \neq 1$, this forces $c_1 = c_2 =: c$, so

$$g(t) = c\,(t\log t - t + 1) + \alpha(t-1), \qquad t > 0.$$

The affine term does not change the divergence, so the normalized convex representative is $f(t) = c(t\log t - t + 1)$. Convexity forces $c \geq 0$ and faithfulness excludes $c = 0$, since $c = 0$ gives $\mathcal{E} \equiv 0$; hence $c > 0$ and $\mathcal{E} = c\,D_{\mathrm{KL}}$. See [18, Sec. 2.1] and [1, Ch. 3] for the classical characterization within the finite-alphabet f-divergence class. □

Remark 15.3 (Smoothness of the Generator Is Derived, Not Assumed)**.** The proof assumes only convexity and derives regularity from the chain rule: the displayed formula shows that every convex generator satisfying the P-weighted chain rule is C^∞ on $(0,\infty)$, with matching one-sided derivatives at $t = 1$. A C^1 hypothesis, sometimes imposed in characterizations of this kind, is therefore a conclusion here rather than a prerequisite.

Two steps carry the weight. The passage to (‡) is a one-sided limit in the probability weight β, which is legitimate for any convex g because $\beta \downarrow 0$ only; differentiating (††) at $\beta = 0$ would not be, since convexity gives differentiability only off a countable set. The bootstrap then converts (‡$_+$) from a constraint *on* g'_+ into a formula *for* it, and a convex function with continuous right derivative is C^1.

The gluing step is not decorative. A generator equal to $c_1(t\log t - t + 1) + \alpha(t-1)$ on $(0,1]$ and to $c_2(t\log t - t + 1) + \alpha(t-1)$ on $[1,\infty)$ with $c_1 \neq c_2$ is convex, is C^1 at $t = 1$, and satisfies both branch equations; only the identity $0 = (c_1 - c_2)(\log u - u + 1)$ excludes it.

Remark 15.4 (Why the Conditional Structure Is Indispensable)**.** The proof above uses a conditional $P_{Y|X}$ that differs across the two branches of X. This is not a matter of convenience. Restricting the chain rule to product distributions yields only additivity over products, and that property does not characterize the logarithmic generator: reverse Kullback–Leibler divergence, $g(t) = -\log t$, satisfies

$$\mathcal{E}(p \otimes p' \| q \otimes q') = \mathcal{E}(p\|q) + \mathcal{E}(p'\|q')$$

for all factors, and is an f-divergence, but is not proportional to D_{KL}. Substituting $g(t) = -\log t$ into the exact identity (†), by contrast, fails immediately: with $u = 2.3$, $q' = (0.3, 0.7)$ and $p' = (0.55, 0.45)$ the two sides of (†) differ by -0.1657, whereas for $g(t) = t\log t - t + 1$ they agree to machine precision.

The separation is systematic rather than incidental to that triple. Over random (u, q', p') with $|\mathcal{O}| \leq 5$, the maximum discrepancy in (†) is 5.5×10^{-40} in 40-digit arithmetic for the forward generator, while for reverse Kullback–Leibler, χ^2, Hellinger, the Jensen–Shannon generator and total variation it is 12.6, 2.0×10^4, 3.1, 3.2 and 4.5 respectively. Sweeping the α-divergence family

$$g_\alpha(t) = \frac{t^\alpha - 1 - \alpha(t-1)}{\alpha(\alpha-1)}$$

through $\alpha \in \{-1, 0, \frac{1}{2}, 0.9, 0.99, 1, 1.01, \frac{3}{2}, 2, 3\}$, the defect in (†) vanishes only at $\alpha = 1$ and is bounded away from zero on either side, so no neighbouring member of the family survives the conditional chain rule. The subsequent symmetry step is equally decisive: the quotient $(u\,g'(u) - g(u))/(u-1)$ evaluated at $u = 1.3, 2.1, 3.7, 5.9, 9.4$ is constant to 0 for $t \log t - t + 1$ and varies by 0.61, 8.1, 0.22 and 0.27 for the other differentiable generators listed. The P-weighting of the conditional term — the factor p_x rather than q_x in front of $\mathcal{E}(P_{Y|x} \| Q_{Y|x})$ — is what breaks the symmetry between forward and reverse divergence, and it is visible only when $P_{Y|x}$ genuinely depends on x.

Remark 15.5 (The Sum-Form Hypothesis Cannot Be Dropped)**.** Faithfulness, regularity, and data processing alone do *not* characterize f-divergences. The Rényi divergences

$$D_\alpha(P\|Q) = \frac{1}{\alpha - 1} \log \sum_i p_i^\alpha q_i^{1-\alpha}, \qquad \alpha \in (0, \infty) \setminus \{1\},$$

are faithful, continuous on full-support pairs, and satisfy the data processing inequality, yet they are not f-divergences: the outer logarithm destroys the sum form. Concretely, every f-divergence is jointly convex in the pair (P, Q), whereas D_α fails joint convexity for $\alpha > 1$. More generally, any strictly increasing reparametrization of an f-divergence inherits faithfulness and data processing while leaving the f-divergence class. The f-divergence hypothesis is therefore not removable, and clause (e) of Theorem 15.1 is an additional assumption rather than a consequence of the others. Sum-form decomposability alone is likewise insufficient without further symmetry and expansibility axioms; we therefore assume membership in the f-divergence class outright, which is the weakest formulation that makes the chain-rule argument valid.

Remark 15.6 (Role of the Lemma)**.** The lemma separates two logical steps, which have different status. The first—membership in the f-divergence class—is an *assumed* structural axiom, hypothesis (d). It does *not* follow from faithfulness, regularity, and data processing: the Rényi divergences satisfy all three and are not f-divergences (Remark 15.5). The second—the chain rule forces the logarithmic generator—is a genuine characterization *within* the f-divergence class, and is what the lemma proves. Neither step is derivable from the enriched-category schema alone: the schema supplies the finite-state syntax, while the analytical axioms supply the cost functional.

15.3 Proof of Clause I

Proof of Clause I. By clause (e), $\mathcal{E}$ is an f-divergence:

$$\mathcal{E}(P\|Q) = \sum_i q_i f\left(\frac{p_i}{q_i}\right),$$

with f convex and $f(1) = 0$.

The exact chain rule forces f to be logarithmic. Applying the chain rule to finite alphabets and to binary refinements yields the standard functional equation characterizing those f-divergences that are additive under conditioning. Under continuity, the solutions are exactly

$$f(t) = c(t \log t - t + 1), \qquad c > 0.$$

Faithfulness excludes $c = 0$. Substitution gives

$$\mathcal{E}(P\|Q) = c \sum_i p_i \log \frac{p_i}{q_i} = cD_{\mathrm{KL}}(P\|Q).$$

$\square$

Remark 15.7 (Why Faithfulness Is Necessary)**.** Without faithfulness, the identically zero functional

$$\mathcal{E}(P\|Q) = 0$$

satisfies data processing, joint convexity, lower semicontinuity, and the chain rule, but corresponds to $c = 0$. Clause I therefore requires faithfulness, or an equivalent nontriviality assumption.

Remark 15.8 (The Chain Rule Is External)**.** The chain rule in Clause I is an analytical axiom on the cost bimodule. It is not derivable from the **V**-category axioms alone. The enriched schema supplies the finite-state syntax; the chain rule supplies the information geometry that selects KL divergence.

Remark 15.9 (Retention Vertex)**.** Clause I explains why retention is the exact $\alpha = 1$ vertex in the exponent correspondence. The cost is KL / relative entropy. The spectral converse for the quadratic restricted retention gap is a separate Schatten-1 / trace-norm statement and is not derived from Clause I alone.

15.4 Proof of Clause II

Proof of Clause II. Since $\mathcal{E}$ is $\{0,\infty\}$-valued and faithful, the only possible zero value occurs at equality:

$$\mathcal{E}(x, y) = 0 \iff x = y.$$

If $x \neq y$, faithfulness forces $\mathcal{E}(x, y) \neq 0$. Because the only other available value is ∞, one must have

$$\mathcal{E}(x, y) = \infty.$$

Thus

$$\mathcal{E}(x, y) = \begin{cases} 0, & x = y, \\ \infty, & x \neq y. \end{cases}$$

$\square$

Remark 15.10 (Normalized 0/1 Version)**.** The structural Boolean cost uses $\{0, \infty\}$. The normalized operational cost replaces ∞ by 1:

$$\mathcal{E}_{0/1}(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

The Myhill–Nerode threshold is unchanged.

Remark 15.11 (Commitment Vertex)**.** Clause II is the cost-face justification for the Booleanized symmetrized $\alpha = 0$ vertex. It does not identify commitment with ordinary Schatten rank or with raw one-sided D_0. The relevant rank is deterministic Boolean realization rank, i.e. the Myhill–Nerode index.

15.5 Proof of Clause III

Proof of Clause III. In a C^*-algebra, the norm is uniquely determined by the C^*-identity:

$$\|T\|^2 = \|T^*T\|.$$

For a positive element T^*T, its norm equals its spectral radius:

$$\|T^*T\| = r(T^*T).$$

In a concrete operator algebra, the spectral radius of the positive operator T^*T is

$$r(T^*T) = \sup_{\|v\|=1} \langle v, T^*Tv \rangle = \sup_{\|v\|=1} \|Tv\|^2 = \|T\|_{\mathrm{op}}^2.$$

Therefore

$$\|T\|^2 = \|T\|_{\mathrm{op}}^2,$$

and hence

$$\|T\| = \|T\|_{\mathrm{op}}.$$

□

Remark 15.12 (Grounding Cost)**.** Clause III justifies using operator-norm distance as the canonical grounding cost when residual objects form a concrete C^*-algebra. The exact spectral converse for the unrestricted linear Hankel relaxation then uses Eckart–Young–Mirsky:

$$\Delta_{\mathrm{grd}}^{\mathrm{unres}}(M) = \sigma_{M+1}(H_\nu).$$

The Hankel-structured finite-rank realization gap satisfies

$$\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) \geq \sigma_{M+1}(H_\nu),$$

with equality only under additional structural hypotheses. The C^*-norm identification does not by itself prove Hankel domination, Hardy-space embedding, or exact symbolic deterministic grounding.

Remark 15.13 (What Clause III Supports at the Grounding Face)**.** Clause III supports the operator-norm / Schatten-∞ face of the grounding regime. The Rényi $\alpha = \infty$ vertex, max-divergence, is related to this face under bounded-overlap hypotheses but is not identical to the C^*-norm distance.

15.6 Scope of the Conditional Representation Theorem

Remark 15.14 (Not an Unconditional Derivation)**.** The Conditional Representation Theorem is conditional by design. It says that if a regime's cost satisfies the relevant analytical axioms, then the canonical cost form follows. It does not derive KL divergence, Boolean equality, or operator norm from the enriched-category schema alone.

Remark 15.15 (Regime-Specific Consequences)**.** The theorem supports the following regime-specific identifications:

(1) Retention costs satisfying data processing, convexity, and the chain rule are scaled KL divergences.

(2) Faithful $\{0, \infty\}$ commitment costs are Boolean equality costs.

(3) Grounding costs arising from concrete C^*-norms are operator-norm costs.

The approximation gaps built from these costs remain governed by the separate meta-theorems: information bottleneck for retention, Myhill–Nerode for commitment, and Eckart–Young–Mirsky for the unrestricted linear Hankel grounding relaxation.

16 Type Discipline for Approximation Statements

The regimes of this manuscript share one combinatorial resource, the index of a finite right congruence on histories, but they do *not* share a semantic category. Most of the errors that a comparative theory of this kind invites are not false lemmas but *type errors*: a quantity proved in one regime, protocol, or aggregation is silently reused in another. This section fixes a type signature that every approximation statement in this article carries, and records the signatures of the principal results, so that a mismatch between regimes is visible in the statement itself.

Definition 16.1 (Type signature of an approximation claim)**.** A claim about a finite-state approximation gap is **well typed** when the following seven coordinates are fixed and stated:

$$\Big\langle \underbrace{\mathsf{R}}_{\text{regime}}, \underbrace{\mathsf{M}}_{\text{machine type}}, \underbrace{\mathsf{F}}_{\text{feasible set}}, \underbrace{\mathsf{A}}_{\text{aggregation}}, \underbrace{\mathsf{S}}_{\text{support}}, \underbrace{\mathsf{P}}_{\text{protocol}}, \underbrace{\mathsf{B}}_{\text{analytical bridge}} \Big\rangle,$$

with admissible values

(1) $\mathsf{R} \in \{\text{commitment}, \text{retention}, \text{grounding}, \text{joint}\}$;

(2) the machine type

$$\mathsf{M} \in \big\{\text{deterministic Mealy},\ \text{stochastic lumped predictor},\\ \text{unifilar controlled } \epsilon\text{-machine},\ \text{linear finite-rank realization}\big\};$$

(3) the feasible set

$$\mathsf{F} \in \big\{\text{right congruences},\ \text{support-relative right congruences},\\ \text{lumpable quotients},\ \text{unifilar-lumpable quotients},\ \text{history factors},\\ \text{unrestricted rank-}M,\ \text{Hankel-restricted}\big\};$$

(4) the aggregation

$$\mathsf{A} \in \big\{\text{worst case, stationary Cesàro, discounted prefix,} \\ \text{finite-horizon cumulative}\big\};$$

(5) $\mathsf{S} \in \{\text{all histories, stationary support } \mathcal{S}^+, \text{positive-mass reachable}\}$;

(6) $\mathsf{P} \in \{\text{none, reset word, persistent stream, active query, active identification}\}$;

(7) B the analytical engine actually invoked, e.g. Myhill–Nerode, information bottleneck, Ky Fan, Eckart–Young–Mirsky, Adamjan–Arov–Krein, Schur test, Le Cam.

Remark 16.2 (Role of Each Coordinate)**.** Each coordinate marks a hypothesis whose omission changes the truth value of a statement in this setting.

- A and S: the Boolean zero-error dichotomy is a biconditional under worst-case aggregation, but under an average-case law it holds only with the Myhill–Nerode relation restricted to $\operatorname{supp}\mu$ (Meta-Theorem 4.7 and Remark 4.9).
- R and the meaning of "predictive state": equality of one-step laws is strictly coarser than causal-state equivalence and need not be a congruence (Definition 3.17, Example 3.24).
- F: "rank $\leq M$" is three different feasible sets — algebraic, bounded-operator, and Hankel-restricted — with different values (Remark 7.2, Theorems 7.3 and 7.9).
- M for retention: the input-driven model of Definition 3.14 is a proper subclass of the unifilar model of Definition 3.15, and lumpability is output-aware in general (Definition 3.21). Conflating the two changes the feasible set, and with it the value of the gap and the zero-retention threshold (Remarks 3.22 and 5.43).
- P: reset-word and persistent-stream mistake bounds are different theorems; the transfer required an explicit construction (Theorem 13.5), and active bounds require an identification objective (Remark 13.50).
- B: Eckart–Young–Mirsky and Adamjan–Arov–Krein solve different problems, and the latter needs a shift-intertwining embedding, not merely a unitary (Theorem 7.9). That theorem is scalar: for $|\Sigma| > 1$ the equality $\Delta^{\mathrm{Hank,str}}_{\mathrm{grd}}(M) = \sigma_{M+1}(H_\nu)$ is conditional on a multi-shift Nehari/AAK theorem that is not proved here (Theorem 7.11, Open Problem 7.13).

Remark 16.3 (How to read a gap symbol)**.** Where ambiguity is possible, a gap carries its distinguishing coordinates explicitly. Thus

$$\Delta^{\sup}_{\mathbb{T}}(M), \qquad \Delta^{\mu}_{\mathbb{T}}(M), \qquad \Delta^{\mathrm{unres}}_{\mathrm{grd}}(M), \qquad \Delta^{\mathrm{Hank,str}}_{\mathrm{grd}}(M), \qquad \mathrm{Mistakes}^{\mathrm{stream}}_{\mathrm{realizable}}(M)$$

are distinct objects, and no inequality between symbols of different type is asserted without an explicit bridge. In particular, the absence of a universal cross-regime ordering (Theorem 8.22) is a statement about incomparable types, not a gap in the analysis.

Table 1: Type signatures of the principal approximation results.

Result	**Regime**	**Feasible set**	**Aggregation / support**	**Bridge**
Myhill–Nerode threshold, Theorem 6.2	commitment	right congruences	worst case, all histories	Myhill–Nerode
Boolean dichotomy (i), Meta-Theorem 4.7	any separated	right congruences	worst case, all histories	separatedness
Boolean dichotomy (ii), Meta-Theorem 4.7	any separated	right congruences	discounted-prefix μ-average, $\operatorname{supp}\mu$	separatedness
IB identity, Meta-Theorem 4.14	retention	lumpable quotients	stationary, $\mathcal{S}^+$	information bottleneck
Zero retention, Theorem 5.2	retention	lumpable quotients	stationary, $\mathcal{S}^+$	Z-predictive distinctness
Controlled IB identity, Theorem 5.32	retention	unifilar-lumpable quotients	stationary, $\mathcal{S}^+$, i.i.d. input	fiberwise information bottleneck
Zero controlled retention, Theorem 5.42	retention	unifilar-lumpable quotients	stationary, $\mathcal{S}^+$, i.i.d. input	stable kernel refinement
Quadratic converse, Theorem 5.8	retention	lumpable quotients	stationary, $\mathcal{S}^+$	Ky Fan / ANOVA
Unrestricted grounding, Theorem 7.3	grounding	unrestricted rank-M	operator norm on H_ν	Eckart–Young–Mirsky
Structured grounding, Theorem 7.9	grounding	Hankel-restricted, $\lvert\Sigma\rvert = 1$	worst case, all histories	Adamjan–Arov–Krein, given a shift-intertwining embedding
Determinism gap, Theorem 7.25	grounding	*algebraic* rank	worst case, all histories	PFA succinctness, rate $\mathcal{S}(k)$
Safe retention, Corollary 12.12	joint	history factors	stationary, $\mathcal{S}^+$	information bottleneck

Table 2: Type signatures of the temporal results. Protocol is the distinguishing coordinate.

Result	**Protocol**	**Alphabets**	**Status**
Reset-word complexity	reset word	$\lvert\mathcal{I}\rvert, \lvert\mathcal{O}\rvert \geq 2$	$\Theta(M \log M)$, exact
Persistent-stream upper bound, Theorem 13.4	persistent stream	$\lvert\mathcal{I}\rvert, \lvert\mathcal{O}\rvert \geq 2$	$O(M \log M)$, unconditional
Persistent-stream lower bound, Theorem 13.5	persistent stream	$\lvert\mathcal{I}\rvert \geq 4$	$\Omega(M \log M)$, deterministic; expectation and high-probability for randomized
Binary-input lower bound, Theorem 13.8	persistent stream	$\lvert\mathcal{I}\rvert = 2$	$\Omega(M \log M)$, constant $c < \frac{1}{3}$
Agnostic regret, Theorem 13.56	both	$\lvert\mathcal{I}\rvert, \lvert\mathcal{O}\rvert \geq 2$	$\Theta(\sqrt{TM \log M})$
Active upper bound, Theorem 13.27(I)	active query	fixed finite	$O(E_{\text{sync}})$ under (RI), unconditional
Active E_{sync} lower bound, Theorem 13.27(II)	active identification	fixed finite	conditional on operational equivalence
Active additive lower bound, Theorem 13.27(III)	active identification	$\lvert\mathcal{I}\rvert \geq 4$	conditional on direct-sum saturation; witnessed by the two-component construction of Theorem 13.48
Active complexity, Corollary 13.47	active identification	$\lvert\mathcal{I}\rvert \geq 2$	$\Theta(M \log M)$, unconditional

17 Scope of Results and Open Problems

The schema provides a common structural syntax. It does not derive the Myhill–Nerode theorem, the Eckart–Young–Mirsky theorem, or the information-bottleneck objective from categorical axioms alone. The analytical engines remain domain-specific.

The model distinguishes:

- shared combinatorial syntax;
- regime-specific semantics;
- exact theorems;
- conditional theorems;
- local or surrogate theorems;
- heuristics;
- open problems.

The classification below states, for each principal result, the hypotheses under which it holds, and lists the problems that remain open.

17.1 Status Categories

The lists below use the following status labels.

- **Exact.** A theorem proved under the stated hypotheses, with no additional domination or achievability assumption required.
- **Conditional.** A theorem proved under an explicit additional hypothesis, such as Schatten domination, stable decay, bounded overlap, or spectral-rate achievability.
- **Local.** A theorem valid in a small-radius or second-order regime under explicit regularity assumptions.
- **Heuristic.** An organizing principle or diagnostic, not a theorem without further hypotheses.
- **Open.** A problem left unresolved here, stated explicitly as an open problem.

17.2 Exact Results

The following results are exact under the hypotheses stated in the body of the manuscript.

- Retention zero iff $M \geq |\mathcal{S}^+|$, assuming distinct predictive distributions on the stationary support.
- Full-KL information-bottleneck identity

$$\Delta^{\mathrm{KL}}_{\mathrm{ret}}(\phi) = I(S; Z \mid K_\phi) = I(S; Z) - I(K_\phi; Z).$$

- Quadratic restricted spectral converse

$$\Delta_{\mathrm{ret}}^{\mathrm{quad}}(M) \geq \frac{1}{2} \sum_{i \geq M} \lambda_i(\mathbf{\Sigma}_\pi).$$

- Unifilar retention: the controlled information-bottleneck identity (Theorem 5.32), the zero-retention threshold at the index of the stable kernel refinement (Theorem 5.42), and the fiberwise quadratic, probability-coordinate, and interior-Fisher spectral converses (Corollaries 5.49–5.51).
- Gaussian quadratic restricted retention NP-completeness for reset machines of synchronization depth one.
- Tagged depth-two NP-hardness in the same regime.
- Commitment exact threshold

$$\Delta_{\mathrm{com}}(M) = 0 \iff M \geq \kappa_{\mathrm{det}}(F).$$

- Safety controller complexity $\kappa_{\mathrm{ctrl}}(\Lambda)$.
- Strategic-spread lower bound with explicit myopic adversary.
- Stateless-game exact corollary $\Delta_{\mathrm{com}}^{\mathrm{game}}(M) = \alpha/(1-\gamma)$.
- Booleanized symmetrized $\alpha = 0$ commitment vertex.
- Unrestricted linear Hankel relaxation converse

$$\Delta_{\mathrm{grd}}^{\mathrm{unres}}(M) = \sigma_{M+1}(H_\nu).$$

- Hankel-structured lower bound

$$\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) \geq \sigma_{M+1}(H_\nu).$$

- Stochasticity floor for deterministic Mealy grounding, and its exact observable refinement: $\sigma_\gamma(\nu) \leq F_\gamma(\nu)$ with $F_\gamma(\nu)$ the exact input-only deterministic floor and $\Delta_{\mathrm{grd}}(M;\gamma) \to F_\gamma(\nu)$ as $M \to \infty$ (Theorem 7.29).
- Finite-section operator certificates for linear grounding: computable two-sided bounds on $\sigma_{M+1}(H_\nu) = \Delta_{\mathrm{grd}}^{\mathrm{unres}}(M)$ from finite Hankel truncations H_N, with an explicit lower certificate $\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) \geq \sigma_{M+1}(H_N)$ for the structured problem (Proposition 7.7).
- Exact zero threshold for the Hankel-structured linear problem: $\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) = 0 \iff \mathrm{rank}(H) \leq M$ (Proposition 7.8).
- Finite-horizon symbolic certificates: a two-sided sandwich $\Delta_{\mathrm{grd}}^{(T)}(M;\gamma) \leq \Delta_{\mathrm{grd}}(M;\gamma) \leq \Delta_{\mathrm{grd}}^{(T)}(M;\gamma) + \gamma^T/(1-\gamma)$, decidable and rational under rational finite-state and rational-γ hypotheses (Proposition 7.31).

- Refinement monotonicity of the tracking deficit: ϕ refining ψ implies $D(\phi) \leq D(\psi)$, so $\min_{|\mathcal{K}|\leq M} D(\phi)$ is nonincreasing in M (Proposition 7.36).
- Joint zero-error memory theorem via meet of right congruences.
- Comparative state-complexity sandwich.
- Independence of regime obstructions.
- Interaction theorem for exact commitment and approximate retention.
- Price of Safety in mutual-information form.
- Reset-word realizable mistake complexity $\Theta(M \log M)$.
- **Persistent-stream realizable mistake complexity** $\Theta(M \log M)$, with the lower bound proved by an explicit transport-plus- readout adversary on a single never-reset stream (Theorem 13.5).
- Active upper bound $O(E_{\text{sync}}(M))$ under objective (RI) (Theorem 13.27(I)), and $O(M \log M + L_{\text{sync}}^{\text{univ}}(M))$ in universal length form (Proposition 13.28).
- **Active realizable mistake complexity** $\Theta(M \log M)$**, unconditionally**: an active halving learner attains objective (RI) within $\log_2 |\mathcal{H}_M \times Q| = O(M \log M)$ mistakes (Theorem 13.41), matching the gated-family lower bound, so $E_{\text{sync}}(M) = \Theta(M \log M)$ with no synchronizability, operational-equivalence, or direct-sum hypothesis (Corollary 13.47).
- **Explicit gated family** $\mathcal{G}_M^{\text{act}}$ with $E_{\text{sync}}(M) = \Omega(M \log M)$, giving an active lower bound $\Omega(M \log M)$ relative to the residual-identification objective (Theorem 13.45). The unknown datum is a transition map, so the bound is for model identification; it does not witness a state-identification component.
- Conditional additive active lower bound under direct-sum saturation (Theorem 13.52), which separates the two terms; the unconditional active lower bound itself is Corollary 13.47.
- **Agnostic minimax regret** $\Theta(\sqrt{TM \log M})$ in both the reset-word and persistent-stream protocols (Theorem 13.56, Corollary 13.57).
- Fixed-budget bias–variance decomposition.
- Adaptive agnostic oracle inequality under bounded losses.
- Realizable 0/1 and mixable log-loss oracle inequalities under stated conditions.
- Linear pinching Price-of-Safety surrogate nonnegativity and Ky Fan majorization.
- Retention as exact $\alpha = 1$ vertex.
- Commitment as Booleanized symmetrized $\alpha = 0$ vertex.
- Grounding as $\alpha = \infty$ supremum/operator-norm face, related but not identical to max-divergence.

17.3 Conditional, Local, and Surrogate Results

The following results are conditional, local, or surrogate statements.

- Local Fisher asymptotic lower bound for full KL under regularity assumptions.
- Conditional unified Schatten converse under response representation and domination.
- Frobenius / Schatten-2 converse under explicit Schatten-2 domination.
- Stable-decay residual-norm spectral bound.
- Max-divergence / operator-norm local equivalence under bounded-overlap hypotheses.
- Superpolynomial determinism gap for formal Hankel rank: the reduction is exact, but the statement is conditional on the external bounded-error succinctness family (Theorem 7.25).
- Budget laws under spectral-rate assumptions.
- Linear block-local Price-of-Safety surrogate. It is unconditionally nonnegative and bounded by the Ky Fan coupling term, but its identification with the discrete Price of Safety requires a two-sided bound on the signed discrepancy $\rho_{\mathrm{safe}} - \rho_{\mathrm{free}}$, neither term of which has a determined sign (Section 10.5).
- Spectral-tail behaviour, which is a heuristic organizing principle requiring a regime-specific response operator and domination bridge rather than following from aggregation (Heuristic 14.1).
- The vertex correspondence $\alpha = p$, which is a labelling consistency between Rényi order and Schatten exponent and carries no derivation of the Mirsky converses (Section 11).
- The commitment vertex, which is a Boolean-rank and Myhill–Nerode threshold rather than an ordinary Schatten $p \to 0$ limit (Theorem 11.11).
- The Hankel-structured converse, which attains the Eckart–Young–Mirsky value only under a Hardy-space embedding (Theorems 7.9 and 7.11).
- The block-mode tracking construction, which realizes a deterministic Mealy machine of the claimed stationary error only under the explicit input-driven synchronization and right-congruence implementability hypothesis of Proposition 7.37; absent that hypothesis the same error formula is only the loss of a genie that observes the latent block label.

17.4 Open Problems

The following open problems remain.

1. **Exact symbolic deterministic grounding gap (partially resolved).** *Hypotheses.* $0 < \gamma < 1$, finite input alphabet, target channel ν. *Established baseline.* Theorem 7.29 identifies the exact input-only deterministic floor $F_\gamma(\nu)$ and proves $\lim_{M\to\infty} \Delta_{\mathrm{grd}}(M;\gamma) = F_\gamma(\nu)$, with $\sigma_\gamma(\nu) \le F_\gamma(\nu)$ possibly strict. Proposition 7.31 supplies, for every finite horizon T, a two-sided certified interval for $\Delta^{\mathrm{symb}}_{\mathrm{grd}}(M) = \Delta_{\mathrm{grd}}(M;\gamma)$

of width at most $\gamma^T/(1-\gamma)$, decidable and rational for rational finite-state channels and rational γ. What remains open is a *closed-form value at finite* M, not merely a limit or a certified interval, and its relation to the unrestricted linear finite-rank Hankel gap $\Delta^{\text{unres}}_{\text{grd}}(M) = \sigma_{M+1}(H_\nu)$, to the stochasticity floor σ_γ, and to the superpolynomial determinism gap of Theorem 7.25 (see Open Problem 7.19). *Success criterion.* A formula or algorithm computing $\Delta^{\text{symb}}_{\text{grd}}(M)$ exactly for every finite M (not just $M \to \infty$ or a fixed horizon T), together with either a proof or a counterexample to $\Delta^{\text{symb}}_{\text{grd}}(M) \geq \sigma_{M+1}(H_\nu)$ for all M.

2. **Hankel-structured equality conditions.** Characterize necessary and sufficient conditions under which
$$\Delta^{\text{Hank,str}}_{\text{grd}}(M) = \sigma_{M+1}(H_\nu).$$
Theorem 7.9 settles the sufficient direction: a Hardy-space symbol with $\varphi \in H^\infty + C(\mathbb{T})$ forces equality for *every* M, with an attained optimum of rational symbol degree at most M. Corollary 7.14 converts this into a necessary condition on the operator: a strict gap at any M rules out Hardy-space representability. What remains open is the intrinsic characterization – an $\ell^2((\mathcal{I} \times \mathcal{O})^*)$-side criterion, stated directly in terms of ν, deciding when such a symbol exists.

3. **Relaxation gaps for the Price of Safety.** Prove explicit bounds relating the linear block-local Price-of-Safety surrogate
$$\mathrm{PoS}^{\text{loc}}_{\text{lin}}(M)$$
to the discrete right-congruence Price of Safety
$$\mathrm{PoS}_{\text{quad}}(M)$$
or the full-KL Price of Safety.

4. **Exact and fixed-alphabet complexity of full-KL retention.** The unrestricted full-KL decision problem is: given $\{P_s\}$, $\{\pi_s\}$, a budget $M \in \mathbb{N}$, lumpability constraints, and τ, decide whether
$$\min_{\substack{\phi \text{ lumpable} \\ |\mathcal{K}| \leq M}} \sum_k \sum_{s:\phi(s)=k} \pi_s D_{\text{KL}}\big(P_s \| \bar{P}_k\big) \leq \tau.$$
The budget M is present because it is what makes the problem a genuine finite-state compression decision rather than a bare clustering feasibility question at the natural number of distinct rows; the case $M \geq |\mathcal{S}^+|$ is trivial by Theorem 5.2, so the decision problem is of interest exactly for $M < |\mathcal{S}^+|$. Because the optimal block representative is the mixture centroid $\bar{P}_k$ (Lemma 5.1), the objective is exactly the weighted *Jensen–Shannon* clustering cost, so the problem is *constrained Jensen–Shannon clustering with lumpability constraints*. The Gaussian quadratic regime of Theorem 5.56 is its second-order Fisher special case, where the centroid degenerates to a mean and the cost to squared Euclidean distance, which is how the k-means reduction applies there. That reduction is carried out in Theorem 5.58: the obstruction — that KL centroids are not

rational in the inputs, so the promise gap cannot be produced by denominator clearing — is met by embedding k-means into a shrinking interior neighbourhood of the uniform law, where the Jensen–Shannon cost is a controlled perturbation of the Euclidean one and the gap is preserved analytically. Four questions remain.

(a) **Membership in NP.** The objective compares a rational threshold against a sum of terms $p \log(p/q)$ with p, q rational. Deciding such inequalities in polynomial time is not known, so exact NP-completeness is open even though hardness is proved.

Proposition 5.61 locates the difficulty exactly. For a fixed partition the whole sum collapses, $\Delta^{\mathrm{KL}}_{\mathrm{ret}}(\phi) = D^{-1} \log R$ with D a positive integer and R a positive rational presented as a product of powers, so the comparison with τ is equivalent to deciding $R \geq e^{D\tau}$. The number of logarithmic terms is therefore not the obstruction; the single rational-versus-exponential comparison is. That comparison is decidable, by Lindemann–Weierstrass as recorded in Remark 5.63, and what is missing is a polynomial bound on the precision needed to resolve it, equivalently a polynomial lower bound on $|R - e^{D\tau}|$. Remark 5.62 shows that R must be kept in factored form for this to be a meaningful question, its expanded numerator and denominator being of exponential bit size. Consequently the problem lies in NP as soon as this PosSLP-type comparison admits polynomial-time verifiable certificates.

(b) **Fixed output alphabet.** The embedding of Theorem 5.58 uses $|\mathcal{O}| = 2d$, growing with the dimension of the k-means instance. Whether the problem is hard for a fixed alphabet, binary in particular, is open.

Remark 5.64 determines how far the alphabet can be reduced within this scheme, and the answer is arithmetic rather than a universal constant: $|\mathcal{O}|$ may be taken as small as the least $n \geq d+1$ whose zero-sum subspace in $\mathbb{Q}^n$ contains d pairwise orthogonal rational vectors of equal norm. The value $n = 2d$ always qualifies, by the doubling map; $n = d+1$ qualifies for $d = 3, 7, 15, \ldots$ — the Sylvester orders — and for every other Hadamard order, the smallest being $d = 11$ of order 12, by the non-constant rows of a Hadamard matrix of order $d+1$; and $n = d+1$ fails for $d = 2$, by a discriminant obstruction. What no choice of scheme can avoid is $|\mathcal{O}| \geq d+1$, since the zero-sum subspace of $\mathbb{R}^{|\mathcal{O}|}$ has dimension $|\mathcal{O}| - 1$ and must accommodate d independent directions. A fixed alphabet therefore caps the embeddable dimension, and a fixed-alphabet hardness proof would have to start from a source problem hard in bounded dimension rather than from Euclidean k-means.

(c) **Hardness without a promise.** The reduction separates instances whose values differ by at least the amplified gap; exact comparison against a rational threshold is a different question.

(d) **Nonvacuous lumpability.** The reduction makes every partition lumpable. The complexity under genuinely restrictive right-congruence constraints is not addressed.

5. **Domination bridges.** Identify task-theoretic conditions under which contextwise ag-

gregation implies Schatten domination, or under which frame-type lower bounds hold.

6. **Intermediate exponent regimes.** Determine whether intermediate Rényi orders

$$\alpha \in (0, \infty) \setminus \{1\}$$

can be made operational as finite-state approximation regimes with matching rank-control and domination structures.

7. **Active learning without synchronization.** Develop active realizable and agnostic bounds when no output-aware synchronizing word exists, or when synchronization is only partial.

8. **Hardy-space embeddings for grounding.** Identify when a Hankel operator admits a Hardy-space embedding making Adamjan–Arov–Krein applicable, and compare the resulting converses with the default Eckart–Young–Mirsky converse.

9. **Deterministic commitment rate-distortion.** *Hypotheses.* A fixed history law μ on $\mathcal{I}^{\mathbb{N}}$ and a causal length-preserving specification F (Definition 6.1). *Established baseline.* Theorem 6.6 gives the exact fixed-budget value of $\Delta^{\mathrm{RD}}_{\mathrm{com}}(M)$ over right-congruence quotients of a one-step observable, and its zero threshold at $M = \kappa_{\mathrm{pair}}(F, \mu)$, the one-step determination index; Remark 6.8 shows this is generically distinct from, and no more than, the worst-case Boolean gap $\Delta_{\mathrm{com}}(M)$. What is open is the *full multi-step* rate-distortion law: the exact value of the M-state quotient loss for approximating the entire residual process $(F_u)_{u \in \mathcal{I}^*}$ under μ-average loss, not just its one-step marginal, together with a converse comparable in strength to the Blahut–Arimoto-type characterization available for classical rate-distortion. *Success criterion.* A formula (or a provably tight two-sided bound with matching rate) for the M-state μ-average multi-step commitment distortion, together with a decidable finite-horizon certificate analogous to Proposition 7.31.

10. **Typed joint quotients across the linear/symbolic boundary.** *Hypotheses.* Two tasks, one governed by a right-congruence quotient budget (commitment, retention, or symbolic grounding under Proposition 7.37) and one governed by the unrestricted linear finite-rank Hankel relaxation of Theorem 7.3. *Established baseline.* Theorem 12.2 and Remark 12.3 prove the product sandwich *only* when every component's budget is a right-congruence index, and explicitly exclude the unrestricted linear coordinate: no representative map is known that combines an M-state quotient with a rank-M' Hankel truncation into one object of controlled joint budget. What is open is whether *any* nontrivial joint budget notion M_{joint}, mixing a quotient index and an operator rank, admits even a one-sided analogue of the sandwich of Theorem 12.2. *Success criterion.* Either (a) a bridge construction – a map from rank-M' Hankel approximants to $O(f(M'))$-state quotients (or conversely) that preserves admissibility and control loss by a stated factor, from which a mixed sandwich would follow by composing with Theorem 12.2; or (b) a proof, e.g. via an explicit family of channels, that no such bridge with subexponential f can exist.

11. **Strategic memory versus information.** *Hypotheses.* A commitment task admitting both the distributional gap $\Delta^{\mathrm{RD}}_{\mathrm{com}}(M)$ and the simultaneous-move strategic gap

$\Delta^{\mathrm{game}}_{\mathrm{com}}(M)$ of Section 6.5. *Established baseline.* Remark 6.8 proves these two quantities are posed on different protocols (averaging under a fixed law versus play against an online adversary) and asserts no general inequality between them; Section 6.5 computes $\Delta^{\mathrm{game}}_{\mathrm{com}}(M)$ exactly in the stateless-game case. What is open is whether, under an explicit correspondence between the adversary's input distribution and a history law μ (e.g. the adversary's minimax-optimal mixed strategy), $\Delta^{\mathrm{game}}_{\mathrm{com}}(M)$ and $\Delta^{\mathrm{RD}}_{\mathrm{com}}(M)$ become comparable – whether strategic memory (states needed to play well against an adaptive adversary) is provably bounded above or below by informational memory (states needed to match a fixed one-step predictive law). *Success criterion.* Either a proved inequality $\Delta^{\mathrm{game}}_{\mathrm{com}}(M) \leq g(\Delta^{\mathrm{RD}}_{\mathrm{com}}(M'))$ or $\Delta^{\mathrm{RD}}_{\mathrm{com}}(M) \leq g(\Delta^{\mathrm{game}}_{\mathrm{com}}(M'))$ for an explicit function g and budget map $M \mapsto M'$, under a stated correspondence between the adversary's strategy and μ; or an explicit task family showing the two quantities are order-independent (each can be made arbitrarily larger than the other at matching budgets), closing the question negatively.

12. **Necessity of the aggregation penalty for nested classes.** Proposition 9.18 settles necessity for arbitrary experts; what remains is the nested case. Theorem 9.34 proves a two-sided minimax characterization of the adaptive oracle envelope under the explicit approximation and estimation floors of Assumption 9.24, matching the adaptive upper bound up to the aggregation penalty $\Psi_M(T)$ and universal constants. The floors are discharged for the realizable persistent-stream regime by Proposition 9.32 and assumed in the stochastic regimes. What remains open is whether $\Psi_M(T)$ is itself necessary, i.e. whether adapting to an unknown budget is strictly harder than the oracle problem, and whether horizon-efficient packings exist for the retention, grounding, and commitment regimes. See Remark 9.36.

17.5 Classification of Results

The schema supplies a shared finite-state syntax, while the analytical engines remain regime-specific. Results are classified as exact, conditional, local, heuristic, or open according to the hypotheses under which they are proved.

18 Conclusion

Finite-index right congruences on histories provide a common combinatorial resource for the three regimes, while the analytical engines governing them remain regime-specific. The principal conclusions are the following.

- **Shared syntax, regime-specific semantics.** Regime-typed histories, finite-index right-action quotients, and variational gaps form a common finite-state syntax: input histories for commitment, feasible joint histories for unifilar retention, the joint alphabet for grounding. Commitment, retention, and grounding attach different semantic objects to states: residual functions, predictive distributions or kernels, and operators or kernels.
- **Stationary aggregation.** The average-case theory uses stationary Cesàro aggregation or discounted prefix measures, finite prefixes not defining a probability measure over all

lengths (Remark 2.5).

- **Retention has two distinct gaps.** The full-KL gap satisfies an exact finite-state information-bottleneck decomposition over stationary lumpable quotients:

$$\Delta_{\mathrm{ret}}^{\mathrm{KL}}(M) = I(S;Z) - \max_{\substack{\phi \text{ lumpable} \\ |\mathcal{K}|\le M}} I(K_\phi;Z).$$

  Zero retention occurs exactly when M is at least the stationary support size of distinct predictive states. The Gaussian quadratic restricted gap satisfies a spectral lower bound with effective rank $M-1$. Full KL has a precise local Fisher asymptotic lower bound under regularity assumptions.

  The same analysis holds for general unifilar controlled machines: the controlled information-bottleneck decomposition conditional on the input, and fiberwise quadratic, probability-coordinate, and interior-Fisher spectral converses, with the input-driven theory as the output-independent specialization. The zero-retention threshold, by contrast, does not transfer verbatim: it is the index of the coarsest unifilar-lumpable refinement of the predictive-kernel partition, which can strictly exceed the number of distinct kernels (Theorem 5.42).

- **Grounding spectral converses are linear finite-rank.** The unrestricted linear finite-rank relaxation satisfies

$$\Delta_{\mathrm{grd}}^{\mathrm{unres}}(M) = \sigma_{M+1}(H_\nu)$$

  by Eckart–Young–Mirsky. The Hankel-structured finite-rank realization gap satisfies

$$\Delta_{\mathrm{grd}}^{\mathrm{Hank,str}}(M) \ge \sigma_{M+1}(H_\nu),$$

  with equality only under additional structural hypotheses, and finite Hankel truncations supply computable two-sided certificates for both quantities (Proposition 7.7). The structured problem has an exact zero threshold at $\mathrm{rank}(H) \le M$ (Proposition 7.8). For the symbolic deterministic (Mealy-machine) grounding gap, the exact input-only deterministic floor $F_\gamma(\nu)$ and its finite-state limit are now theorems (Theorem 7.29), with decidable finite-horizon certified intervals (Proposition 7.31); a closed-form value at finite M, and its relation to the linear relaxation, remain open (Open Problem 7.19). AAK is used only under separate Hardy-space embeddings.

- **Commitment is Boolean rank.** Exact commitment is governed by Myhill–Nerode index. The exponent correspondence identifies commitment with the Booleanized symmetrized $\alpha = 0$ vertex, not with raw D_0 and not with ordinary Schatten rank.

- **The spectral-tail picture is a heuristic.** Spectral tails arise from regime-specific response operators, effective rank budgets, and domination bridges. Aggregation alone does not determine the tail. No unconditional cross-regime ordering follows.

- **The Schatten template is conditional.** Mirsky's theorem unifies the analytic converse form once a Schatten domination bridge is supplied. Grounding is the $p = \infty$ instance for the unrestricted linear finite-rank relaxation; retention is the $p = 1$ instance with effective rank $M-1$; commitment is the Boolean-rank vertex.

- **The oracle bridge is adaptive and type-correct.** Offline approximation and online estimation compose into a bias–variance decomposition. Adaptive aggregation attains a penalized minimum without knowledge of the optimal budget. Spectral tails govern online rates only under explicit achievability assumptions.

- **The oracle floors are discharged in one regime only.** Assumption 9.24 is a minimax separation hypothesis. It is established unconditionally for the realizable persistent-stream regime by Proposition 9.32, where the packing is the gated family $\mathcal{G}_M^{\mathrm{act}}$ and $\Delta_M = M \log M$; there Theorem 9.34 is a theorem rather than a reduction. For the retention, grounding, and commitment regimes the floors are not established, and there the theorem remains a conditional reduction. The discharged case is realizable, so it certifies the estimation axis only; by Remark 9.25 the sum-form envelope follows from the minimum form through the discrete sandwich, and by Lemma 9.26 the resulting constant $(1+\kappa)^{-1}$ is absolute rather than instance-dependent. See Section 17.4.

- **The matching minimax lower bound is conditional on explicit floors.** An adaptive oracle upper bound and a conditional continuous bias–variance sandwich hold in general, and under the floors of Assumption 9.24 a matching minimax lower bound holds as well (Theorem 9.34); the envelope is therefore optimal up to the aggregation penalty $\Psi_M(T)$. The constant in the sum-form version is absolute, since Lemma 9.26 bounds the jump ratio by $2^\alpha(1+\log 2)^\beta$ for a regularly varying estimation envelope. The floors themselves are proved for the realizable persistent-stream regime (Proposition 9.32) and assumed elsewhere. Whether the penalty $\Psi_M(T)$ is necessary, and whether the floors hold in the stochastic regimes at horizons T of the order at which the envelopes cross, remain open.

- **The linear Price-of-Safety surrogate is exact but distinct.** Pinching gives an exact majorization law for a linear second-order, state-indexed, block-local surrogate. It does not compute the discrete right-congruence Price of Safety without relaxation-error control.

- **The exponent correspondence is a vertex correspondence.** Rényi order, Schatten exponent, and aggregation label share vertex labels $\{0, 1, \infty\}$, but they are distinct layers. The equality $\alpha = p$ is a formal consistency, not a derivation.

- **Temporal claims are protocol-specific.** For the reset-word protocol, realizable finite-automaton learning has $\Theta(M \log M)$ mistake complexity. The same $\Theta(M \log M)$ holds for a single *persistent* Mealy stream that is never reset: the upper bound is finite-class counting, and the matching lower bound is proved in Theorem 13.5 by an adversary that makes state navigation free and extracts $\Omega(M \log M)$ bits through forced-mistake readouts. Finite-class aggregation gives agnostic regret

  $$O\big(\sqrt{TM \log M}\big).$$

  For active learning, all bounds are stated against the residual-identification objective, an identification requirement being unavoidable in a protocol where the learner selects its own inputs. The sharp upper bound is

  $$\mathrm{Mistakes}_{\mathrm{active,RI}}(M) \le O(E_{\mathrm{sync}}(M)),$$

with the universal length form $O(M \log M + L_{\text{sync}}^{\text{univ}}(M))$ when the learner must fix its synchronization experiment in advance. On the full class $\mathcal{H}_M$ the complexity is settled outright,

$$\text{Mistakes}_{\text{active,RI}}(M) = \Theta(M \log M),$$

the upper bound by an active halving learner over the version space of machine-state pairs, the lower bound by the gated adversarial family. The $\Omega(M \log M)$ lower bound comes from that explicit construction, not from ordinary Littlestone dimension, which does not transfer to a protocol in which the learner chooses its own inputs.

A direct-sum instance forces its two components on disjoint rounds, giving

$$\text{Mistakes}_{\text{active,RI}}(M) \;\geq\; S_M + C_M,$$

but no non-degenerate two-term decomposition of the active complexity is available in the mistake currency: state identification in a known minimal skeleton costs only $O(\log M)$ mistakes (Proposition 13.19). Whether a genuine decomposition exists in the experiment-length currency is Open Problem 13.49.

- **NP-hardness is regime-qualified.** The Gaussian quadratic restricted retention decision problem is NP-complete already for minimal reset machines of synchronization depth one, and a tagged depth-two construction also gives NP-hardness. For unrestricted full-KL retention over a finite output alphabet, Theorem 5.58 gives NP-hardness under a promise and Corollary 5.59 upgrades this to APX-hardness, so no polynomial-time approximation scheme exists unless $\mathsf{P} = \mathsf{NP}$. Two qualifications attach to those two results, and both have been located precisely. The output alphabet produced by the embedding has size $2d$, so hardness for a *fixed* alphabet is not established; by Remark 5.64 the size can be reduced to $d+1$ exactly when the zero-sum subspace of $\mathbb{Q}^{d+1}$ admits an orthogonal rational basis of equal norms, which holds for $d = 3, 7, 15, \ldots$ and fails for $d = 2$, but it can never be taken below $d+1$, so the alphabet grows with the dimension under any variant of the construction. Membership in NP is open; by Proposition 5.61 the objective is $D^{-1} \log R$ for a factored rational R, so the question is the single comparison $R \geq e^{D\tau}$, decidable by Lindemann–Weierstrass but lacking a polynomial precision bound.

Each result above is stated under the hypotheses given in the corresponding section, and its type signature in the sense of Definition 16.1 fixes the regime, feasible set, aggregation, support, protocol, and analytical bridge to which it applies.

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Notation Index

The tables below collect the principal recurring symbols of the manuscript, grouped by layer, with a cross-reference to the site at which each is fixed. Symbols used only locally — the witness machines of the explicit constructions, the forcing alphabets $\{\mathtt{r}, \mathtt{e}, \mathtt{d}, \mathtt{c}\}$, the block witnesses, the game-theoretic state variables — are defined at their site of use and are not repeated here. Macro families indexed by a regime superscript r (as in A_T^{r}, $\mathsf{E}_M^{\mathsf{r}}$, $\mathcal{H}_M^{\mathsf{r}}$) share one meaning across regimes; only the base entry is listed.

Data and Code Availability

No empirical dataset was used. The results are theoretical, and the computational material reported here is of two kinds, held to different standards.

Machine-checked statements are formalized in Lean 4 against Mathlib and are verified by the compiler; these are stated as such in Remark 5.26.

Exhaustive enumerations, numerical evaluations and search results are *computational observations*. They are reported with the enumeration convention, the arithmetic used, and the resulting counts, and they illustrate or falsify statements rather than proving them; where a theorem depends on one, the dependence is stated in the theorem. The corresponding programs, the extremal machine tables, and the exact outputs accompany this manuscript as supplementary material: the package contains the numerical verification suite together

Table 3: Shared schema objects.

Symbol	Meaning	Fixed at
$\mathbf{V}$	Lawvere cost poset $([0,\infty], \geq, +, 0)$	2.1
$\mathbf{H}$	history $\mathbf{V}$-category on $\mathcal{I}^*$	2.3
$\widehat{\mathbf{H}}$	profinite completion of $\mathbf{H}$	2.17
H	history system $(H, h_\varnothing, \mathcal{A}, \cdot)$, pruned or total	2.6
$\mathbb{T}$	task theory $(\mathbf{H}, \mathbf{R}, \mathcal{E}, \delta, \mathrm{Agg}, \mathcal{L})$	2.2
$\mathcal{E}$	cost profunctor (reflexive; separated where stated)	2.2
$\sim_\delta$	Myhill–Nerode equivalence of the trajectory	3.2
$\mathrm{index}(\sim)$	number of classes of a right congruence	2.11
$\Delta_{\mathbb{T}}(M)$	budget-M variational gap of a task theory	2.13
$\mu_\gamma(u)$	discounted prefix law $(1-\gamma)\gamma^{\lvert u\rvert}\,\Pr[\text{prefix } u]$	2.4
$\kappa_{\mathrm{obs}}(\delta, \mu)$	observable support index	4.3
$\Sigma = \mathcal{I} \times \mathcal{O}$	joint (grounding) alphabet; $\lvert\Sigma\rvert$ alphabet size	3.3

with the exact outputs of every check reported above, the enumeration programs implementing the conventions of Remark 5.29, the machine tables of the extremal witnesses, and the Lean 4 development of Remark 5.26 itself: the seventeen machine-checked statements in their seven modules, together with the build script and the axiom-audit gate that recompiles the development and verifies every statement against Lean's standard three axioms.

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Competing Interests

The author declares no competing interests.

Acknowledgments and Declaration of Generative AI

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Table 4: Regime gaps and thresholds.

Symbol	**Meaning**	**Fixed at**
$\Delta_{\mathrm{com}}(M)$, $\Delta_{\mathrm{com}}(M)$	exact (worst-case Boolean) commitment gap	6.2
$\kappa_{\mathrm{det}}(F)$	Myhill–Nerode index of a specification F	6.1
$\Delta^{\mathrm{RD}}_{\mathrm{com}}(M)$	distributional rate–distortion commitment gap under μ	6.4
$\kappa_{\mathrm{pair}}(F, \mu)$	one-step determination index	6.5
$\Delta^{\mathrm{game}}_{\mathrm{com}}(M)$	simultaneous-move strategic commitment gap	6.5
$\kappa_{\mathrm{ctrl}}(\Lambda)$	controller complexity of a safety property	6.9
$\Delta^{\mathrm{KL}}_{\mathrm{ret}}(M)$	full-KL retention gap (input-driven)	3.30
$\Delta^{\mathrm{KL,ctrl}}_{\mathrm{ret}}(M)$, $\Delta^{\mathrm{KL,gen}}_{\mathrm{ret}}(M)$	controlled and fiberwise controlled gaps	5.31, 5.34
$\Delta^{\mathrm{KL},(r)}_{\mathrm{ret}}$	full-KL gap of the r-perturbed family	5.13
$\Delta^{\mathrm{quad}}_{\mathrm{ret}}(M)$, $\Delta^{\mathrm{quad,ctrl}}_{\mathrm{ret}}(M)$	quadratic restricted gaps (input-driven, controlled)	3.31
N^*	index of the stable kernel refinement (zero-retention threshold, unifilar)	5.40
$\Delta^{\mathrm{unres}}_{\mathrm{grd}}(M)$	unrestricted linear finite-rank Hankel relaxation gap	3.7
$\Delta^{\mathrm{Hank,str}}_{\mathrm{grd}}(M)$	Hankel-structured finite-rank gap	3.8
$\Delta^{\mathrm{lin}}_{\mathrm{grd}}(M)$	linear finite-rank residual-cost gap	7.4
$\Delta_{\mathrm{grd}}(M; \gamma)$	discounted symbolic grounding gap over Mealy machines	7.27
$\sigma_\gamma(\nu)$, $F_\gamma(\nu)$	stochasticity floor; observable deterministic floor	7.5, 7.29
$D(\phi)$	tracking deficit above the one-step floor	7.35
$\mathrm{PoS}^{\mathrm{loc}}_{\mathrm{lin}}(M)$, $\mathrm{PoS}_{\mathrm{quad}}(M)$, $\mathrm{PoS}(M)$	Price-of-Safety surrogate, discrete quadratic, mutual-information forms	10.7, 10.5, 12.12

Table 5: Divergences, operators, and spectral objects.

Symbol	Meaning	Fixed at
D_{KL}, D_α	Kullback–Leibler; classical Rényi order α	11.1
$\widetilde{D}_\alpha$	sandwiched (operator) Rényi divergence	11.2
$D_{\max}$, $D_{\max}^{\mathrm{sym}}$	max-divergence and symmetrization	7.20
H_ν	Hankel operator of the channel ν on $\ell^2(\Sigma^*)$	3.5
$\sigma_{M+1}(\cdot)$	$(M+1)$-st singular value (indexed from $\sigma_1 = \lVert\cdot\rVert_{\mathrm{op}}$)	7.9
$\Phi_r^{(p)}(A)$	Schatten-p tail beyond rank r	8
$\lVert\cdot\rVert_{\mathcal{S}_p}$	Schatten-p norm of a compact operator	8
A_δ, $r_{\mathbb{T}}(M)$, c_p	response operator, effective rank budget, domination modulus	8.9
$\mathbf{\Sigma}_\pi$	stationary Fisher (or feature) covariance	5.3
$\mathbf{\Sigma}_p$, $\mathbf{\Sigma}_\eta$	covariances of predictive probability vectors / natural parameters	5.16, 5.20
G	state-indexed centered Gram matrix	10.1
$\mathrm{Pinch}_{\mathcal{A}}$	pinching (block conditional expectation) $\mathrm{Pinch}_{\mathcal{A}}$	10.1
$\lVert X\rVert_{(r)}$	Ky Fan r-norm	10.1
N_Λ, $\mathrm{rank}_{\mathbb{B}}^{\det} N_\Lambda$	Boolean Hankel matrix and deterministic Boolean realization rank	8.19

Table 6: Temporal, online-learning, and strategic quantities.

Symbol	**Meaning**	**Fixed at**
$\mathcal{H}_M$	deterministic Mealy transductions with at most M states	13.1
Ldim	Littlestone dimension	13.1
$\mathrm{Mistakes}_{\mathrm{active,RI}}(M)$	active minimax mistakes to objective (RI)	13.15
$E_{\mathrm{sync}}(M)$, $E^{\mathrm{SI}}_{\mathrm{sync}}$	synchronization mistake complexities for objectives (RI), (SI)	13.18
$L^{\mathrm{adapt}}_{\mathrm{sync}}(M)$, $L^{\mathrm{univ}}_{\mathrm{sync}}(M)$	machine-specific and universal adaptive synchronization depths	13.17
$\mathcal{G}^{\mathrm{act}}_M$	gated active family (free/read modes f, r)	13.44
$V_{\mathrm{alt}}(q)$, $V_{\mathrm{sim}}(M)$	alternating-move value; simultaneous-move value at budget M	6.4
$\alpha(q)$	strategic spread at reward state q	6.5
$L^{\star,\mathsf{r}}_T$	unrestricted finite-state benchmark loss	9.1
$A^{\mathsf{r}}_T(M)$	budget-M approximation deficit	9.5
$\mathsf{E}^{\mathsf{r}}_M(T)$	online estimation rate for budget M	9.6
Reg_T	cumulative regret against the benchmark	9.7
$\Psi_M(T)$	adaptive aggregation penalty over the budget index	9.8